



GE Medical Systems

Technical Publications

Direction 5133324

Revision 2

Signa® LCC Pre–EXCITE Non–*TwinSpeed* to EXCITE HD *TwinSpeed* Upgrade

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003



GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
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1	Apr 25, 2005	Initial Release.
2	Nov 20, 2010	Updated Ferrous Material Warnings for Dock, Blower Box, Rear Pedestal, Tools, and other components throughout manual. Section 1: Added 200 gauss zone illustration and notice about safety training. Section 6: Updated Dock handling warning when removing Dock from room. Section 12-1: Added check for proper blower box anchoring. (CAPA 1859293).

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
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Direction 2128126	1*	Section 3		ENCLOSURE &		2164904-10	-		
A	2	POWER OFF		CABLES		10-1 to 10-3	1		
i	1	2164904-3	-	2164904-6	-			Section 11	
		3-1	1	6-1 to 6-11	1	Section 11		TAC CABINET	
Section 1				6-12	2	2164904-11	-		
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		4-45 to 4-68	1	6-24	2			Section 13	
Section 2		4-69	2	6-25 to 6-47	1	APPENDIX			
UPGRADE		4-70	1	6-48	2	2164904-13	-		
INSTALLATION				6-49 to 6-66	1	13-1 to 13-15	1		
PROCESS		Section 5		6-67	2	13-16	2		
2164904-2	-	PENETRATION PANEL		6-68 to 6-73	1	13-17 to 13-18	1		
2-1	1	2164904-5	-						
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* This revision number/letter corresponds to the indicated document's revision control system.

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SECTION 1 – GETTING STARTED

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1-1 UPGRADE INTRODUCTION

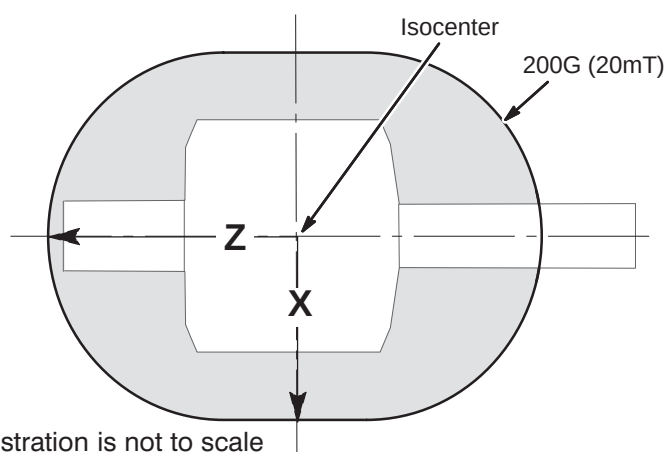
This publication documents the mechanical installation information for upgrading a Signa® 1.5T pre-EXCITE system with an LCC magnet and a WideOpen or Horizon style enclosure to Signa EXCITE HD 1.5T TwinSpeed system. Refer to illustration 1-1 for pictorial overview of upgraded system.

NOTICE:

During this hardware upgrade, the magnet may be ramped to full field strength. Individuals working in the magnet room must be MR safety trained.

WARNING!**FERROUS MATERIAL HAZARD!**

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE. THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.



Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
Unshielded Magnet	Measure ²	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 1.5T EXCITE system are listed in the *Signa 1.5T EXCITE HD Upgrade Pre-Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-3 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 1.5T EXCITE HD Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. The IIS key will need to be re-generated. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-7-1 MATERIAL POLICY OVERVIEW

1-7-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-7-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-7-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the EXCITE HD upgrade to GoldSeal which includes:

- All removed cabinets
- Penetration Panel
- SRI module
- PAC module
- All removed Magnet Enclosure parts
- All removed Rear Pedestal parts
- LPCC, Carriage, and/or removed Bridge and related parts
- Octane or Indigo or Operator Console, and Computer cabinet with all associated cables and modules

1-7-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-7-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-7-5 REGION SPECIFIC RETURN INFORMATION**1-7-5-1 Sites in Americas**

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

1-7-5-1 Sites in Americas (Continued)

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)-747-6997 or Dial Comm 8*579-6997

Fax: (414)-747-6855

Field ASPEN: 1-800-525-1516, Box #26041

1-7-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS-Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-7-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS-E – GOLD SEAL OPERATION & RECYCLING CENTER

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL

Quai 45 à 49 – 91090 LISSES – FRANC

Tel: (+33) 1-60-91-57-35 or (+33) 1 64 97 55 41

Fax: (+33) 1-60-86-12-31

For non-Gold Seal returns in Europe:

European Renewable Resources Center

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2

Quai 45 à 49 – 91090 LISSES – FRANCE

TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1-7-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1-8.

1-8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
[Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-10 UPGRADE SYSTEM CONFIGURATION

Illustration 1-1 shows the major equipment installed for a Signa 1.5T EXCITE HD TwinSpeed Upgrade.

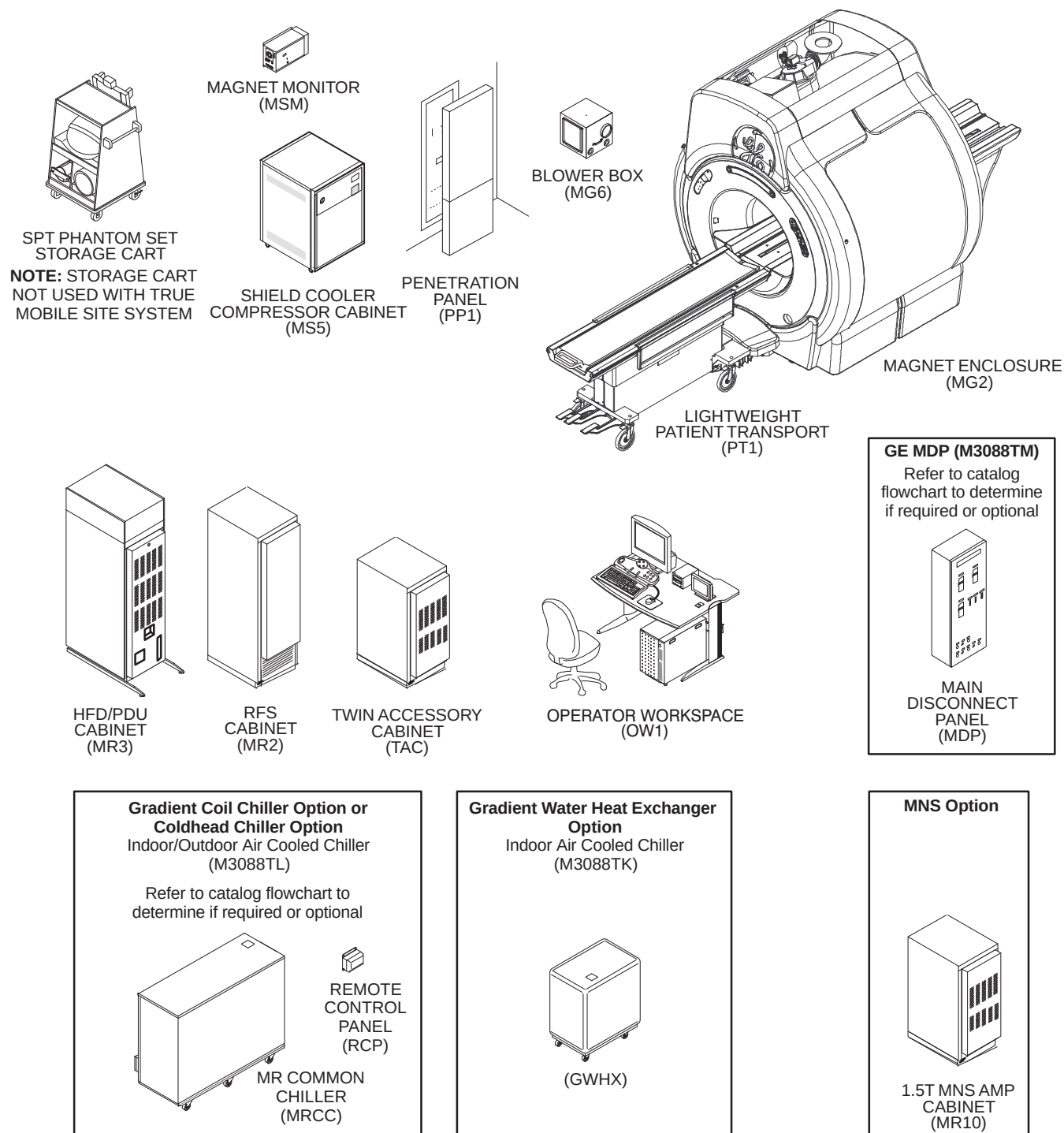


ILLUSTRATION 1-1
 LCC PRE-EXCITE 1.5T UPGRADED TO EXCITE HD TWIN SPEED SYSTEM MAJOR EQUIPMENT

SECTION 2 – UPGRADE INSTALLATION PROCESS

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2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

2-2 UPGRADE PROCEDURE

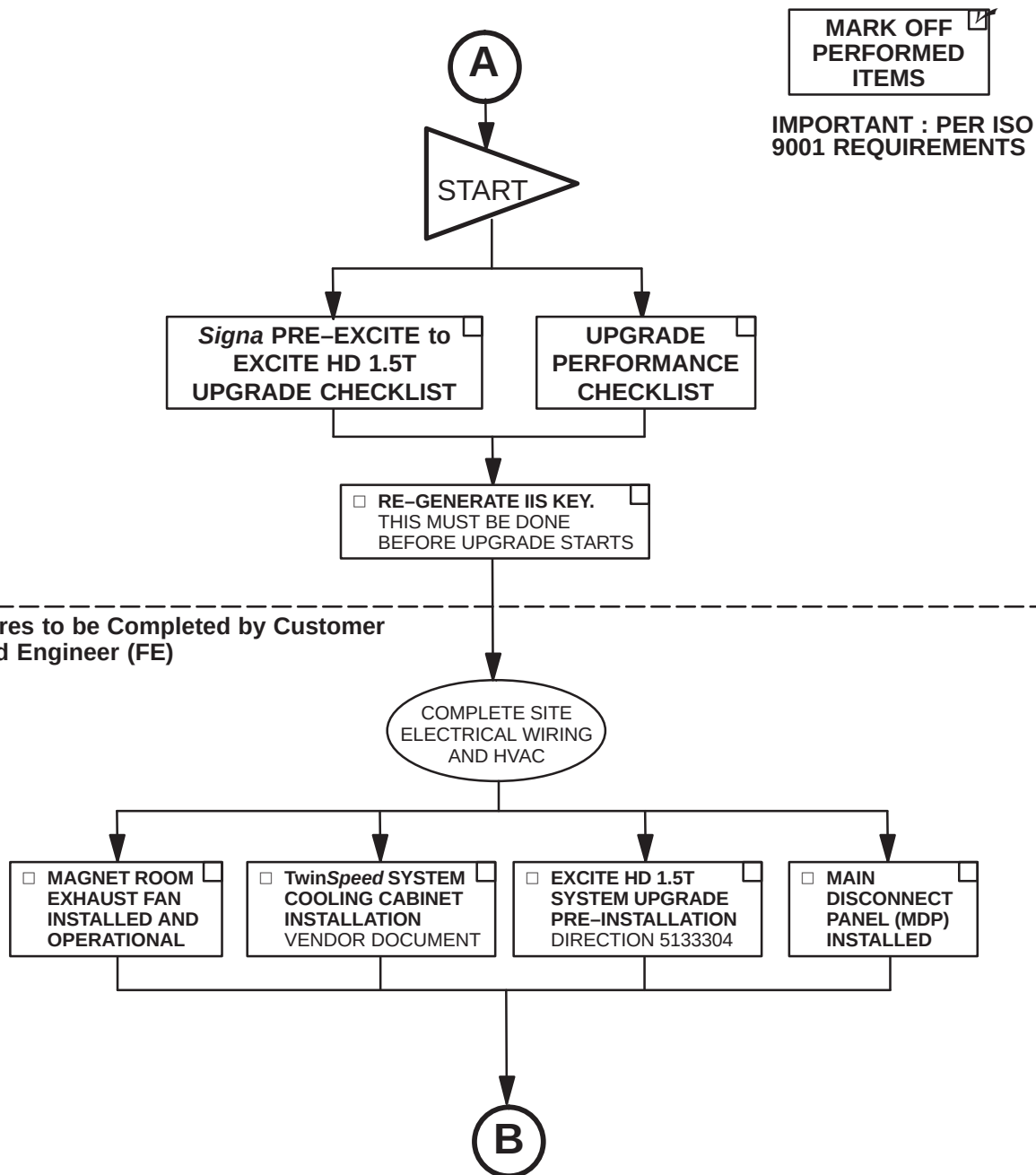
The following upgrade flow charts contain information for upgrading to Signa EXCITE HD 1.5T TwinSpeed.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa EXCITE HD equipment has been delivered together. **Make sure that every part required is available before starting.**

2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

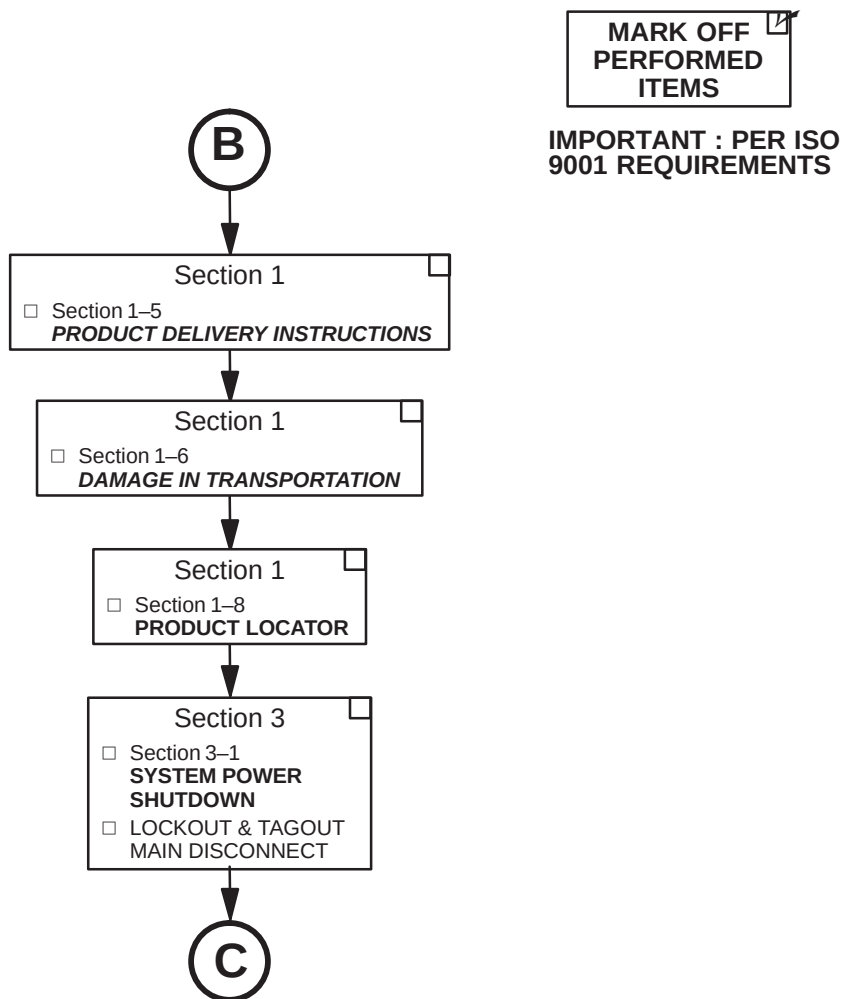
Procedures to be Completed by Field Engineer



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2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

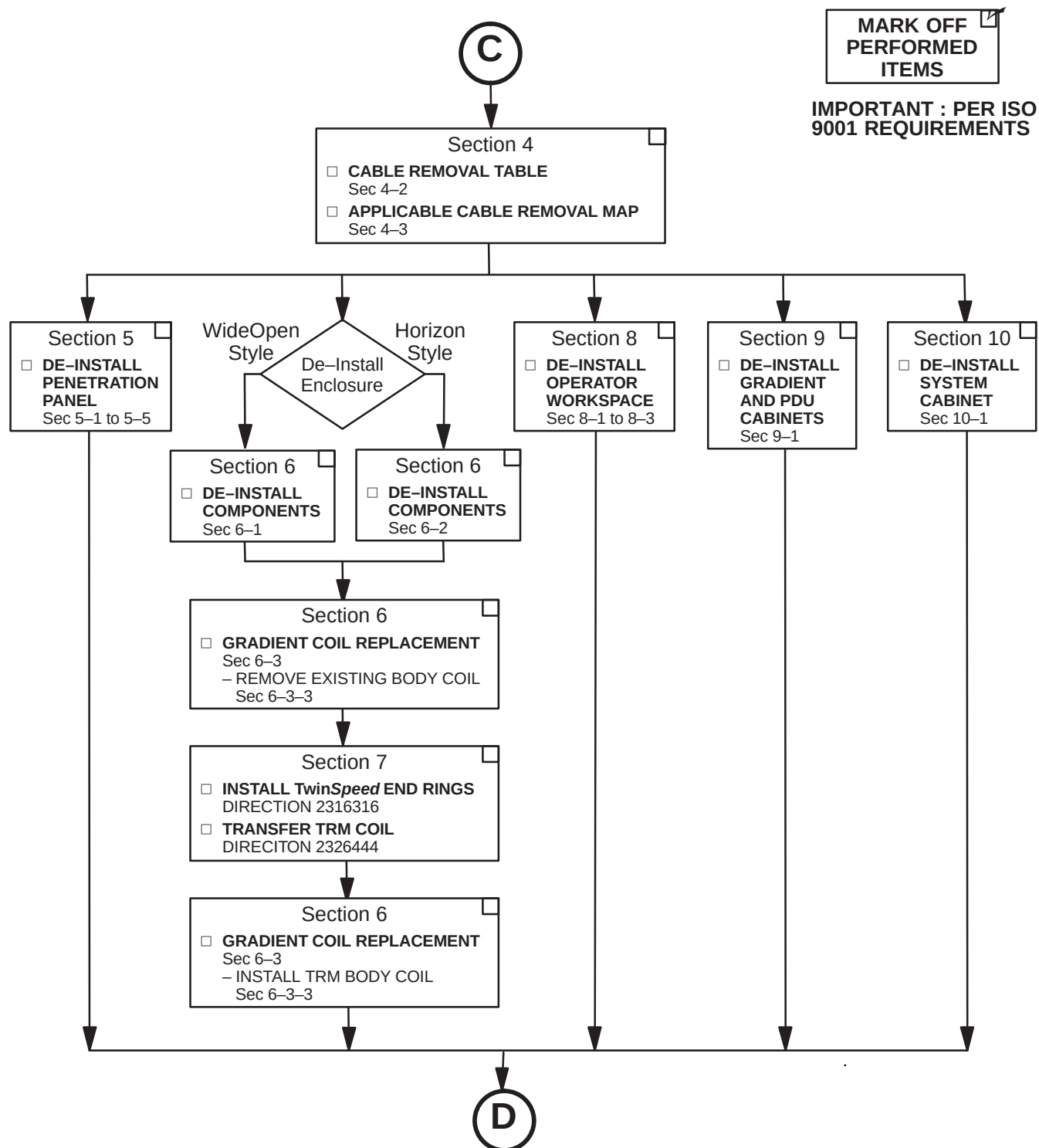
Procedures to be Completed by FE and/or Mechanical Installers



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2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

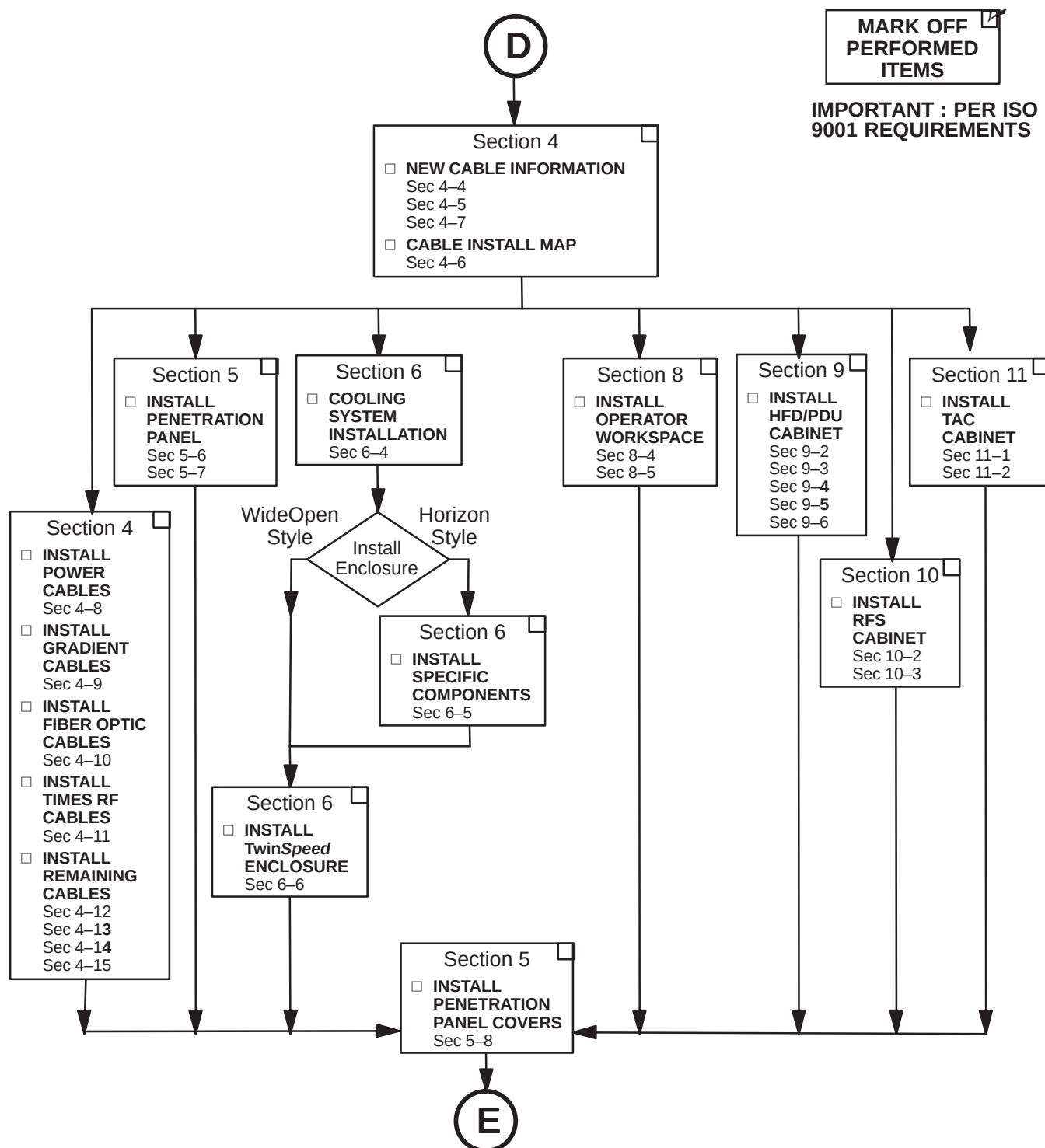
De-Installation Procedures to be Completed by FE and/or Mechanical Installers



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2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

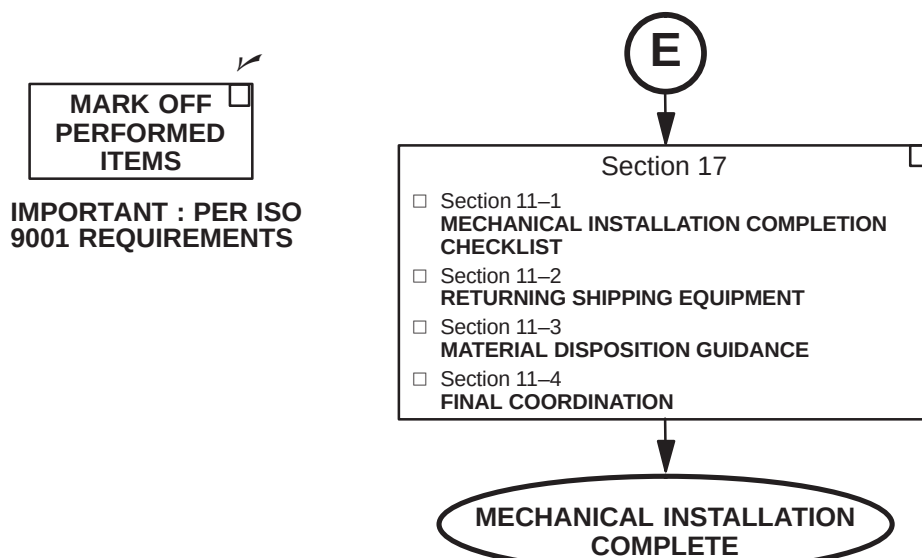
Installation Procedures to be Completed by FE and/or Mechanical Installers



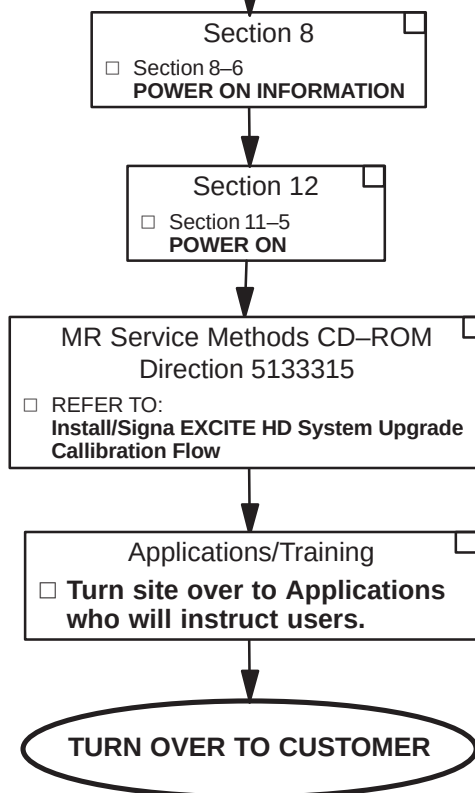
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2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



Procedures to be Completed by FE and Applications



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

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4-1 INTRODUCTION

This section provides information on removing existing cables and installing new cables for upgrading a Signa Pre-EXCITE system with an LCC magnet to an EXCITE HD 1.5T TwinSpeed system.

4-2 CABLE DE-INSTALLATION TABLE

Table 4-1 lists all cables to be removed from various WideOpen and Horizon Box enclosures (on LCC magnet) system configurations when upgrading to 1.5T EXCITE HD TwinSpeed. The letters on the right side of the table are defined by the following cable removal explanations:

- A** – Existing system is Signa WideOpen enclosure with ACGD/PDU
- B** – Existing system is Signa Horizon Box enclosure with ACGD/PDU
- C** – Existing system is Signa WideOpen enclosure with RF/PDU and SGD
- D** – Existing system is Signa Horizon Box enclosure with RF/PDU and SGD or Gradient/GRAM
- E** – Existing system is Signa WideOpen enclosure with pre-RF/PDU and SGD or Gradient/GRAM

If upgrading to 1.5T TwinSpeed with EXCITE Technology:

- F** – Additional cables to be removed if upgrading from Indigo Computer to 1.5T TwinSpeed with EXCITE
- G** – Additional cables to be removed if upgrading from Octane Computer to 1.5T TwinSpeed with EXCITE

Note

Discard means to remove and return to GE for recycling

4-2 CABLE DE-INSTALLATION TABLE (continued)

TABLE 4-1
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	COMMENTS
030	PDI-A2-TS1-16,17,18,NEUTGND	MR2-A12-J1 AC IN			X	X	X			
037	PD1-GND BUS	MR2-A11 GND	X	X	X	X	X			
038	PD1-GND BUS	MR1-A7 GND	X	X			X			
041	PD1-TS1-1,2,3,GND	MR3-CABI/FJ6			X	X	X			
042	PD1-TS1-4,5,6,GND	MR3-CABI/FJ7			X	X	X			
043	MR7-A4-J7	PD1-A2			X	X	X			SEE NOTE 3
044	PP1-GND	MR1-A7-GND	X	X	X	X	X			
045	PP1-GND	MG6-GND	X	X	X	X	X			
047	PD1-TS1-X,X,NEUTGND	OW1-A2-A5-J1						X		
048	PD1-GND BUS	OW1-A3-A5-GND						X	X	
049	OW1 A1 A1	OW1 A2 A5 J2						X	X	
065	PD1-TS1-26,27,GND	WC1			X	X	X			
229	MR2-A11-J1	MR1-A7-J3	X	X	X	X	X			
231	MR2-A11-J2	PP1-J8	X	X	X	X	X			
235	MR1-A7-J13	PP1-J44	X	X	X	X	X			
236	MR1-A7-J12	PP1-J4	X	X	X	X	X			
257	MG3-A3-J1	PP1-J44	X	X	X	X	X			
258	MG3-A17-J1	PP1-J4	X	X	X	X	X			
262	PP1-J8	MG3-A3-A5-OUTPUT	X	X	X	X	X			
282	MG3-A2-J8	PP1-J12	X	X	X	X	X			
296	PP1-J18	EO2								SEE NOTE 5
297	PP1-J18	EO1								SEE NOTE 5
300	PP1-J15	MG2-A15								
318	PP1-J11	MG3-A7-LONG DRIVE	X	X	X	X	X			
385	PP1-J20	MG2-A30-TS1,2,3								
387	MG2-A29-P1	PP1-J47	X	X	X	X	X			
404	MG3 A17 J1	MG2 A16 A5 J6	X	X	X	X	X			
457										SEE NOTE 2
458										SEE NOTE 2
485	MG3 A17 J2	MG2 A16 A3 OUT	X	X	X	X	X			
486	PP1-A14-J85	MR1-A7-J22	X	X	X	X	X			
487	PP1-A15-J78	MR1-A7-J20	X	X	X	X	X			
488	PP1-J80	MR2-A11-J8	X	X	X	X	X			
489	PP1-J81	MR2-A11-J9	X	X	X	X	X			
490	PP1-J82	MR2-A11-J10	X	X	X	X	X			
491	PP1-J80	MG3-A36-J7	X	X	X	X	X			
492	PP1-J81	MG3-A36-J8	X	X	X	X	X			
493	PP1-J82	MG3-A36-J9	X	X	X	X	X			
494	PP1-A14-J87	MG3-A36-J6	X	X	X	X	X			
495	PP1-A15-J78	MG3-A36-J10	X	X	X	X	X			
499	PP1-A14-J84	MR2-A11-J5	X	X	X	X	X			

4-2 CABLE DE-INSTALLATION TABLE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	COMMENTS
605	MR2-A11-J11	PP1-J10		X		X	X			
605	MS1-A3-A1-P403	PP1-J10		X		X	X			
700	PP1 A14 J86	MG2/MG3 A17 J2	X	X	X	X	X			
701	MR2-J11-J15	RF DOOR SWITCH								SEE NOTE 1
702	MR2-J11-J18	MR1-A7-J5	X	X	X	X	X			
703	MR2 A11 J23	PD1 A14 A1 J2	X	X	X	X	X			
706	MR2 A11 J14	PD1 A14 A1 J3 / MR1 PD1 J3	X	X	X	X	X			
708	MR2-A11-J16	MR1-A7-J27	X	X	X	X	X			
709	MR1-A7-J26	MR3-A9-J1	X	X	X	X	X			
710	MR2-A11-J17	MR3-A9-J2	X	X	X	X	X			
711/712	PP1-A17-J91	MR2-A11-J13	X	X	X	X	X			
711/712	PP1-A17-J91	MG2-A33-J11,J12,J13 MG2-A11-J9,J10								
715	PP1-J14	MG2-A33-J1								
716	PP1-J89	MG2-A11-J2								
726	PP1-A16-J77	MR1-A7-J18	X	X	X	X	X			
727	PP1-A16-J77	MG3-A36-J5	X	X	X	X	X			
728	MG3 A36 J11	MG2 A16 OUT J1	X	X	X	X	X			
729	MG3 A36 J12	MG2 A16 OUT J2	X	X	X	X	X			
730	MG3 A36 J13	MG2 A16 OUT J3	X	X	X	X	X			
731	MG3 A36 J14	MG2 A16 OUT J4	X	X	X	X	X			
732	MG3 A36 J15	MG2 A16 OUT J5	X	X	X	X	X			
733	MG3 A36 J16	MG2 A16 OUT J6	X	X	X	X	X			
734	MG3 A36 J17	MG2 A16 OUT J7	X	X	X	X	X			
735	MG3 A36 J18	MG2 A16 OUT J8	X	X	X	X	X			
736	MG3 A36 J19	MG2 A16 A8-J9	X	X	X	X	X			
743	MG2-A12-A2-J2	MG3-A3-J3	X	X	X	X	X			
744	MG2-A12-A2-J1	MG3-A3-J4	X	X	X	X	X			
749	MG2-A33-J10	MG2-A36	X	X	X	X	X			
753	MR7-A4-J1	MR3-A11-J5				X	X			SEE NOTE 3
754	MR7-A4-J2	MR3-A11-J4				X	X			SEE NOTE 3
755	MR7-A4-J3	MR3-A11-J3				X	X			SEE NOTE 3
756	MR7-A4-J4	MR3-A11-J8				X	X			SEE NOTE 3
757	MR7-A4-J5	MR3-A11-J7				X	X			SEE NOTE 3
758	MR7-A4-J6	MR3-A11-J6				X	X			SEE NOTE 3
759	MR3-11-J12	MR7-A4-J8				X	X			SEE NOTE 3
760	MR3-11-J11	MR7-A4-J9				X	X			SEE NOTE 3
761	MR3-11-J10	MR7-A4-J10				X	X			SEE NOTE 3
762	MR3CABI/FJ3	PP1-A7-1/2	X	X	X	X	X			
762	MG2-A12-A1-1/2	PP1-A7-1/2	X	X	X	X	X			
763	MR3CABI/FJ4	PP1-A7-3/4	X	X	X	X	X			
763	MG2-A12-A1-3/4	PP1-A7-3/4	X	X	X	X	X			

4-2 CABLE DE-INSTALLATION TABLE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	COMMENTS
764	MR3CABI/FJ5	PP1-A7-5/6	X	X	X	X	X			
764	MG2-A12-A1-5/6	PP1-A7-5/6	X	X	X	X	X			
768	MR1-A7-J31	PP1-J20	X	X	X	X	X			
769	MR1-A7-J34	PP1-J15	X	X	X	X	X			
770	MR1-A7-J41	PP1-J11	X	X	X	X	X			
771	MR1-A7-J36	PP1-J47	X	X	X	X	X			
772	MR1-A7-J40	PP1-J14	X	X	X	X	X			
773	MR1-A7-J35	PP1-A17-J1	X	X	X	X	X			
774	MR1-A7-J32	MR2-A11-J25	X	X	X	X	X			
775	MR1-A7-J42	PP1-A11-J92	X	X	X	X	X			
776	MR1-A7-J44	PP1-A11-J93	X	X	X	X	X			
777	MR1-A7-J43	PP1-A11-J94	X	X	X	X	X			
778	MG2-A12-A2-J3	PP1-A11-J72	X	X	X	X	X			
779	MG2-A12-A2-J4	PP1-A11-J73	X	X	X	X	X			
780	MG3-A3-J6	PP1-A11-J74	X	X	X	X	X			
781	MG2-A12-A2-J5	PP1-A11-J75	X	X	X	X	X			
782	MG2-A12-A2-J6	PP1-A11-J76	X	X	X	X	X			
783	MR1-A7-J37	PP1-F5,F6	X	X	X	X	X			
784	MG6-J1	PP1-F5,F6	X	X	X	X	X			
785	MR1-A7-J47	PP1-F7,F8	X	X	X	X	X			
786	MG6-J2	PP1-F7,F8	X	X	X	X	X			
788	PP1-J12	OW1-A1-A4-J7	X	X	X	X	X			
789	MR2-A11-J26	OW1-A1-A2-J18	X	X	X	X	X			
790	OW1-A2-A6-XO,RO,X1,R1	MR2-A11-J21	X	X	X	X	X			
791	MR2-A11-J30	OW1-A2-A4-EXP1	X	X	X	X	X			
792	OW1-A2-A1-MIC/AUDIO	OW1-A1-A4-J11,J13						X		
793	OW1-A2-A1-COM 1	OW1-A1-A4-J9						X		
794	OW1-A2-A3-MIC/AUDIO	OW1-A1-A4-J12,J14						X		
795	OW1-A2-A3-PWR	OW1-A1-A4-J17						X		
796	OW1-A2-A3-KBD/MS	OW1-A1-A4-J8						X		
797	OW1 A2 A1 PC	OW1 A3 LCD						X	X	
799	OW1-A15-MONITOR (OR 820)	OW1-A4 (B/W) (OR 820)						X		
805	OW1-A1-A4-J6	OW1-A7 (NON-SCIM KYBRD)						X	X	
807	OW1-A16	OW1-A5 (DASM)						X	X	
809	OW1-A15-MIC	OW1-A1-A4-J12,J14						X		
810	OW1-A15-MD,KYBD	OW1-A1-A4-J8						X		
811	OW1-A2-A4-PORT2	OW1-A15-ETHERNET						X		
812	OW1-A15-COM1	OW1-A13 (MODEM)						X		
813	OW1-A2-A1-COM2	OW1-A15-COM2						X		
814	OW1-A1-A4-J9	OW1-A2-A1-KBD,MOUSE						X		
815	OW1-A16-SCSI	OW1-A15-SCSI (PCI)						X		

4-2 CABLE DE-INSTALLATION TABLE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA 1.5T EXCITE HD

RUN#	FROM	TO	A	B	C	D	E	F	G	COMMENTS
816	OW1-A2-A6-J1	OW1-A15-BIT3						X	X	
817	OW1-A1-A2-J17	OW1-A2-A5-J3							X	
818	MR2-A11-J31	OW1-A16-PORT6,7,8							X	
819	OW1-A2-A7-SCSI	OW1-A15-SCSI							X	
820	OW1-A15-MONITOR	OW1-A4 (B/W)						X		
836	MR2-A11-J21	OW1-A15-BIT3	X	X	X	X	X			
848	MG2-A4-J5	MG2-A11-J4								
861	MG2-A4-J4	MG2-A33-J5,6,7,8,9	X	X	X	X	X			
862	MG2-A4-J1	MG2-A5-J1								
863	MG2-A4-J2	MG2-A5-J2								
897	MG2-A4-J3	MG2-A33-J4								
966	MR1-A7-J1	MR3-PD1-7,8,9,GND	X	X						
967	MR1-A7-CB1	MR3-PD1-10,11,12,GND	X	X						
968	MR2-A11-J1	MR3-PD1-4,5,6,GND	X	X						
969	OW1-A2-A5-GND	MR3-PD1-18,19,GND,N							X	
970	WC1	MR3-PD1-16,17,GND	X	X						
971	MR2-A11-J23	MR3-PD1-J2	X	X	X	X	X			
972	MR2-A11-J14	MR3-PD1-J3	X	X	X	X	X			
	MG3-A2-J7	MG2-A11-A1-MIC	X	X	X	X	X			SEE NOTE 4

Table 4-1 Notes:

1. A new RF Door Switch cable, Run 701, is provided for upgrade. If existing cable is routed within wall, it may be reused.
2. If optional Oxygen Monitor is installed on site, retain Runs 457 and 458 for connection to new Penetration Panel.
3. These cable Runs are removed only if GRAM Cabinet is part of existing site being deinstalled.
4. Existing microphone cable will be replaced with upgrade.
5. Existing Emergency Off Switch cables, Runs 296 and 297, may be retained and reused.

4-3 CABLE AND COMPONENT REMOVAL MAPS FOR UPGRADE TO TwinSpeed

Cables and components that are removed, relocated, and/or disconnected and reused when upgrading an existing Pre-EXCITE system with an LCC magnet that has a WideOpen or Horizon Box enclosure are identified on the following "Affected Cables for Upgrading to Signa 1.5T EXCITE HD TwinSpeed" interconnect maps.

The computer will be upgraded to Linux PC. Select one of the following for upgrading the Operator Workspace area.

<u>Description</u>	<u>Page</u>
--------------------	-------------

- Signa LX with Indigo Computer to 1.5T EXCITE HD TwinSpeed 4-9
- Signa LX with Octane Computer to 1.5T EXCITE HD TwinSpeed 4-8

For the remainder of the cables, cabinets, and modules being upgraded, select one of the following cable removal maps.

<u>Description</u>	<u>Page</u>
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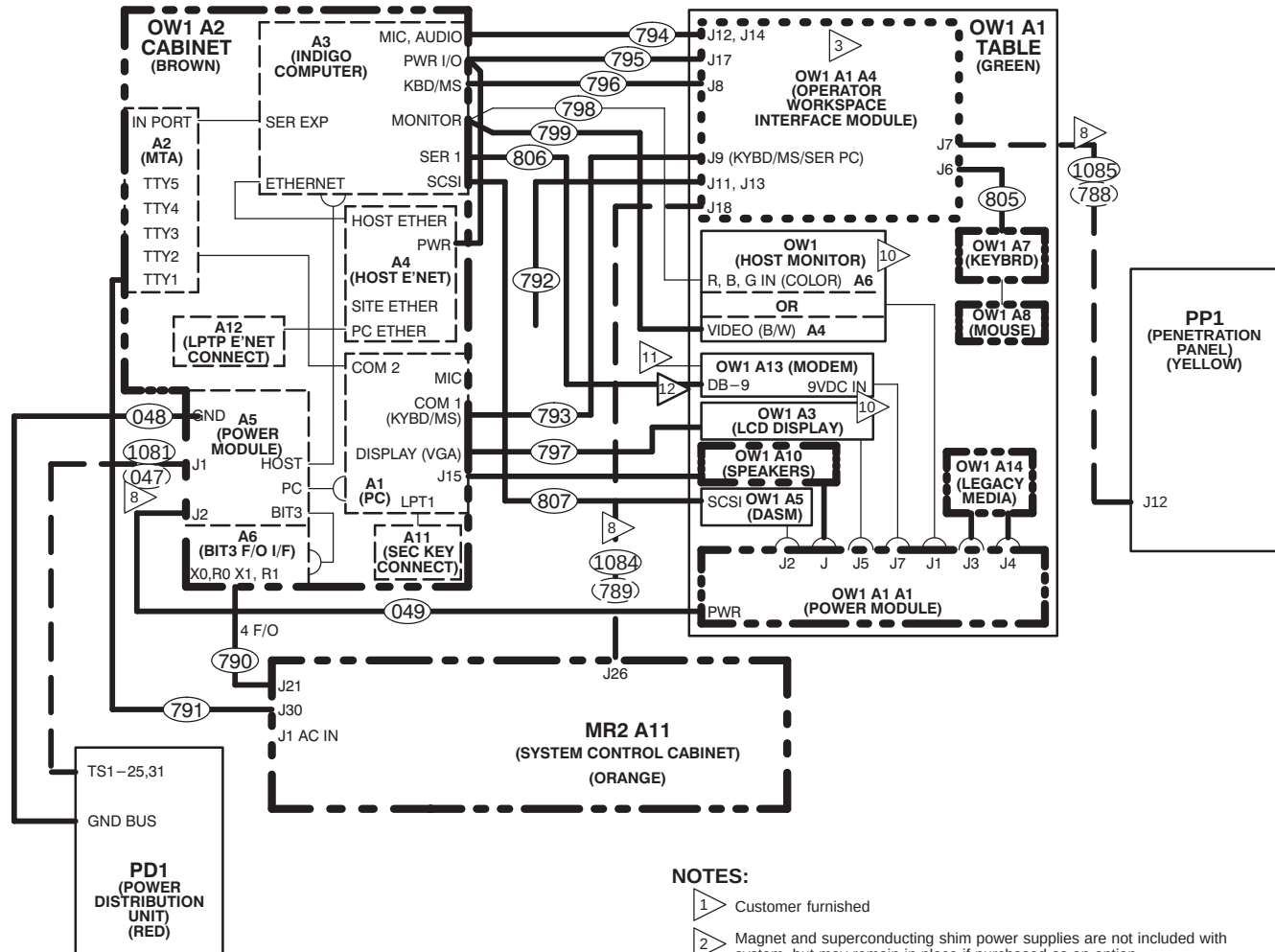
- Signa WideOpen enclosure with ACGD/PDU to Signa 1.5T EXCITE HD TwinSpeed 4-11
- Signa Horizon Box enclosure with ACGD/PDU to Signa 1.5T EXCITE HD TwinSpeed 4-12
- Signa WideOpen enclosure with RF/PDU and SGD to Signa 1.5T EXCITE HD TwinSpeed 4-13
- Signa Horizon Box enclosure with RF/PDU and SGD to Signa 1.5T EXCITE HD TwinSpeed 4-14
- Signa WideOpen enclosure with pre-RF/PDU and SGD to Signa 1.5T EXCITE HD TwinSpeed 4-15

For the convenience of the mechanical installer, a duplicate System "cable removal" map is provided. This will allow the install team to post one map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation when marking off cables that have been de-installed.

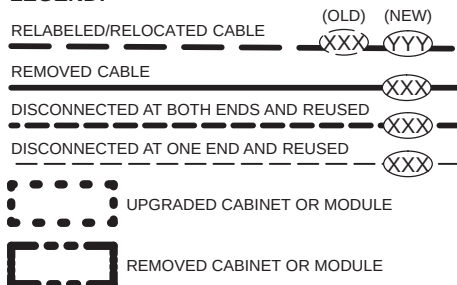
Duplicates of the above maps are on the following pages:

<u>Description</u>	<u>Page</u>
--------------------	-------------

- Signa WideOpen enclosure with ACGD/PDU to Signa 1.5T EXCITE HD TwinSpeed 4-16
- Signa Horizon Box enclosure with ACGD/PDU to Signa 1.5T EXCITE HD TwinSpeed 4-17
- Signa WideOpen enclosure with RF/PDU and SGD to Signa 1.5T EXCITE HD TwinSpeed 4-18
- Signa Horizon Box enclosure with RF/PDU and SGD to Signa 1.5T EXCITE HD TwinSpeed 4-19
- Signa WideOpen enclosure with pre-RF/PDU and SGD to Signa 1.5T EXCITE HD TwinSpeed 4-20

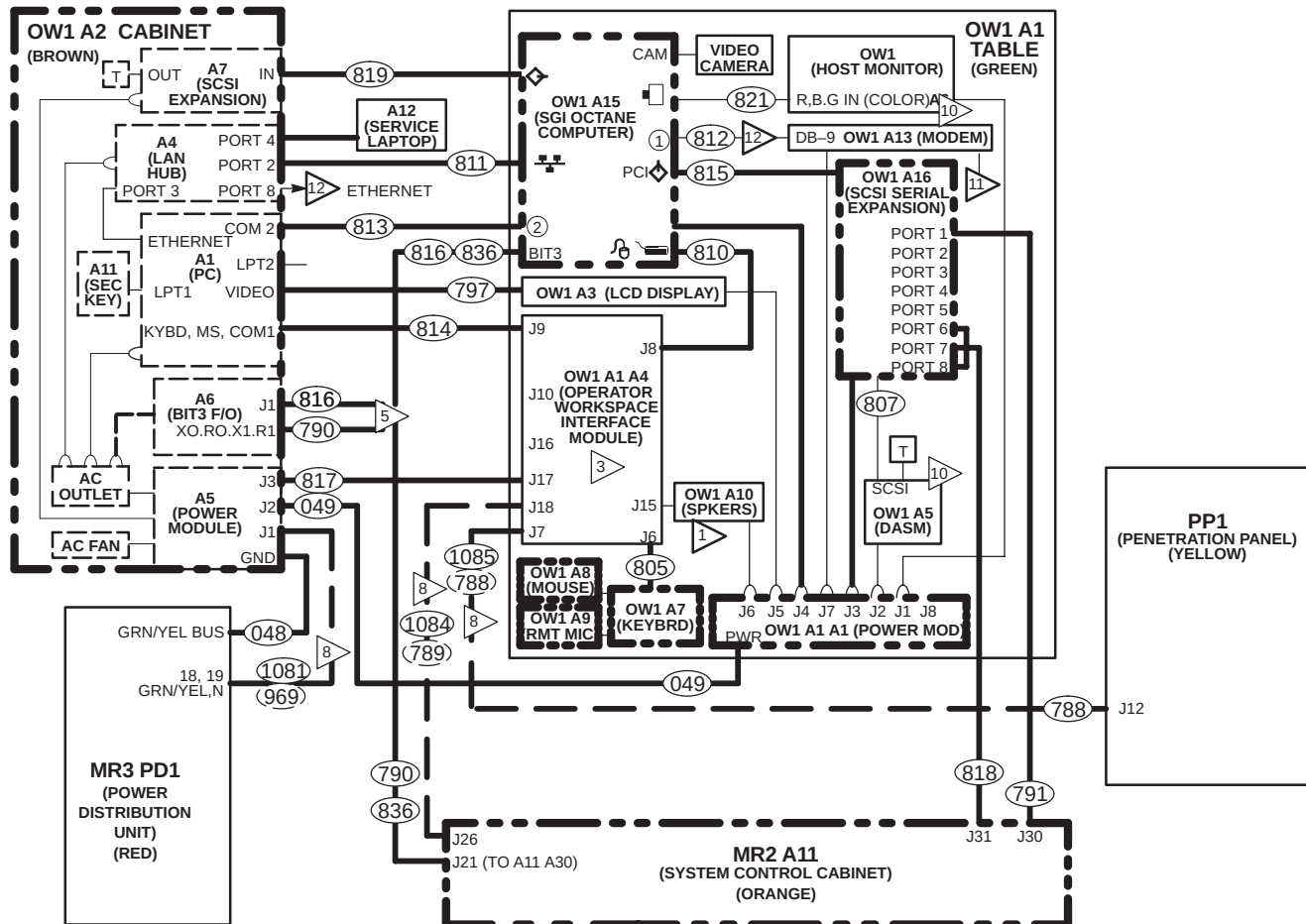
AFFECTED CABLES FOR UPGRADING A
SIGNA LX SYSTEM with Indigo Computer
to TwinSpeed EXCITE

LEGEND:



NOTES:

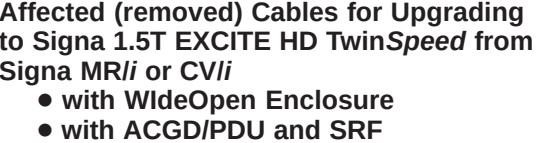
- 1 Customer furnished
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 For most LX upgrades, the WIM will be transferred to the new GOC Cabinet. Refer to Tab 13.
- 4 Typical connection for normally closed emergency off switch
- 5 Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. If so for this site, remove Runs 790 & 816 and disconnect Bit3 power cable from AC Outlet in Computer Cabinet.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 Existing Runs 708, 710 and 711/712 can be relabeled and reused if conduit is shortened at System Cabinet. Refer to Section NO TAG for instructions.
- 8 Refer to Section NO TAG for instructions on relabeling cables.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 DASM, Modem, LCD Display, and Color Monitor may be used with EXCITE. Refer to Site plans for verification.
- 11 Customer supplied telephone line for Insite Modem.
- 12 Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

AFFECTED CABLES FOR UPGRADING A
SIGNA LX SYSTEM with Octane Computer
to TwinSpeed EXCITE

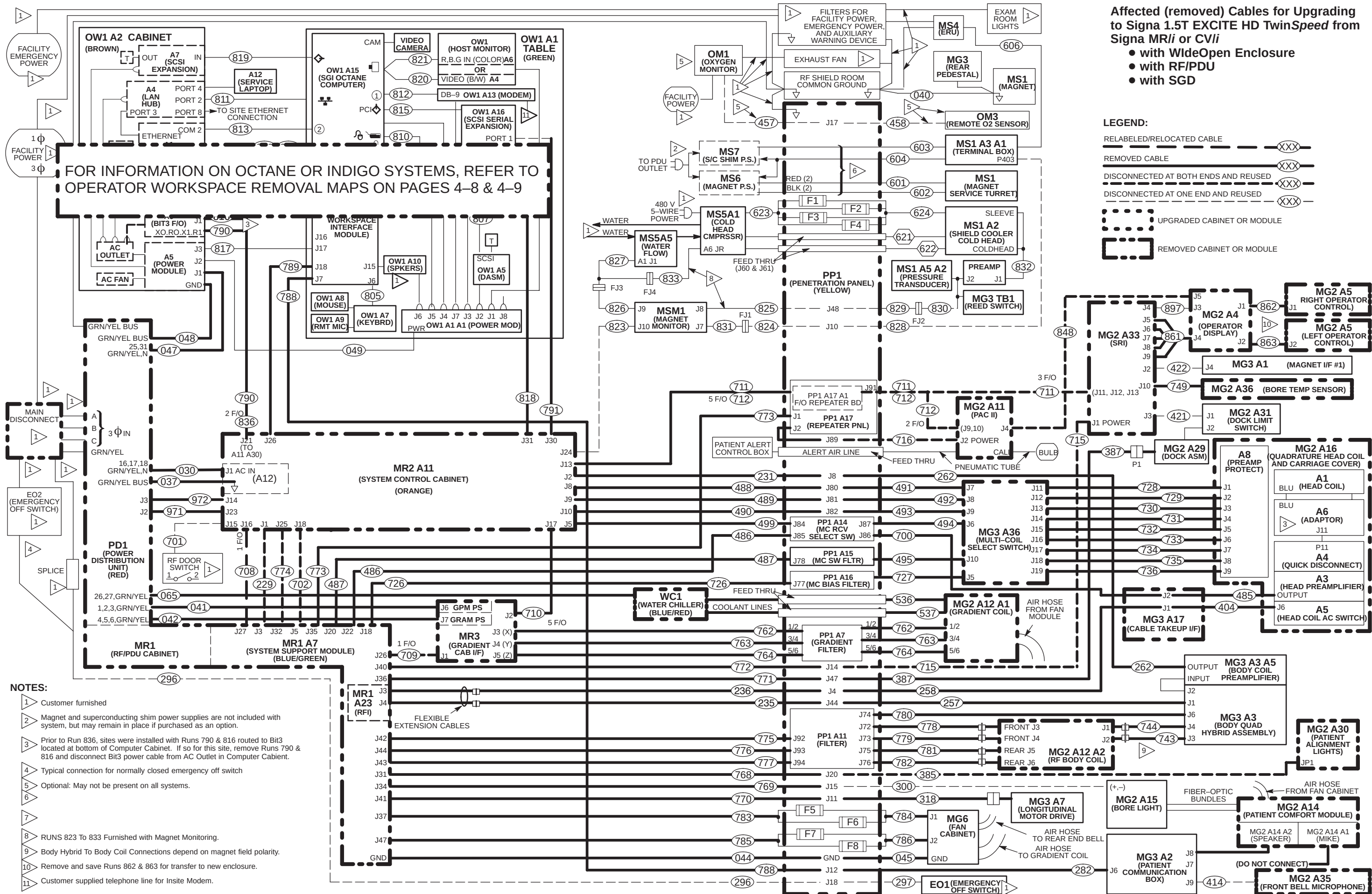
NOTES:

- 1 Customer furnished
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 For most LX upgrades, the WIM will be transferred to the new GOC Cabinet. Refer to Tab 13..
- 4 Typical connection for normally closed emergency off switch
- 5 Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. If so for this site, remove Runs 790 & 816 and disconnect Bit3 power cable from AC Outlet in Computer Cabinet.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 Existing Runs 708, 710 and 711/712 can be relabeled and reused if conduit is shortened at System Cabinet. Refer to Section NO TAG for instructions.
- 8 Refer to Section NO TAG for instructions on relabeling cables.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 DASM, Modem, LCD Display, and Color Monitor may be used with EXCITE. Refer to Site plans for verification.
- 11 Customer supplied telephone line for Insite Modem.
- 12 Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

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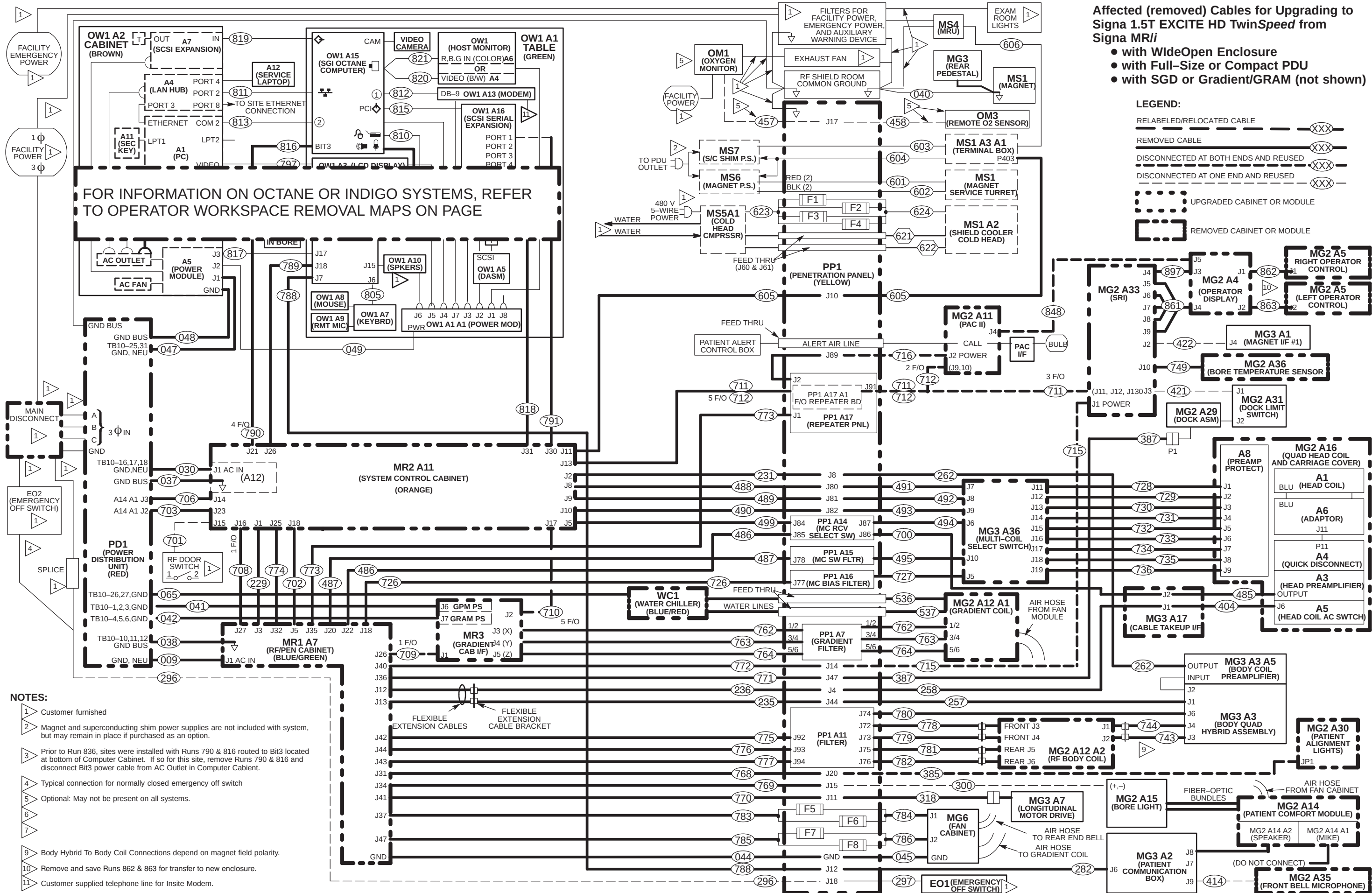


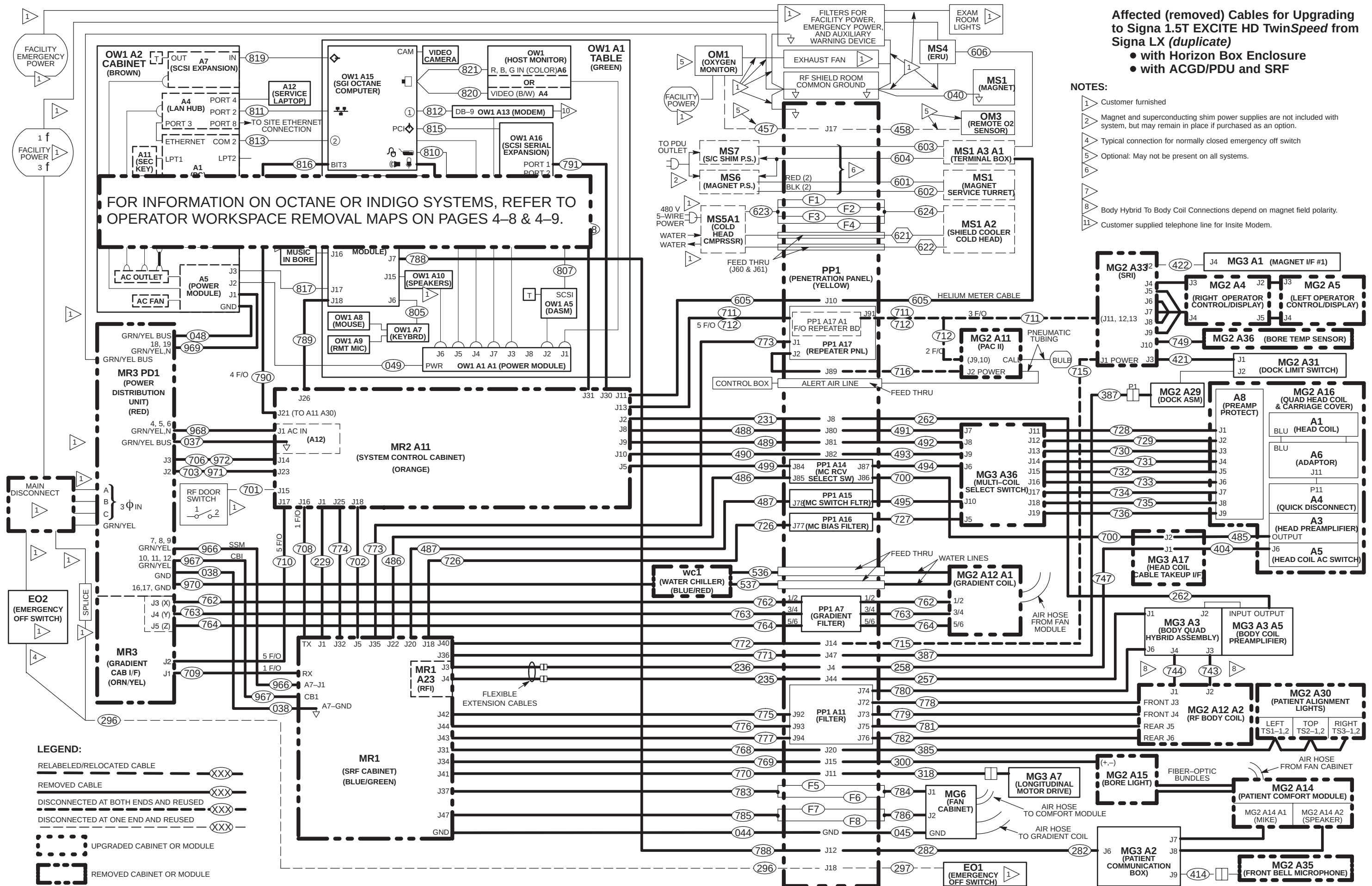
- 1 Customer furnished
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. If so for this site, remove Runs 790 & 816 and disconnect Bit3 power cable from AC Outlet in Computer Cabinet.
- 4 Typical connection for normally closed emergency off switch
- 5 Optional: May not be present on all systems.
- 6
- 7
- 8 RUNS 823 To 833 Furnished with Magnet Monitoring.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 Remove and save Runs 862 & 863 for transfer to new enclosure.
- 11 Customer supplied telephone line for Insite Modem.

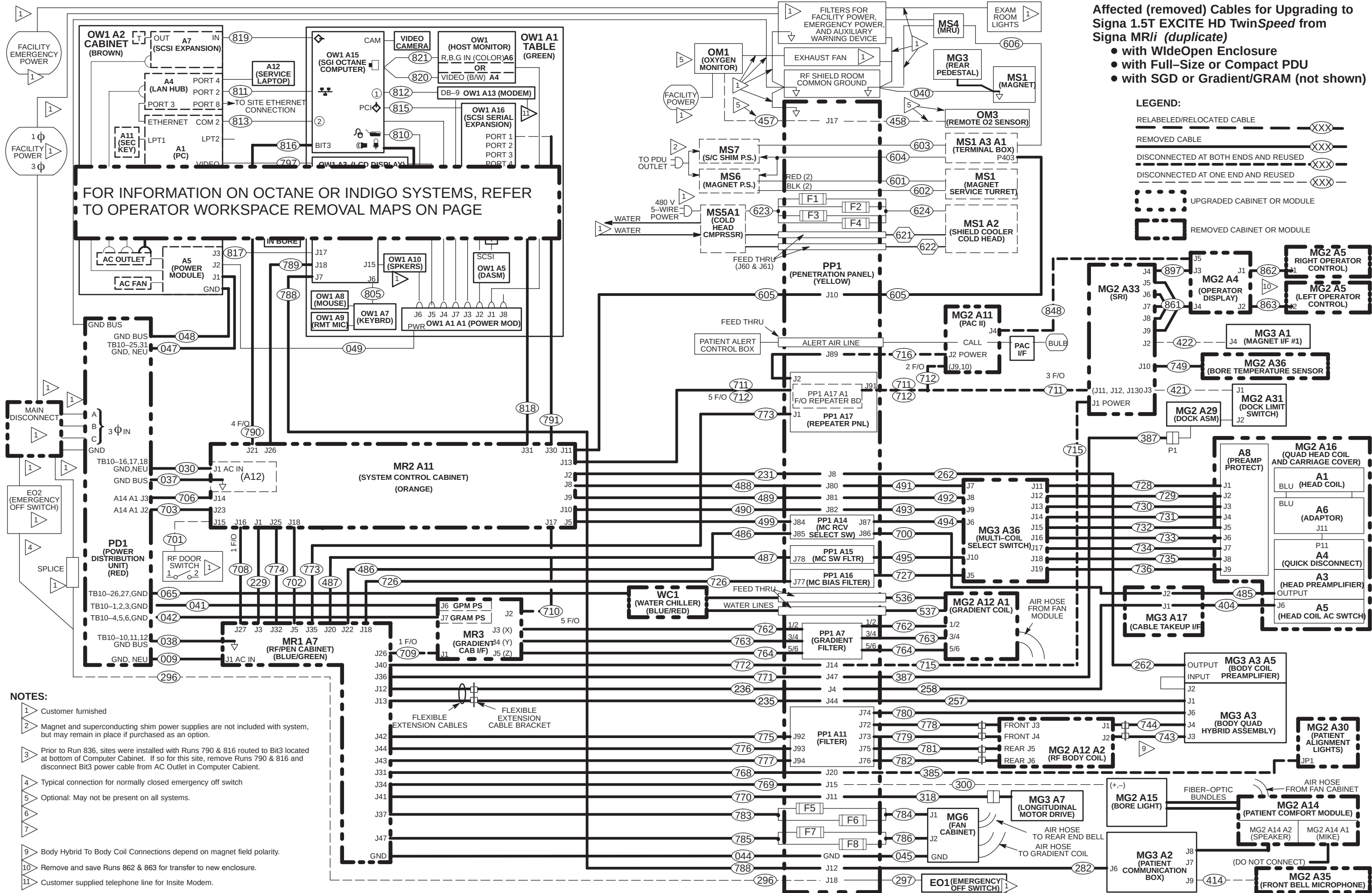


Affected (removed) Cables for Upgrading to Signa 1.5T EXCITE HD TwinSpeed from Signa MR/i or CV/i

- with WideOpen Enclosure
- with RF/PDU
- with SGD







4-4 NEW CABLES IDENTIFICATION

The information in this section assumes that Signa 1.5T TwinSpeed upgrade components, appropriate Site Kit, and options are delivered complete with all cables present.

Cable shipment containers have color coded labels which identify the cable routing areas:

- Green – Equipment Room
- Purple – Operator’s Room
- Orange – Magnet Room
- Yellow – Post-installation (Phantom’s, Coils, etc.)

Refer to Section 13, Appendix, for connection designator information and specifications on all cables to be installed for this upgrade.

4-5 CABLE INTERCONNECT DOCUMENTATION

The cable maps illustrate the Signa EXCITE HD 1.5T TwinSpeed system interconnect for all supplied and existing cables and the interface with customer supplied wiring.

- Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.
Instructions for connections for optional laser cameras are supplied with option.
Connection procedures for other options are covered in the installation manual shipped with the option.
Spectroscopy option cabling can be found on page 4-29.
- Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
- Interconnect details for magnet subsystem are covered in the applicable magnet subsystem manual.

4-6 INTERCONNECT MAPS FOR UPGRADED SYSTEM

For the convenience of the mechanical installers, a duplicate System Interconnect is provided. This will allow the install team to post one interconnect map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation when marking off cables when their connections are complete.

Fold-out EXCITE HD 1.5T TwinSpeed System Interconnect Cable Map is provided on the following pages:

Description	Page
Signa EXCITE HD 1.5T TwinSpeed System Interconnect (Fixed Site)	4-23
Signa EXCITE HD 1.5T TwinSpeed System Interconnect (Fixed Site) (DUPLICATE)	4-25
Signa EXCITE HD 1.5T TwinSpeed System Cooling Cabinet Interconnect Options	4-27
Signa EXCITE HD 1.5T TwinSpeed System Interconnect for 2KW MNS Option	4-29

Tape maps in a convenient location in the Magnet Room and the Equipment/Operator Room to check off routed cables. Additional information on each cable is located in the Appendix. Upon completion of connecting cables, return fold-out maps to the binder for future reference.

4-7 SORTING AND ROUTING CABLES



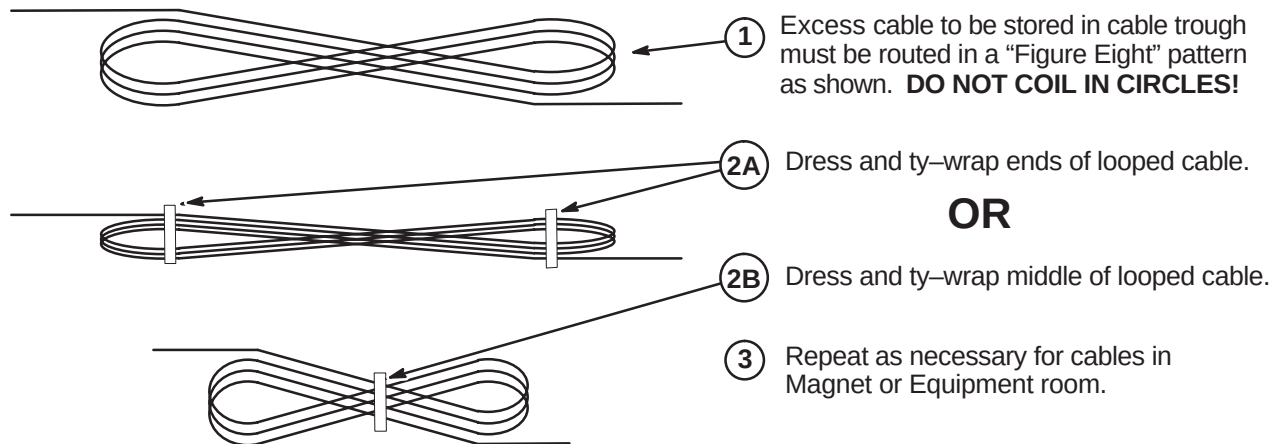
Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

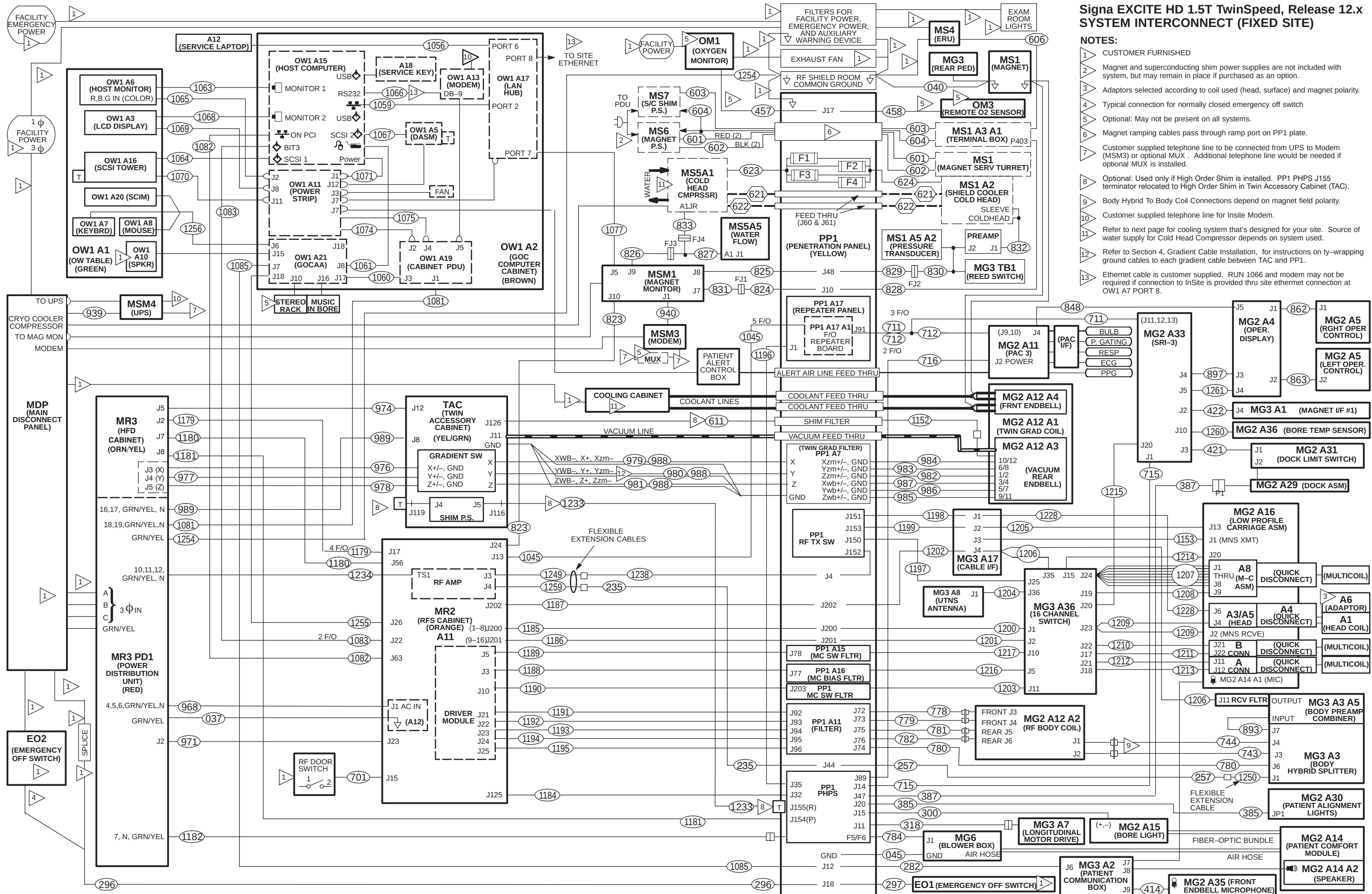
2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4-1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.



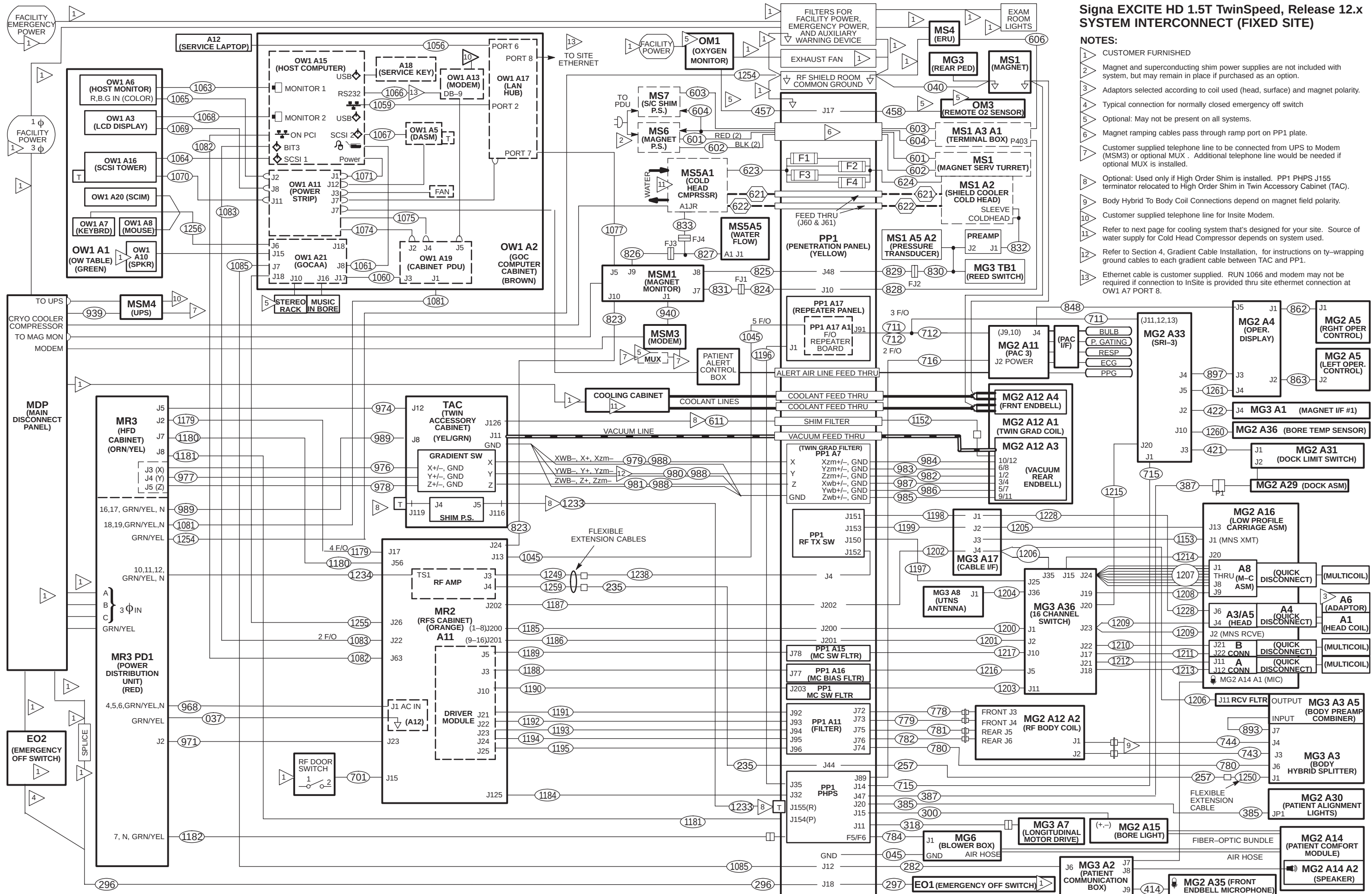
PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 4-1

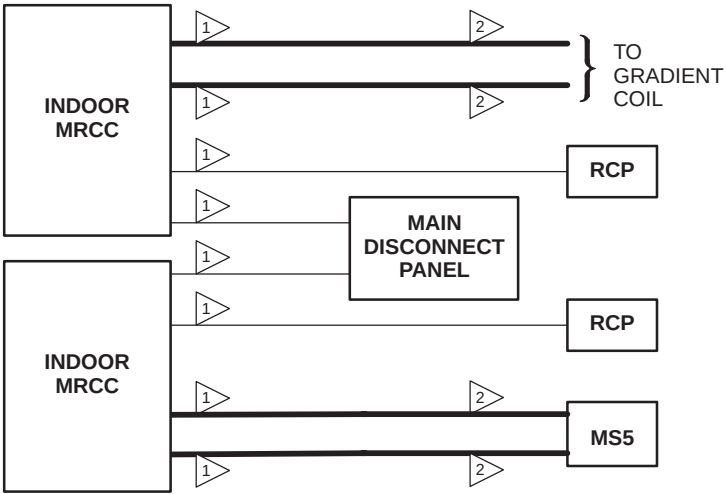
4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.



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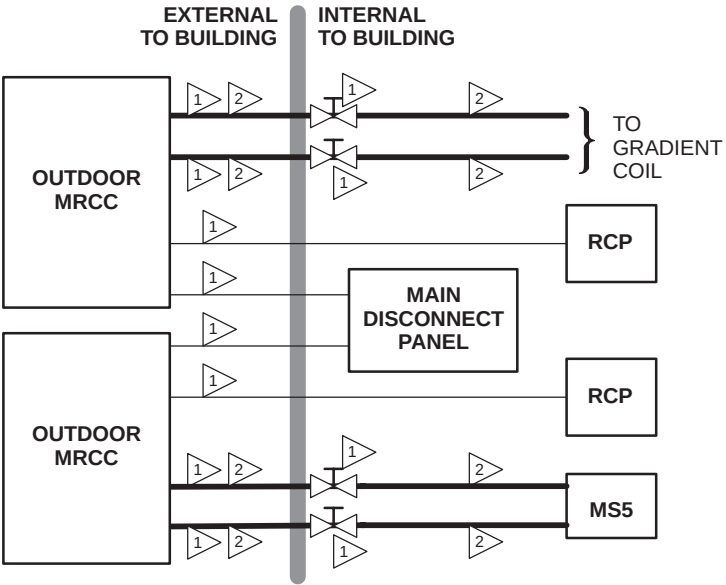


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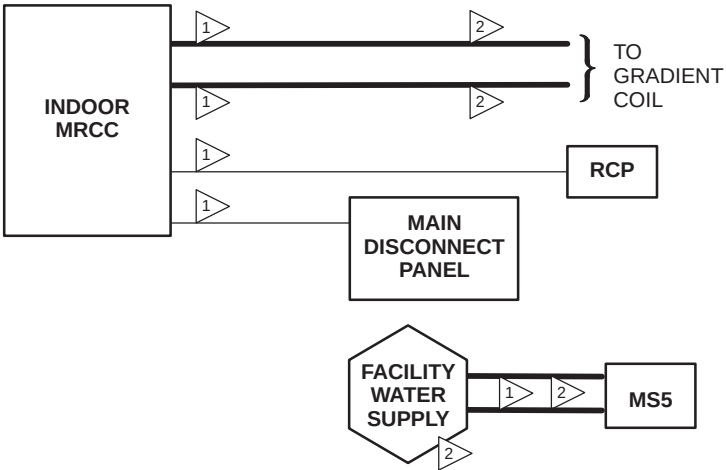
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



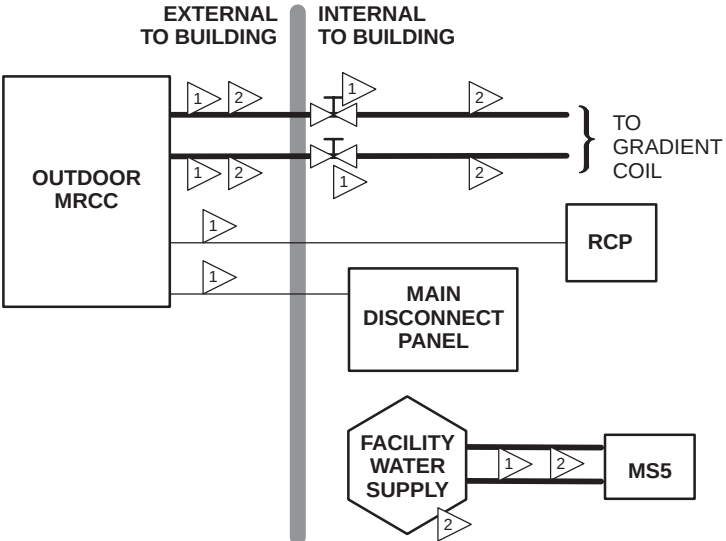
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



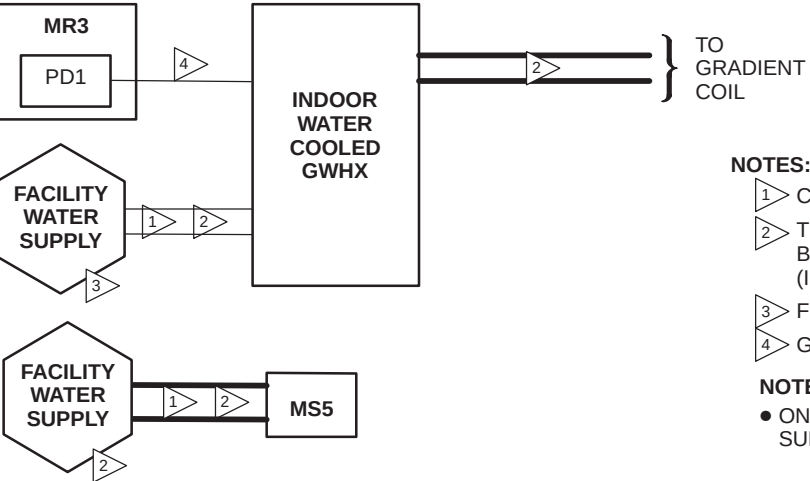
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM

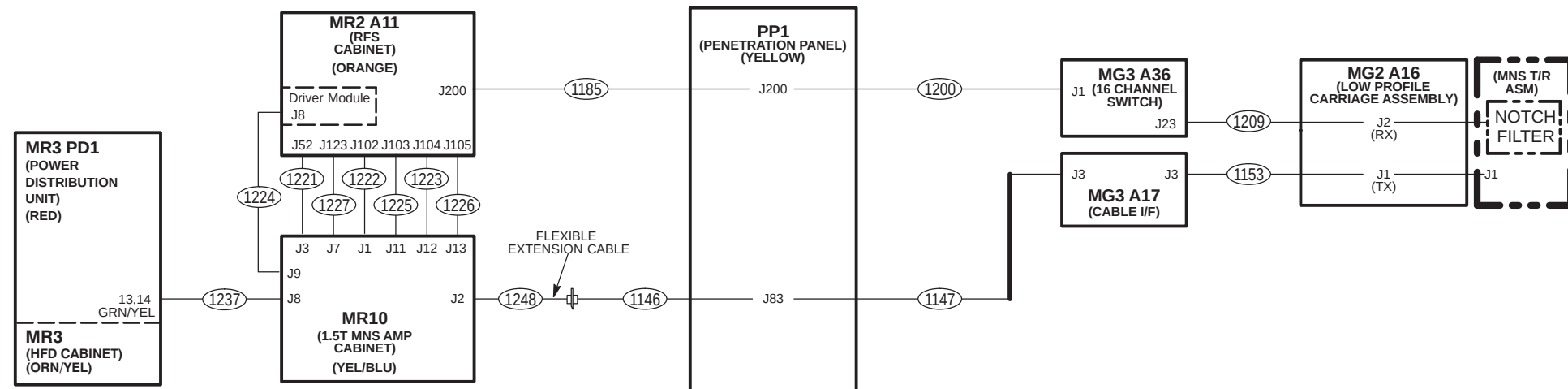


- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
 - 3 FOR INDOOR WATER COOLED GWHX ONLY.
 - 4 GWHX PROVIDED CABLE.
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO GWHX SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR GWHX SUBSYSTEM GROUP INTERCONNECT DIAGRAM

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**Signal EXCITE 1.5T TwinSpeed, Release 12.0: 2KW MNS OPTION
SYSTEM INTERCONNECT (FIXED SITE)**



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4-8 POWER CABLE INSTALLATION

The magnet enclosure and magnet room cables must be installed simultaneously. Cable locations depend on how enclosures are installed.

4-8-1 HFD CABINET CABLE CONNECTIONS

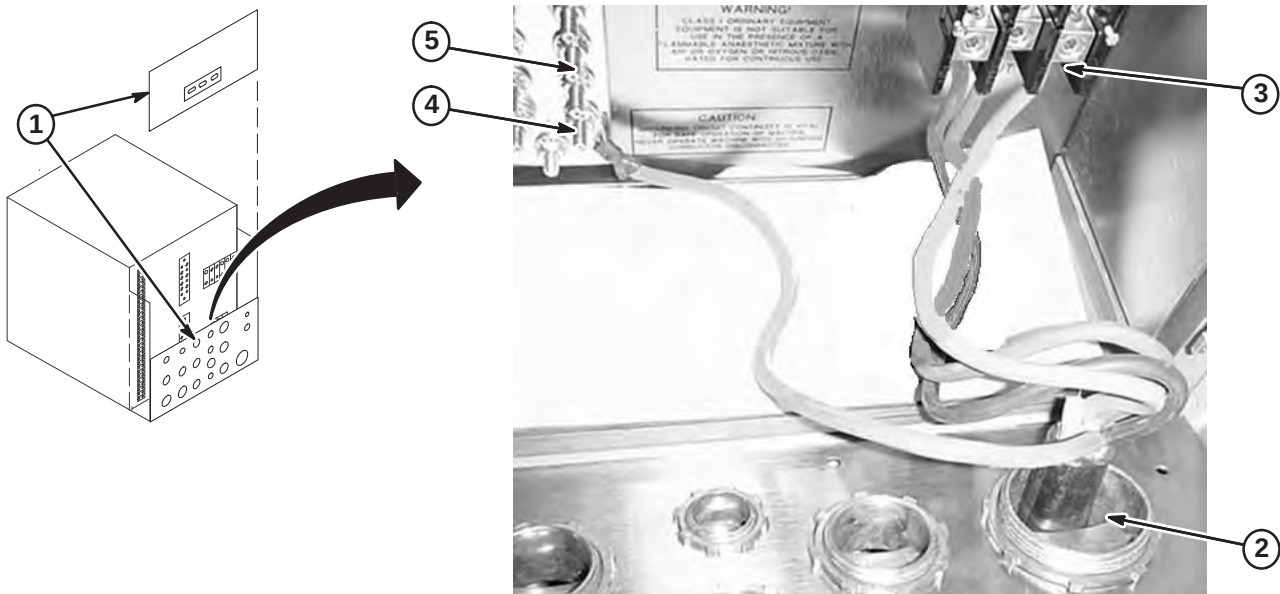
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE HFD CABINET.

4-8-1-1 HFD Cabinet Power and Ground Cable Installation

1. Remove rear PDU module covers.
2. Route facility input power cable and ground wire thru interface panel.
3. Connect L1-L3 conductors to terminal block.
4. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.
5. Connect RF screen room ground conductor to Ground Bus Bar.

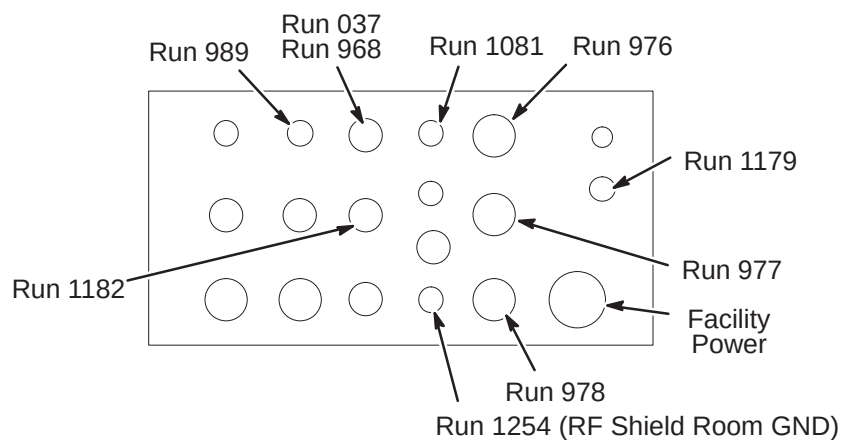


4-8-1-2 Route Power Cables Thru Interface Panel

1. Route Runs thru interface panel as marked.

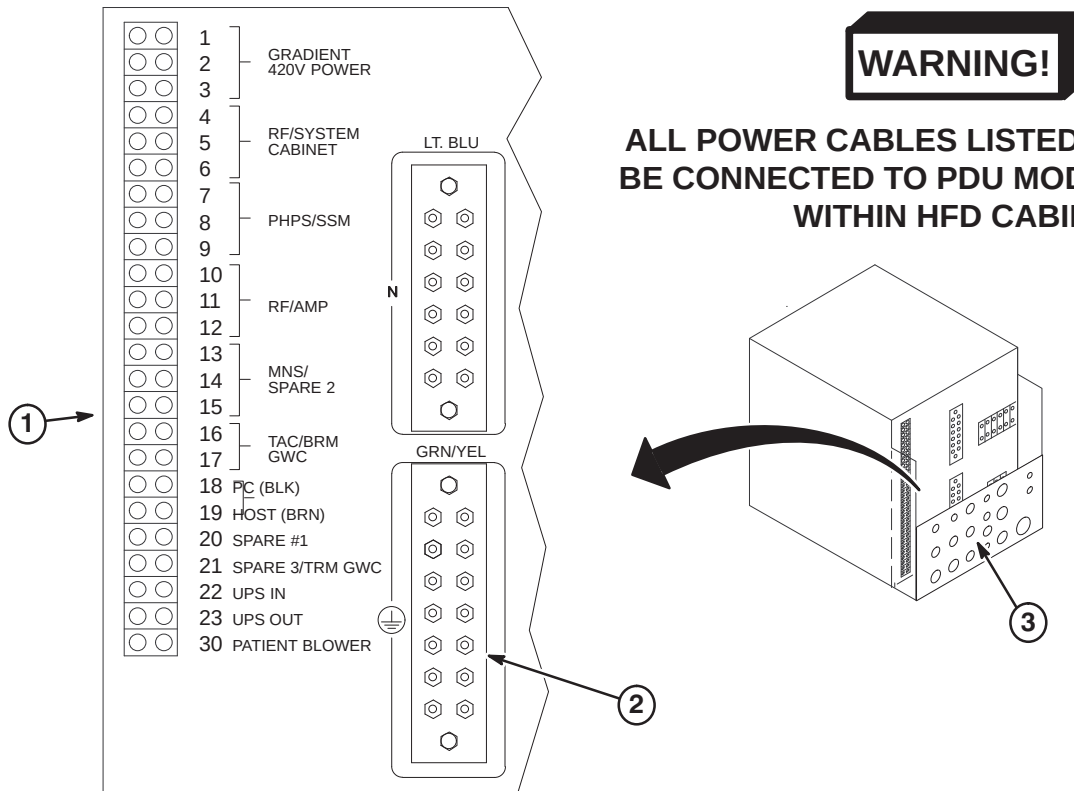
Note: The GRN/YEL ground wire, Run 037, **MUST** be routed with Run 968 thru the same port.

2. Route any option Runs as marked.



4-8-1-3 Connect Power Cables to PDU Module

1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below. Relable PDU connection panel as shown below.
2. Trim and connect ground wire cables (Runs 037 and 1254) to ground bus connectors.
3. Tighten clamps of interface panel to secure cables in place.



CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	ϕ A, B, C	RFS Cabinet
7	1182	ϕ A	PHPS Module
10,11,12	1113	ϕ A, B, C	NB RF Amp Cabinet
16,17	989	ϕ A,C	TAC Cabinet
18,19	1081	ϕ C,A	Operator Workspace
(MDP)* or 30 (PDU)		ϕ A or ϕ A	AirSys Water Cooled Cabinet or AirSys Heat Exchanger
GRN/YEL GRN/YEL	037 1254		System Cabinet Ground Wire RF Shield Room Ground (Customer Supplied)

COLOR CODE: for remainder of cables:
 3 ϕ = Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
 Single ϕ = Brown, Black

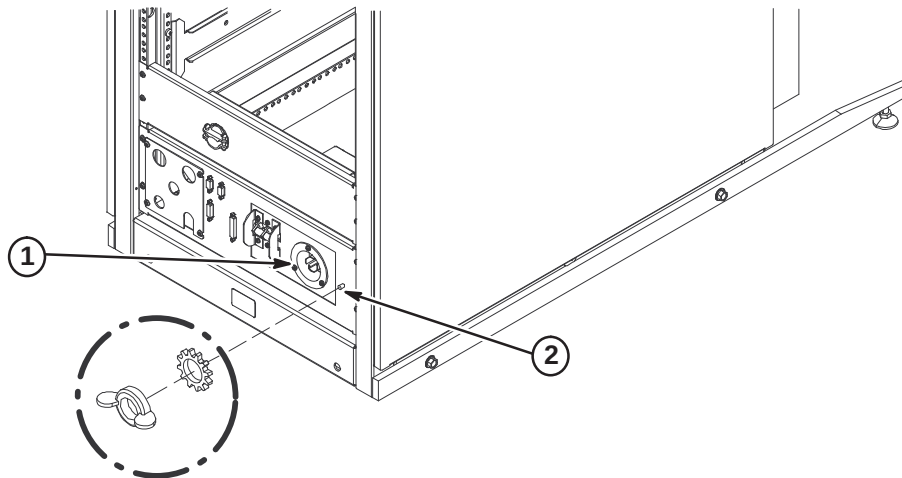
* Connection to Main Disconnect Panel to be completed by customer electrician.

4-8-2 EQUIPMENT ROOM CABINET POWER CABLE CONNECTIONS

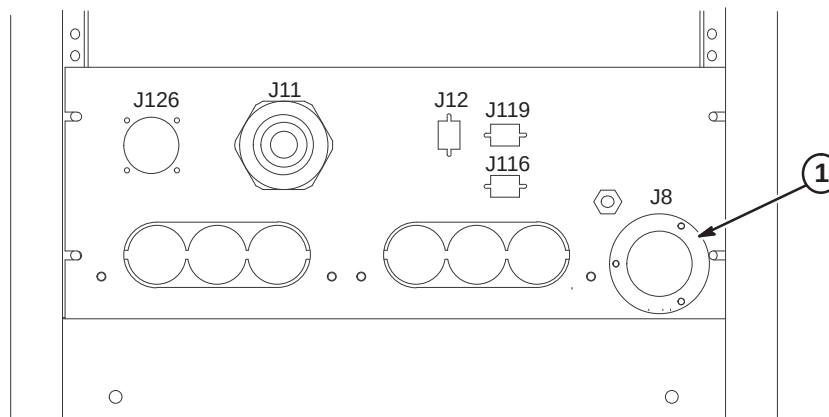
**CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU
MODULE LOCATED WITHIN HFD CABINET.**

4-8-2-1 RF/System Cabinet Power and Ground Cable Connections

1. Connect Run 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.

**4-8-2-2 TAC Cabinet Power Cable Connection**

1. Connect Run 989 cable plug to J8.



4-9 GRADIENT CABLE INSTALLATION

Nine cables have been delivered to site. All have three lead wires with terminals and attached cable labels at one end. The leads on this cable end are connected to the Twin Gradient Filter on the Penetration Panel. The other end is cut flush with no labels or terminals. The nine cables with supplied labels are as follows:

RUN	LABEL	SUPPLIED LABELS
979	PP1-A7-X	(1) RUN 976, (1) RUN 979, (2) X+, (1) X-, (1) GND, (1) XZM-, (1) XWB-
980	PP1-A7-Y	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y-, (1) GND, (1) YZM- (1) YWB-
981	PP1-A7-Z	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z-, (1) GND, (1) ZZM- (1) ZWB-
982	PP1-A7-Z _{ZM}	(1) RUN 982, (1) ZZM- (1) ZZM+
983	PP1-A7-Y _{ZM}	(1) RUN 983, (1) YZM- (1) YZM+
984	PP1-A7-X _{ZM}	(1) RUN 984, (1) XZM- (1) XZM+
985	PP1-A7-Z _{WB}	(1) RUN 985, (1) ZWB- (1) ZWB+
986	PP1-A7-Y _{WB}	(1) RUN 986, (1) YWB- (1) YWB+
987	PP1-A7-X _{WB}	(1) RUN 987, (1) XWB- (1) XWB+

4-9-1 GRADIENT CABLE INSTALLATION IN EQUIPMENT ROOM

Refer to Section **4-9-2** for installation of Gradient cables in Magnet Room.

4-9-1-1 General Steps for Cutting Cables to Length and Routing in Equipment Room

1. Move Run 979, 980, and 981 cables to equipment room. The end of cable with terminals will be connected to Penetration Panel.
2. Route cables from Gradient Filter of Penetration Panel to right side of interface panel at rear of Twin Accessory Cabinet (TAC).
3. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at TAC Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
4. Refer to Procedure **4-9-1-5** to terminate and label TAC end cable Runs 979, 980, and 981
5. Refer to Procedure **4-9-1-6** for connection of Runs 979, 980, and 981 to TAC cabinet.
6. Route remainder of cables from rear of TAC to rear of HFD cabinet. Make sure enough slack is provided for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
7. Refer to Procedure **4-9-1-8** for connection of Runs 976, 977, and 978 to TAC cabinet.
8. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
9. Refer to Procedure **4-9-1-11** for connection of Runs 976, 977, and 978 to HFD cabinet.

4-9-1-2 Connect Runs 979, 980, and 981 to Penetration Panel

1. Remove gradient filter terminal covers.
2. Route Run 981 (Z) to left side and Runs 979 (X) and 980 (Y) to right side of gradient filter on Penetration Panel.

Note: The three X and three Y terminals are routed to the right side and three Z terminals to the left side.

3. Connect +, ZM- (cable marked as TWO), & WB- (cable marked as THREE) wires to gradient filter terminals as marked on each side of gradient filter.

Note: Red tape indicates positive (+) wires.

4. Align cables to obtain maximum spacing from other cables and conductive metal objects.

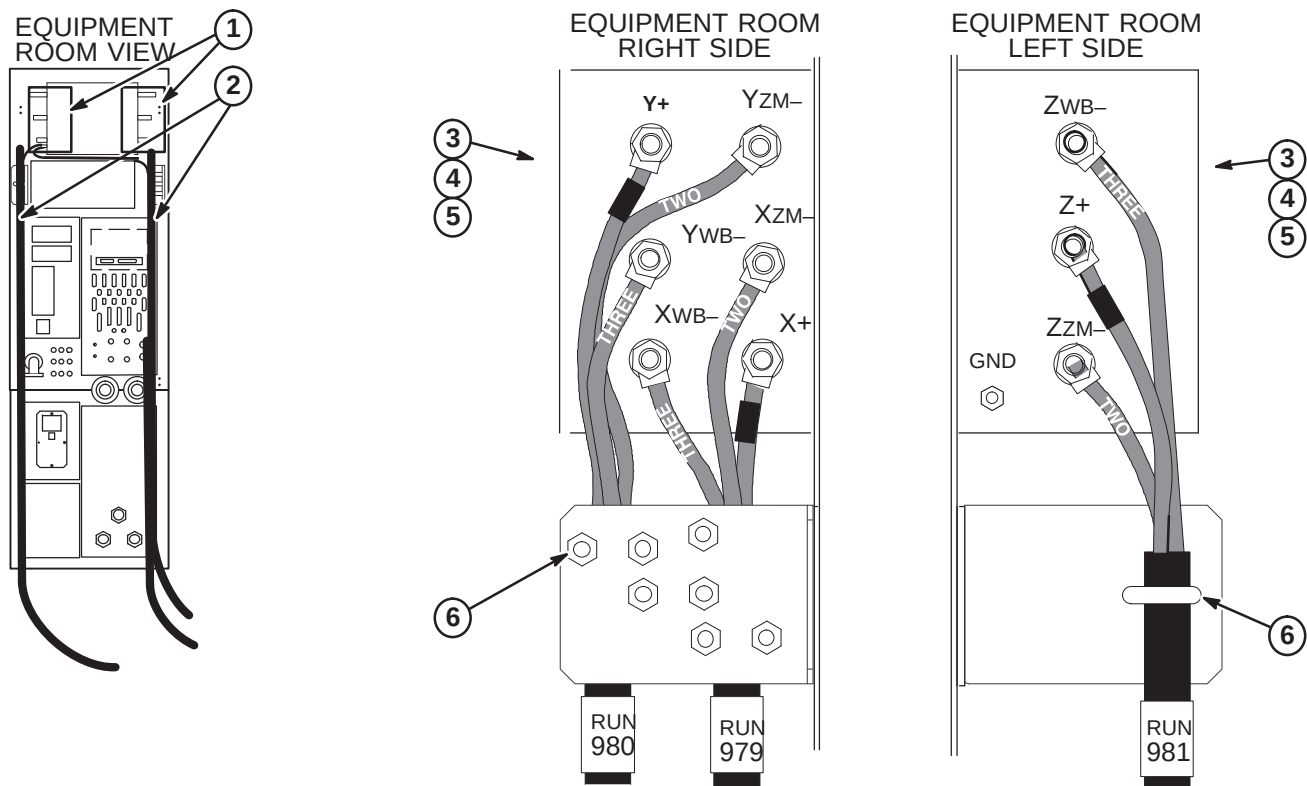
Note: When tightening nuts, it is important to hold screw body.

5. Tighten nuts to a torque of 35 ft-lbs (67-74Nm) or after tightening, verify that nothing is loose.

Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.

6. Route and secure each cable Run to strain-relief bracket with U-bolts.

Note: Tighten U-bolts to strain-relief cables only. Do not over tighten; cables may become damaged.

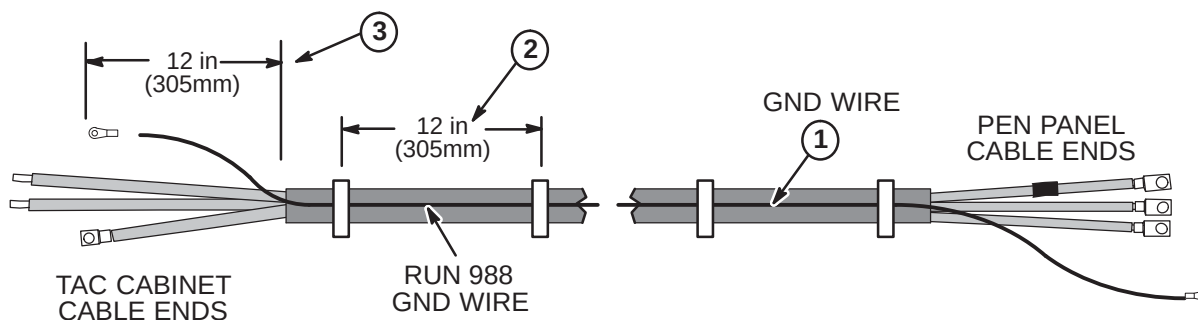


4-9-1-3 Attach Ground Cables to Runs 979, 980, and 981

For each gradient cable routed between the Penetration Panel and TAC Cabinet, a ground wire, Run 988 (2283297), has to be ty-wrapped to each cable.

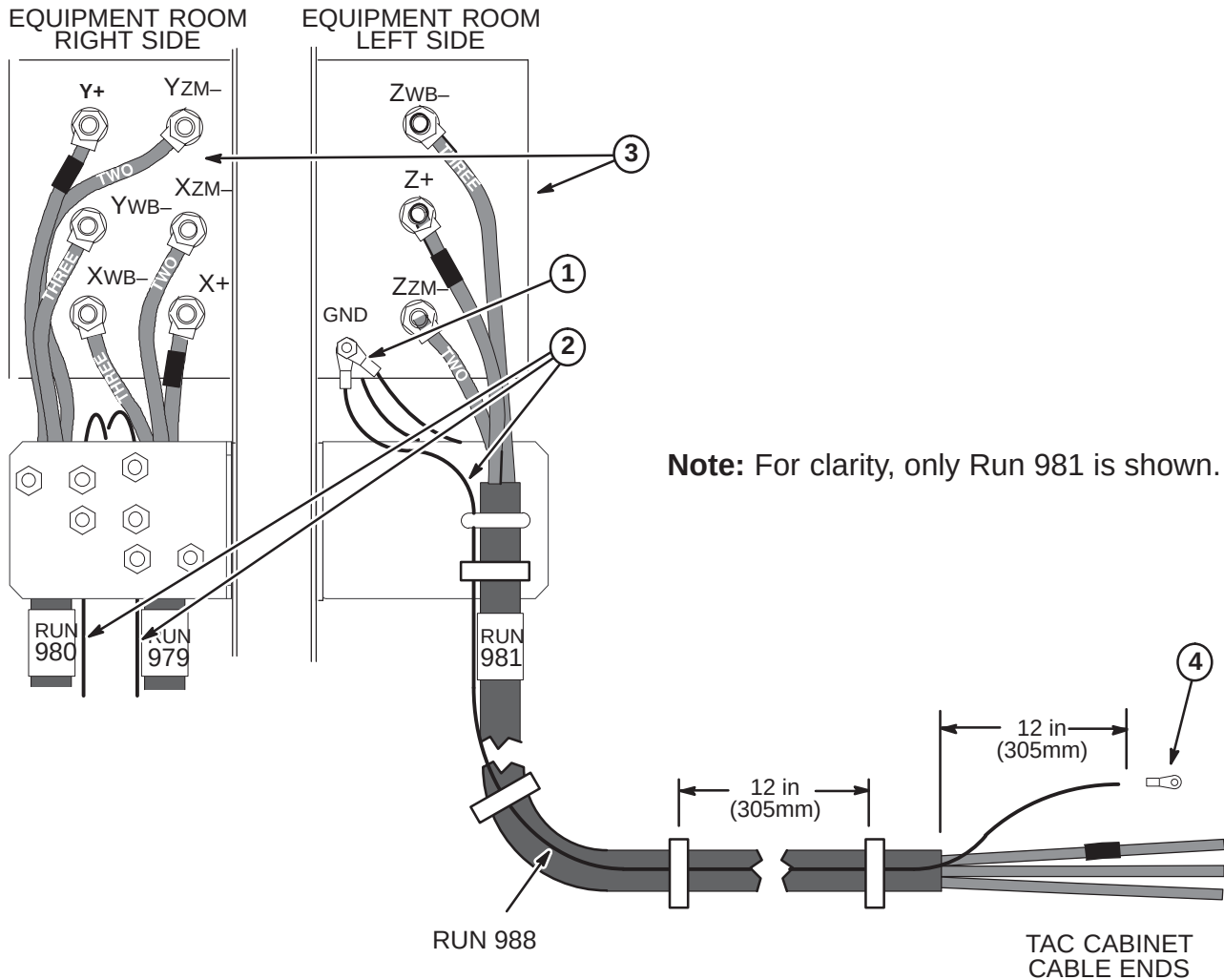
1. One Ground Wire will be routed along each of the Gradient Cables, Runs 979, 980, and 981.
2. Each Run 988 must be ty-wrapped about every 12 inches along entire length of each gradient cable.
Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. End of Run 988 at TAC cabinet should extend about 12 inches (305mm) past end of outer jacket after trimming.
Note: Refer to Procedure **4-9-1-5** for length of gradient cable jacket material to be removed.

Excess ground cable can not be coiled. It must be cut to length and reterminated.



4-9-1-4 Connect Run 988 Cables Ground Wires to Gradient Filter

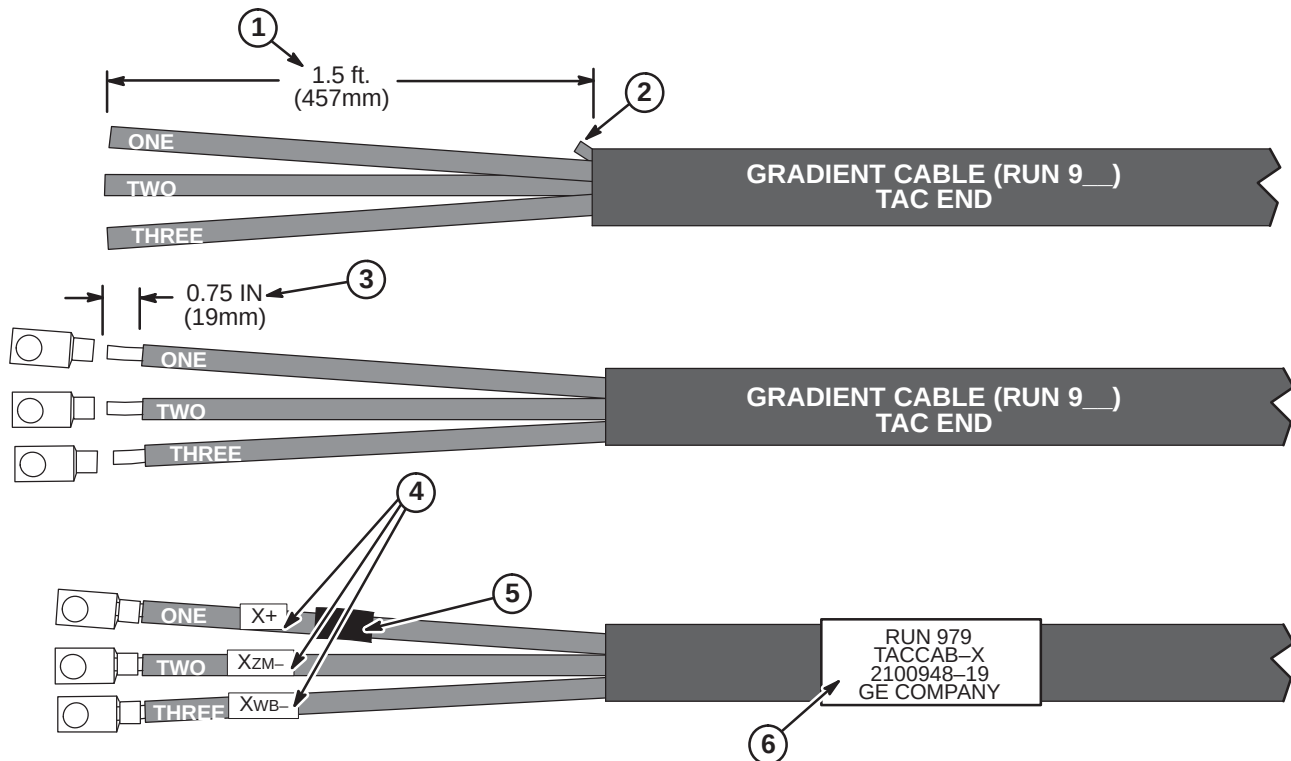
1. Connect the three Run 988 Gound Wires to GND on left side of Gradient Filter.
2. Each Run 988 must be ty-wrapped about every 12 inches along entire length of each gradient cable.
Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. Re-install gradient filter terminal covers.
4. Cut each Run 988 to length, reterminate with 3/8 ring terminal for connection to GND stud on TAC Cabinet.



4-9-1-5 Terminate Runs 979, 980, and 981 for TAC Cabinet Connection

The following steps are required to terminate the end of the cable that was cut in Procedure 4-9-1-1. The other end of the cable is connected to the Penetration Panel.

1. Remove 18 inches of jacket from end of cable Runs 979 (X), 980 (Y), and 981 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires "ONE", "TWO" and "THREE" and crimp a furnished T & B 1/2" (G2-3-12) terminal to each using Ratcheting Crimper with #2 die.
4. Place furnished "X+" label on wire "ONE".
Place furnished "XZM-" label on wire "TWO".
Place furnished "XWB-" label on wire "THREE".
5. Wrap red tape around wire "ONE".
6. Place furnished "RUN 979" label on cable.

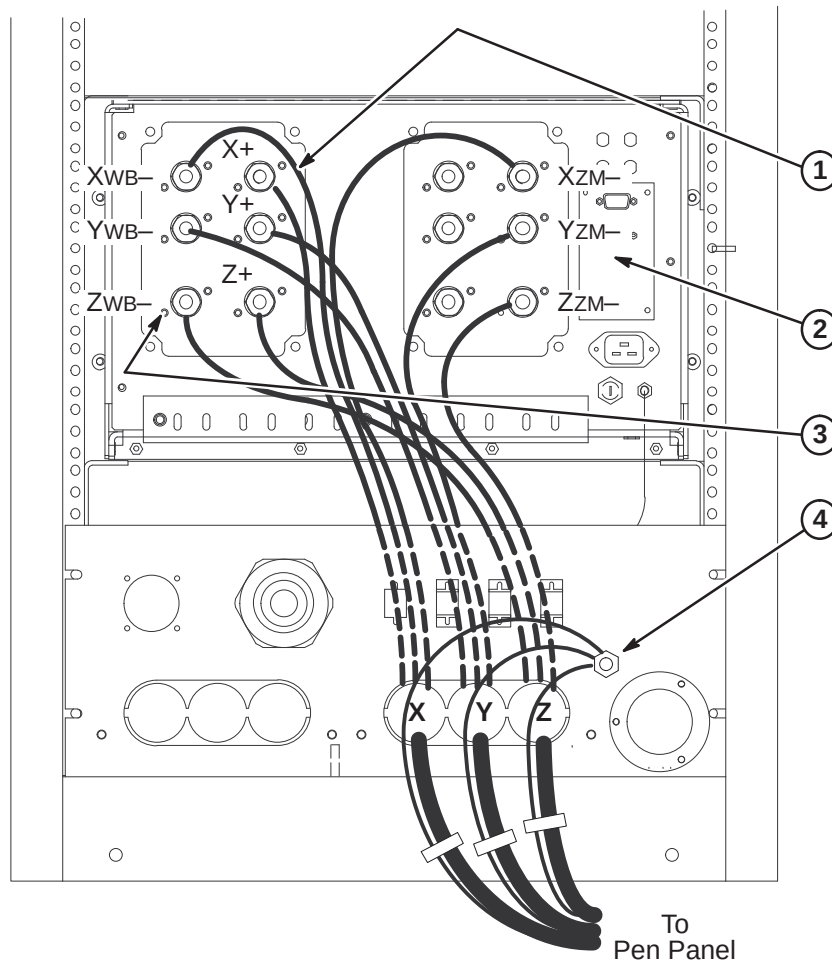


- ⑦ Steps above show completion of Run 979. Repeat same process for Runs 980 (Y+, YZM-, YWB-) and 981 (Z+, ZZM-, ZWB-).

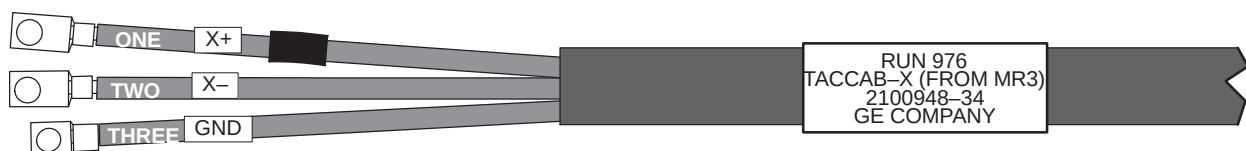
4-9-1-6 Connect Runs 979, 980, and 981 to TAC Cabinet

To make installation easier, route and connect Run 981 (Z terminals) first, then Run 980 (Y), and last Run 979 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect XZM-, YZM-, and ZZM- cables as marked.
3. Connect XWB-, YWB-, and ZWB- cables as marked.
4. Route and connect the three Run 988 Ground wires to GND stud on interface panel.

**4-9-1-7 Pre-terminated cable ends for Runs 976, 977, and 978**

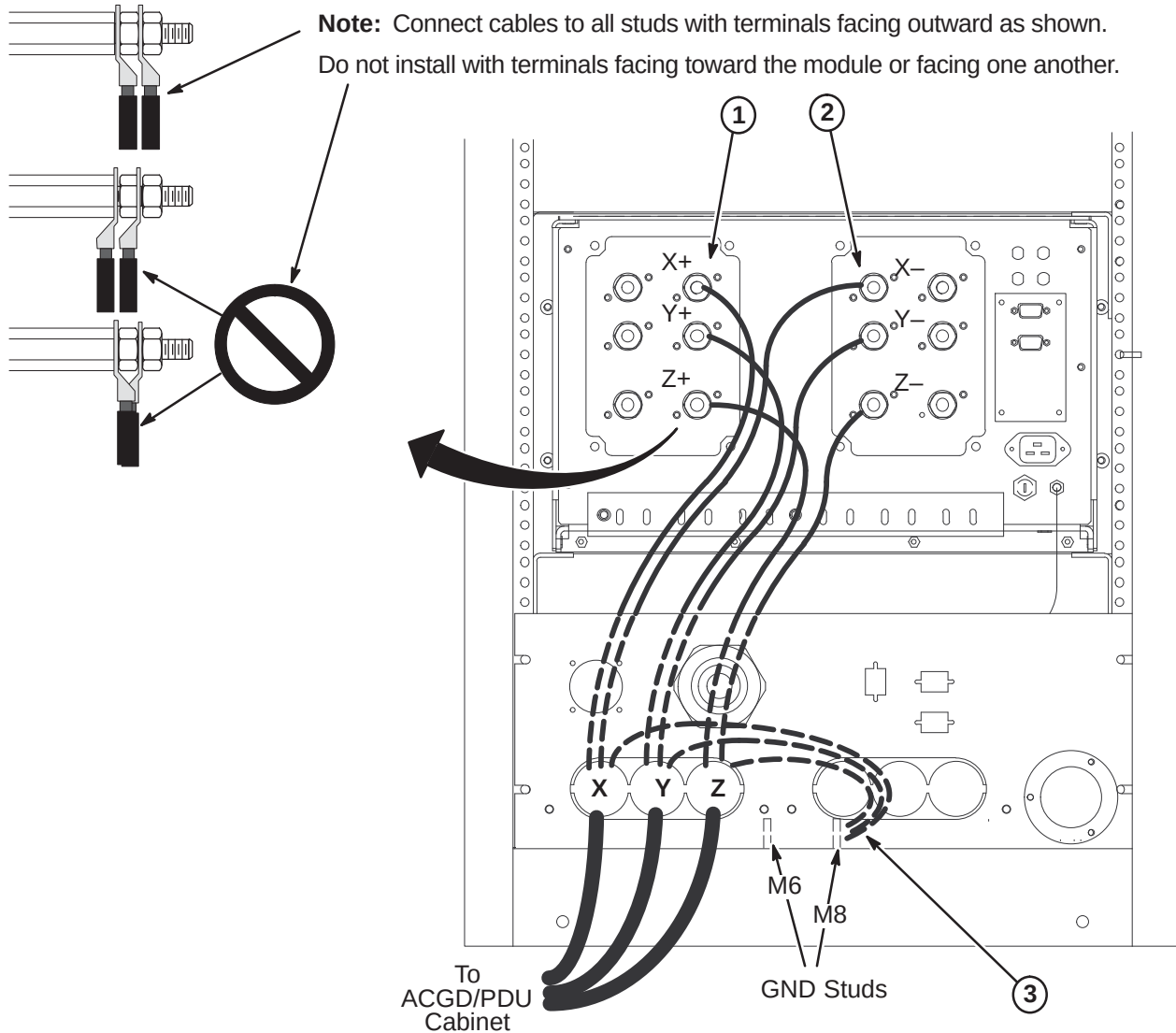
Cables delivered to site will be pre-terminated for the TAC Cabinet connection of Runs 976, 977, and 978. Refer to Procedure 4-9-1-1 for cutting cables to length for connection to HFD Cabinet. A typical pre-terminated end for each cable is shown below.



4-9-1-8 Connect Runs 976, 977, and 978 to TAC Cabinet

To make installation easier, route and connect Run 978 (Z terminals) first, then Run 977 (Y), and last Run 976 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect X-, Y-, and Z- cables as marked.
3. Connect Ground leads from each cable to M8 GND stud located at bottom of cabinet.



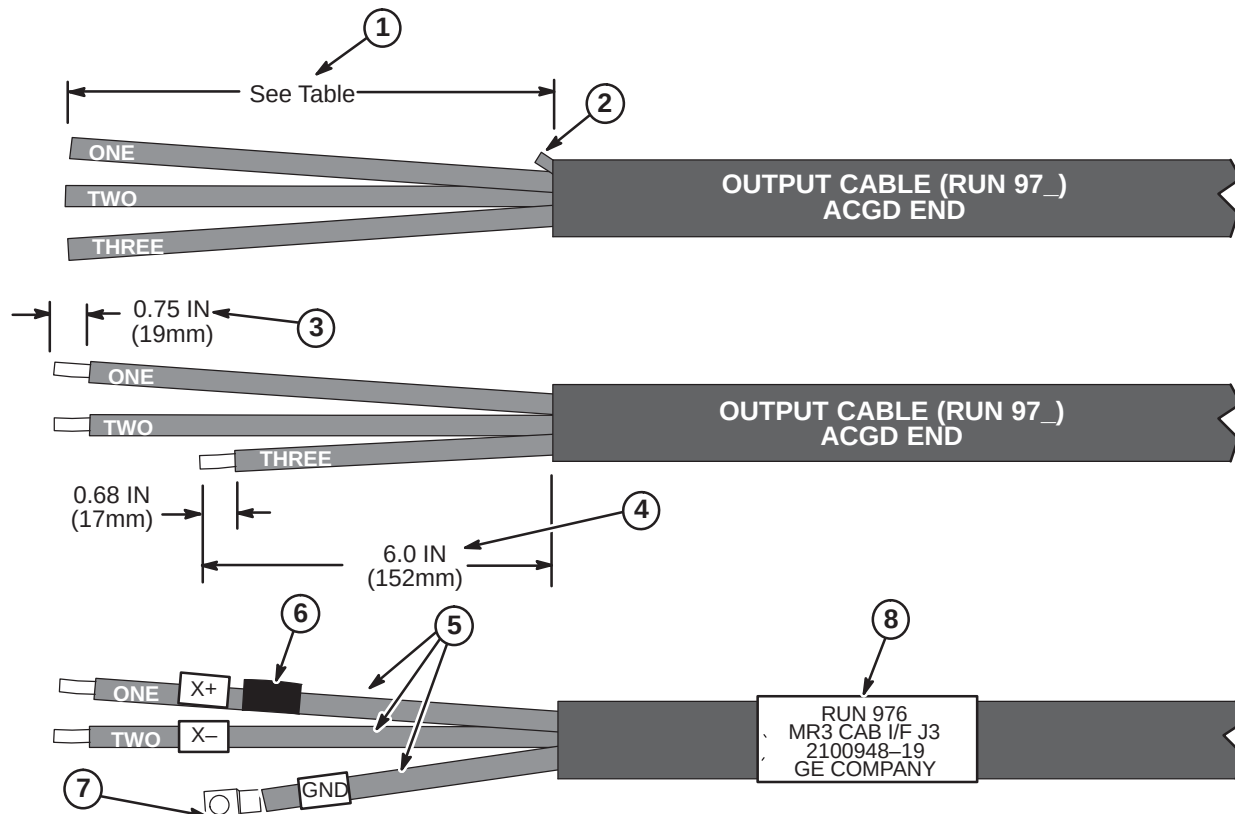
4-9-1-9 Terminate Runs 976, 977, and 978 for Connection to HFD Cabinet

The following steps are required to terminate the end of the cable that is to be connected to the HFD Cabinet. The other end of the cable is connected to the TAC Cabinet.

	Run 976	Run 977	Run 978
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

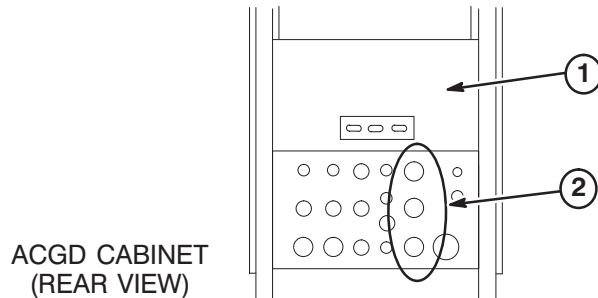
1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 976 (X), 977 (Y), and 978 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires "ONE" and "TWO".
4. Cut wire "THREE" to 6.0 inches (152mm).
5. Place furnished "X+" label on wire "ONE".
Place furnished "X-" label on wire "TWO".
Place furnished "GND" label on wire "THREE".
6. Wrap red tape around wire "ONE".
7. Crimp a furnished T&B #G2-38 terminal on GND wire end using a Ratcheting Crimper with #2 die.
8. Place furnished "RUN 976" label on cable.

Steps in illustration below show completion of Run 976. Repeat same process for Runs 977 (Y+, Y-) and 978 (Z+, Z-). **Red tape, Y+ (RUN 977) and Z+ (RUN 978) labels are attached to wire "ONE".**



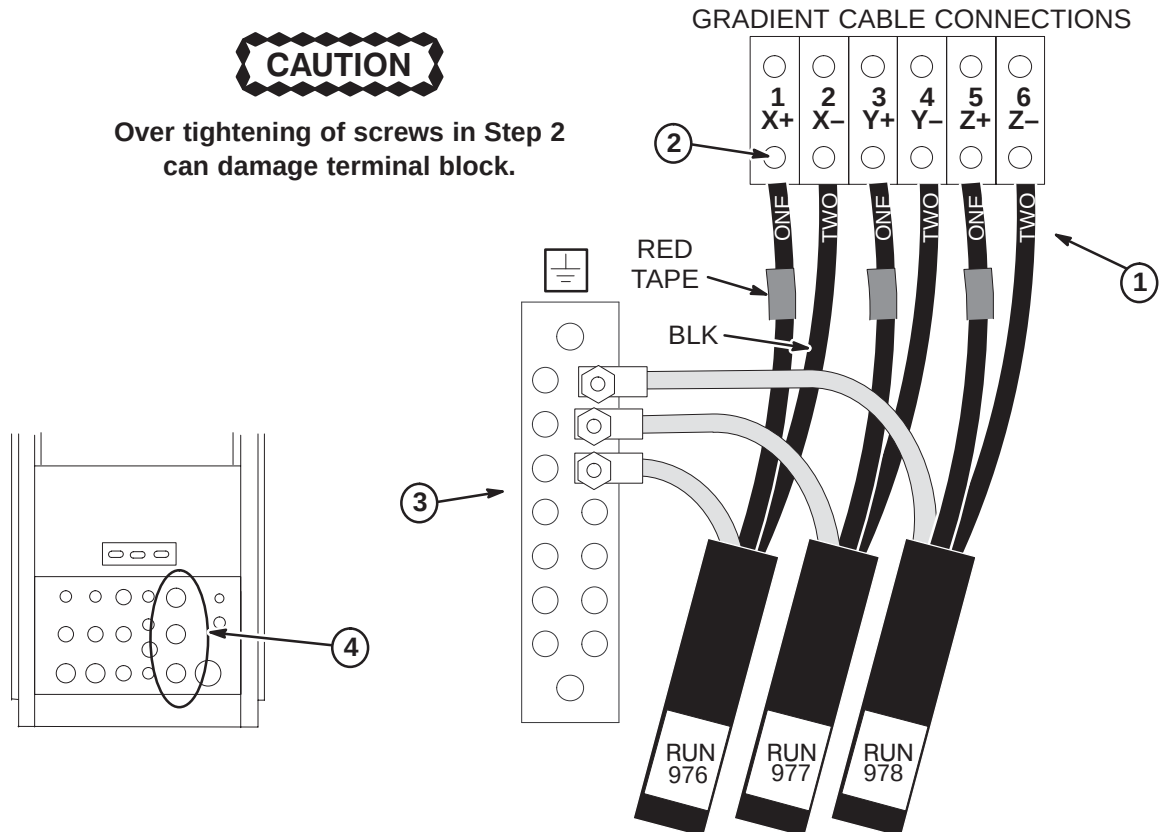
4-9-1-10 Gradient Output Cable Interface Panel Access at HFD Cabinet

1. Remove Rear Cover Plate by removing seven screws that hold it in place.
2. Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 976 into J3, Run 977 into J4, and Run 978 into J5)



4-9-1-11 Gradient Output Cable Routing and Connection in HFD Cabinet

1. Connect gradient output cables with red-taped (+) #1 wires to + output terminals and black (-) #2 wires to - output terminals.
2. Tighten six screws to 52-65 inch-lbs (5.88-7.34 N-M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
3. Connect the GND leads of each run to the Ground Bus Bar studs.
4. Tighten clamps on Interface Panel to secure gradient output cables.



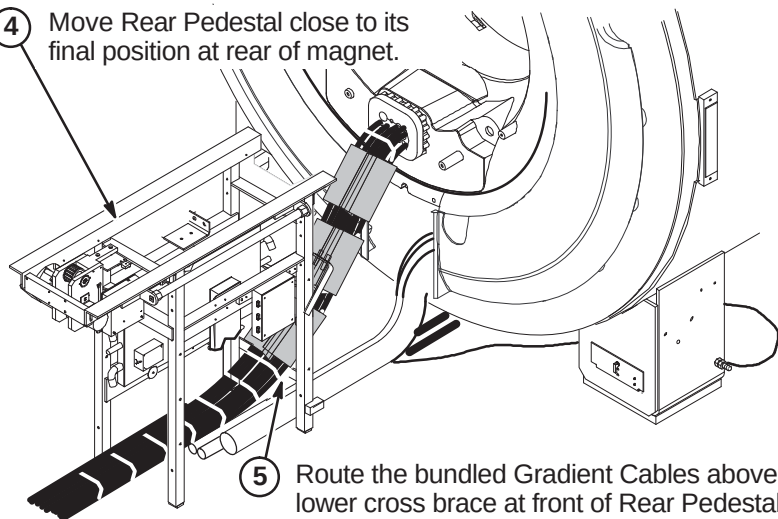
4-9-2 GRADIENT CABLE INSTALLATION IN MAGNET ROOM

Nine cables have been delivered to site. All have three lead wires with terminals and attached cable labels at one end. The leads on this cable end are connected to the Twin Gradient Filter on the Penetration Panel. The other end is cut flush with no labels or terminals. The nine cables with supplied labels are as follows:

Refer to Section 4-9-1 for installation of Gradient cables in Equipment Room.

RUN	LABEL	SUPPLIED LABELS
979	PP1-A7-X	(1) RUN 976, (1) RUN 979, (2) X+, (1) X-, (1) GND, (1) XZM-, (1) XWB-
980	PP1-A7-Y	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y-, (1) GND, (1) YZM- (1) YWB-
981	PP1-A7-Z	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z-, (1) GND, (1) ZZM- (1) ZWB-
982	PP1-A7-Z _{ZM}	(1) RUN 982, (1) ZZM- (1) ZZM+
983	PP1-A7-Y _{ZM}	(1) RUN 983, (1) YZM- (1) YZM+
984	PP1-A7-X _{ZM}	(1) RUN 984, (1) XZM- (1) XZM+
985	PP1-A7-Z _{WB}	(1) RUN 985, (1) ZWB- (1) ZWB+
986	PP1-A7-Y _{WB}	(1) RUN 986, (1) YWB- (1) YWB+
987	PP1-A7-X _{WB}	(1) RUN 987, (1) XWB- (1) XWB+

- ① The following instructions are for installing and connecting the Magnet Room gradient cable Runs 982, 983, 984, 985, 986, and 987.
- ② Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. Do not cut Body Coil gradient cables.
- ③ Locate furnished bag of parts that contain the terminals and labels required to be attached to the cables.
- ④ Move Rear Pedestal close to its final position at rear of magnet.

**WARNING!**

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

Note: Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.

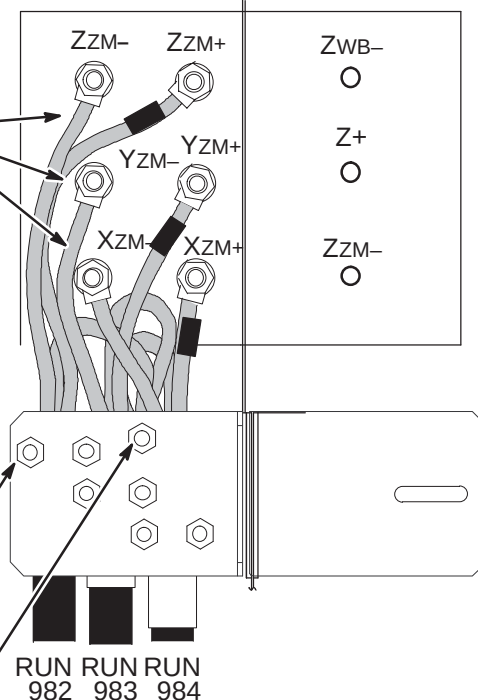
WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

4-9-2-1 Connect ZM (Zoom) Cables to Penetration PanelMAGNET ROOM
VIEW

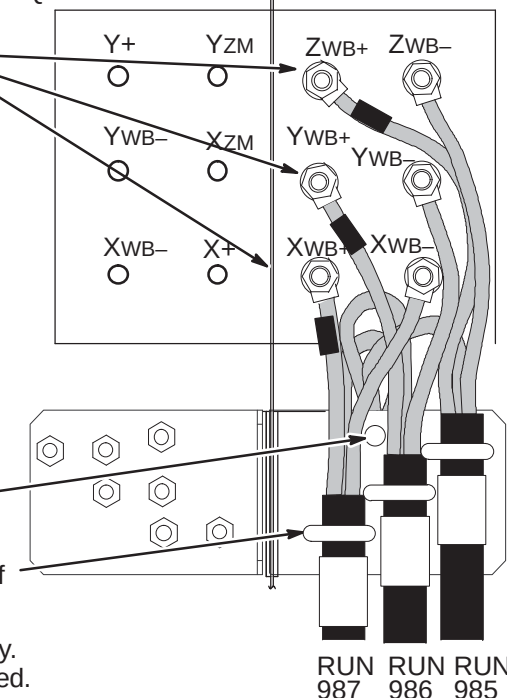
- ① Remove gradient filter terminal covers.
- ② Route Runs 982, 983, and 984 to right side of gradient filter on Penetration Panel.
- ③ Connect six ZM- and ZM+ wires to gradient filter terminals as marked.
Note: Red tape indicates positive (+) wires.
- ④ Align cables to obtain maximum spacing from other cables and conductive metal objects.
Note: When tightening nuts, it is important to hold screw body.
- ⑤ Tighten nuts to a torque of 35 ft-lbs (67-74Nm) or after tightening, verify that nothing is loose.
Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
- ⑥ Route and secure each cable Run to strain-relief bracket with U-bolts.
Note: Tighten U-bolts to strain-relief cables only. Do not over tighten; cables may become damaged.
- ⑦ Secure GND lead wire on Ground Stud for each cable Run.

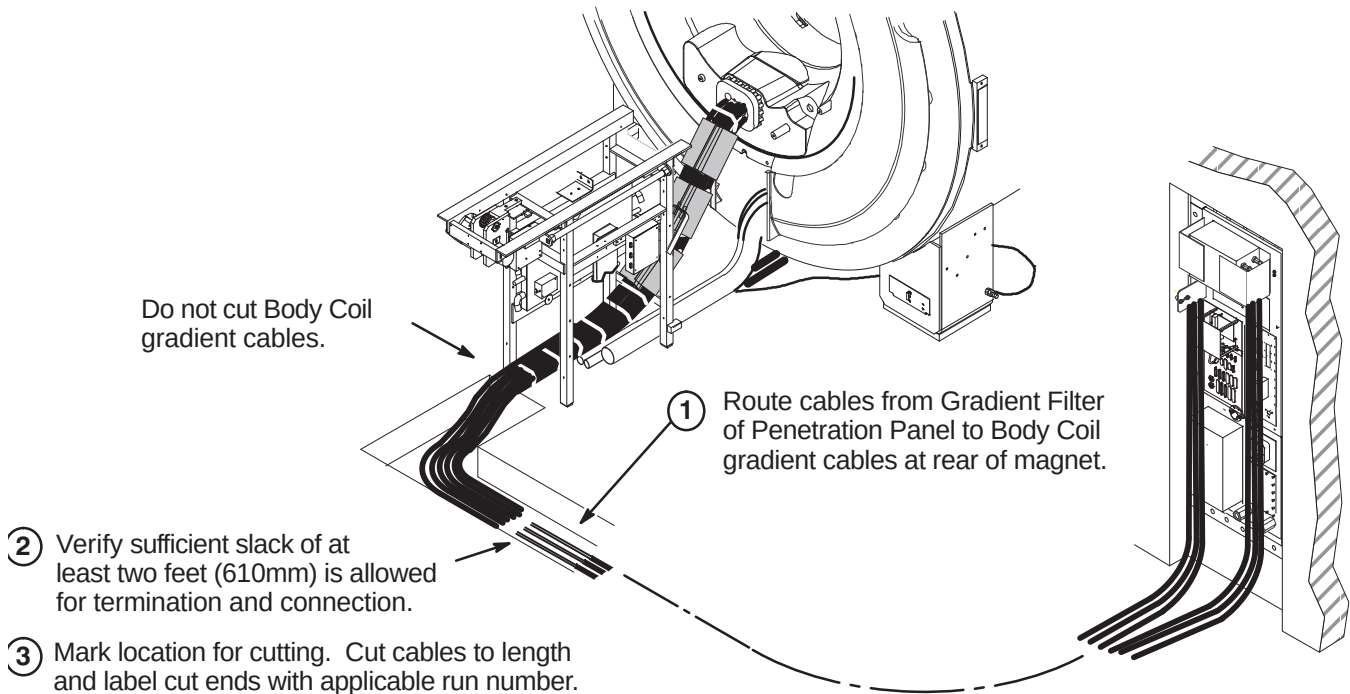
MAGNET ROOMEQUIPMENT ROOM

**4-9-2-2 Connect WB (Whole Body) Cables to Penetration Panel**MAGNET ROOM
VIEW

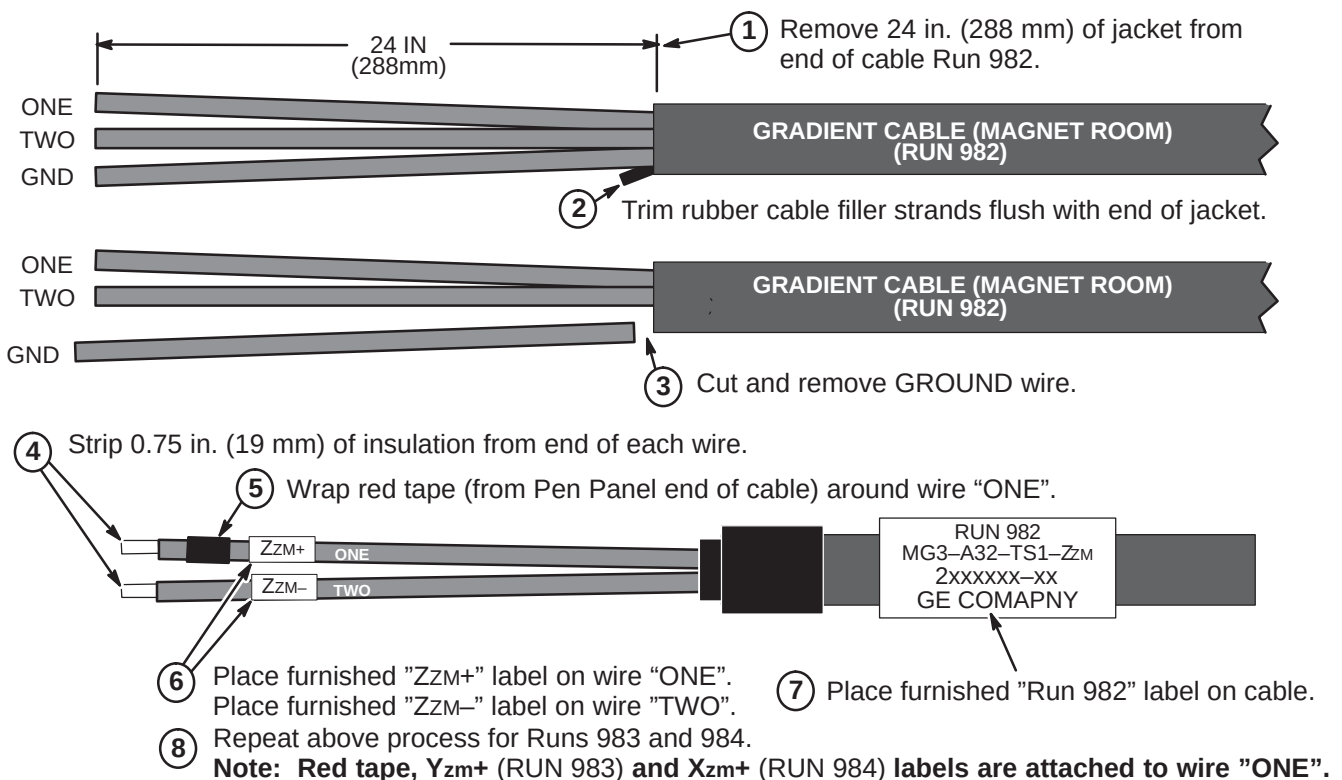
- ① Route Runs 985, 986, and 987 to left side of gradient filter on Penetration Panel.
- ② Connect six WB- and WB+ wires to gradient filter terminals as marked.
Note: Red tape indicates positive (+) wires.
- ③ Align cables to obtain maximum spacing from other cables and conductive metal objects.
Note: When tightening nuts, it is important to hold screw body.
- ④ Tighten nuts to a torque of 35 ft-lbs (67-74Nm) or after tightening, verify that nothing is loose.
Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
- ⑤ Secure GND lead wire on Ground Stud for each cable Run.
- ⑥ Route and secure each cable Run to strain-relief bracket with U-bolts.
Note: Tighten U-bolts to strain-relief cables only. Do not over tighten; cables may become damaged.
- ⑦ Replace gradient filter terminal covers.

EQUIPMENT ROOM MAGNET ROOM



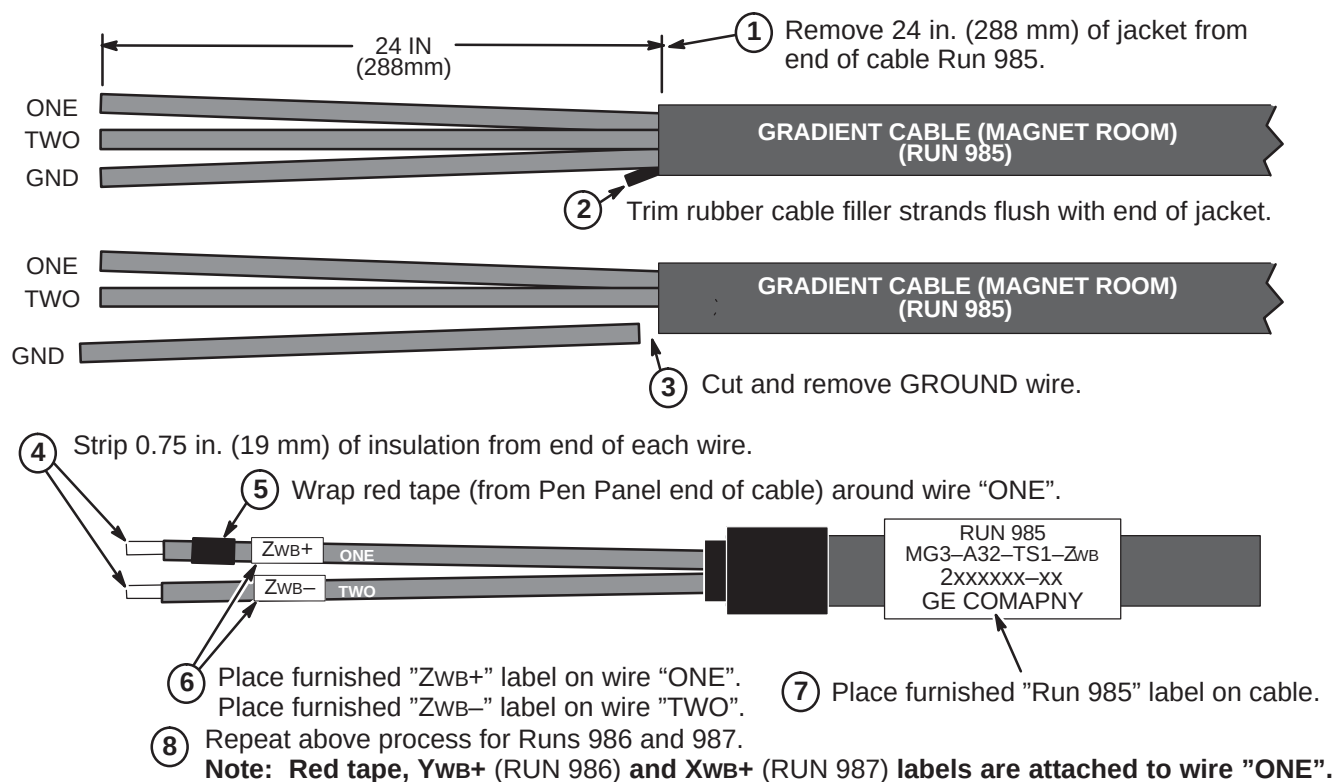
4-9-2-3 Route Gradient Cables to Rear of Magnet**4-9-2-4 Terminate Runs 982, 983, And 984 for Magnet Room**

Note: The following steps are required to terminate the end of the cable that was cut in Section 4-9-2-3. The other end of the cable is connected to the Penetration Panel.



4-9-2-5 Terminate Runs 985, 986, And 987 for Magnet Room

Note: The following steps are required to terminate the end of the cable that was cut in Section 4-9-2-3. The other end of the cable is connected to the Penetration Panel.

**WARNING!**

THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET WHICH COULD CAUSE INJURY OR DAMAGE TO THE EQUIPMENT.

4-9-2-6 Connect Magnet Room Gradient Cables

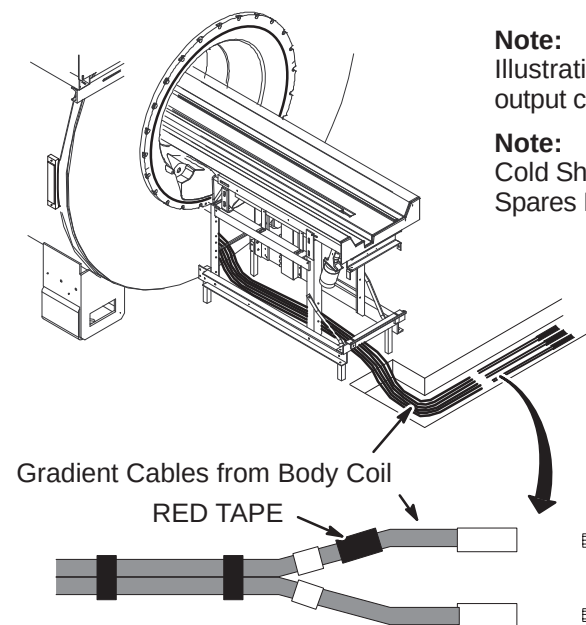
**Note:**

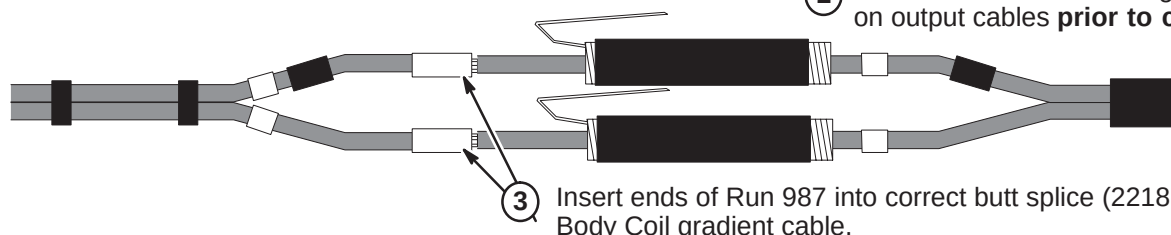
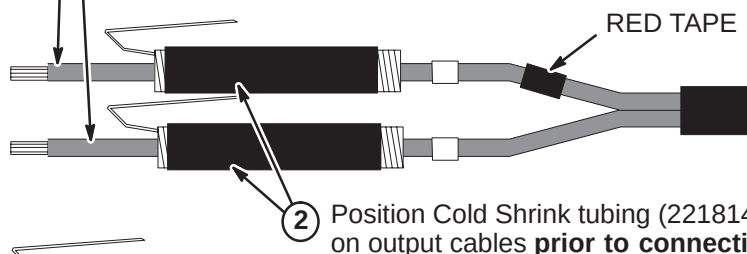
Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

Note:

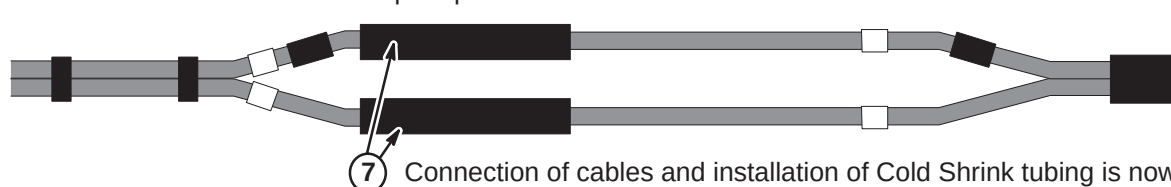
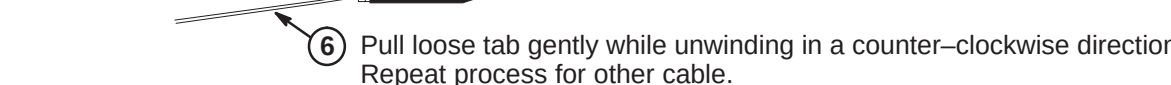
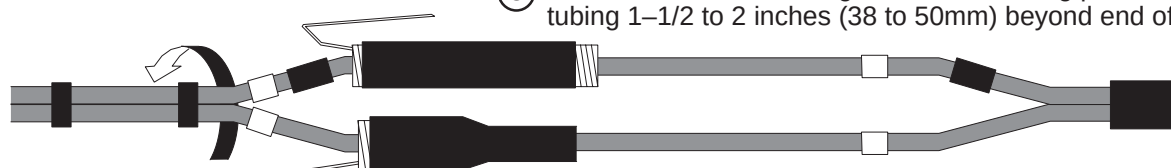
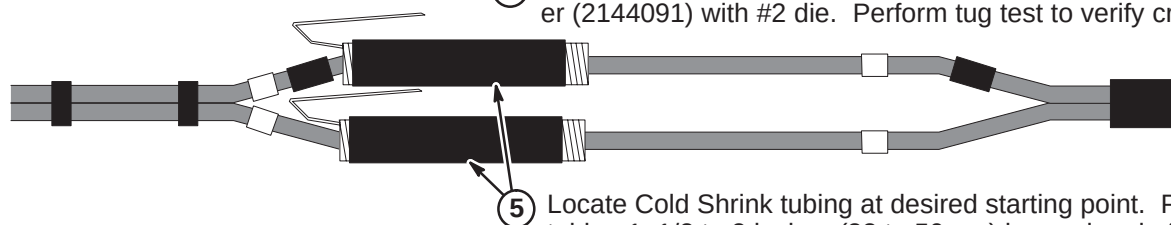
Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet

Gradient Output Cables from Penetration Panel

- ① Cable connection process shown below is for Run 987. Repeat process for Runs 982, 983, 984, 985 and 986.
Note: Red tape, XWB+ (RUN 987), YWB+ (RUN 986), ZWB+ (RUN 985), XzM+ (RUN 984), YzM+ (RUN 983), and ZzM+ (RUN 982) labels are attached to wire "ONE".



- ④ Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die. Perform tug test to verify crimp.

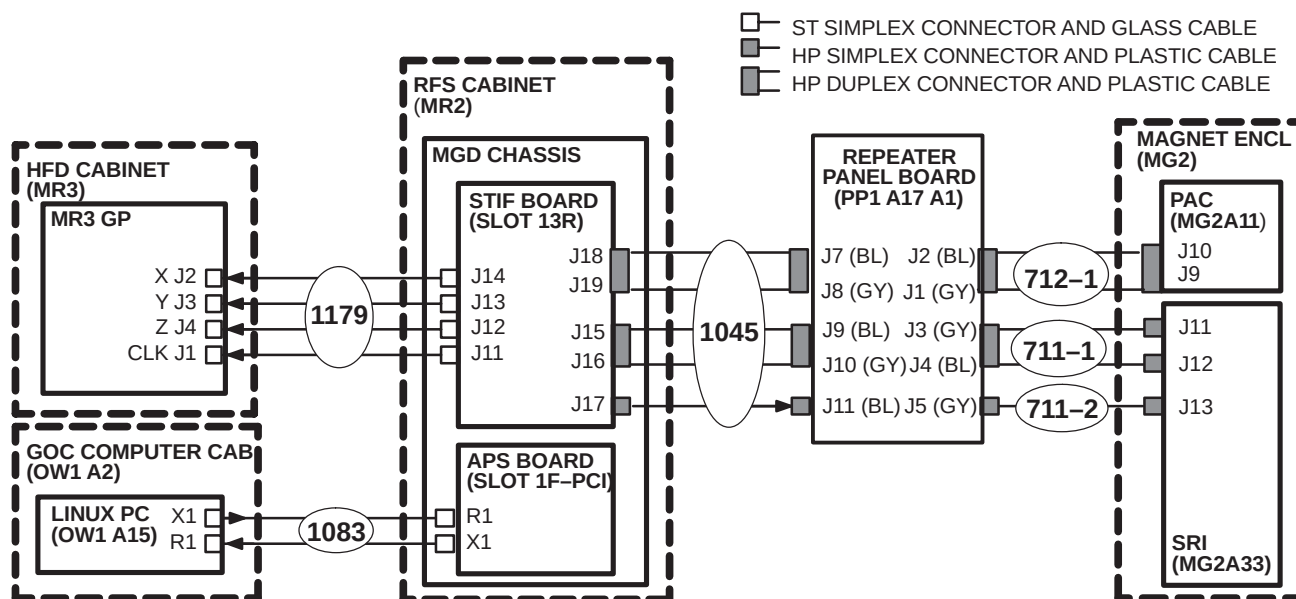


4-10 FIBER OPTIC CABLE INSTALLATION

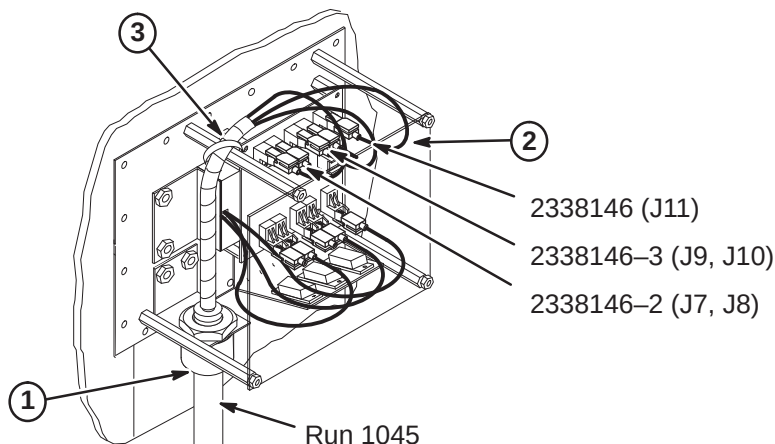
Fiber Optic Run 711/712 is reused and connected to the SRI and PAC modules on the platform assembly mounted to the right side of the magnet between the legs.

If storage of excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are in Appendix.

1. Unroll coiled fiber optic conduits for Runs 711/712, 1045, 1083, and 1179 along intended routes.
2. Connection of Runs is shown in diagram below.

**4-10-1 FIBER OPTIC INSTALLATION IN EQUIPMENT ROOM****4-10-1-1 Connect Run 1045 to Penetration Panel**

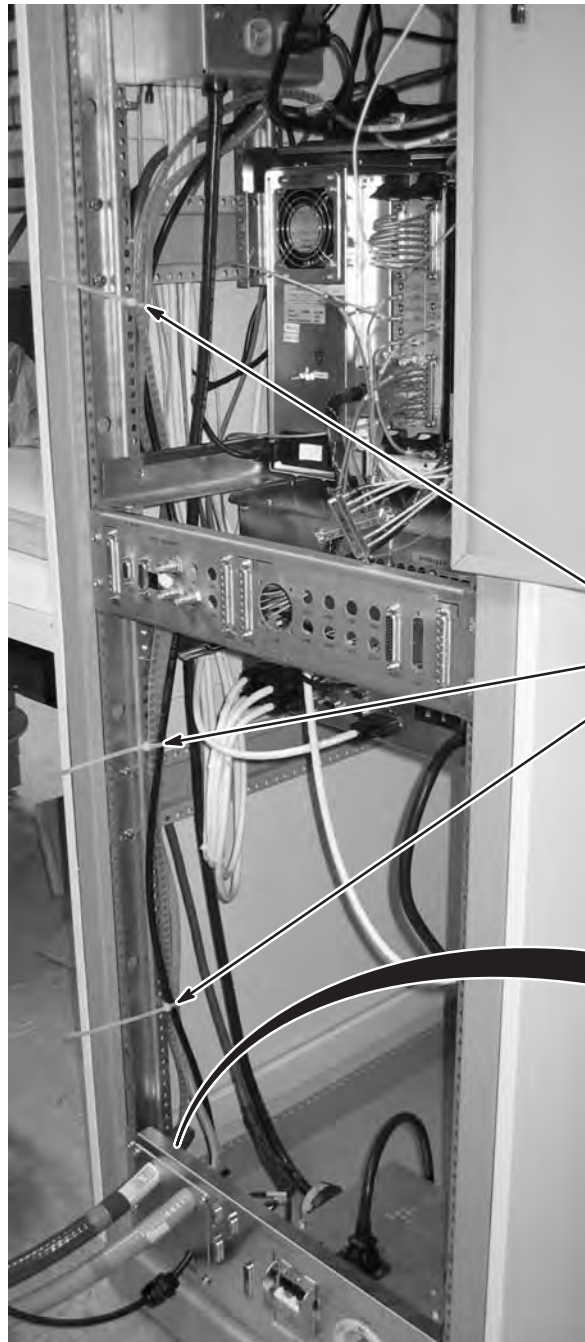
1. Move Run 1045 to the Equipment Room side and attach "PP1-A17" end to access bracket on Penetration Panel. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
2. Adjust route of individual fiber optic runs to maintain large radius and connect Run 1045 as shown.
3. Adjust position of protective spiral wrap and secure with tie-wrap.
4. Route other end of Run 1045 to System Cabinet.



4-10-1-2 Route Fiber Optic Runs 1045, 1083, and 1179 in RF/System Cabinet

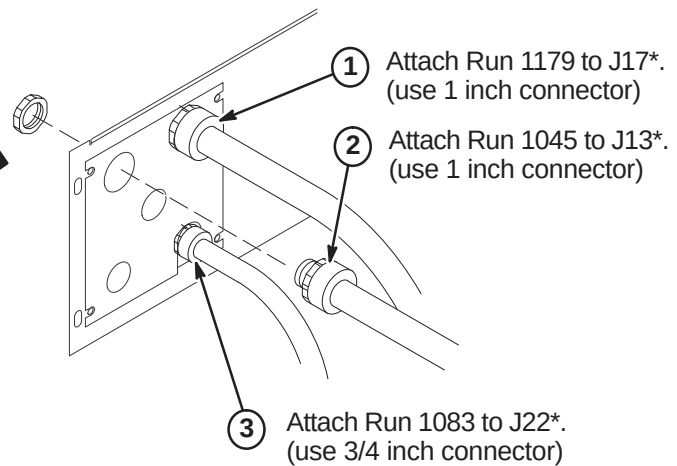
CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



SYSTEM CABINET
(REAR VIEW)

- ④ Route spiral wrapped fiber optic cables behind rails along side of cabinet to MGD Chassis and tie-wrap to vertical rail as shown.



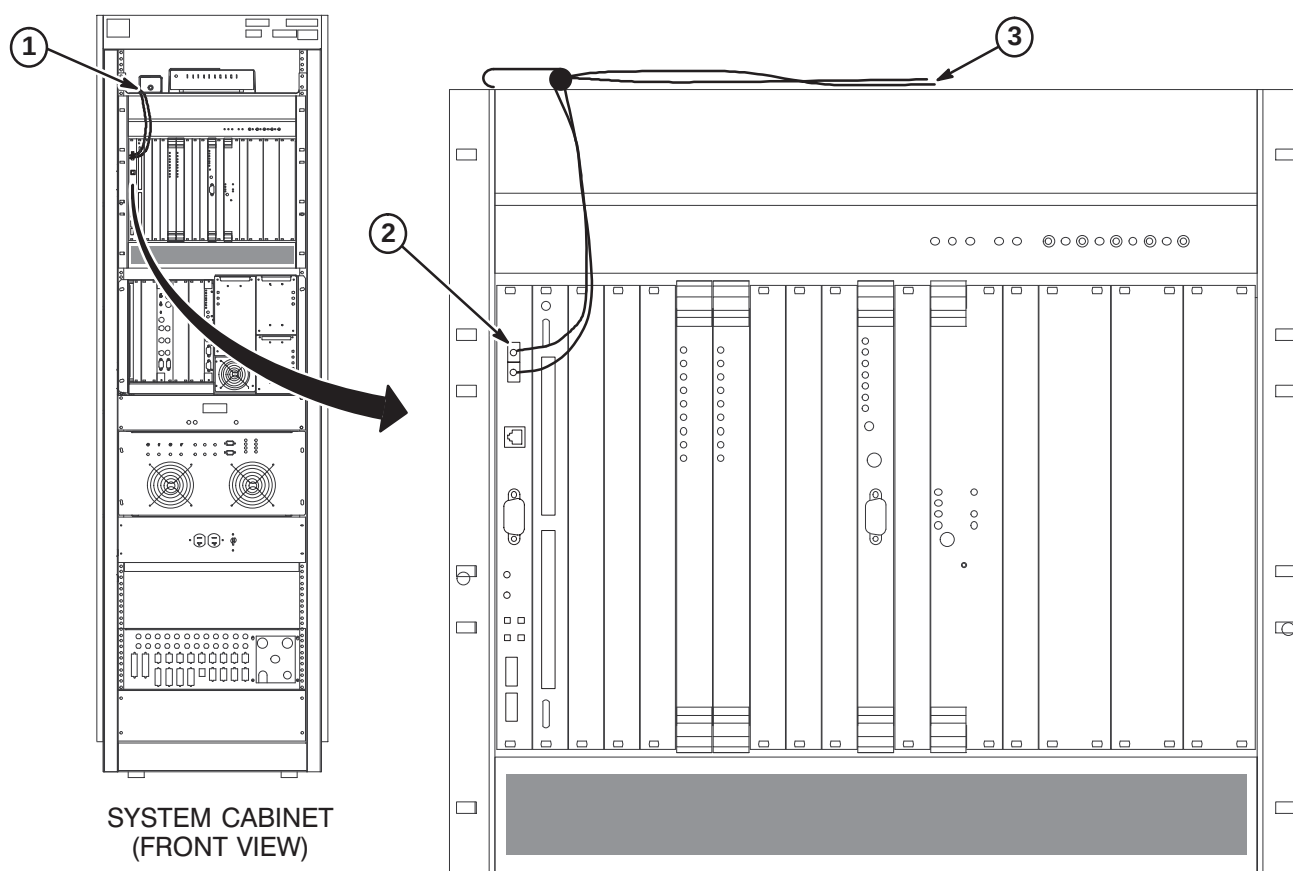
* Instructions provided with connectors.

4-10-1-3 Route/Connect Fiber Optic Run 1083 to APS Board

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route Run 1083 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**
2. Connect the Orange and Blue fiber optic cables to the IT-MGD Daughter Board as indicated.
3. Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.

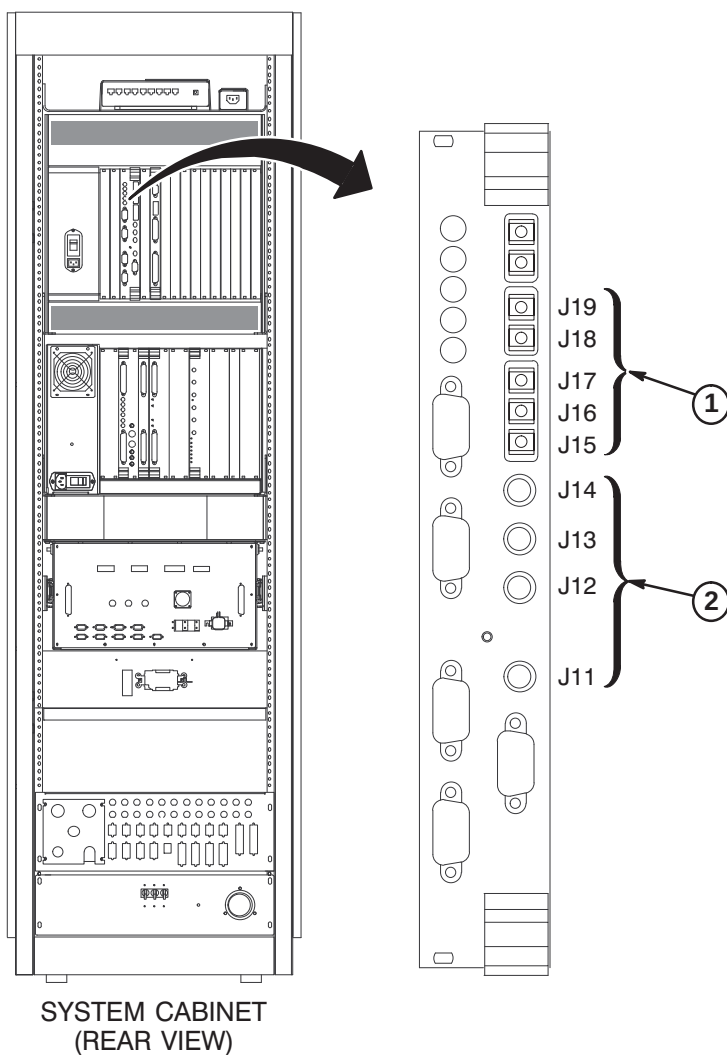


4-10-1-4 Route/Connect Fiber Optic Runs 1179 and 1045 to STIF Board

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
2. Route/Connect Run 1179 fiber optic cables to J11, J12, J13, and J14 on the STIF Board.

**Note:**

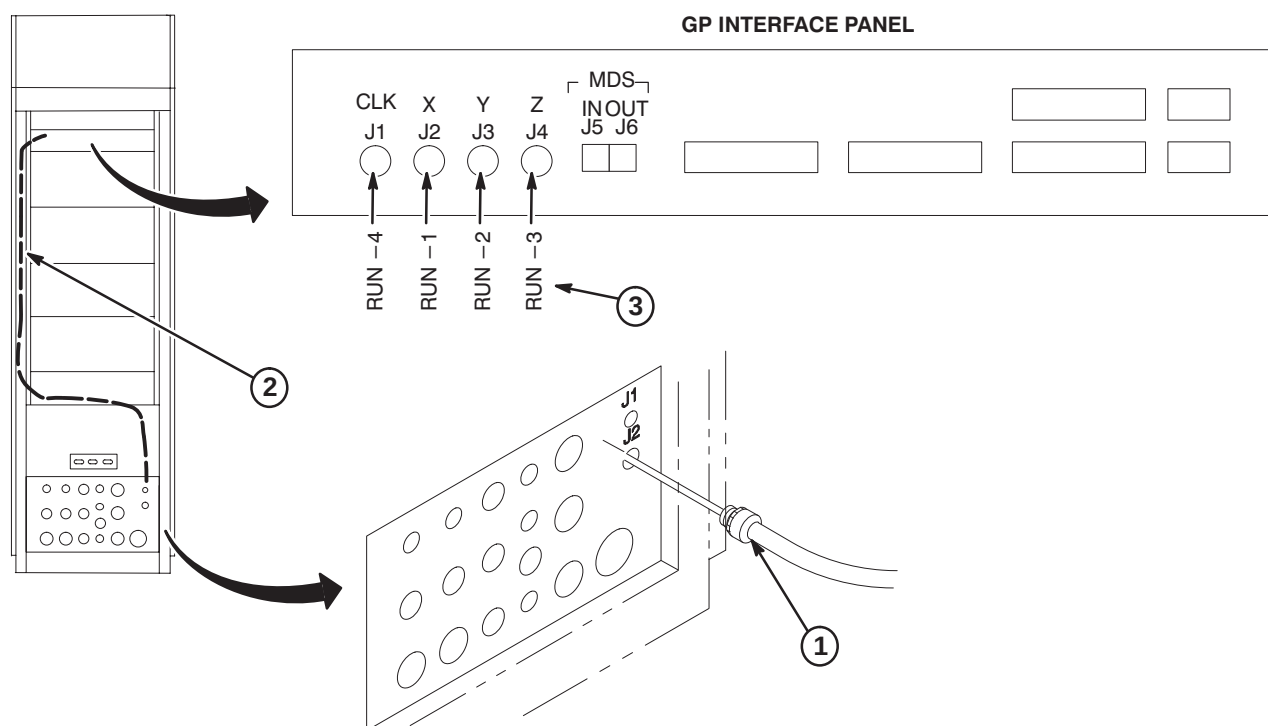
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

4-10-1-5 Route/Connect Fiber Optic Run 1179 to GP3 Module in HFD

CAUTION

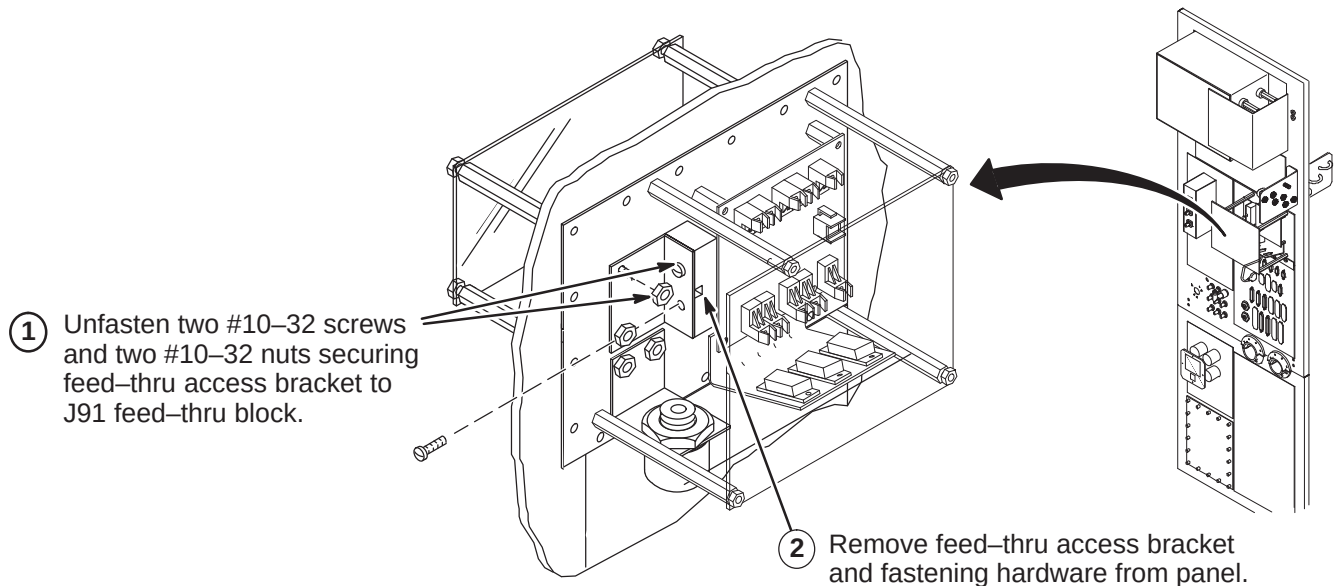
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.

1. Attach Run 1179 to I/F J2.
2. Route fiber optic cables to GP3 Module. Leave enough slack for extending GP module forward.
3. Connect fiber optics to GP3 module according to table.

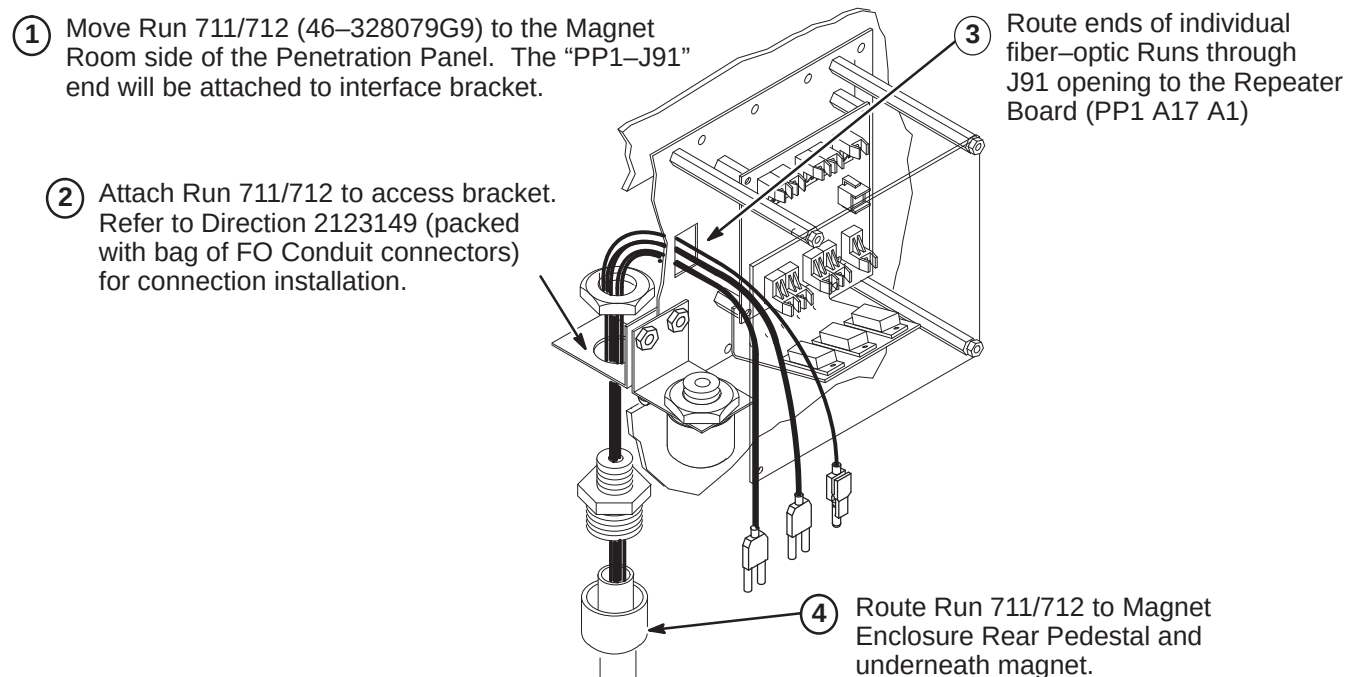


4-10-2 FIBER OPTIC INSTALLATION IN MAGNET ROOM

Fiber Optic Run 711/712 is reused and connected to the SRI and PAC modules on the platform assembly mounted to the right side of the magnet between the legs. The conduit is stored for shipment in the bore of the magnet. Unroll conduit along designated path in cable trough to Penetration Panel.

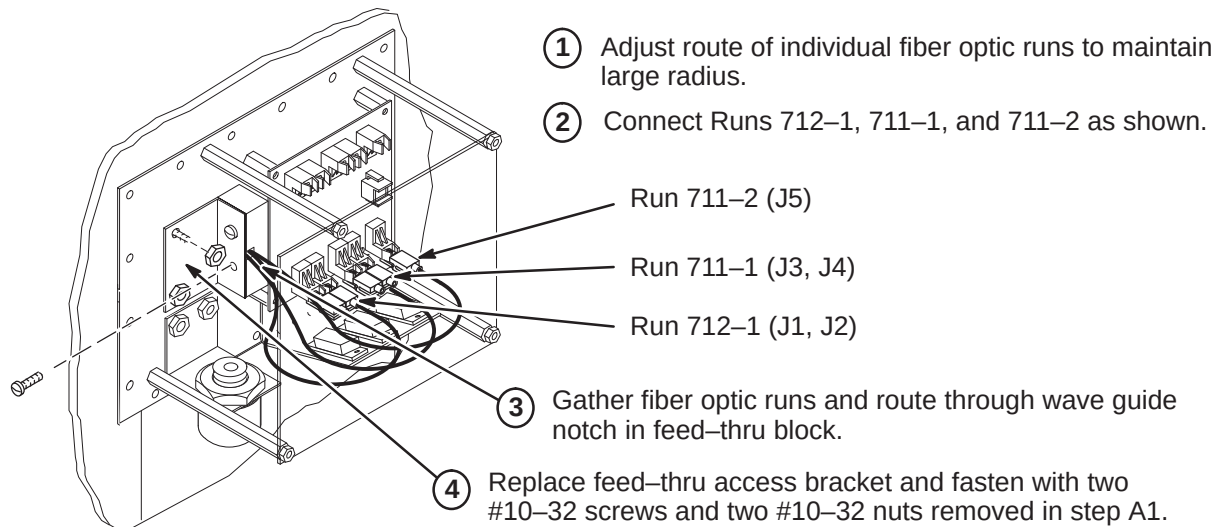
4-10-2-1 Repeater Panel Feed-thru Bracket Removal**4-10-2-2 Routing Run 711/712 in Magnet Room**

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cut and ends terminated with the Fiber Optic Connector Repair Kit. Refer to Appendix for instructions. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

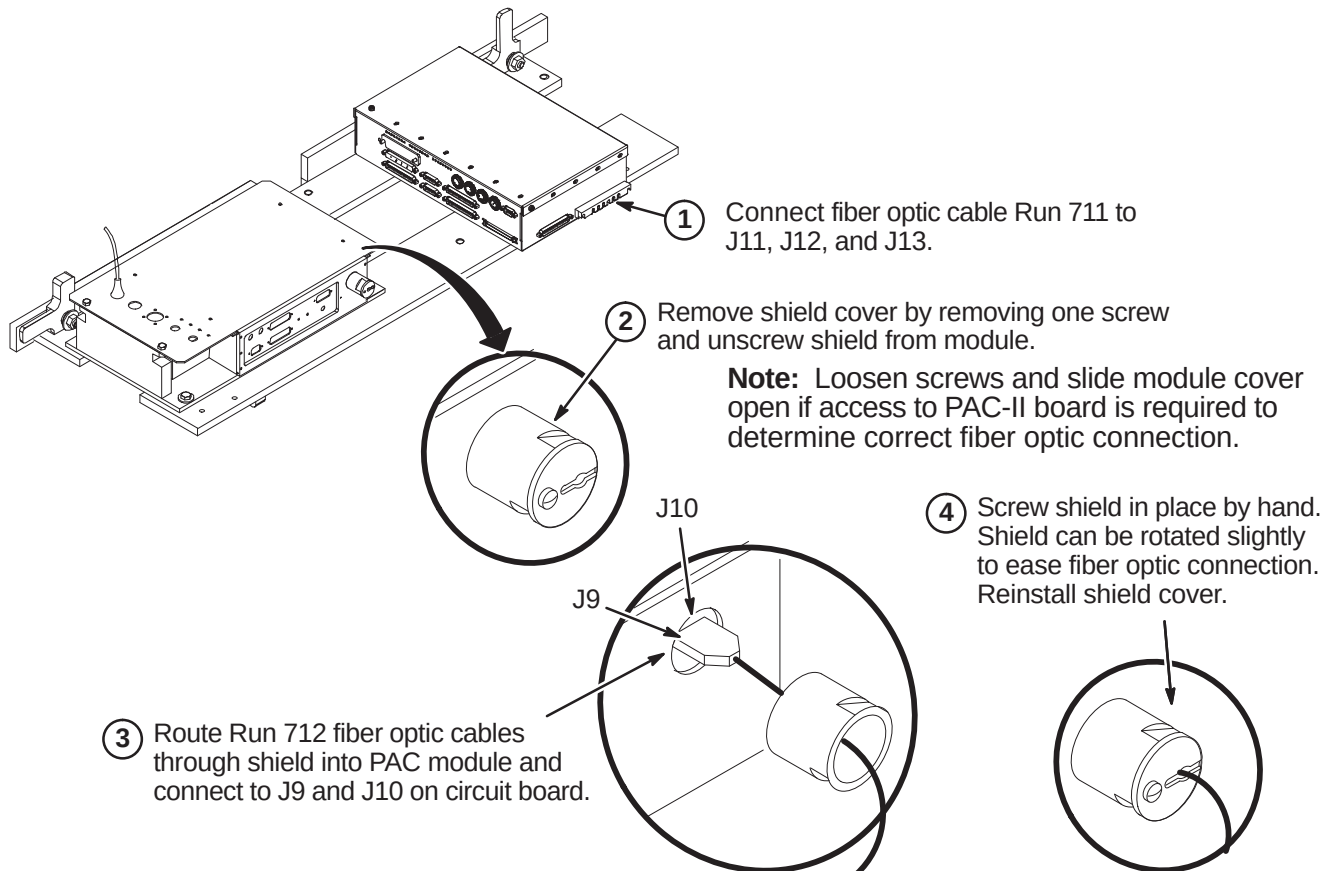


4-10-2-3 Connecting Fiber Optic Run 711/712 to Repeater Panel

Note: Take care to maintain the minimum 2 inch bend radius.

**4-10-2-4 Connect to PAC and SRI Modules**

Refer to Section 19-6-14 for additional information on installing modules and connecting cables.



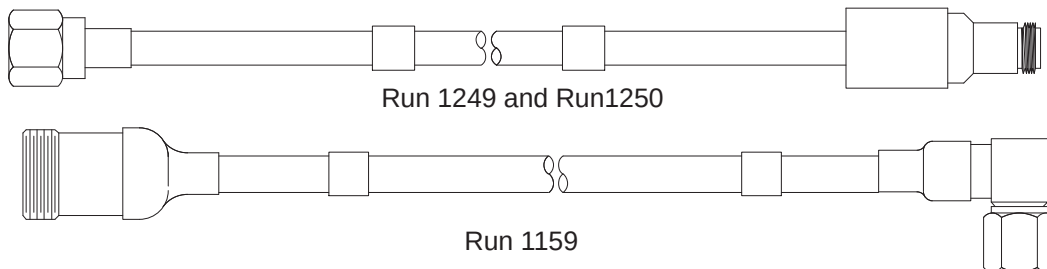
4-11 TIMES MICROWAVE RF CABLE INSTALLATION**4-11-1 CABLE INFORMATION**

Five cables, three extension cables, and other associated parts to be installed are provided for this procedure. Also provided are two Times Microwave stripping tools for LMR 1200 (2384858) and the LMR 600 (2352193), and the Times Microwave cutting tool (5111565).

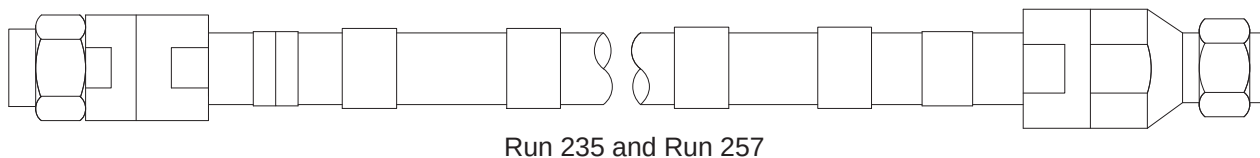


Do not subject LMR-600FR cable (Head) to a bend radius less than 3 inches (75mm). Do not subject LMR-1200FR cable (Body) to a bend radius less than 6.5 inches (165mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

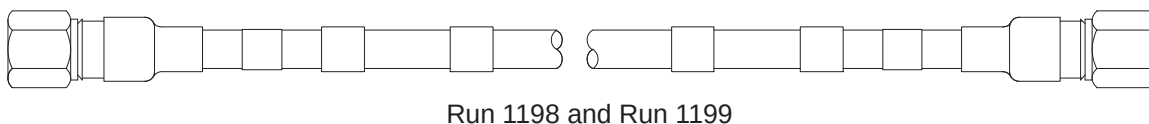
Two extension cables, 5116213-2, Run 1249 (Head) and 5121269, Run 1159 (Body) are connected to the RF Amplifier module in the RF/System Cabinet. The other extension cable, Run 1250, 5121245, is connected to the Body Hybrid Splitter in the Rear Pedestal.



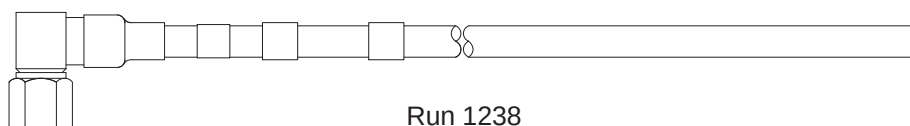
Cable 2316584-2 (Body, Runs 235 & 257), will be cut to length to make two cables and terminated for routing in the equipment and scan rooms.



Cable 5115915 (Head, Runs 1198 & 1199), will be cut to length to make two cables and terminated for routing in the scan room.



Cable 5115916 (Head, Run 1238), will be cut to length and terminated for routing in the equipment room.

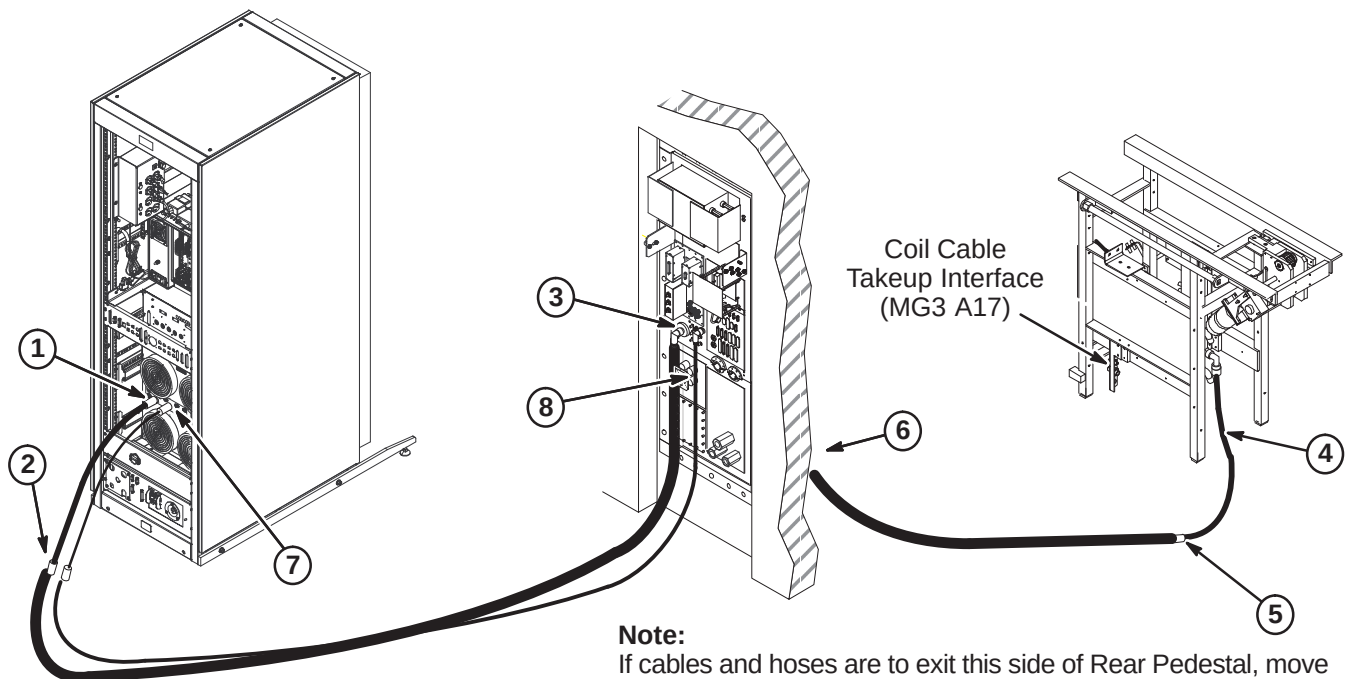


The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

4-11-2 CABLE ROUTING AND TERMINATING INSTRUCTIONS**4-11-2-1 Install Runs 235, 257, 1238, 1249, 1250, and 1259**

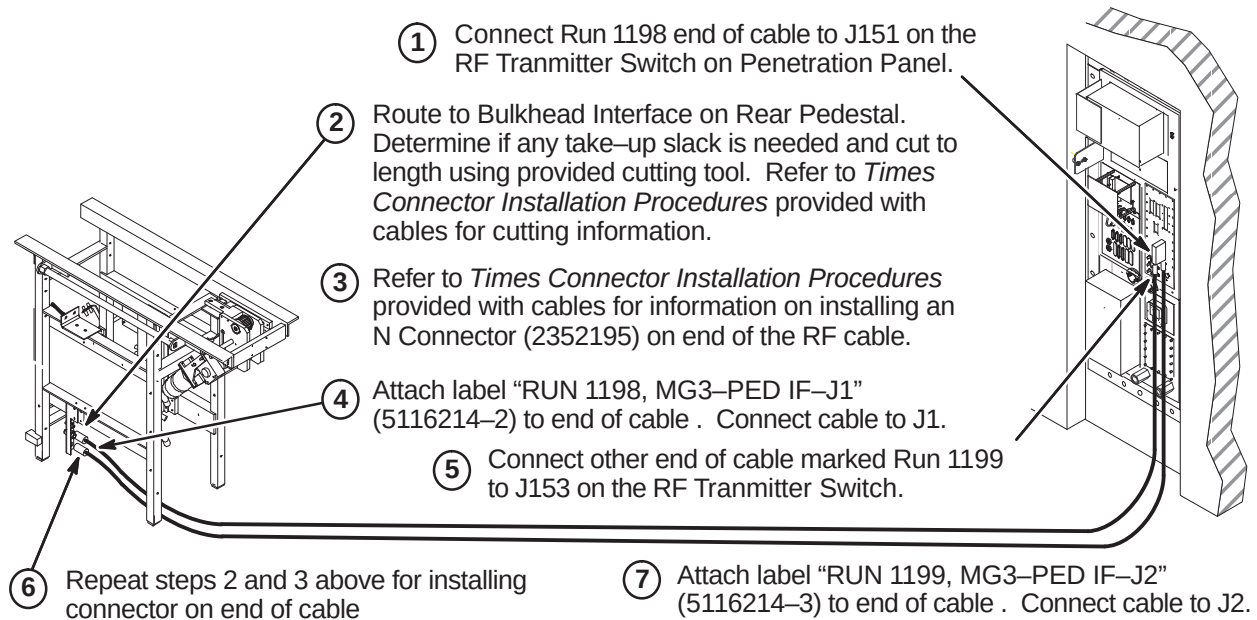
RF cables will be routed and connected as indicated in the following steps.

1. Connect extension cable Run 1259 to J4 Body Output at rear of RF Amp in RFS cabinet.
2. Connect end of Times RF Body cable marked "(RUN 235) FJ5" to Run 1259, Body extension cable. Attach label "RUN 235, Connect to Run 1259" (5121244) over existing label.
3. Route cable to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4-11-2-4** for termination procedure of cut end.
4. Connect Run 1250 to J1 on Hybrid Splitter.
5. Move remainder of RF Body cable to Scan Room and connect other end of cable marked "(RUN 257) MG3A3-J1" to Run 1250 at Rear Pedestal as shown below. Attach label "RUN 257, Connect to Run 1250" (5121244-2) over existing label.
6. Route remainder of cable to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4-11-2-4** for termination instructions of cut end.
7. Connect extension cable Run 1249 to J3 Head Output at rear of RF Amp IN RFS cabinet.
8. Connect end of Times RF Head cable marked "(RUN 1238) PP1-J4" to J4 at Penetration Panel. Route to RF Head extension cable at rear of SRF cabinet. Determine if any take-up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **4-11-2-3** for termination procedure of cut end.

**Note:**

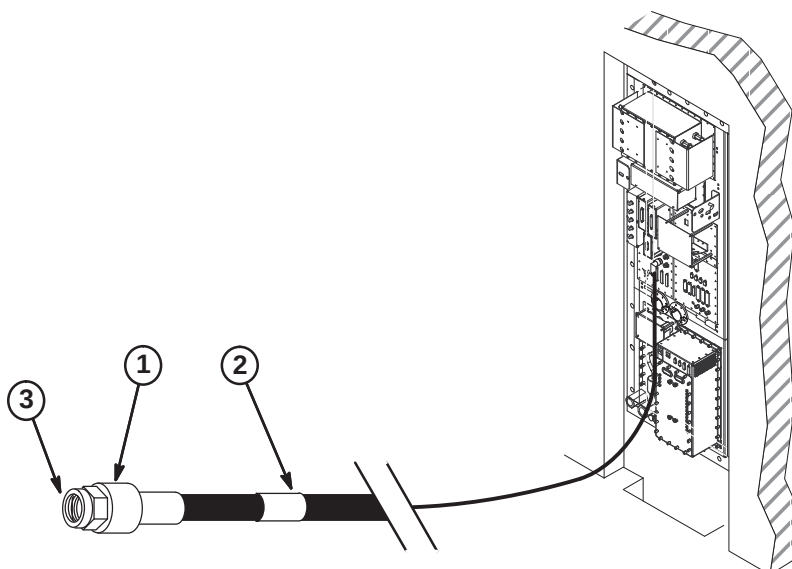
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

4-11-2-2 Install Runs 1198 and 1199**4-11-2-3 Terminate and Connect RF Head Coil Cable**

Perform the following steps for terminating and connecting the RF Head cables in the Equipment room.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing a Straight Male N Connector (2352195) on each end of the RF cable. Use provided cutting tool (5111565) and the provided LMR600 Stripping Tool (2352193).
2. Attach label "RUN 1238 CONNECT TO RUN 1249" (5116214) to cable as shown in Equipment room.
3. Connect cable to Run 1249.



4-11-2-4 Terminate and Connect RF Body Cables at Penetration Panel

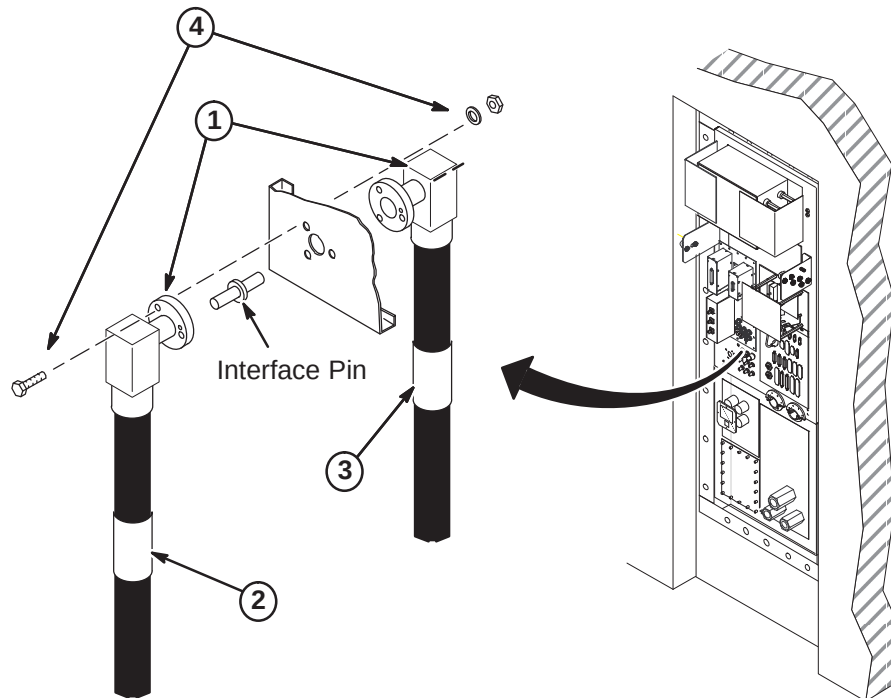
Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2384859) on each end of the RF cables. Use provided cutting tool (5111565) and the LMR1200 Stripping Tool (2384858).

Note

Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

2. Attach label "RUN 235, PP1-J44" (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label "RUN 257, PP1-J44" (2357230-2) to cable as shown on Scan room side of Penetration Panel.
4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
 - 1/4-20 x 1.25 Hex Head Bolt (brass)
 - 1/4 Screw Size Washer (phos bronze)
 - 1/4-20, 7/16" Hex x 3/16" thk Nut (brass)



4-12 PNEUMATIC PATIENT ALERT INSTALLATION

If not already installed, complete the following procedures.

The customer and users should be involved in the following decisions:

- Routing of Pneumatic Tubing
- Location of Control Box

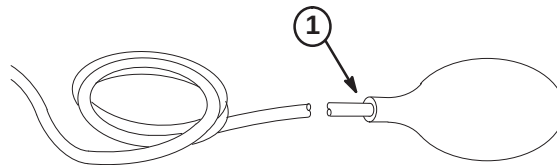
CAUTION

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

4-12-1 ROUTE AND INSTALL PNEUMATIC TUBING

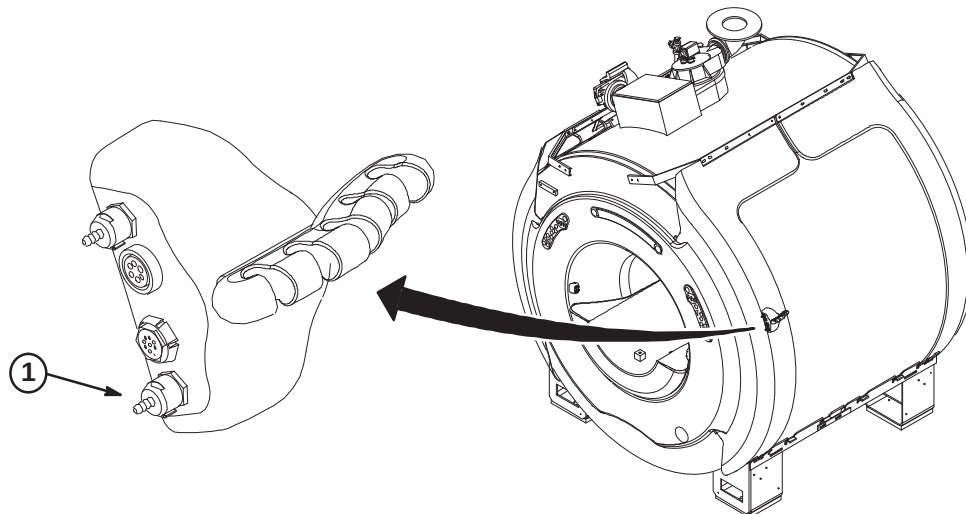
4-12-1-1 Squeeze Bulb Installation

1. Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
2. Before cutting pneumatic tubing from roll to length required, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the "CALL" connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.



4-12-1-2 Connect Tube to Remote PAC Interface Assembly

1. Connect end of tubing to "CALL" connection on Remote PAC Interface Assembly.



4-12-1-3 Magnet Room Routing of Tubing & Wave Guide Installation**CAUTION**

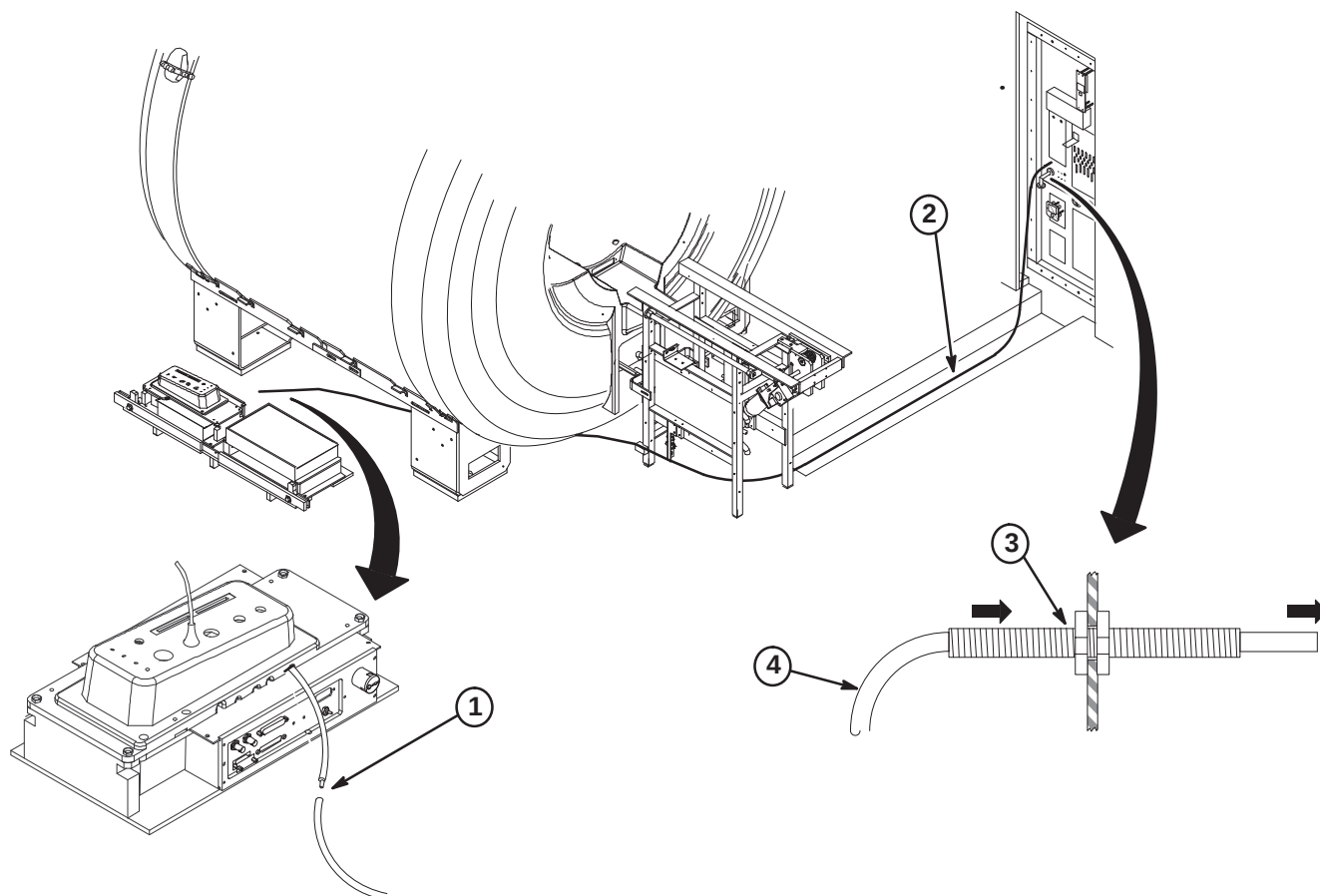
Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

1. Push end of pneumatic tubing onto PAC-II tubing connector.

Note: If the pneumatic tubing is to be pulled thru fiber-optic conduit, coat with baby powder, not grease, to allow easier pulling.

2. Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.
3. Insert Wave Guide thru "ALERT AIR LINE" hole and fasten to Penetration Panel with nut on each side.
4. Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.



4-12-2 DETERMINING NEED OF EXTENDER BOX

If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator's Console), Extender Kit, 46-317758P2, must be ordered.

- **If the pneumatic tubing is long enough to reach control box:**
The extender is not needed. Go to Procedure 4-12-3-1, CONTROL BOX INSTALLATION.
- **If the pneumatic tubing is not long enough to reach control box:**
The extender is needed. Go to Procedure 4-12-2-1, INSTALLATION OF EXTENDER KIT.

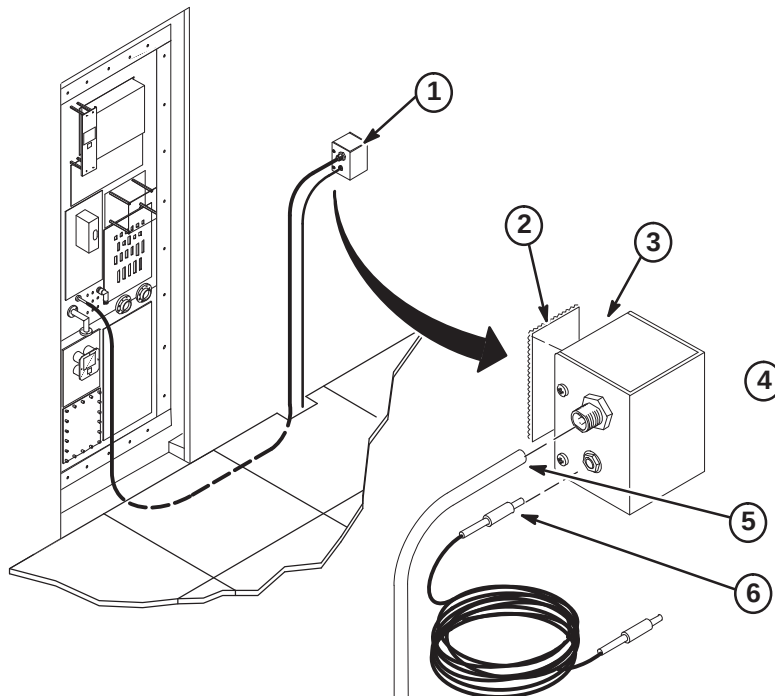
4-12-2-1 Installation Of Extender Kit (46-317758P2)

CAUTION

Do not mount Extender Box in Magnet Room because the box contains ferrous parts.

Install Extender Box

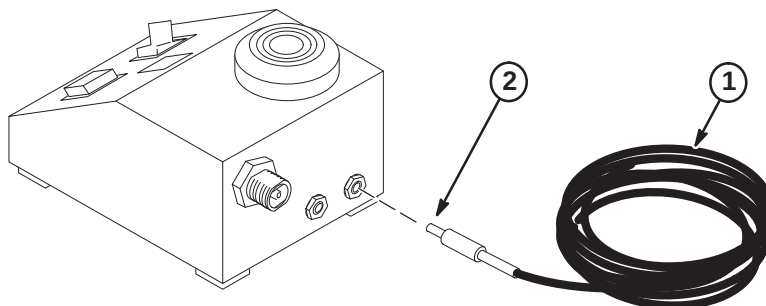
1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed-thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.



4-12-2-1 Installation of Extender Kit (46-317758P2) (Continued)

Connection of Extender Wire to Control Box

1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.



4-12-3 INSTALLATION OF CONTROL BOX

4-12-3-1 Control Box Installation Options

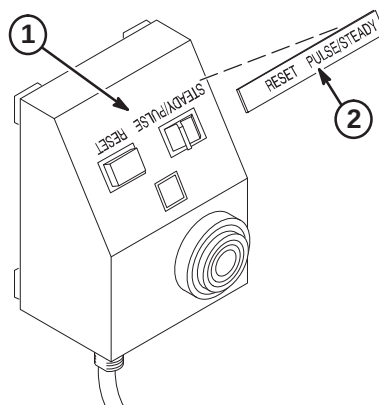
Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

- Mount control box on a wall or other vertical surface per Procedure **4-12-3-2, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF**.
- Mount control box under shelf per Procedure **4-12-3-2, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF**.
- Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Go to Procedure **4-12-3-6, FINAL CONNECTIONS TO CONTROL BOX**.

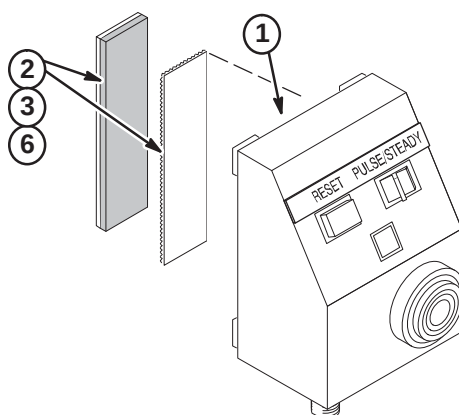
Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

4-12-3-2 Control Box Installation On Wall Or Under Shelf

1. Clean Control Box surface.
2. Remove adhesive backing from metalized nameplate and apply over silk screened lettering as shown.
3. Select one of the following for installation of Control Box:
 - If control box will be mounted to WALL or SHELF with Velcro, go to Procedure **4-12-3-3**
 - If control box will be mounted to WALL with screws, go to Procedure **4-12-3-4**
 - If control box will be mounted to SHELF with screws, go to Procedure **4-12-3-5**

**4-12-3-3 Control Box Mounted With Velcro**

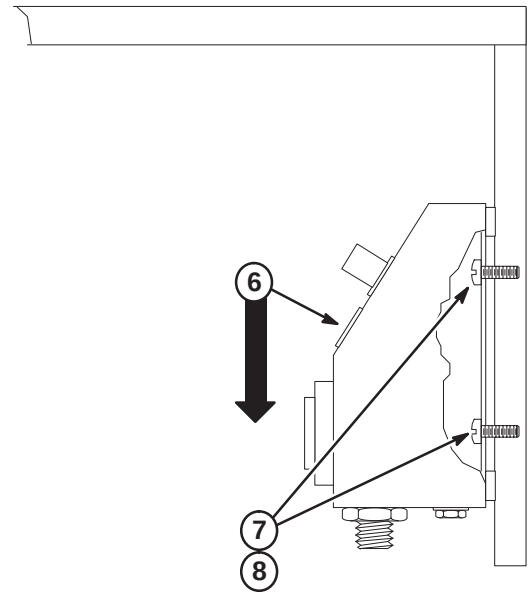
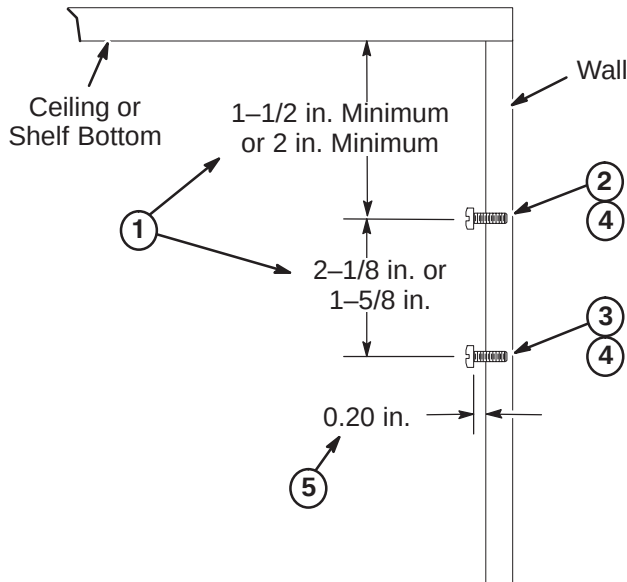
1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. jMount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.
8. Proceed to Procedure **4-12-3-6**, FINAL CONNECTIONS TO CONTROL BOX.



4-12-3-4 Installing Control Box with Mounting Screws on Vertical Surface

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

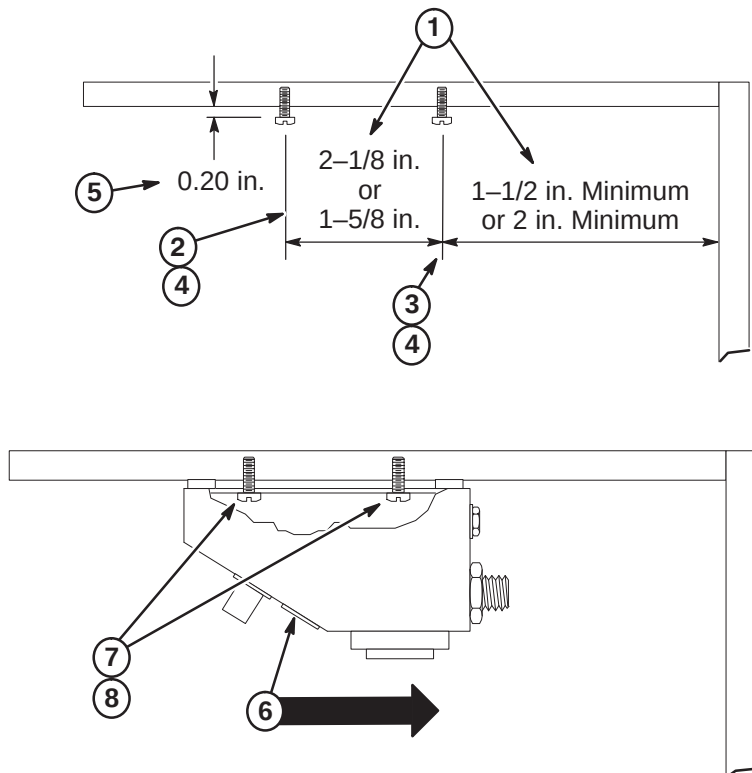
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark top mounting hole position on wall.
3. Mark bottom mounting hole position on wall.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4-12-3-6, FINAL CONNECTIONS TO CONTROL BOX**.



4-12-3-5 Installing Control Box with Mounting Screws Under Shelf

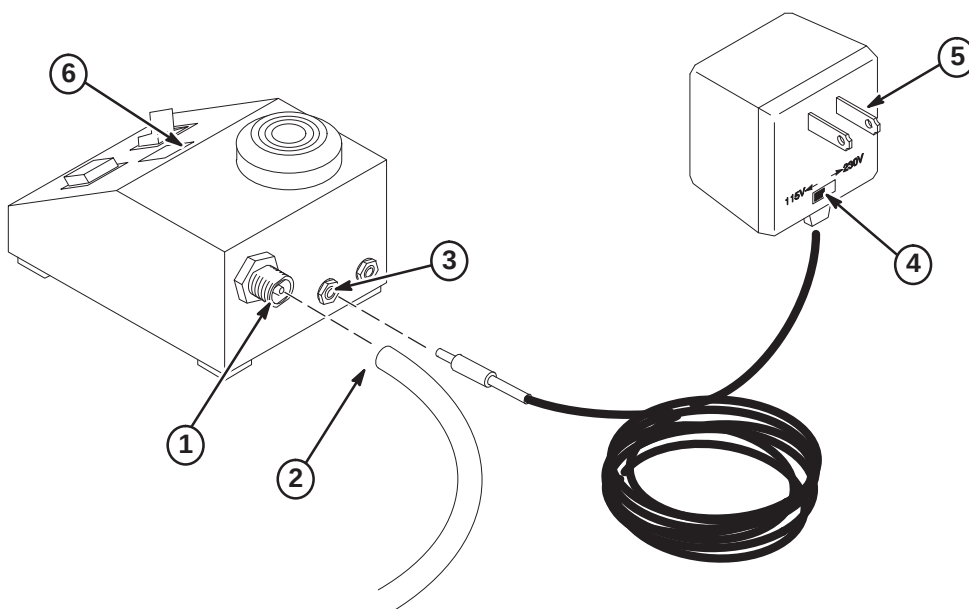
Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark front mounting hole position on underside of shelf.
3. Mark rear mounting hole position on underside of shelf.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push inward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **4-12-3-6, FINAL CONNECTIONS TO CONTROL BOX**.



4-12-3-6 Final Connections to Control Box

1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection.
2. Cut tubing and push fully onto feed-thru connector on back of control box.
3. Plug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is lighted.

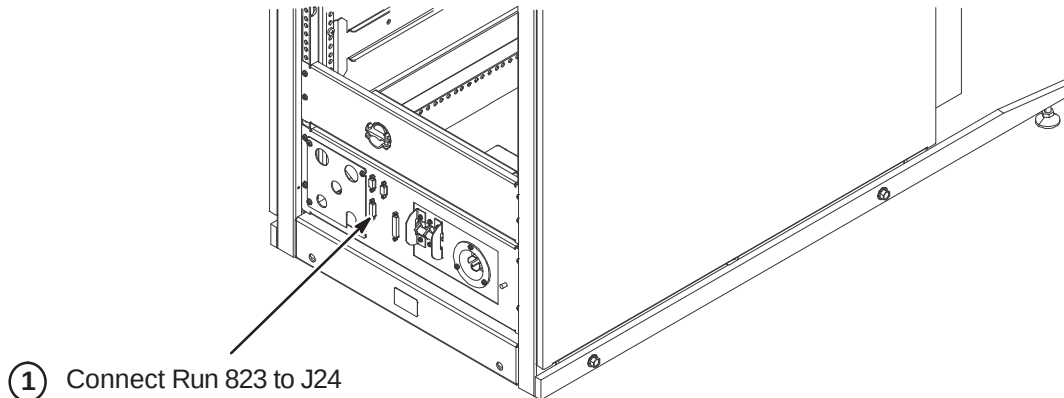


4-13 INSTALLATION OF REMAINING EQUIPMENT ROOM CABLES

Connect remaining Runs within equipment room. Dress and secure all cables after connections are made. Reference interconnect map and Cable Table in Appendix for to/from designator information and Run number reference.

4-13-1 INSTALLATION OF MAGNET MONITORING CABLES

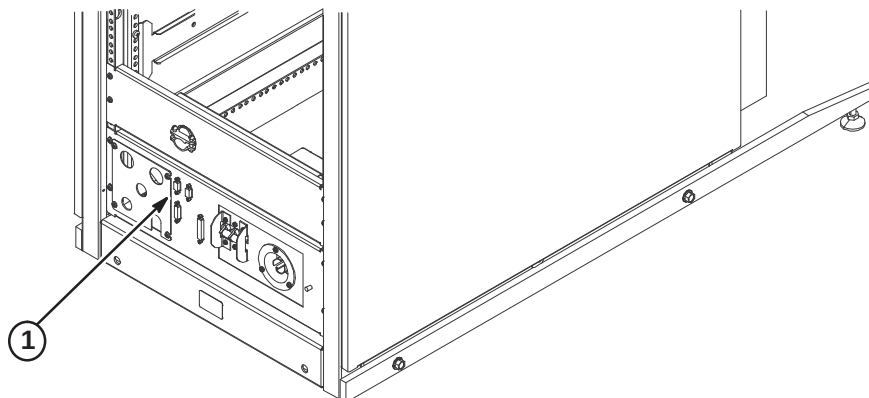
Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual*, for instructions on connecting cables in the RFS Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



4-13-2 INSTALL RUN 701 – RF DOOR SWITCH (If Required)

Run 701 is a 100 ft cable with a 9-pin subminiature "D" plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminiature D plug.

1. Connect Run #701 to J15 on RFS Cabinet and route to RF Door Switch.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.



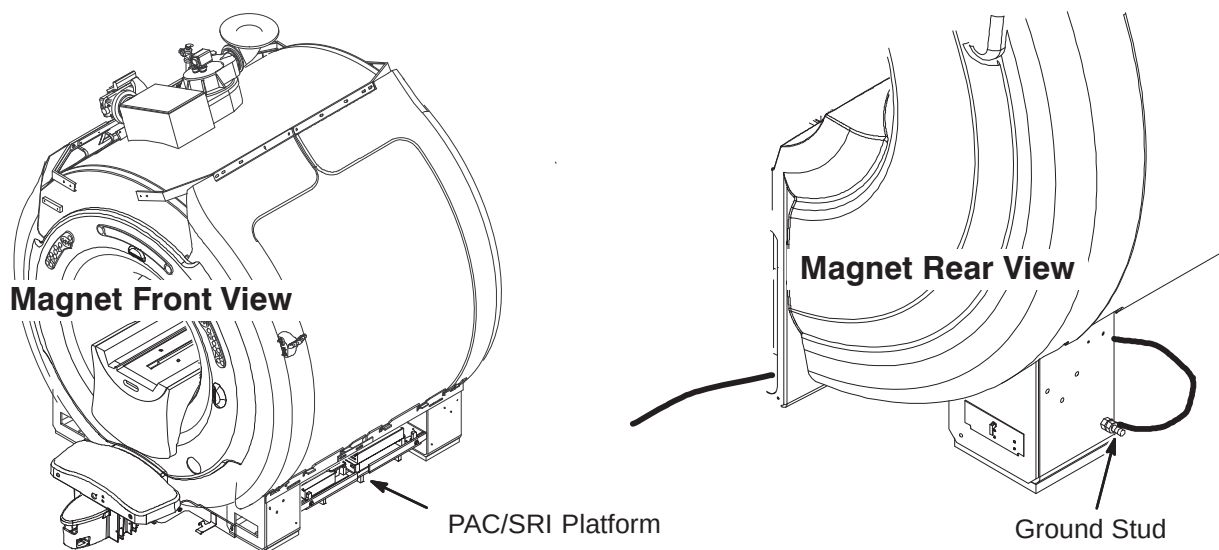
4-14 INSTALLATION OF REMAINING MAGNET ROOM CABLES

The magnet enclosure and magnet room cables must be installed simultaneously. Cable locations depend on how enclosures are installed. Refer to Section 6 for complete magnet room installation of cables.

WARNING!

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.

The PAC/SRI Platform is located on right side of magnet. The Magnet ground cable, RUN 040, has to be connected to the opposite side rear left leg as shown above. The other end of Run 040 is connected to the RF Screen Room common ground.



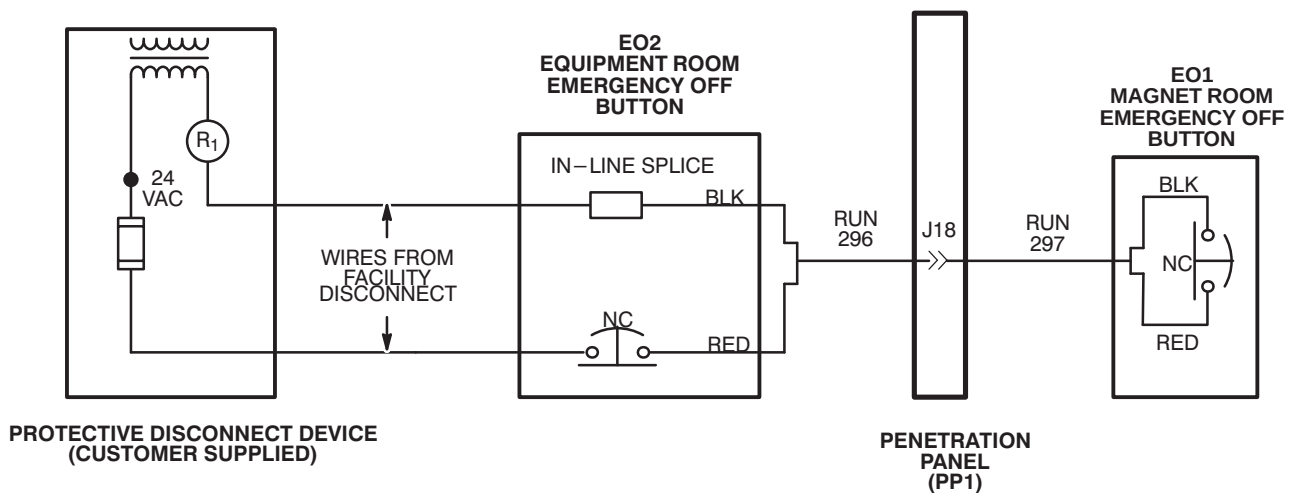
4-15 EMERGENCY OFF CONNECTIONS

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to *Direction 5133302, Signa EXCITE 1.5T TwinSpeed Pre-Installation*, Section 5, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

Note: In Illustration below, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

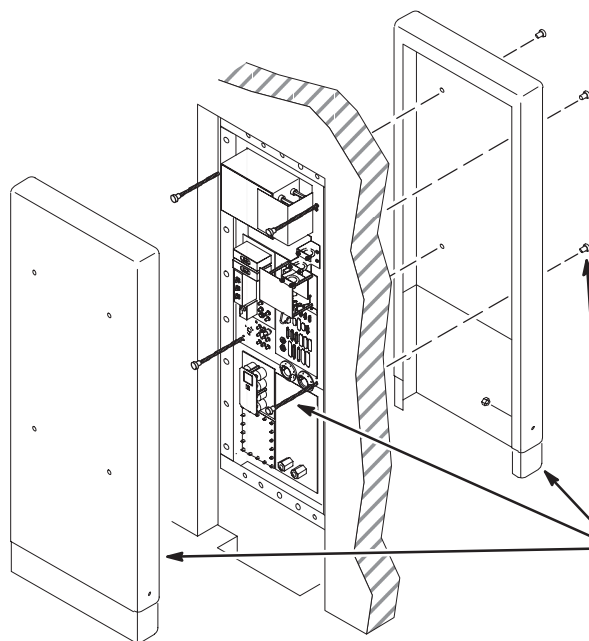
2. At equipment room room "Emergency Off" (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2).
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2).
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push-on terminal.
5. Terminate black wire at end of Run 296 with local supplied "push-on" terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in-line splice.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1).



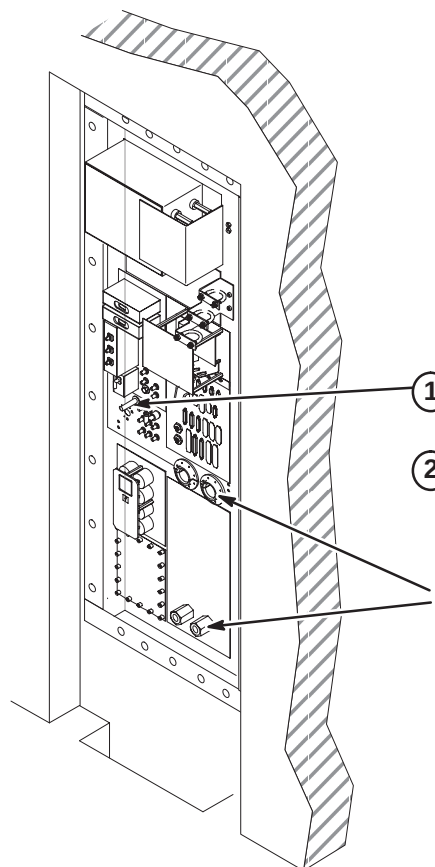
SECTION 5 – PENETRATION PANEL

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5-1 REMOVE PENETRATION PANEL COVERS

- ① If installed, unfasten and save Penetration Panel covers, stand-offs, and mounting hardware.
- ② Move above items to a safe storage area. They will be reinstalled at end of upgrade.

5-2 REMOVE PARTS FOR TRANSFER TO NEW PENETRATION PANEL

- ① Disconnect and remove Patient Alert tubing. Transfer Patient Alert Wave Guide to new Penetration Panel re-using mounting hardware.
- ② If Patient Alert Extender Box is mounted on Penetration Panel, remove and re-install on new Penetration Panel.

CAUTION! - OLDER SYSTEMS USED BERYLIUM PASTE ON WAVE GUIDES AND GAS LINES. WASH HANDS IF YOU COME IN CONTACT WITH THIS MATERIAL.

5-3 DISCONNECT COOLANT HOSES

CAUTION

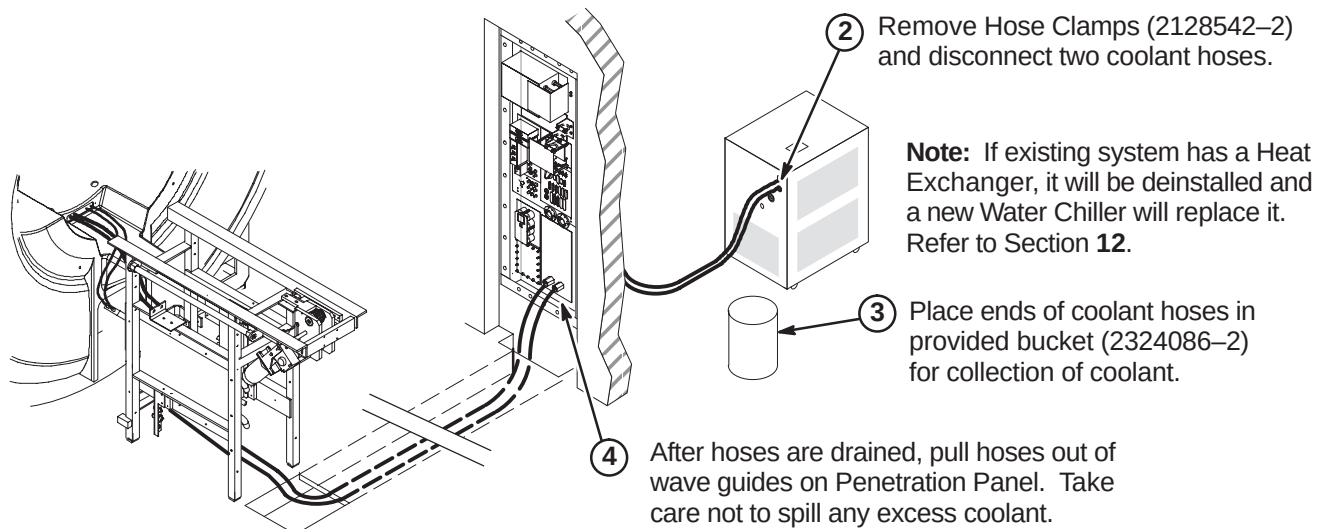
Coolant spillage possibility. The gradient coil cooling system is pressurized. Turn off the heat exchanger or water chiller before disconnecting the hoses. This takes pressure off the system and helps avoid coolant from being sprayed all over the immediate area.

WARNING!

POSSIBLE EYE INJURY! DRAINING OF COOLANT (ANTI-FREEZE) MAKES SPLASHING POSSIBLE. THIS COULD CAUSE EYE INJURY. BE SURE TO WEAR LATEX GLOVES AND SAFETY GLASSES.

Note: Four hands are better than two – While it is possible for one person to disconnect the coolant hoses, it is easier if two people do it. There would be less chance of spilling the coolant.

- ① Prior to disconnection of coolant hoses, calculate the amount of coolant that will be drained from system to determine if the 5 gallon bucket provided with upgrade will be sufficient for collection of coolant to be drained. Approximately two gallons of coolant will remain in gradient coil after disconnection. If more containers are needed for collection of drained coolant, they must be obtained locally.

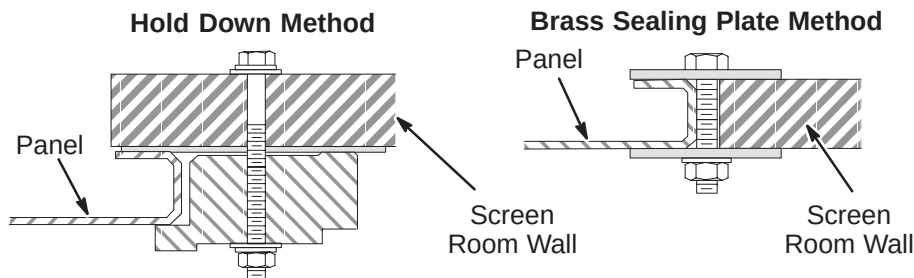


5-4 DEINSTALL PENETRATION PANEL

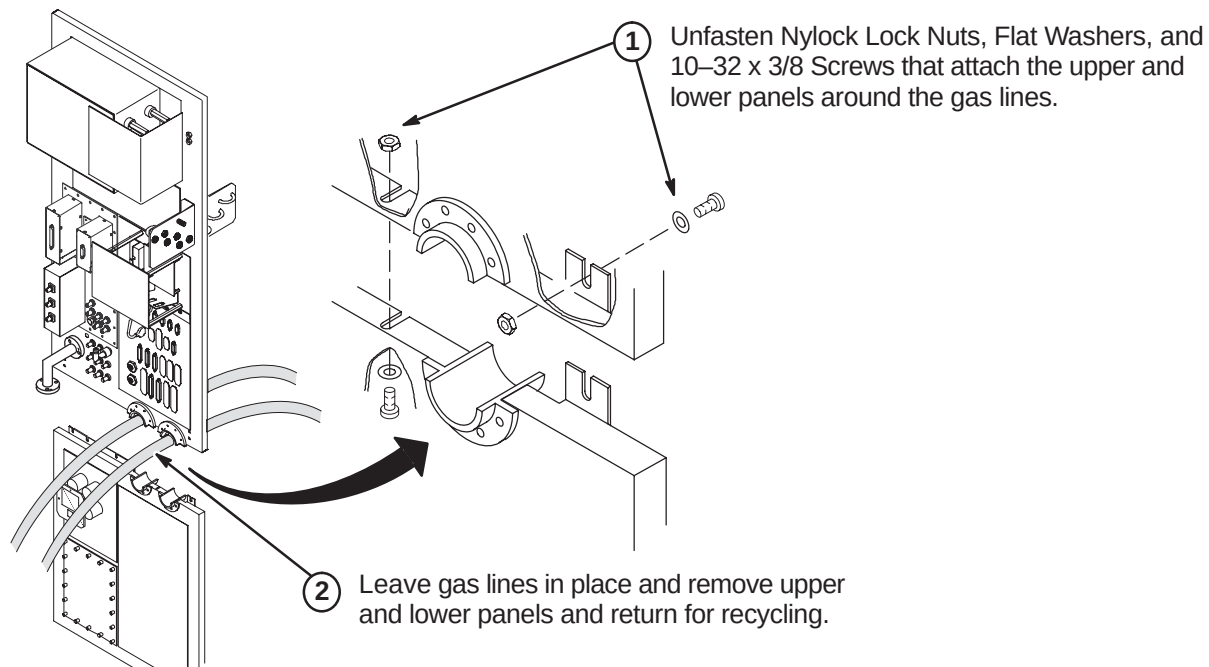
- ① Disconnect all cables attached to both sides of Penetration Panel.

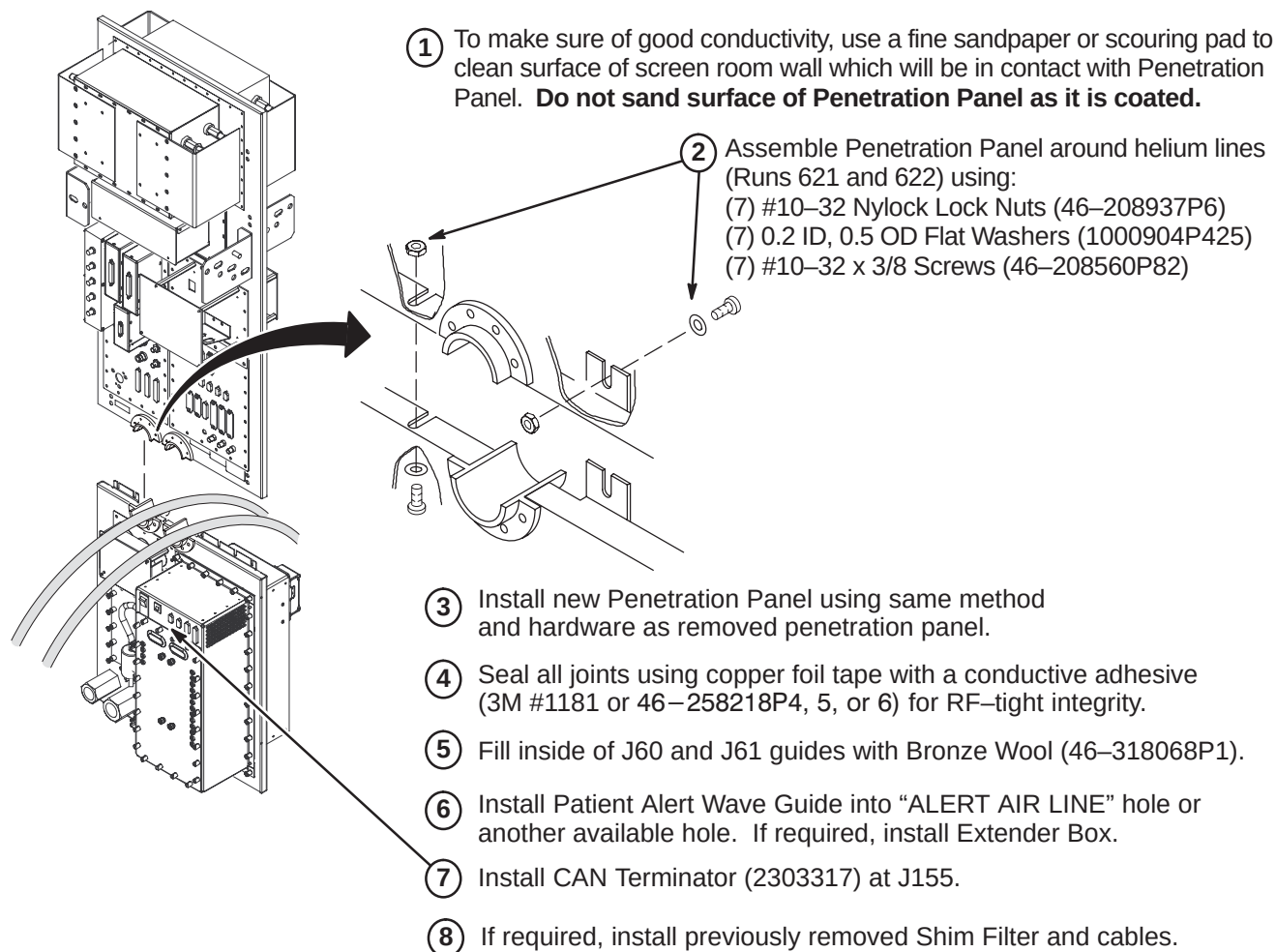
Note: Refer to Cable Section in this manual to determine which cables are retained and which are returned to GE for recycling.

- ② The two helium gas lines will be reused and **MUST NOT** be disconnected. Remove and save Bronze Wool (46-318068P1) from inside of J60 and J61.
- ③ Deinstall Penetration Panel by removing the mounting hardware located on outside edge of panel. Two different methods are shown below.



- ④ Save the grade 8 bolts and related mounting hardware for installing the new Penetration Panel.

5-5 UNFASTEN PENETRATION PANEL AROUND HELIUM GAS LINES

5-6 INSTALL EXCITE HD PENETRATION PANEL**5-7 CONNECT CABLES TO PENETRATION PANEL**

For information and instructions on cable and hose connections at the Penetration Panel, refer to applicable:

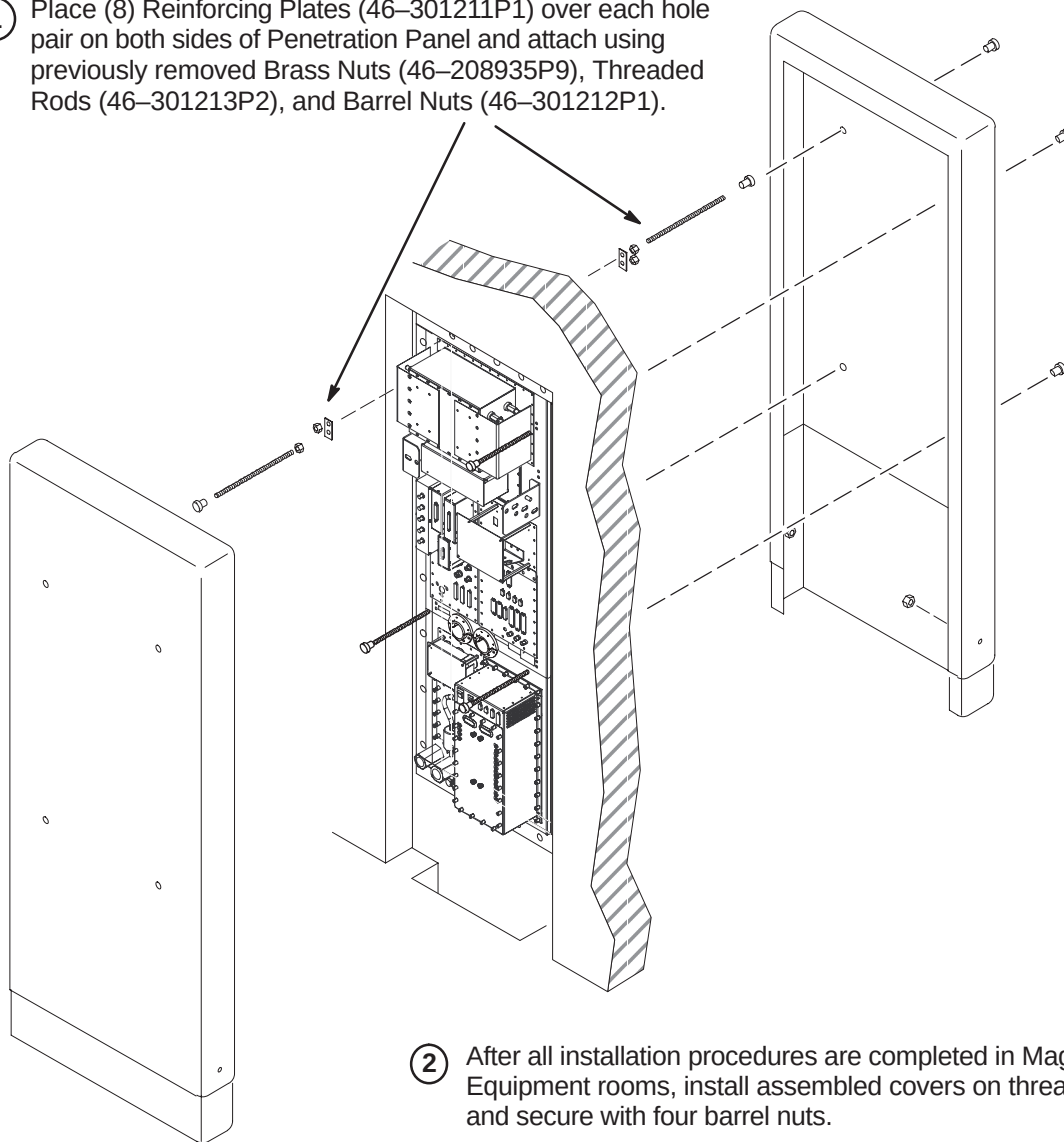
- Installation Flowchart in Section 1
- Cable Interconnect Map in Section 4
- Cable and System Installation in Sections 4 and 6 .
- Coolant Hose installation in Section 6

Make sure to reinstall all customer added option connections.

5-8 INSTALL PENETRATION PANEL COVERS (If Previously Removed)**Note**

The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

- ① Place (8) Reinforcing Plates (46-301211P1) over each hole pair on both sides of Penetration Panel and attach using previously removed Brass Nuts (46-208935P9), Threaded Rods (46-301213P2), and Barrel Nuts (46-301212P1).



- ② After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

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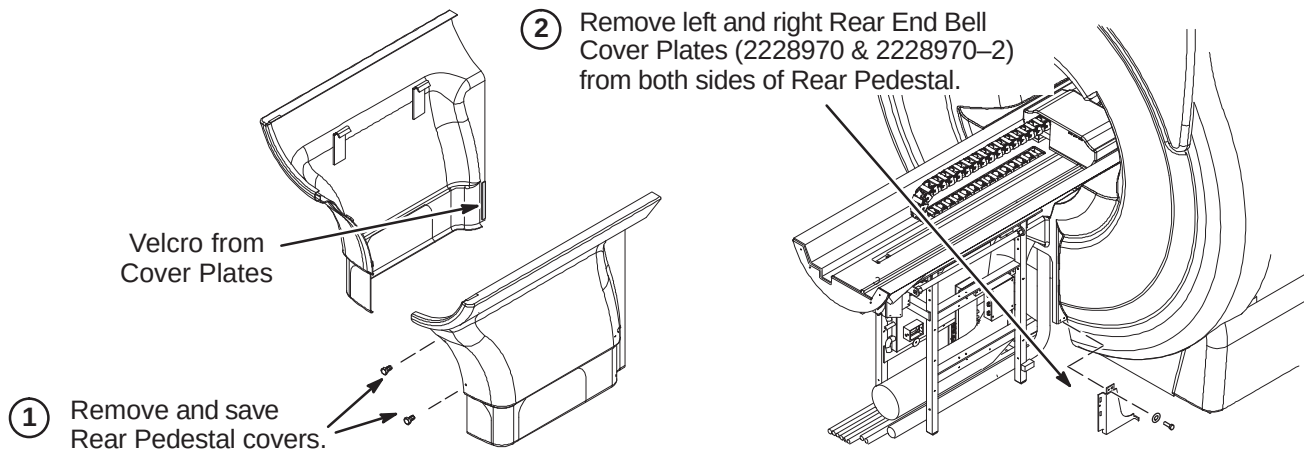
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6-1 DE-INSTALL WideOpen ENCLOSURE

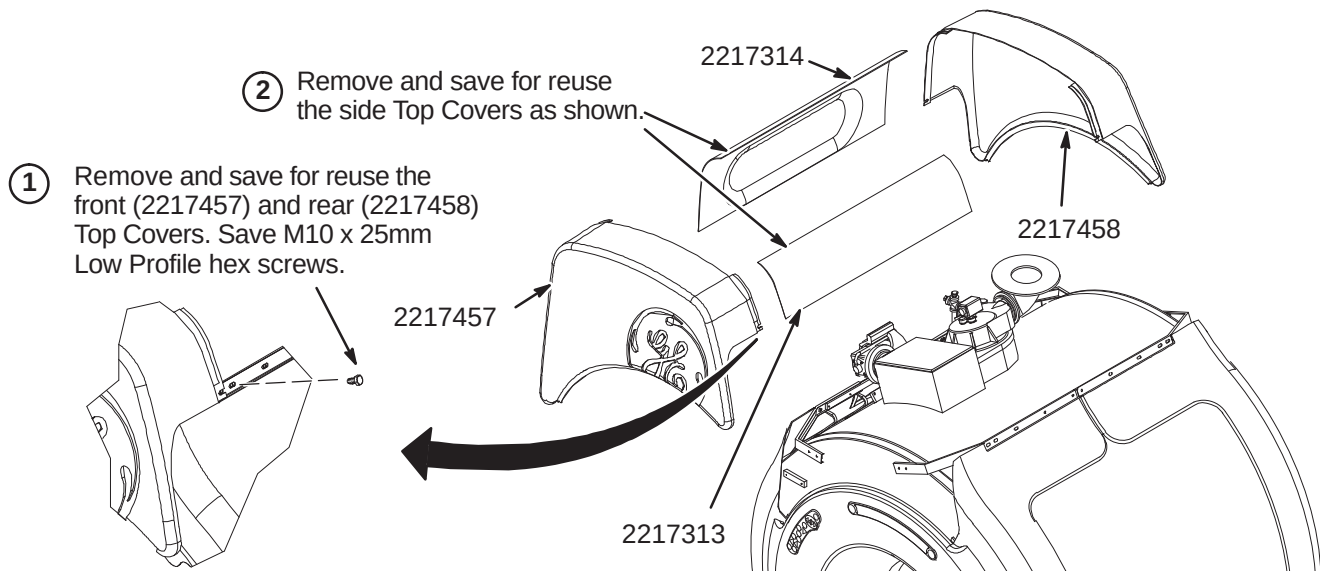
Refer to Section 1-7 for instructions on returning all material and components de-installed and not to be used for the upgrade. Also refer to applicable "Removal" cable map for information on cables to be de-installed and returned for recycling.

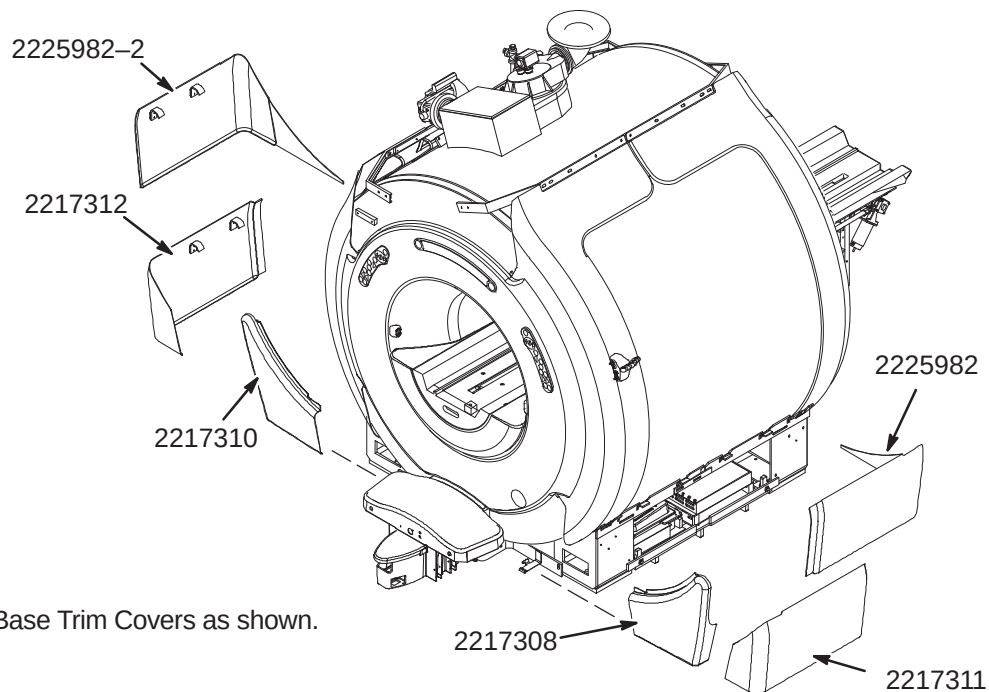
6-1-1 DE-INSTALL ENCLOSURE COVERS

6-1-1-1 Remove Rear Pedestal Covers

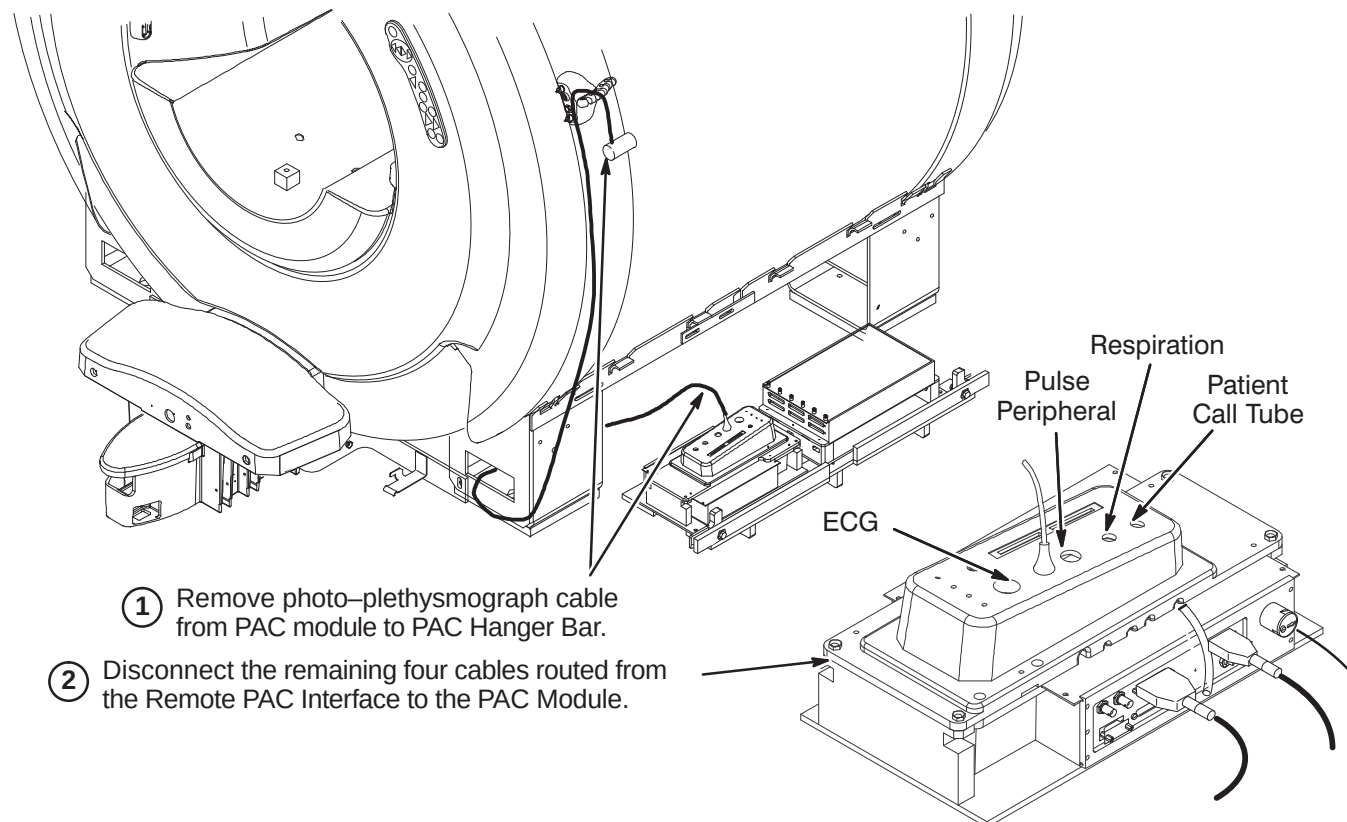


6-1-1-2 Remove Enclosure Top Covers



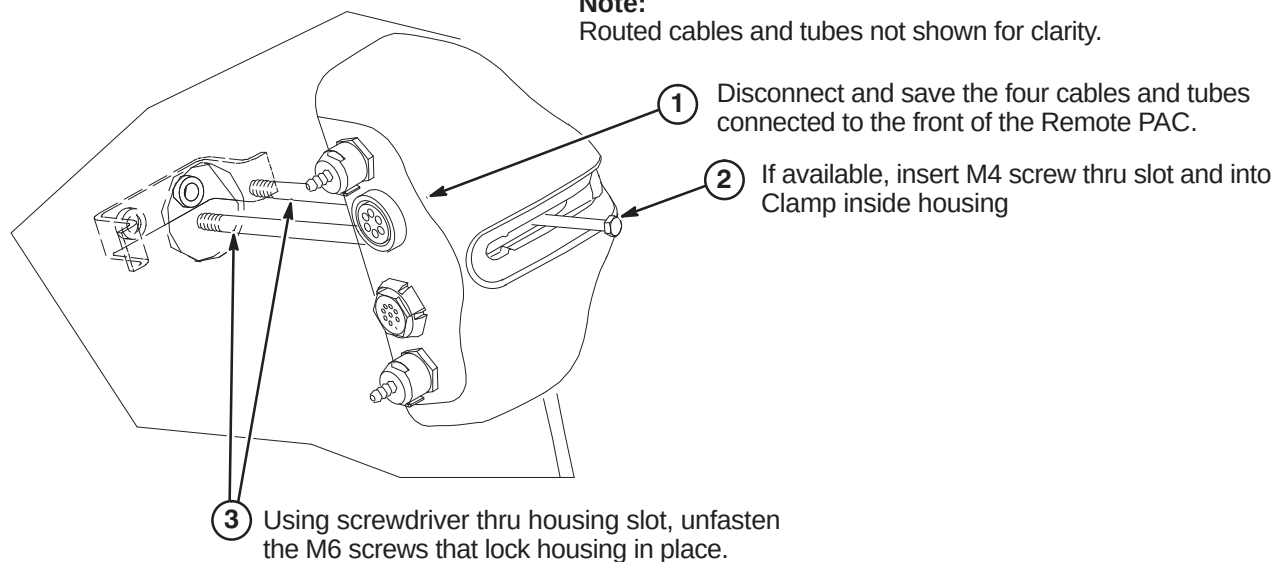
6-1-1-3 Remove Enclosure Base Trim Covers

- ① Remove remaining side Base Trim Covers as shown.

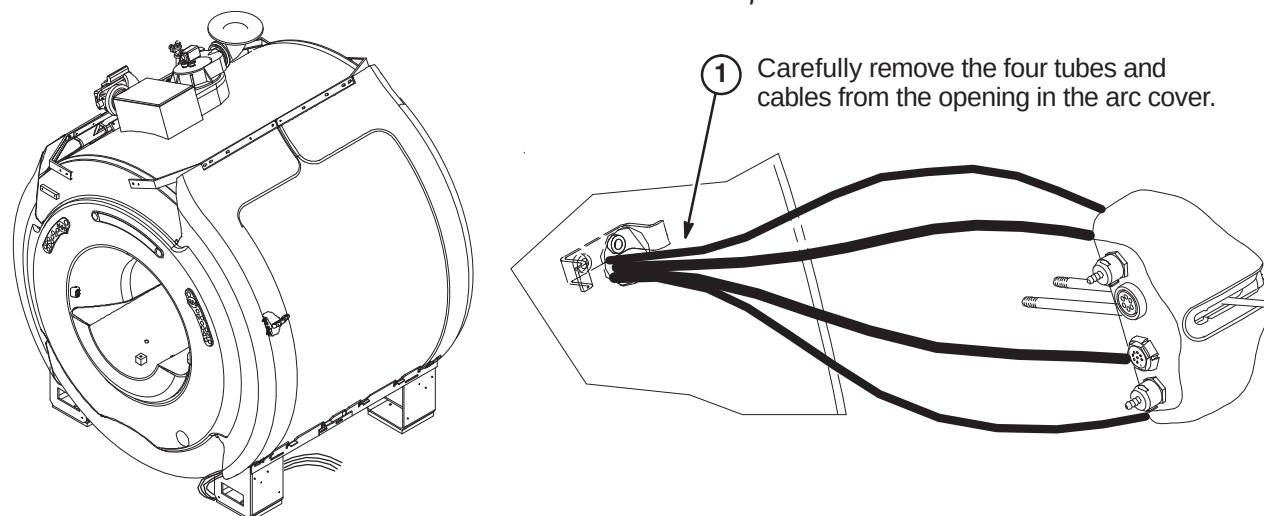
6-1-1-4 Disconnect Cables To PAC Module

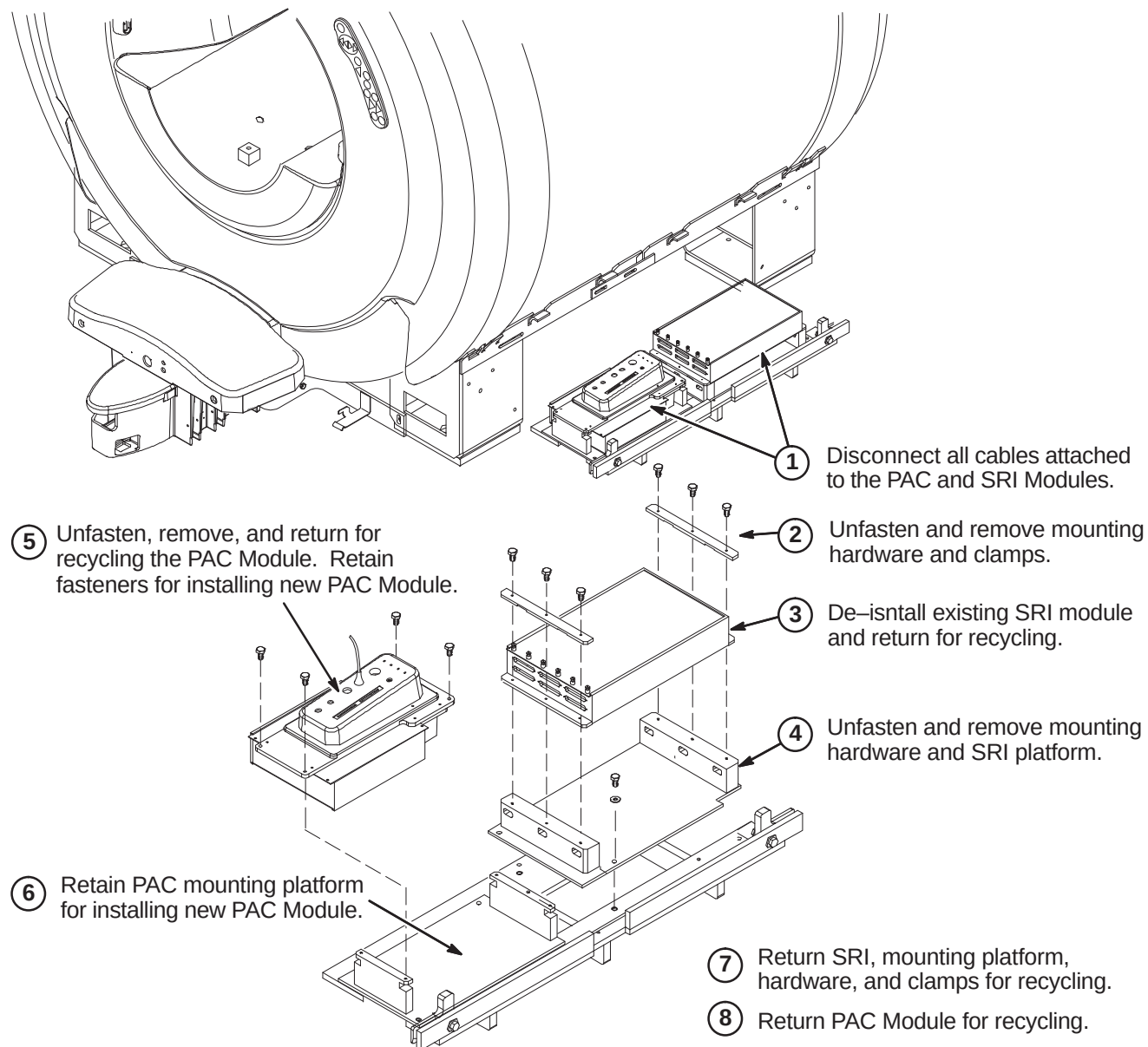
6-1-1-5 Removing Remote PAC interface to Arc Cover**Note:**

Routed cables and tubes not shown for clarity.

**6-1-1-6 Removing Remote PAC Interface Cables And Tubes****Note:**

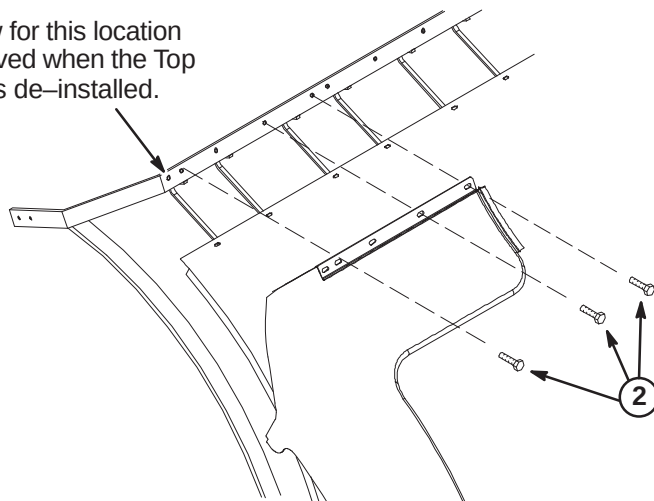
The ECG, Pulse Peripheral, Respiration, and Patient Call Tube will be reused on new TwinSpeed enclosure.



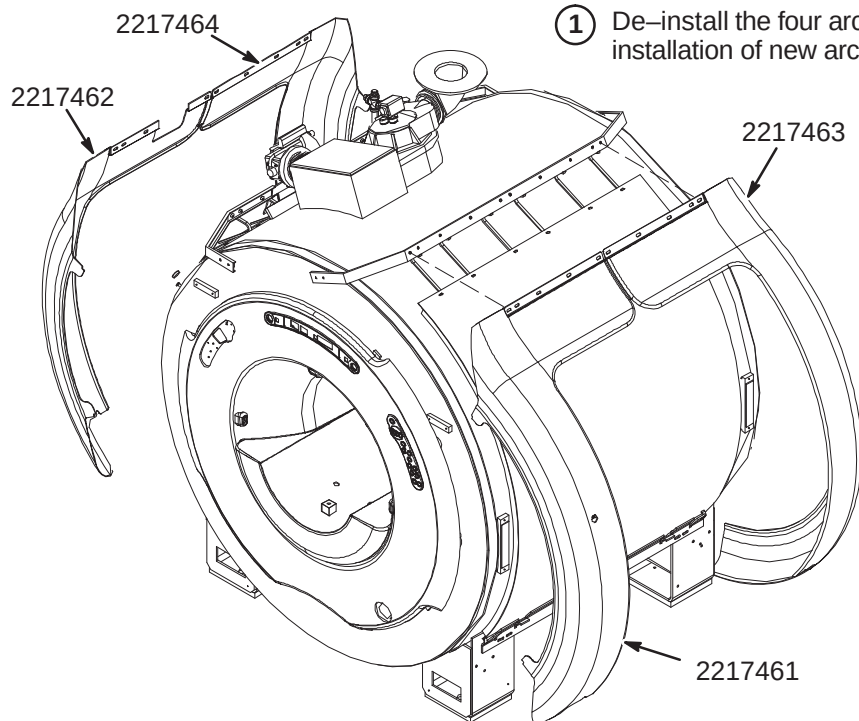
6-1-1-7 De-Install SRI/PAC Module

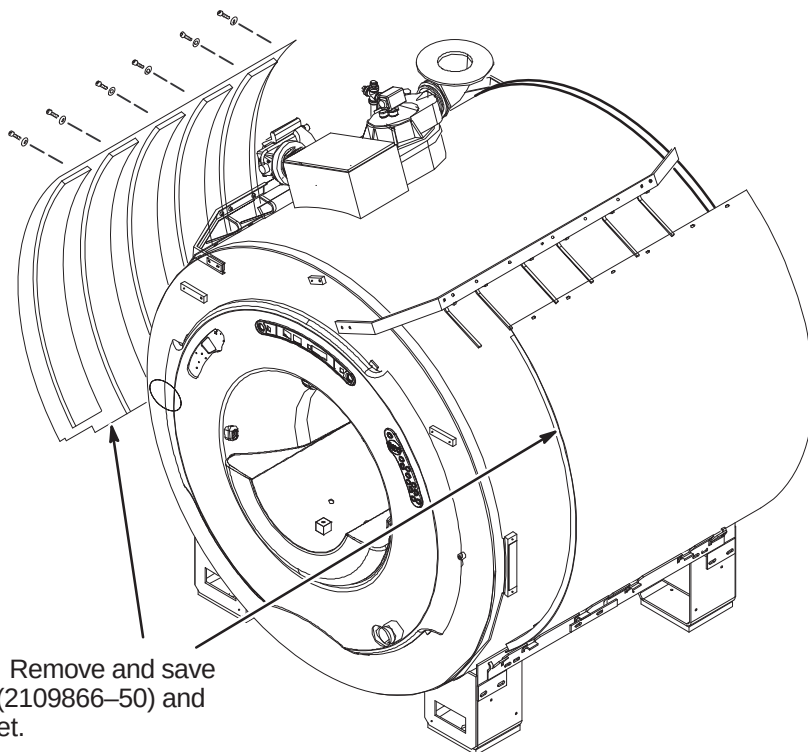
6-1-1-8 Unfasten Top Of Arc Cover

- ① The screw for this location was removed when the Top Cover was de-installed.

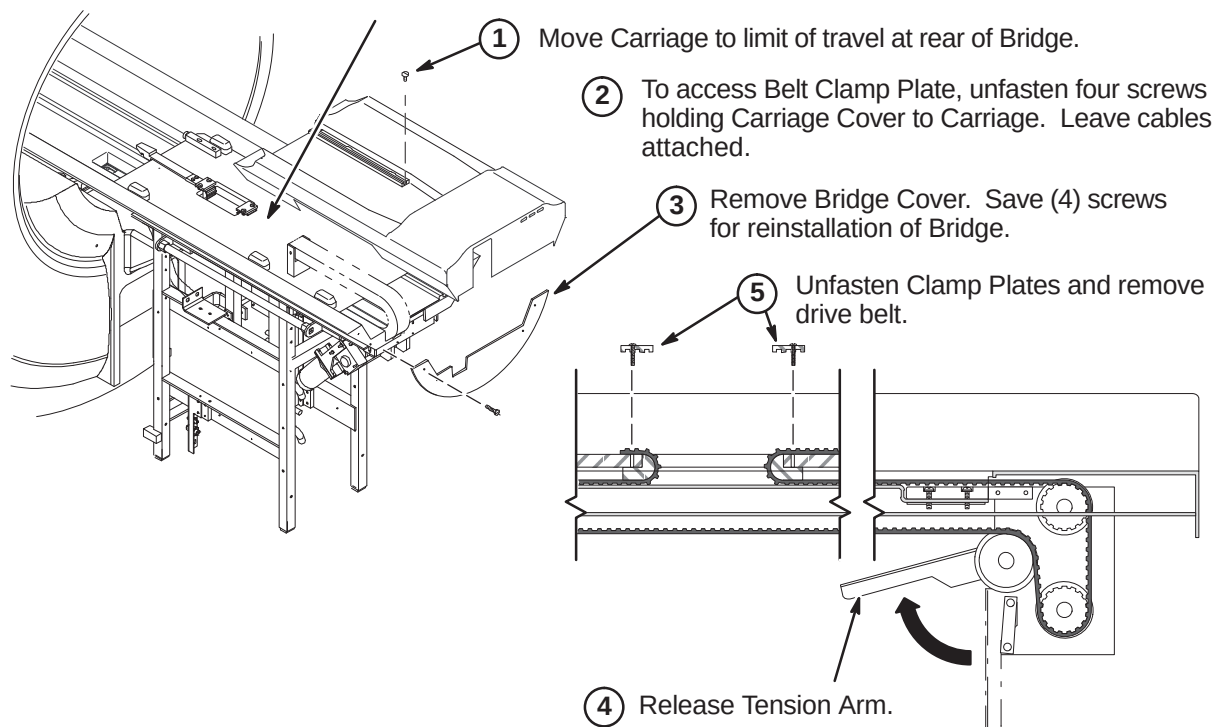
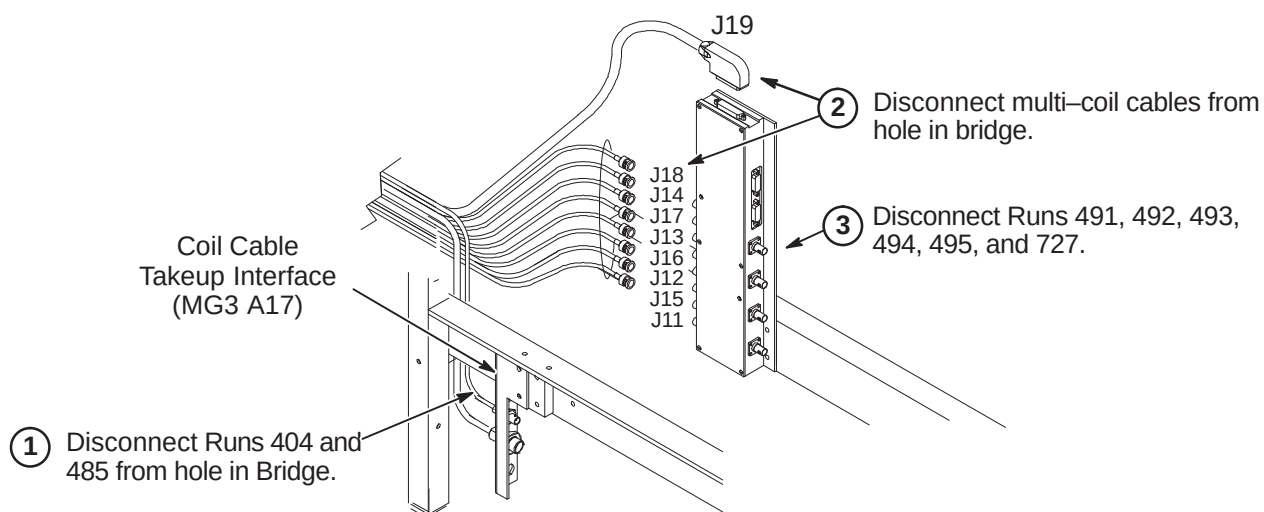
**6-1-1-9 De-Install Arc Cover**

- ① De-install the four arc covers. Save mounting hardware for installation of new arc covers.



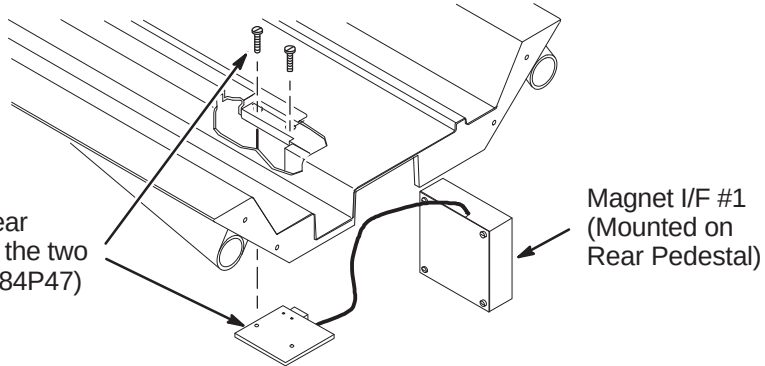
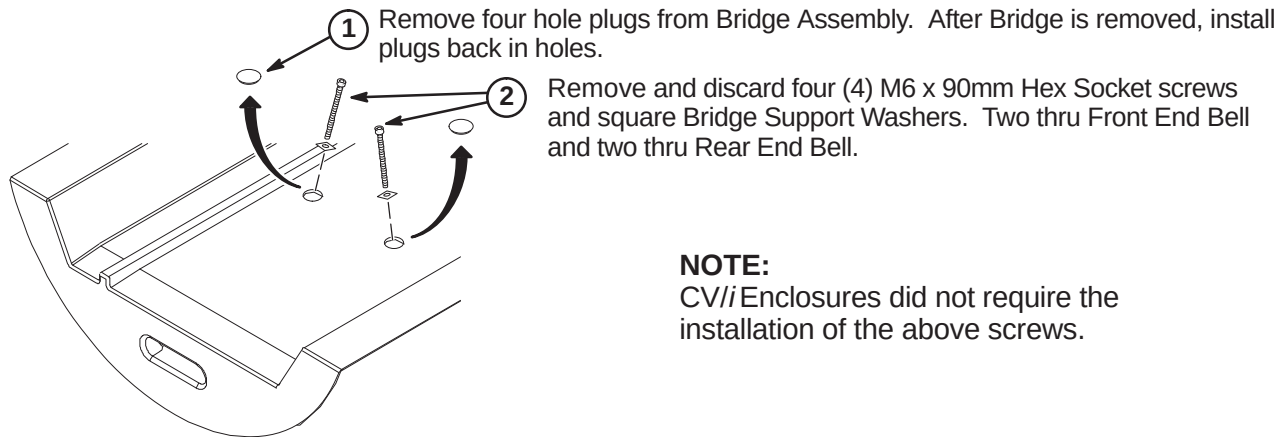
6-1-1-10 De-Install Right And Left Side Wrap Covers

- ① De-install the left and right Side Wraps. Remove and save the six M10 x 14mm Hex Head Screws (2109866-50) and washers that attach each wrap to bracket.

6-1-2 DE-INSTALL BRIDGE**6-1-2-1 Remove Bridge Cover and Drive Belt****6-1-2-2 Disconnect Runs 404, 485 And Multi-Coil Cables**

6-1-2-3 Disconnect Bridge Mount ASM

- ① Unfasten the Bridge, Tongue and Rear Bridge Mount Asm (2132034). Save the two (2) 10-32 x 1-3/8 Screws (46-220184P47) for refastening later.

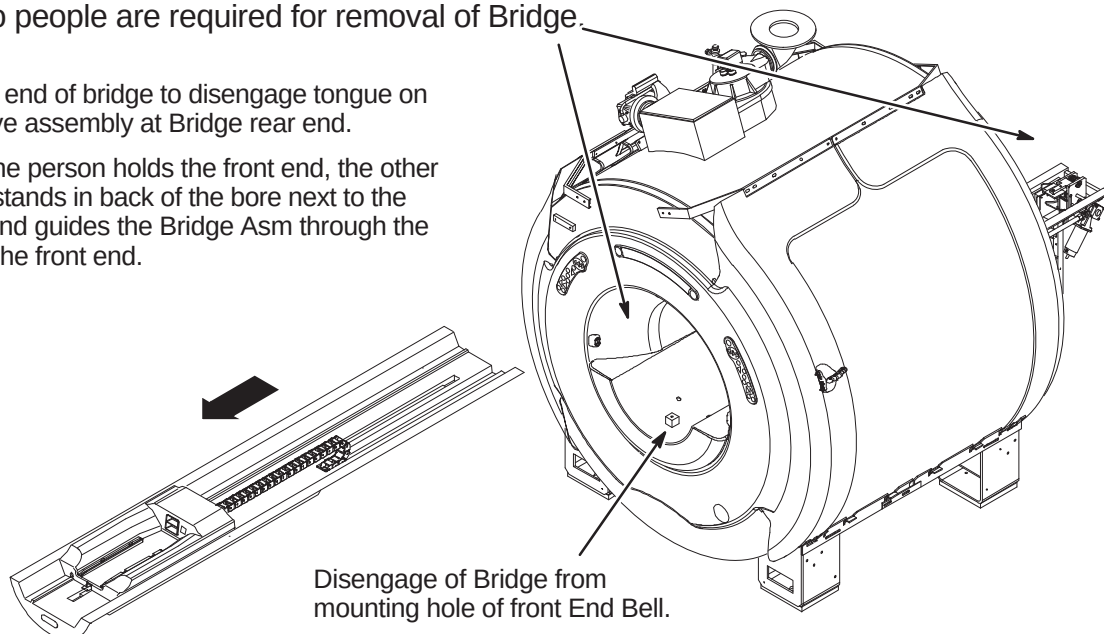
**6-1-2-4 Unfasten Saddle Bracker Screws (MR *li* Only)**

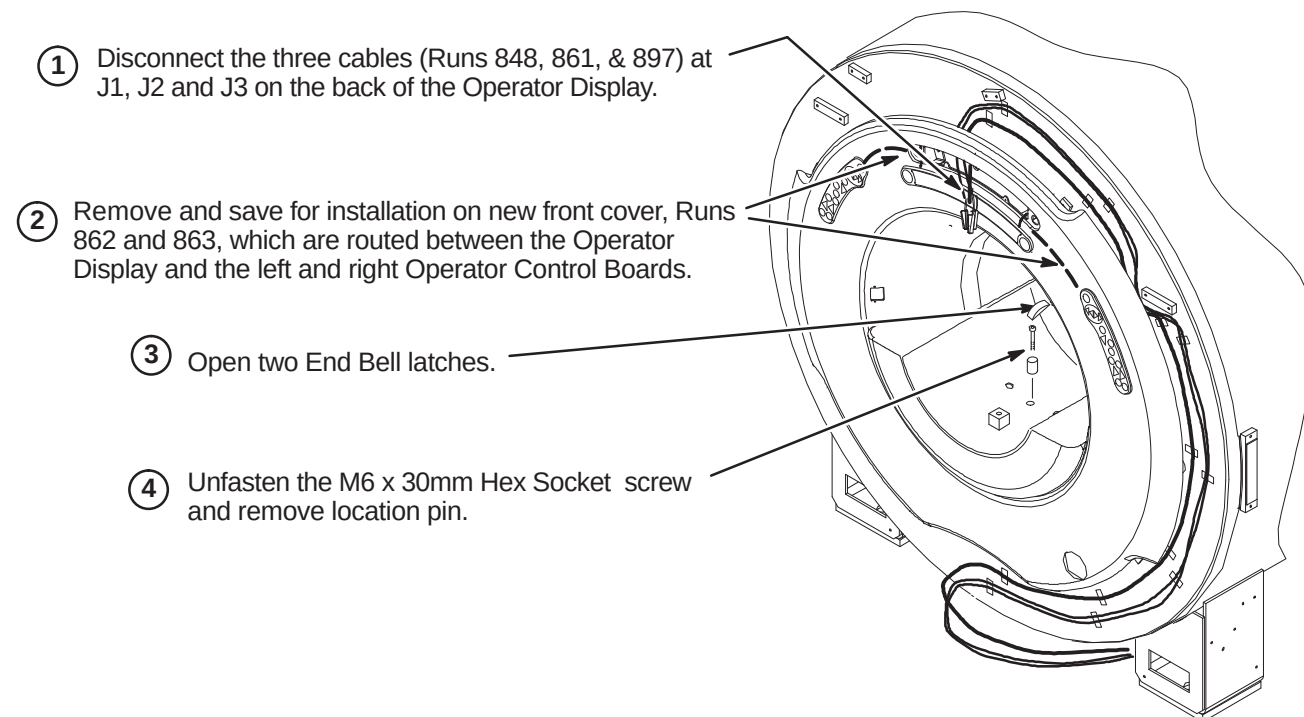
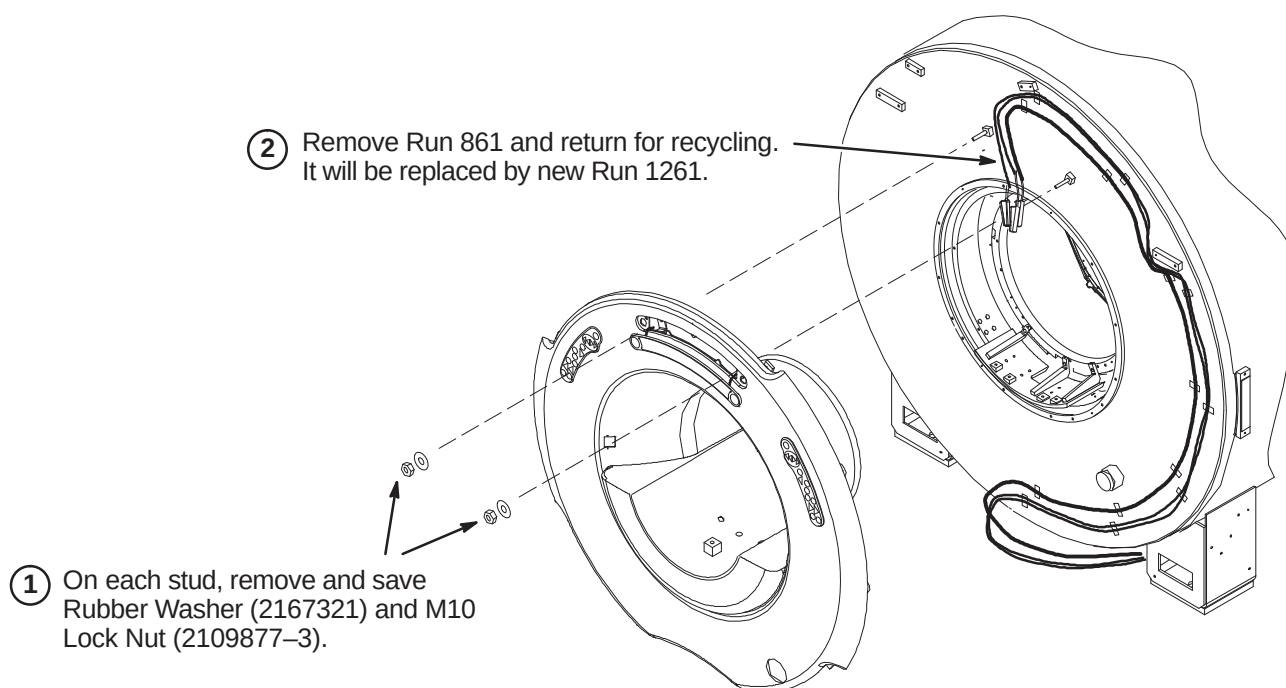
NOTE:
CV/*i* Enclosures did not require the installation of the above screws.

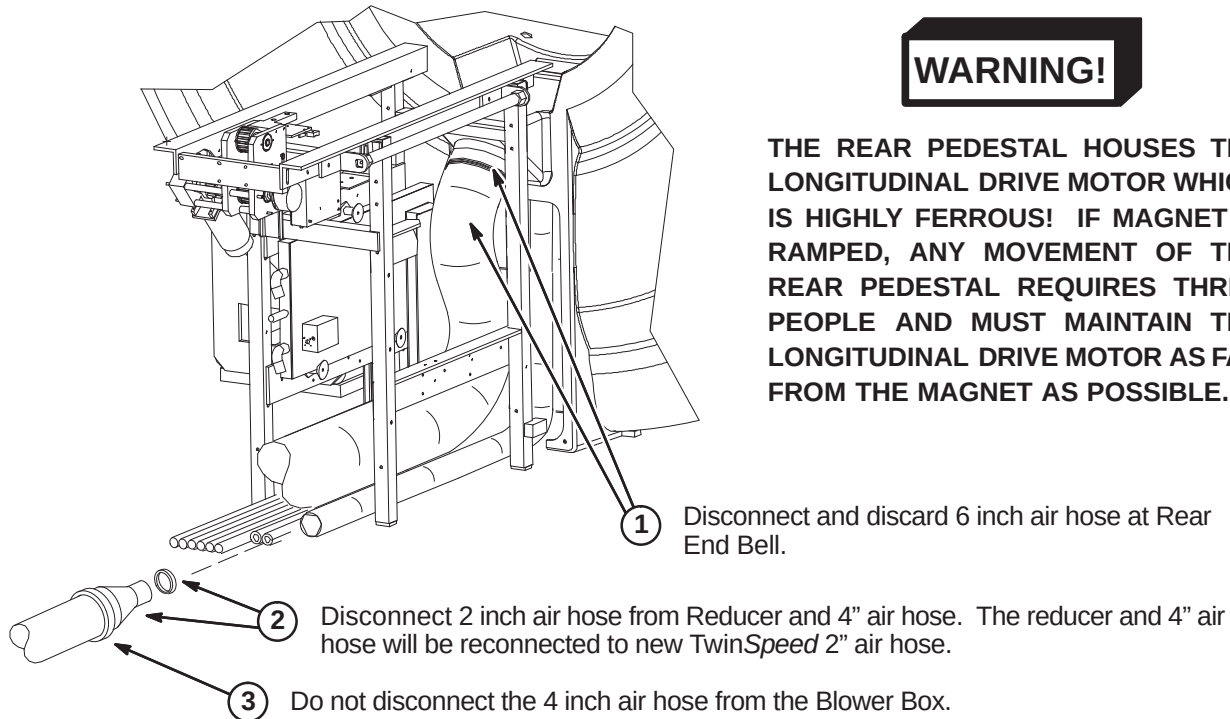
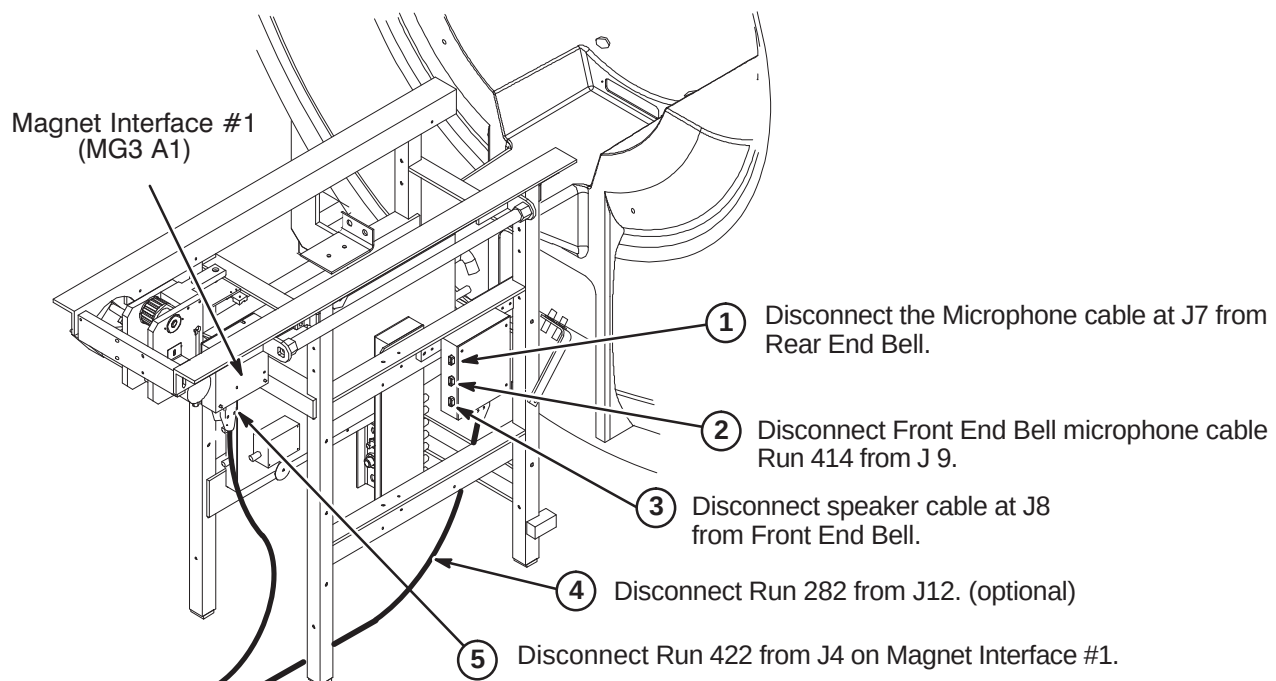
6-1-2-5 Remove Bridge

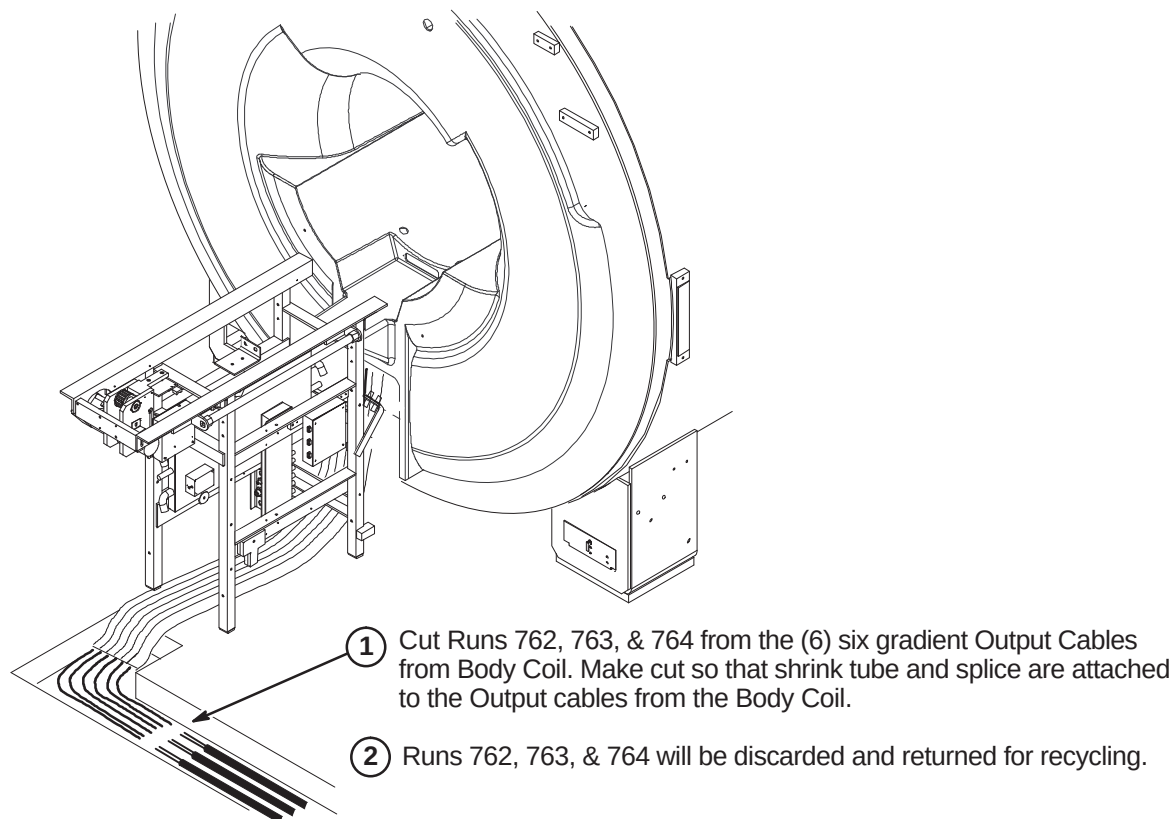
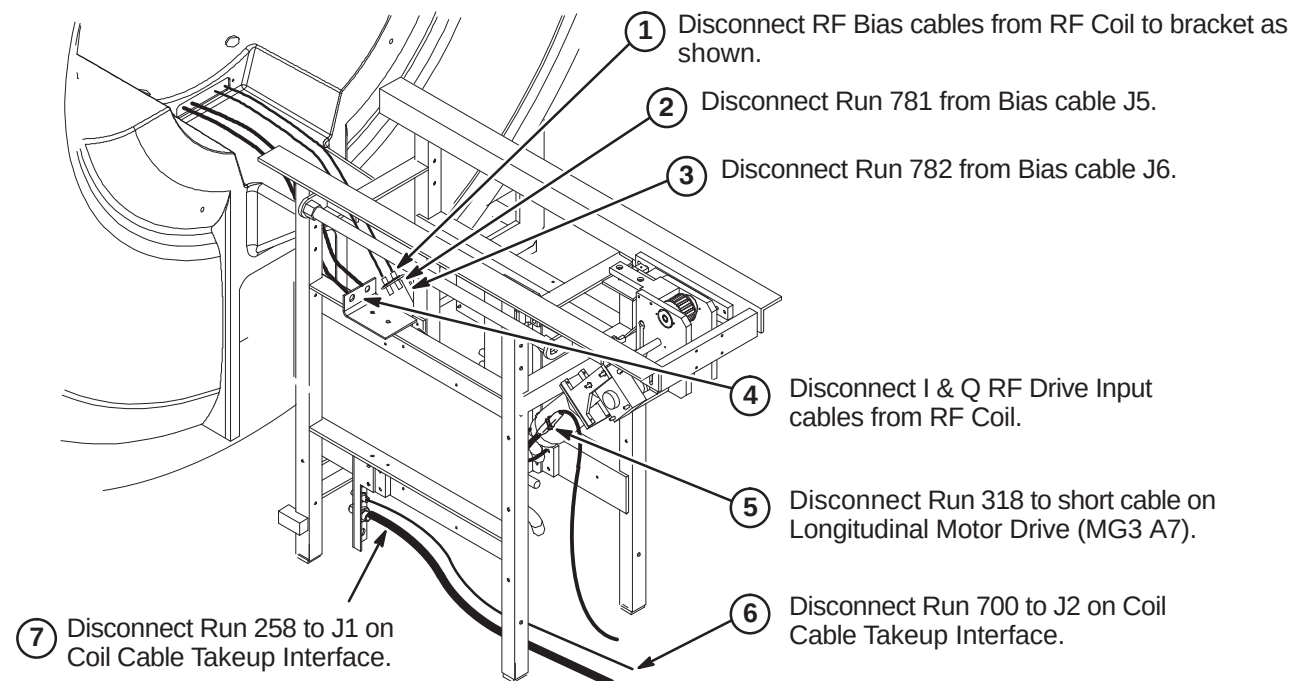
NOTE: Two people are required for removal of Bridge.

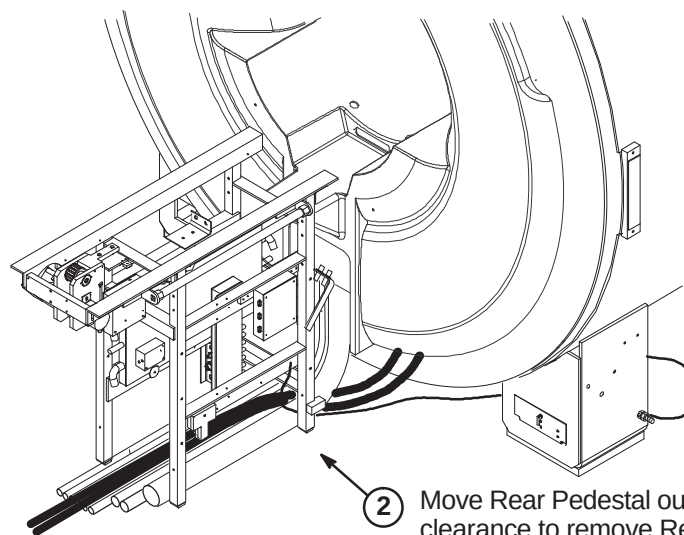
- ① Lift front end of bridge to disengage tongue on long drive assembly at Bridge rear end.
- ② While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore to the front end.



6-1-3 DE-INSTALL FRONT END BELL**6-1-3-1 Prepare Front End Of Bell For Removal****6-1-3-2 Remove Front End Bell From Magnet**

6-1-4 DE-INSTALL REAR PEDESTAL**6-1-4-1 Disconnect Air Hoses At Rear Of Magnet****6-1-4-2 Disconnect Front and Rear End Cables**

6-1-4-3 Disconnect Gradient Cable Runs 762, 763 And 764**6-1-4-4 Disconnect RF Coil BIAS & Drive Cables**

6-1-4-5 Location Of Rear Pedestal**WARNING!**

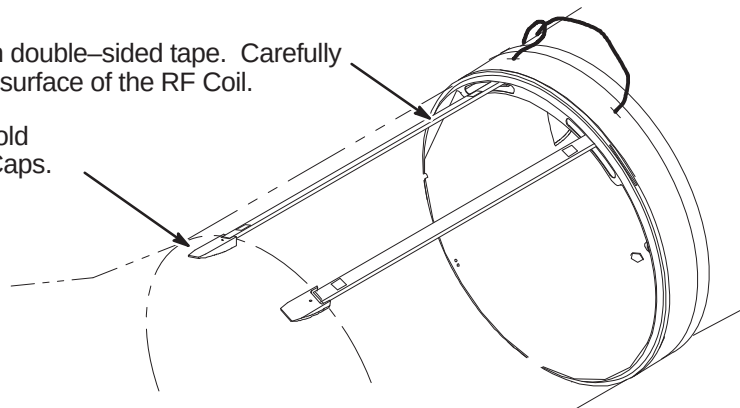
THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

① Disconnect Ground cable from Rear Pedestal.

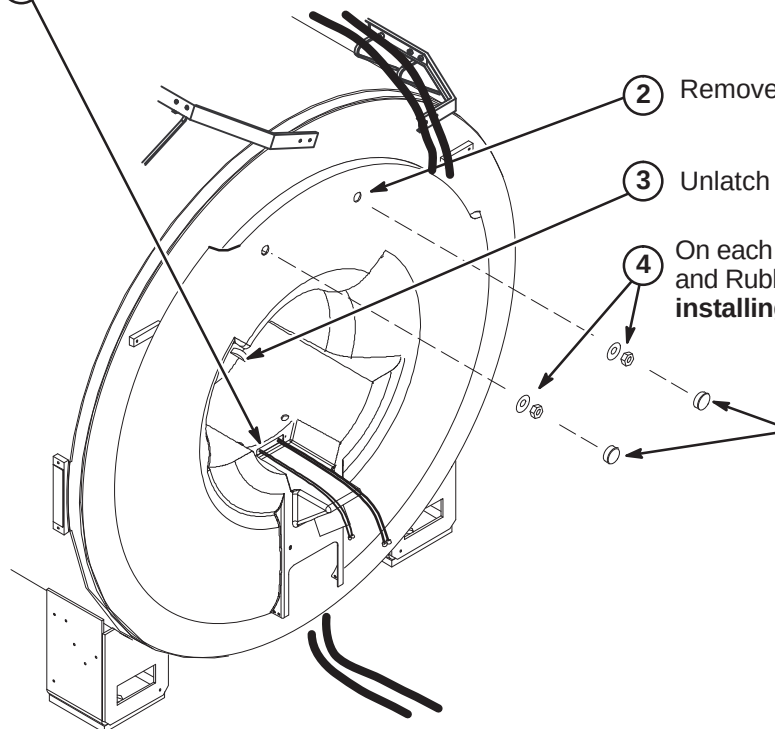
② Move Rear Pedestal out of the way so there is enough clearance to remove Rear End Bell Cover.

6-1-5 DE-INSTALL REAR END BELL**6-1-5-1 Unfasten and De-Install Bore Light Panels**

- ② Bore Light Panels are held in place with double-sided tape. Carefully remove the Bore Light Panels from the surface of the RF Coil.
- ① At front End Bell, remove screws that hold Bore Light Panels into Bore Light End Caps.

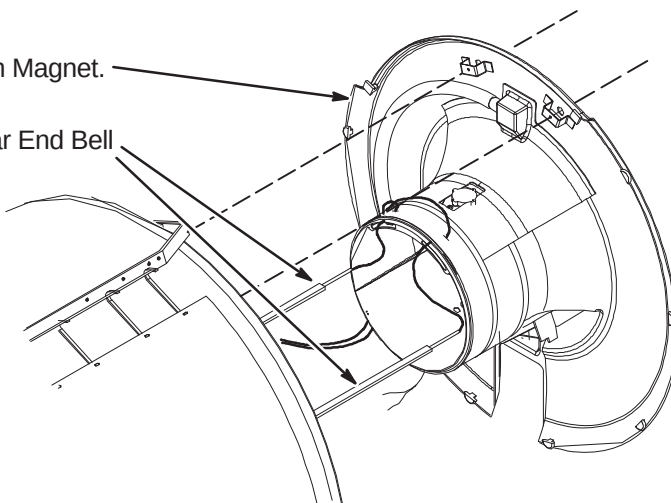
**6-1-5-2 Unfasten Rear End Bell From RF Coil And Magnet**

- ① Remove feed thru cover and gasket over the Body Coil cables.
 - ② Remove and save Access Hole Plugs
 - ③ Unlatch the two latches.
 - ④ On each stud, unfasten M10 Lock Nut (2109877-3) and Rubber Washer (2167321). **Save and reuse for installing new Rear Cover**
- Access Hole Plugs

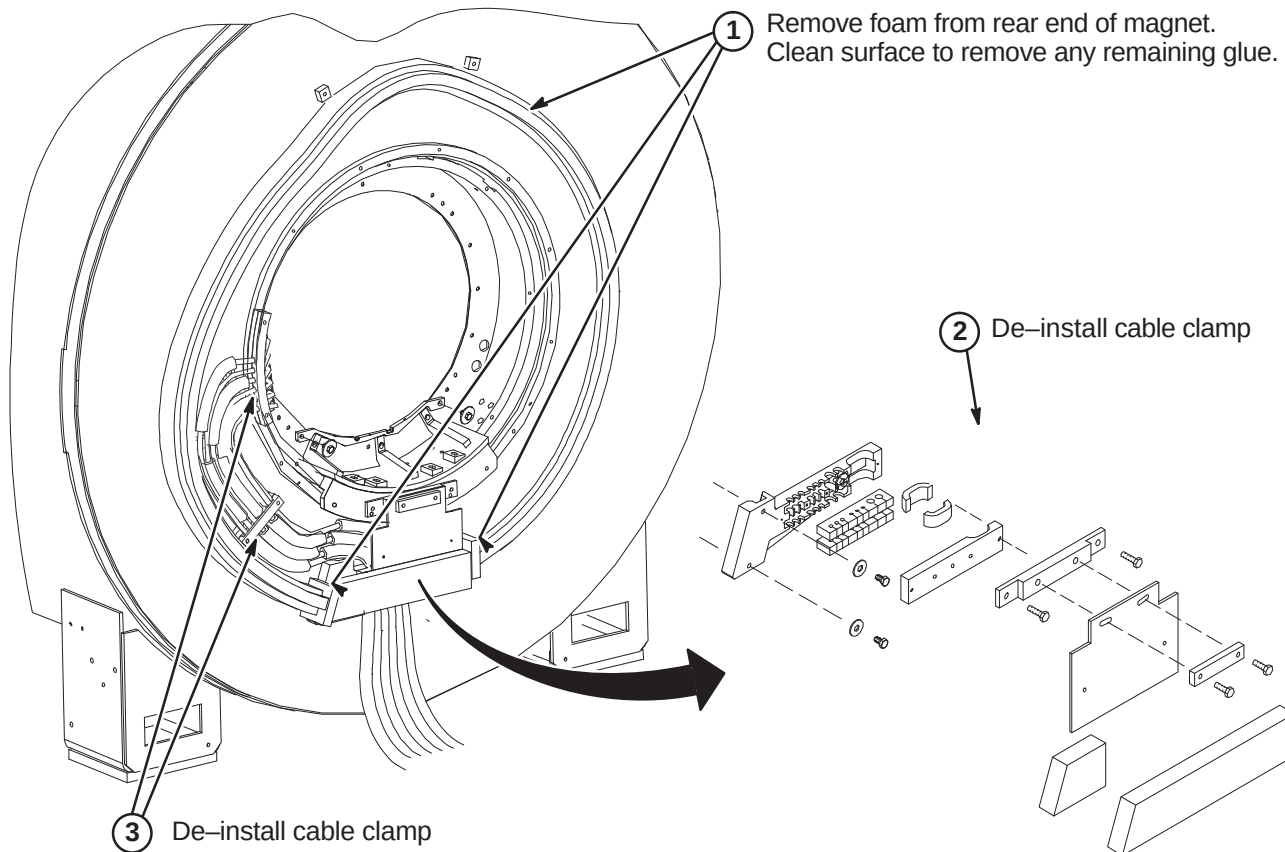


6-1-5-3 Remove Rear End Bell

- ① Carefully remove Rear End Bell from Magnet.
- ② Remove Bore Light Panels from Rear End Bell

**6-1-5-4 Remove Foam And Cable Clamps**

- ① Remove foam from rear end of magnet. Clean surface to remove any remaining glue.
- ② De-install cable clamp
- ③ De-install cable clamp

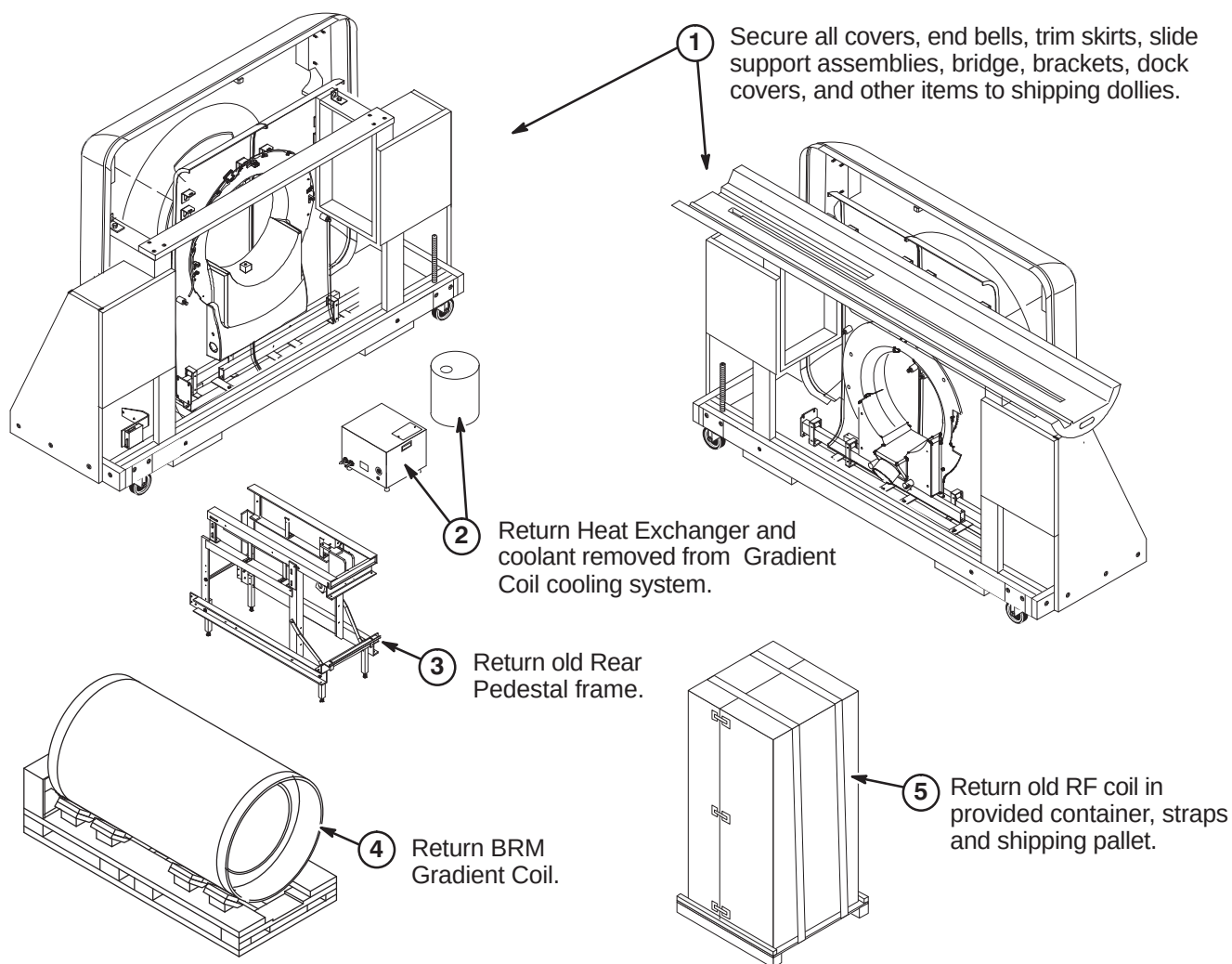


6-2 DE-INSTALL HORIZON ENCLOSURE

Refer to Section 1-7 for instructions on returning all material and components de-installed and not to be used for the upgrade. Also refer to applicable "Removal" cable map for information on cables to be de-installed and returned for recycling.

6-2-1 COMPONENTS TO BE RETURNED FOR RECYCLING AT COMPLETION OF UPGRADE

Note: All items de-installed during this upgrade are to be returned for recycling. If the two cover shipping dollies are provided, secure as many of the items as possible to them for this purpose. Shown below is an example of how the dollies might look when loaded. (not all items are shown)



6-2-2 DE-INSTALL ENCLOSURE COVERS AND DOCK

WARNING!

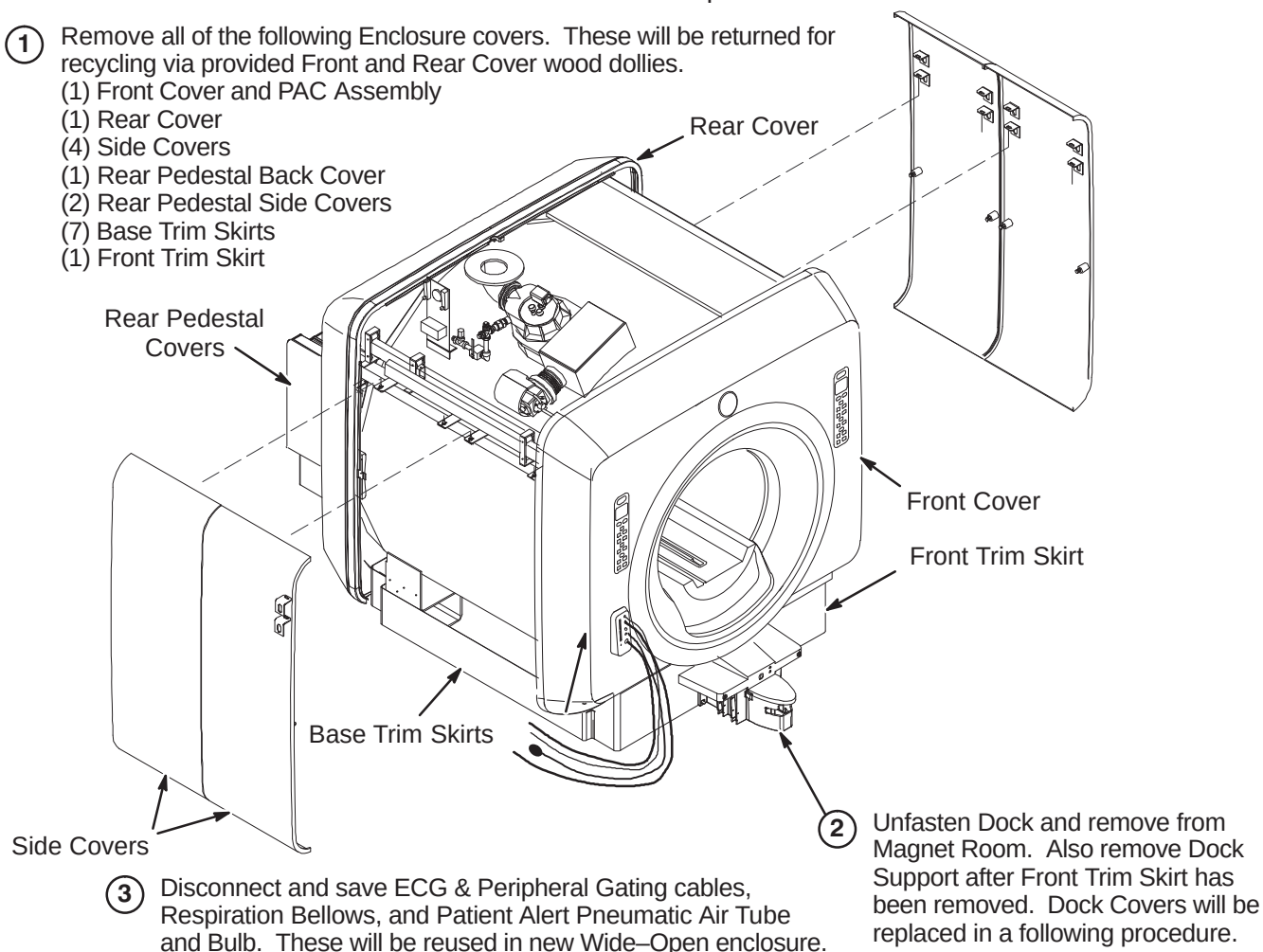
DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY.

DO NOT LIFT THE DOCK MORE THAN TWO INCHES (5 cm) OFF THE FLOOR. If the height (to clear the dock anchor) exceeds two inches (5 cm), the magnet must be ramped down prior to removing or installing the Dock as there is significant risk of the magnet attracting the Dock if it needs to be raised more than two inches (5 cm). For detailed Dock removal and installation procedures, refer to the *Signa EXCITE HD 1.5T Service Methods* CD-ROM, Direction 5133315: Replacements>Magnet Enclosure>LCC Dock Replacement.

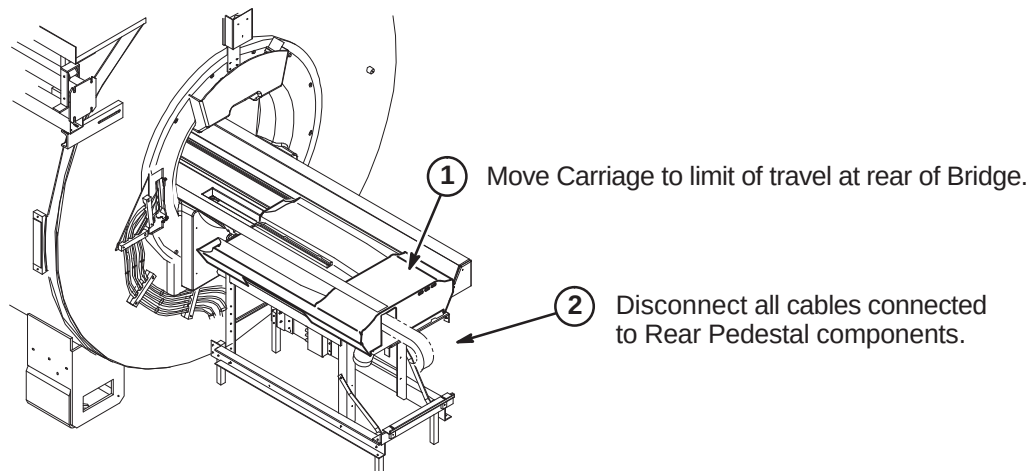
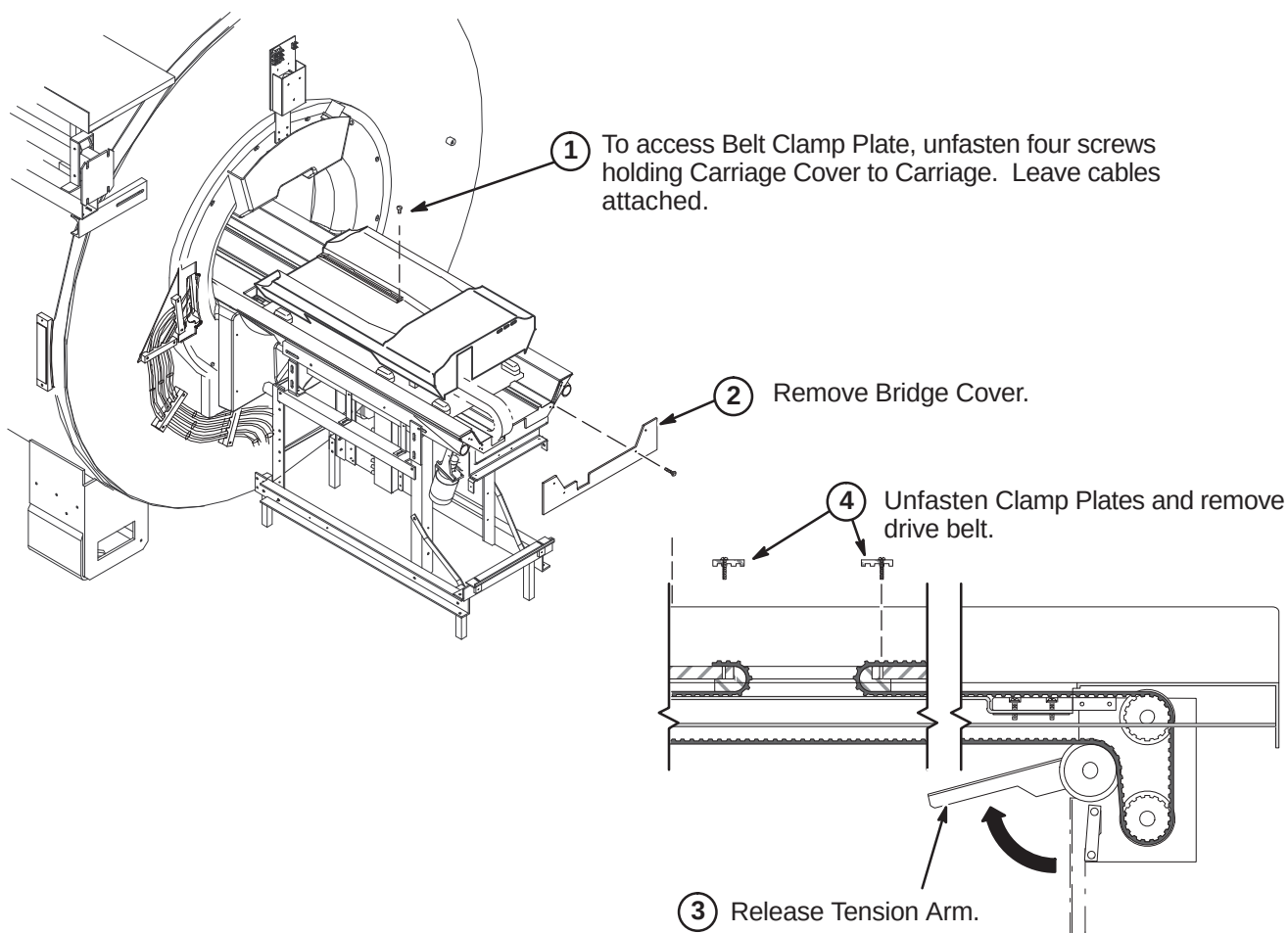
Note: All cables attached to Front Cover to be returned except for Run 385. Disconnect and save. It will be reused in new Wide-Open enclosure.

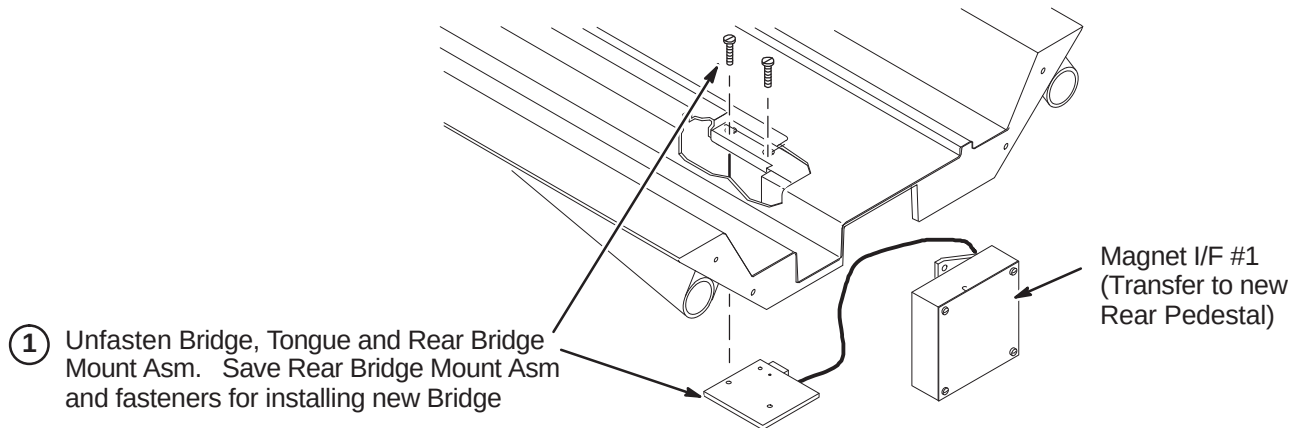
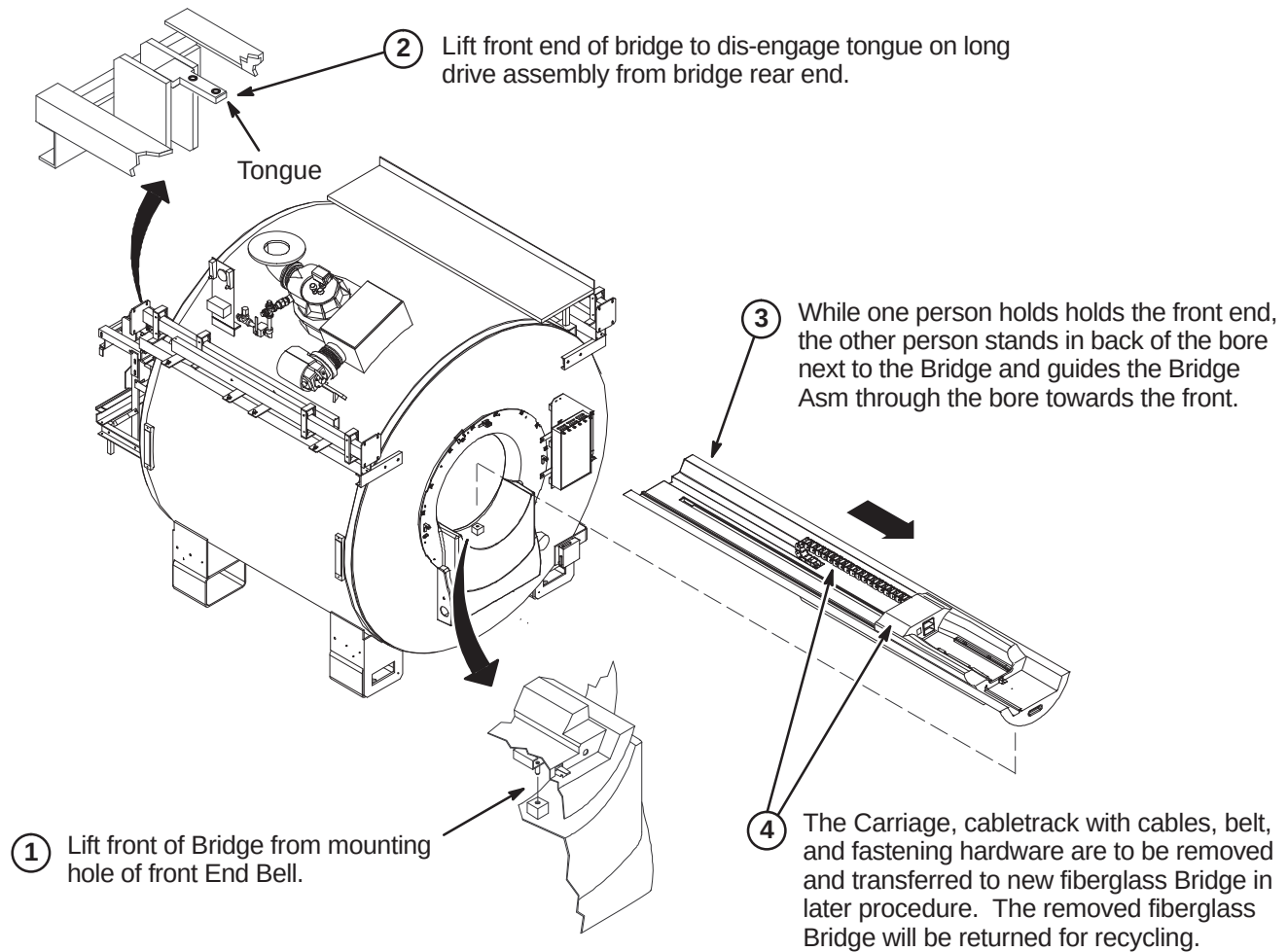
- ① Remove all of the following Enclosure covers. These will be returned for recycling via provided Front and Rear Cover wood dollies.

- (1) Front Cover and PAC Assembly
- (1) Rear Cover
- (4) Side Covers
- (1) Rear Pedestal Back Cover
- (2) Rear Pedestal Side Covers
- (7) Base Trim Skirts
- (1) Front Trim Skirt



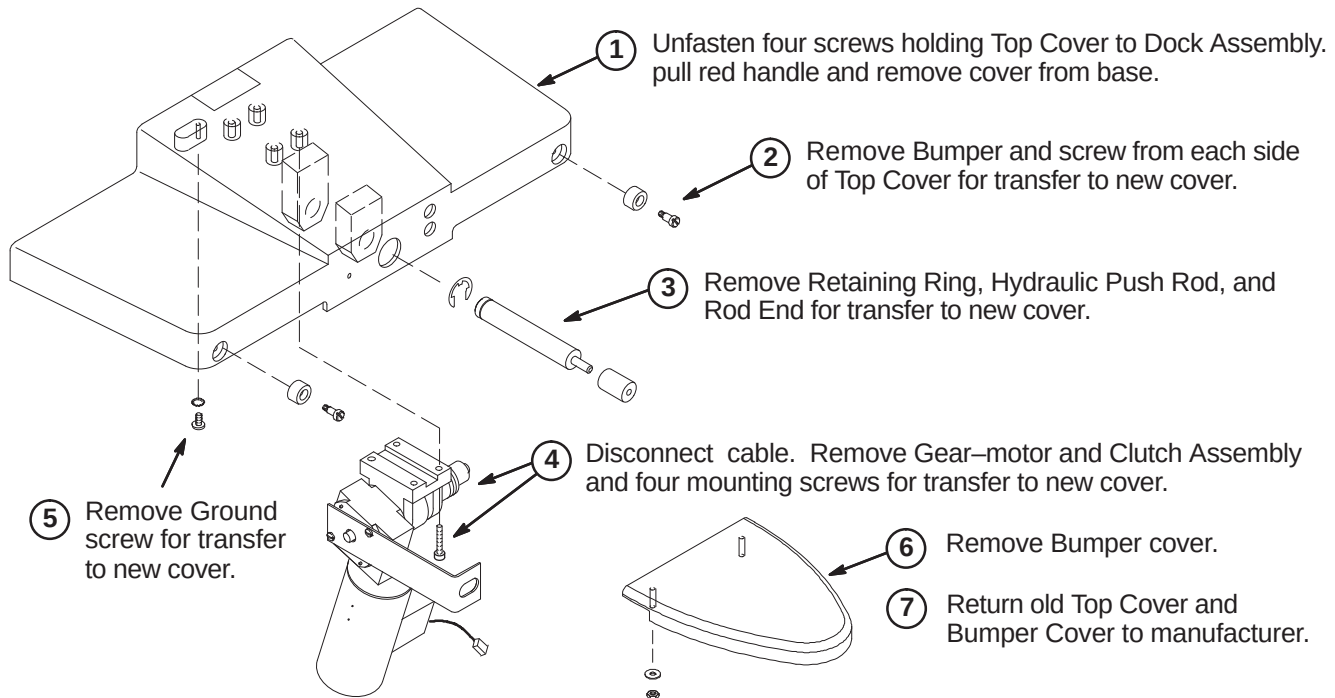
- ③ Disconnect and save ECG & Peripheral Gating cables, Respiration Bellows, and Patient Alert Pneumatic Air Tube and Bulb. These will be reused in new Wide-Open enclosure.

6-2-3 DE-INSTALL BRIDGE AND LPCA**6-2-3-1 Disconnect Rear Pedestal Cables****6-2-3-2 De-Install Drive Belt**

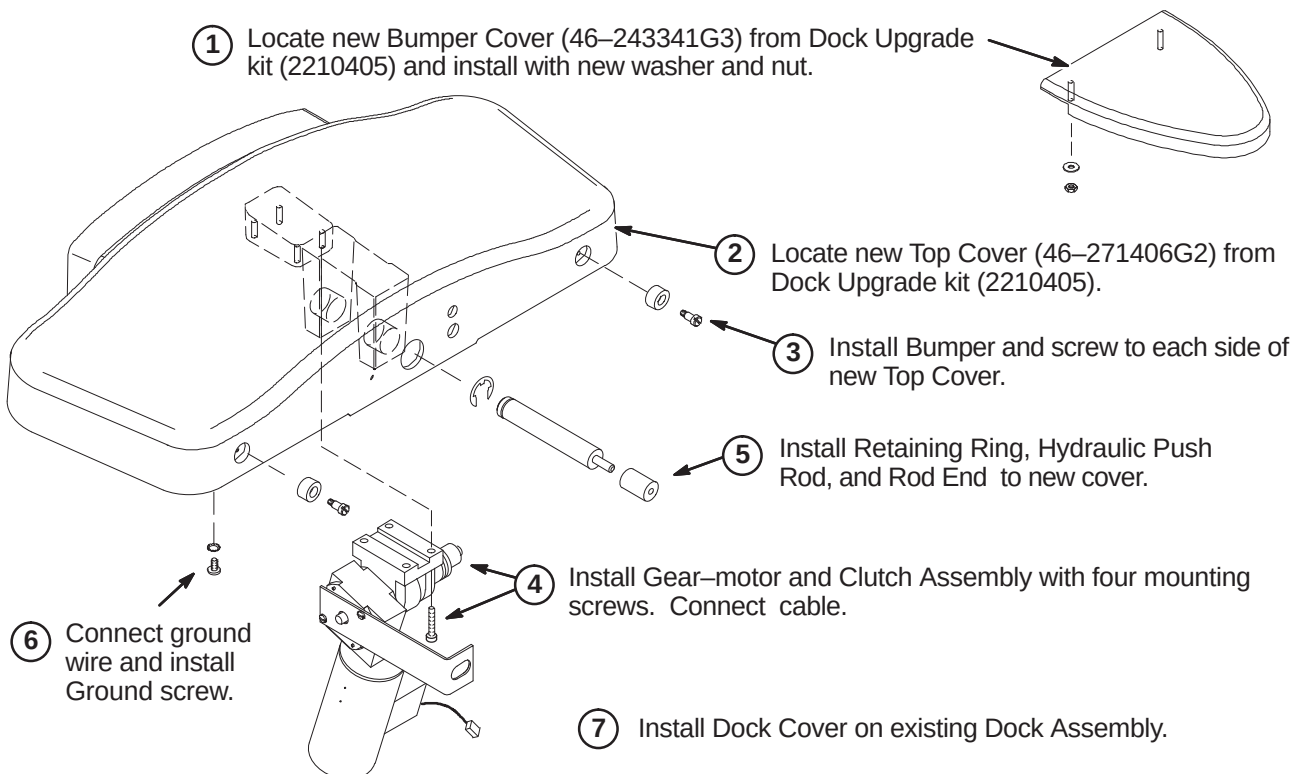
6-2-3-3 Unfasten Bridge Mount Assembly**6-2-3-4 De-Install Bridge**

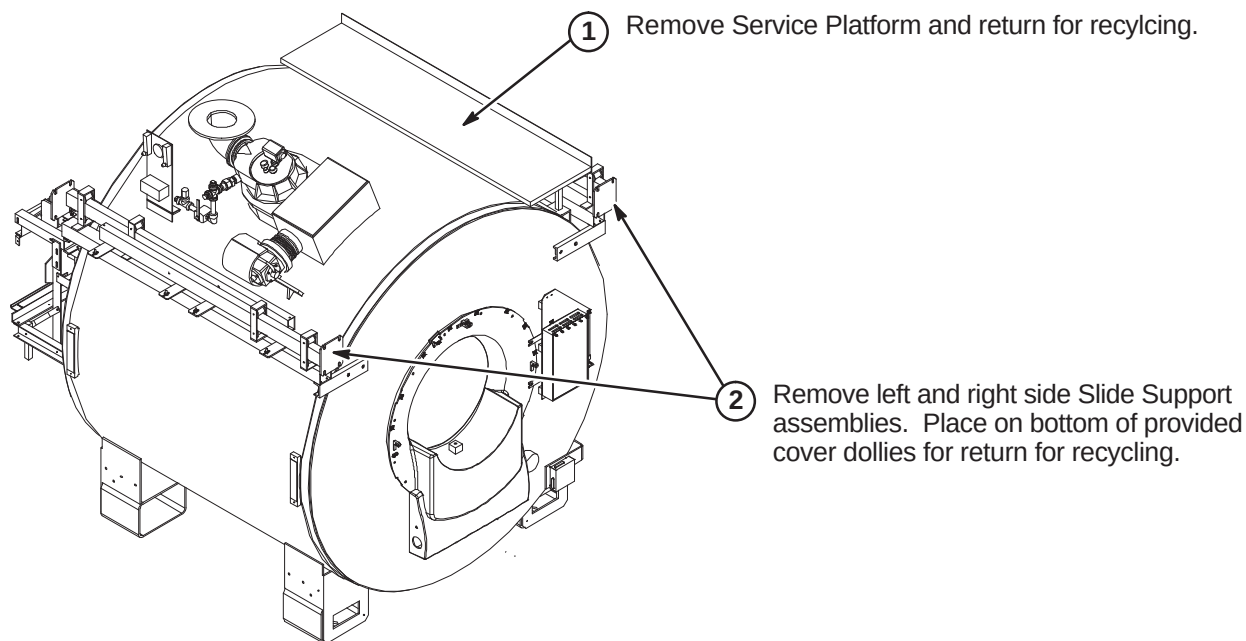
6-2-4 TRANSFER DOCK COMPONENTS

6-2-4-1 Remove Components from Old Dock Cover



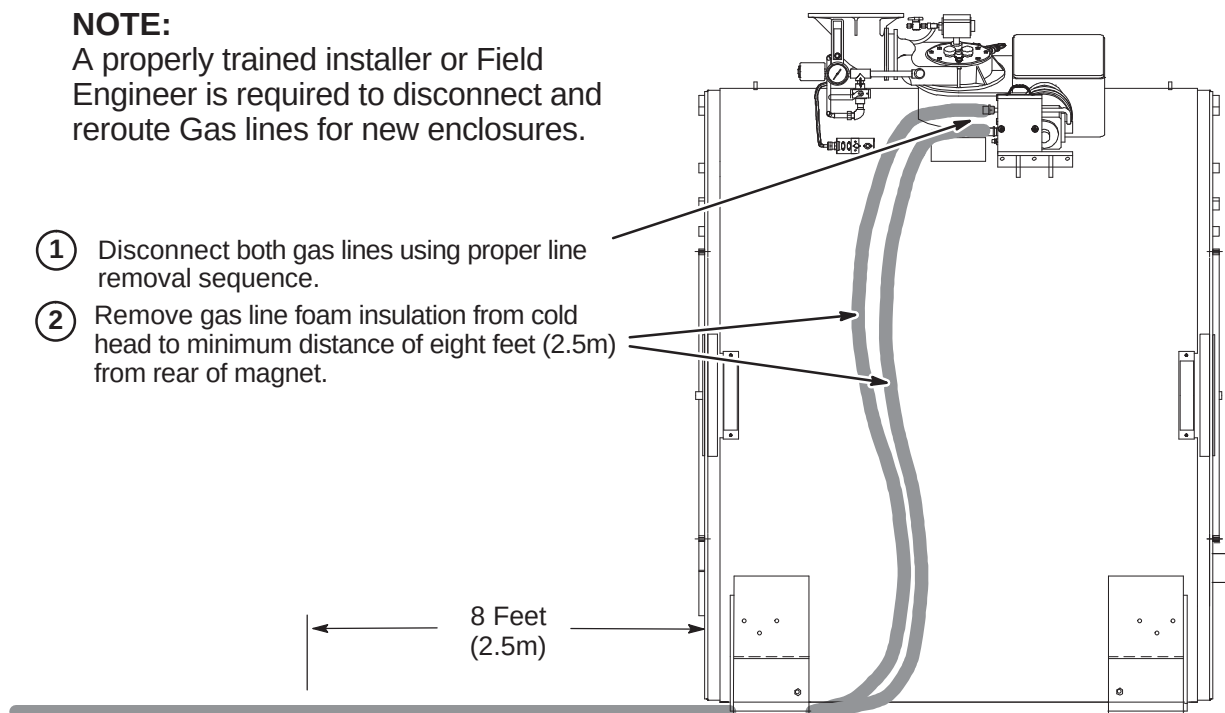
6-2-4-2 Install Components on New Dock Cover

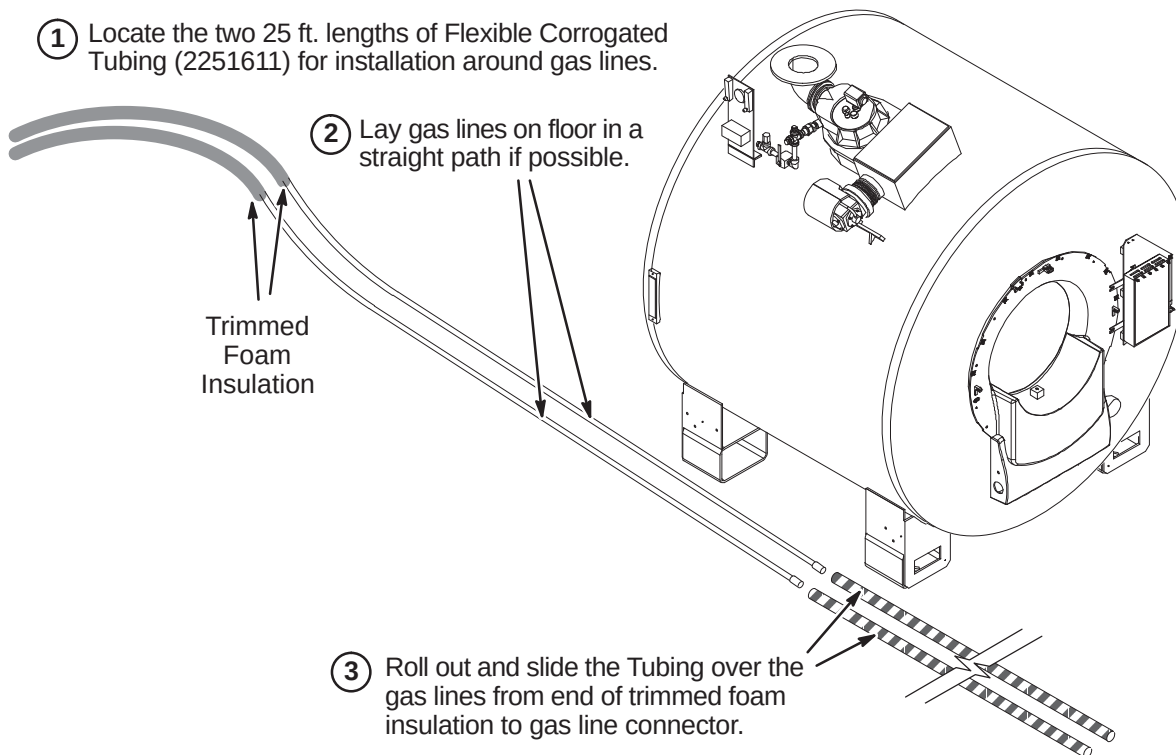


6-2-5 DE-INSTALL SERVICE PLATFORM AND SLIDE SUPPORT ASSEMBLIES**6-2-6 REPLACE GAS LINE FOAM COVER****6-2-6-1 Remove Gas Line Foam Insulation and Disconnect Gas Lines****NOTE:**

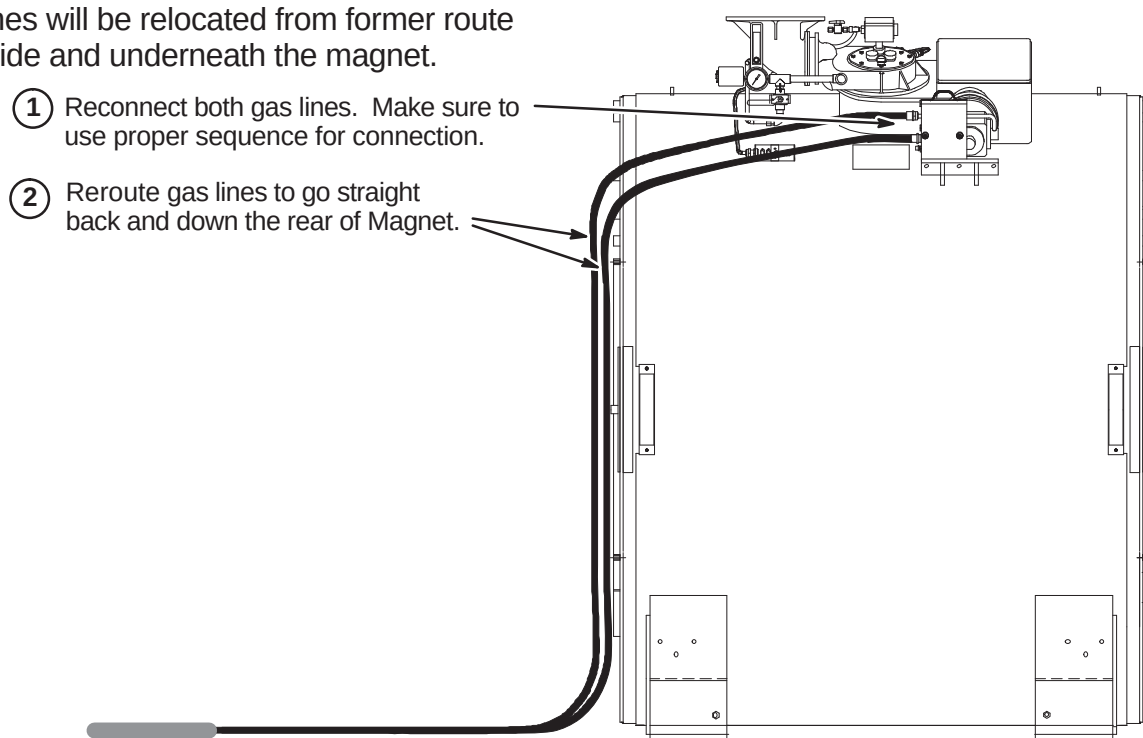
A properly trained installer or Field Engineer is required to disconnect and reroute Gas lines for new enclosures.

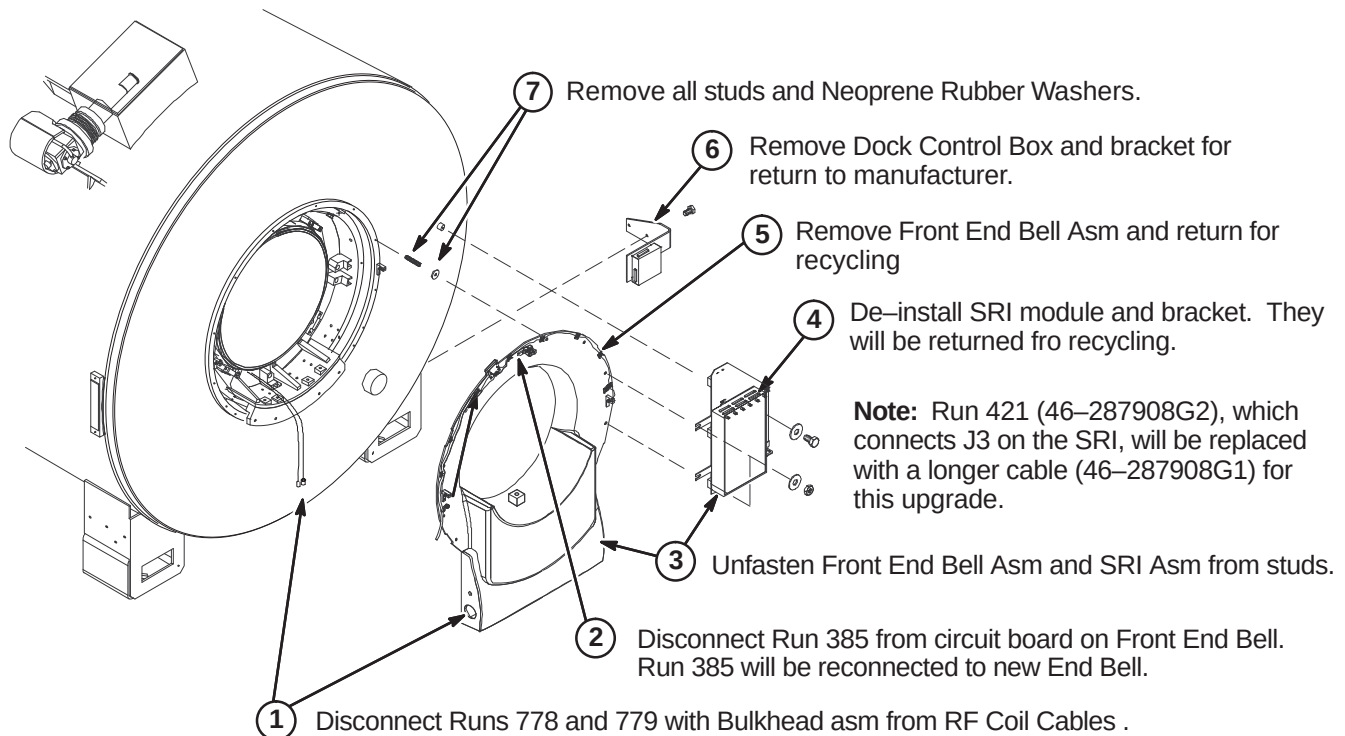
- 1 Disconnect both gas lines using proper line removal sequence.
- 2 Remove gas line foam insulation from cold head to minimum distance of eight feet (2.5m) from rear of magnet.



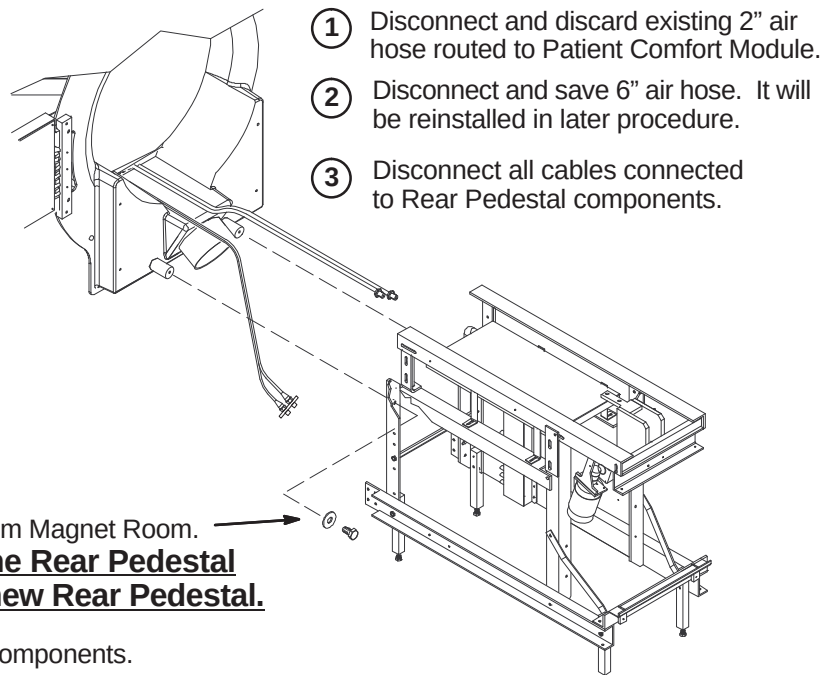
6-2-6-2 Install Flexible Tubing on Gas Lines**6-2-6-3 Connect and Route Gas Lines**

Note: Gas lines will be relocated from former route of down the side and underneath the magnet.



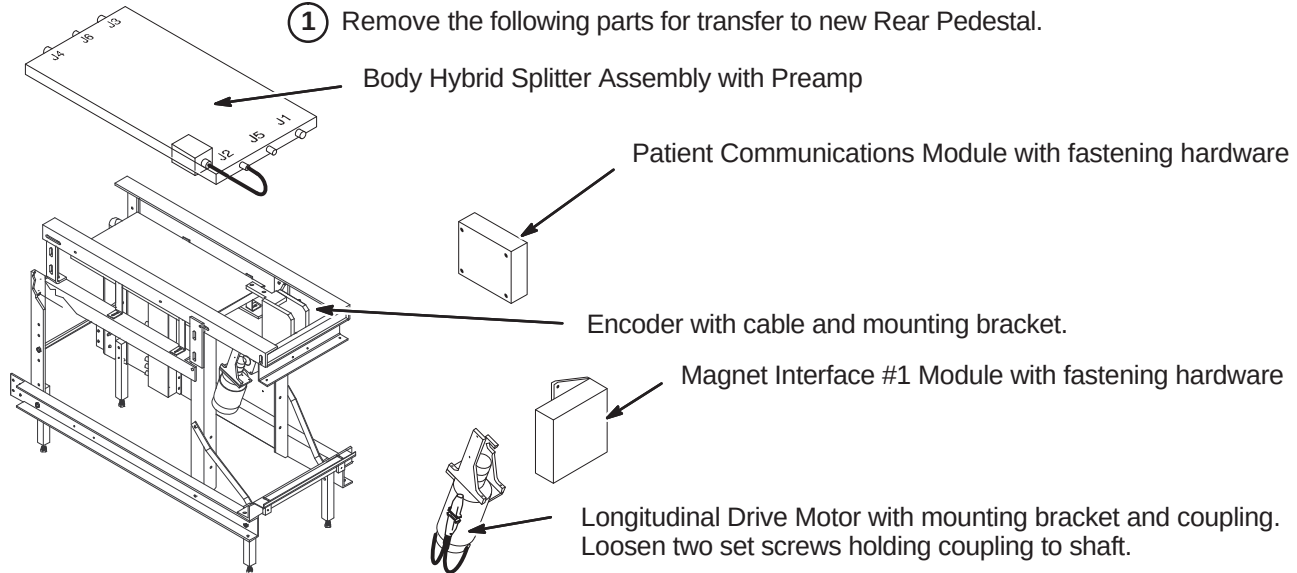
6-2-7 DE-INSTALL COMPONENTS AT FRONT OF MAGNET**6-2-8 DE-INSTALL REAR PEDESTAL AND TRANSFER COMPONENTS****6-2-8-1 Disconnect Air Hoses & Cables and Remove Rear Pedestal****WARNING!**

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.



6-2-8-2 Remove Components from Old Rear Pedestal

- ① Remove the following parts for transfer to new Rear Pedestal.

**6-2-8-3 Install Components on New Rear Pedestal**

- ① Install Magnet Interface #1 Module with fastening hardware.

Note: Connectors are located at bottom.

- ③ Install Encoder with cable and bracket.

- ④ Remove two (2) flat head screws and install Longitudinal Drive Asm.

- ⑤ Connect coupling to shaft while positioning bracket around brass rods.

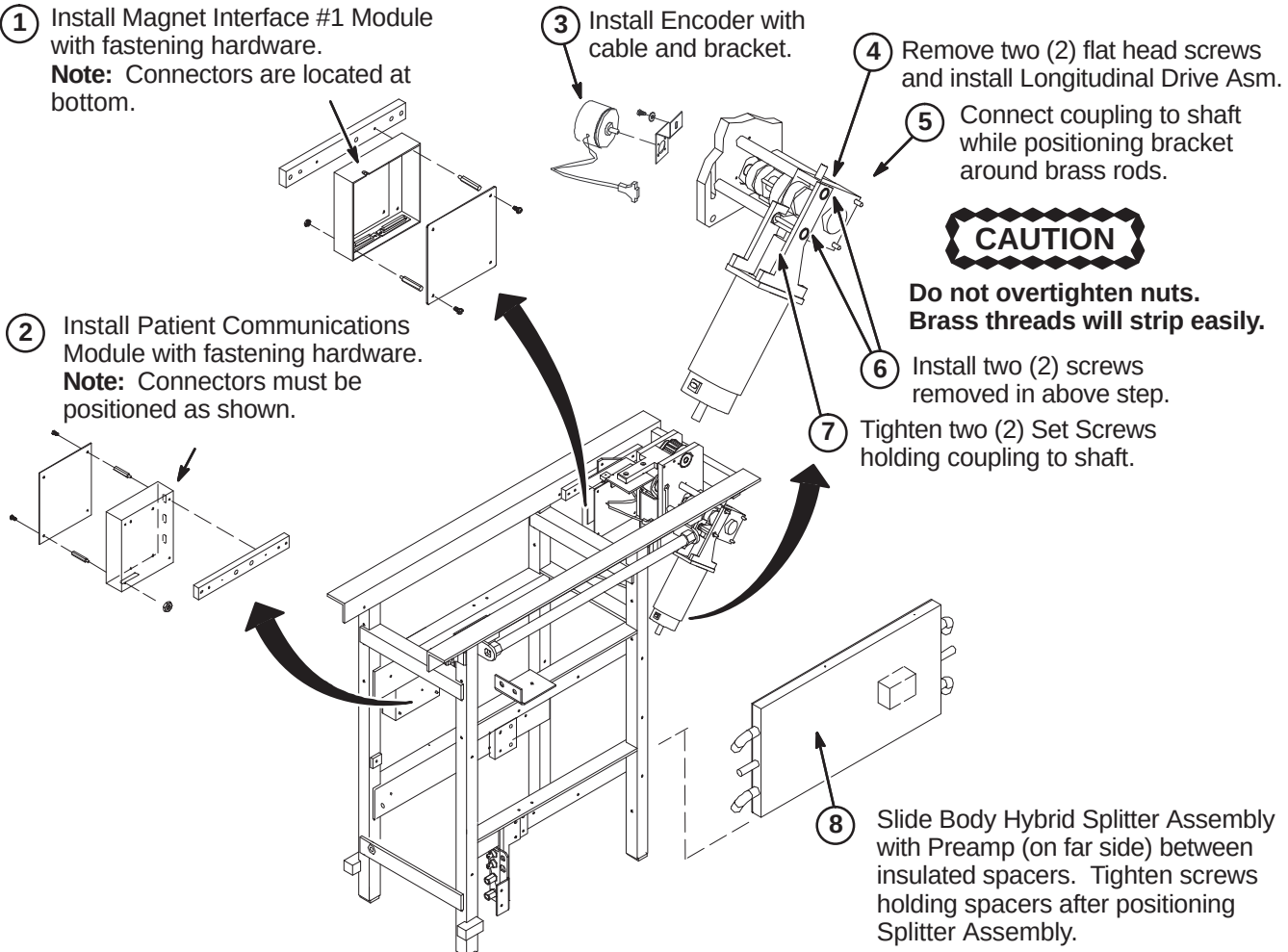
CAUTION

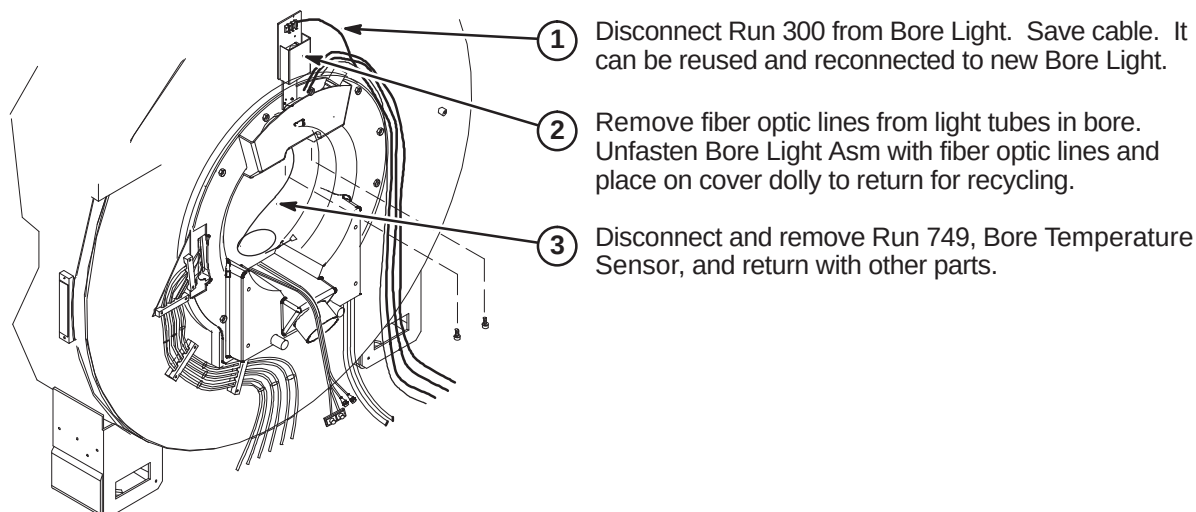
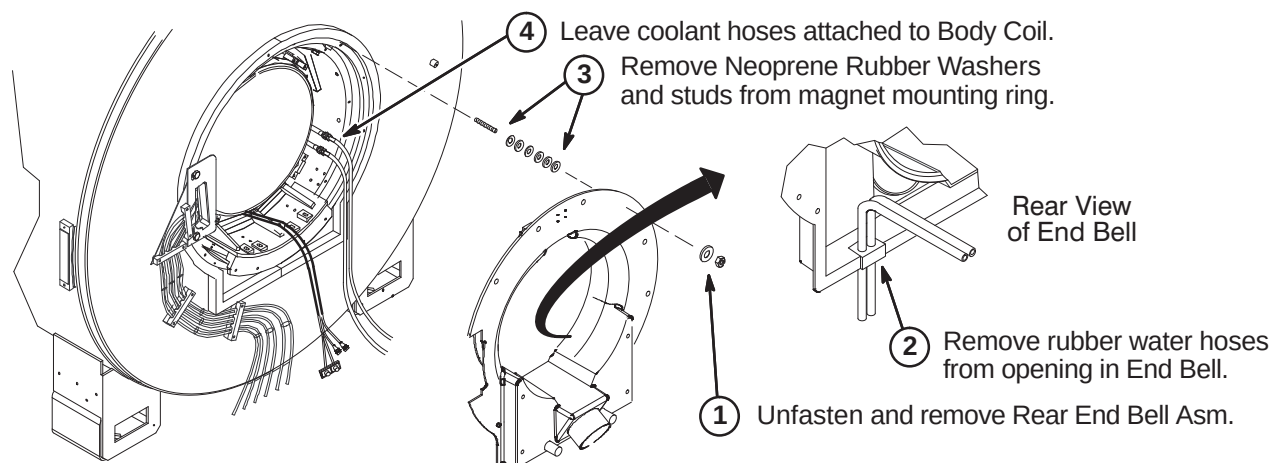
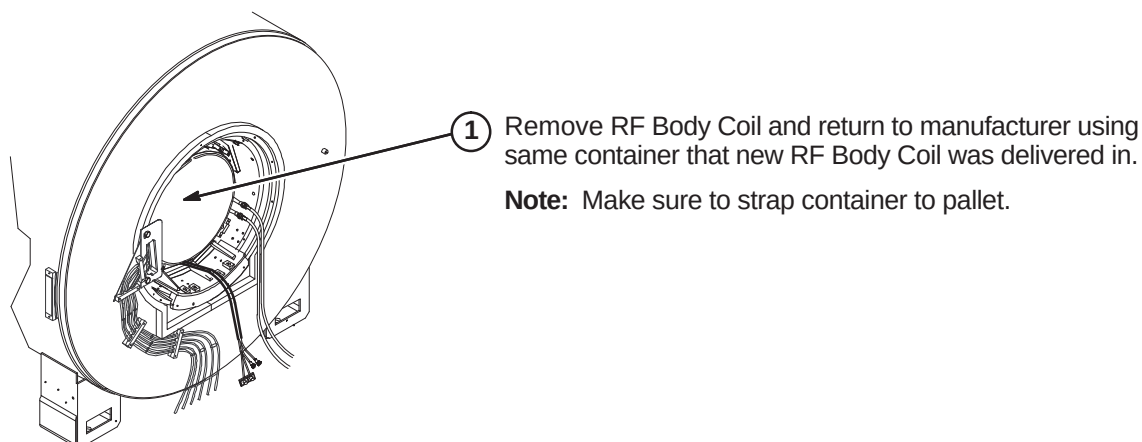
Do not overtighten nuts. Brass threads will strip easily.

- ② Install Patient Communications Module with fastening hardware. **Note:** Connectors must be positioned as shown.

- ⑥ Install two (2) screws removed in above step.

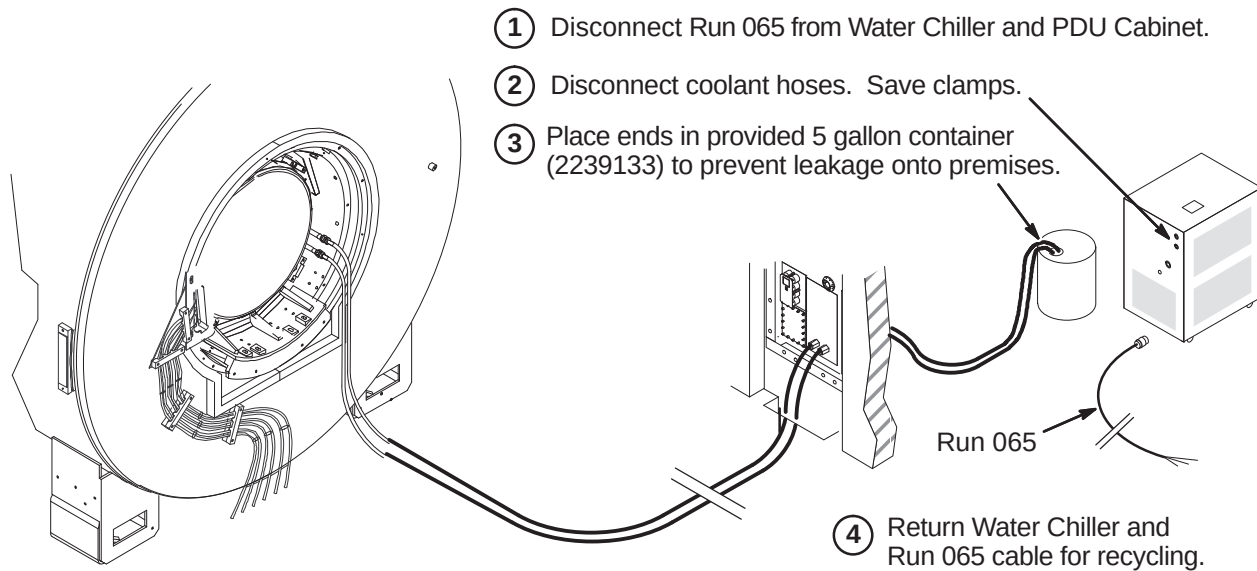
- ⑦ Tighten two (2) Set Screws holding coupling to shaft.



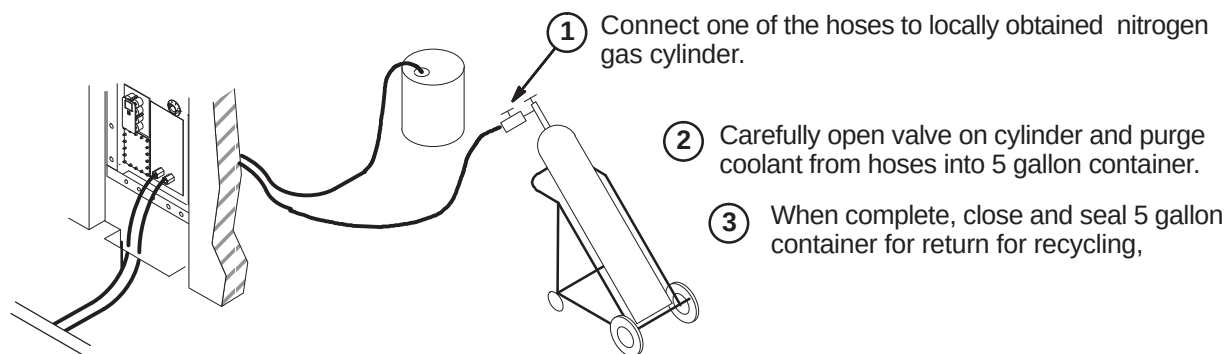
6-2-9 DE-INSTALL COMPONENTS AT REAR OF MAGNET**6-2-9-1 Remove Bore Light Assembly and Run 749****6-2-9-2 Remove Rear End Bell****6-2-10 DE-INSTALL RF BODY COIL**

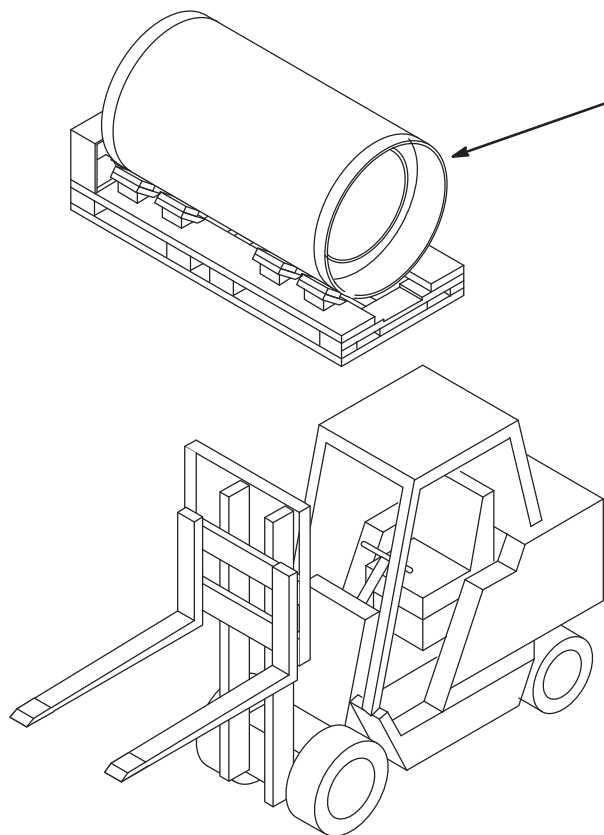
6-3 GRADIENT COIL REPLACEMENT

6-3-1 DISCONNECT WATER CHILLER



6-3-2 DRAIN COOLANT FROM BODY COIL AND COOLANT HOSES

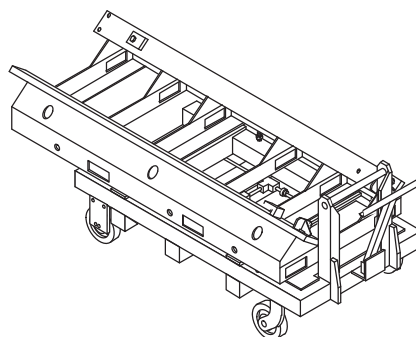


6-3-3 GRADIENT COIL UPGRADE INFORMATION

The new TwinSpeed TRM Body Coil will arrive on site on a wooden pallet only. It will not be on a movable coil cart.

Arrangements should also have been made to:

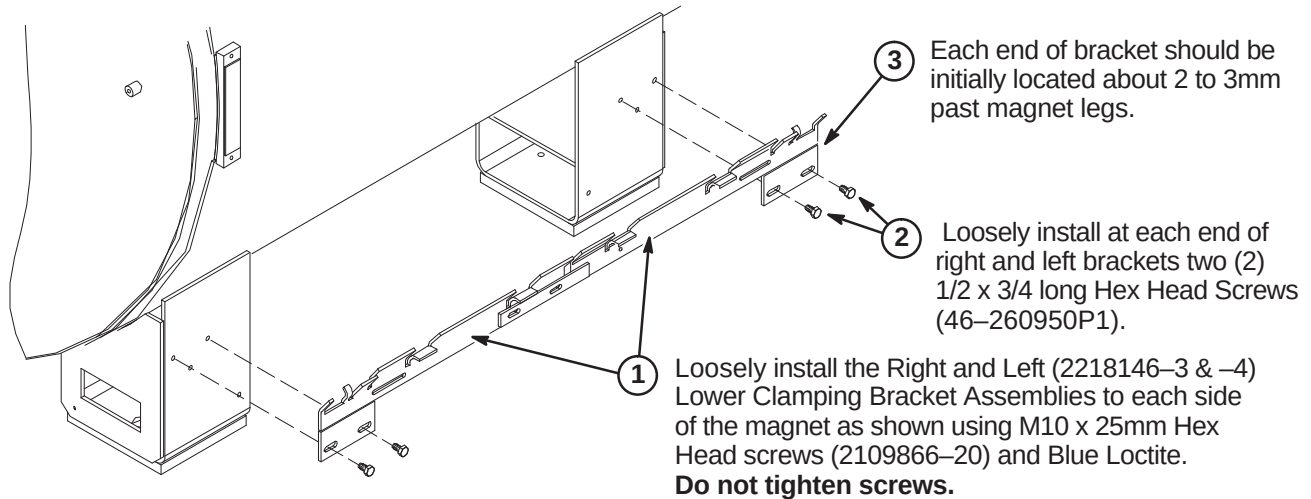
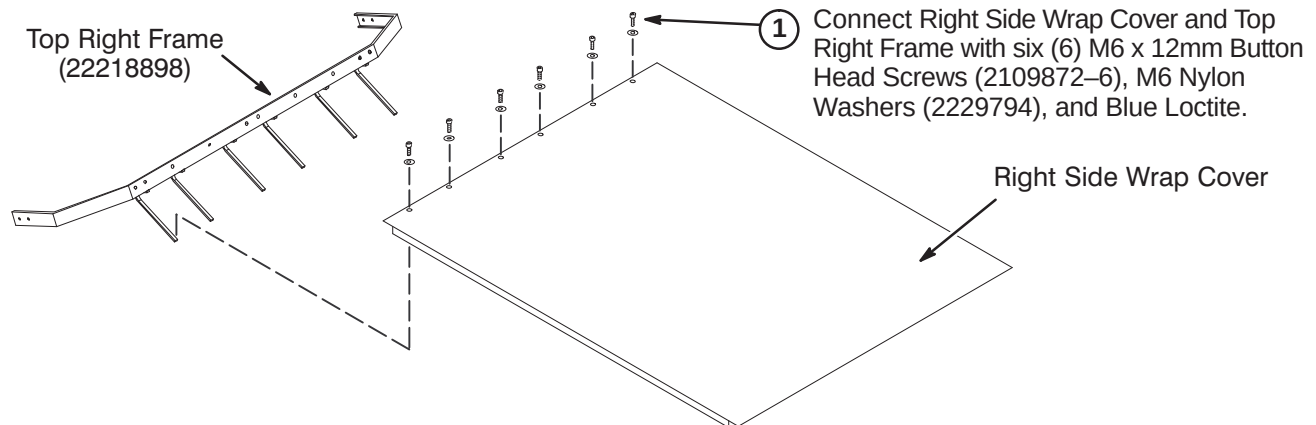
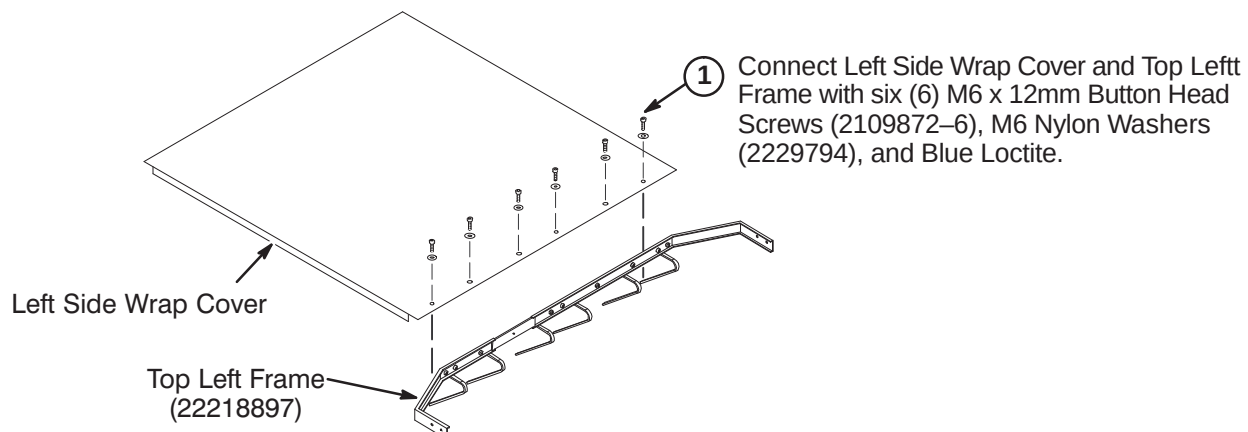
- Have riggers and/or lift truck on site to unload new TRM coil from delivery truck
- Have coil cart and cradle, 2134817 & 2134810-2, on site to de-install existing BRM or CRM Body Coil
- Have BRM/TRM Insertion Tool, 2164744-3 on site to remove the existing BRM and install the new TRM Body Coil

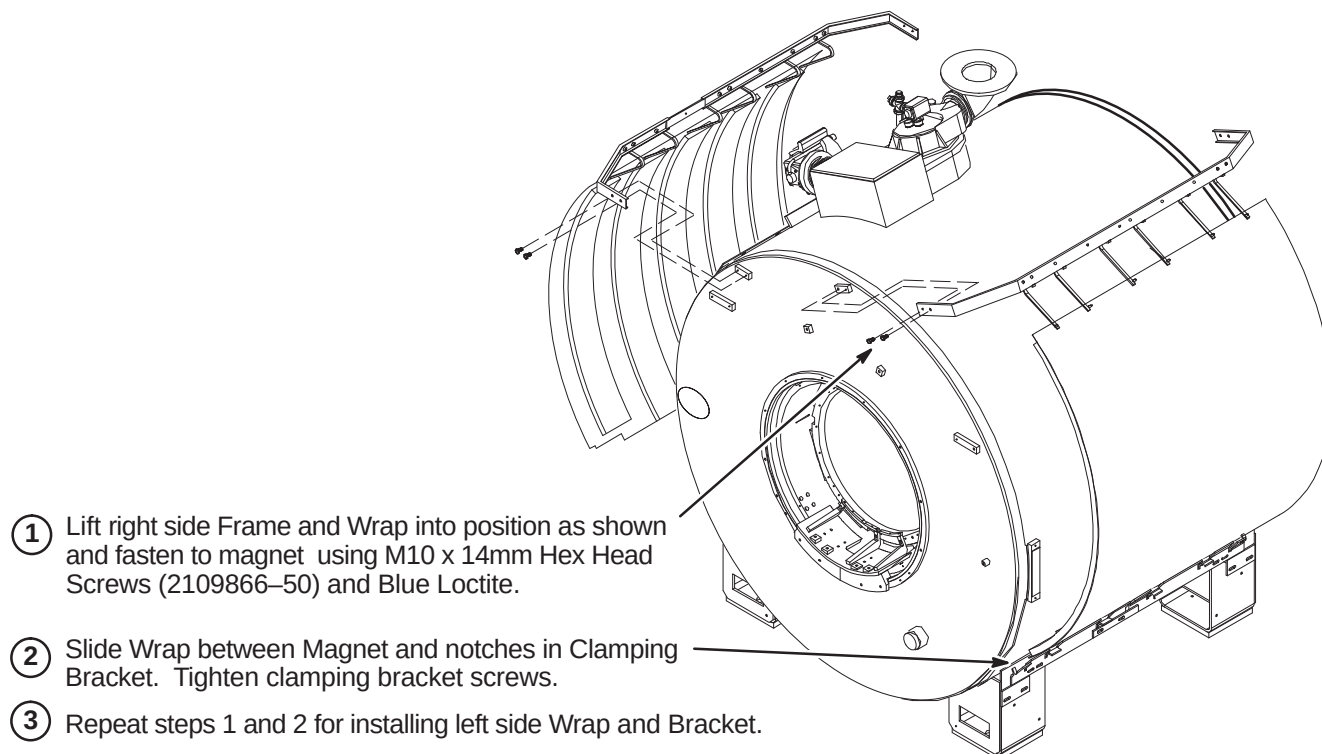
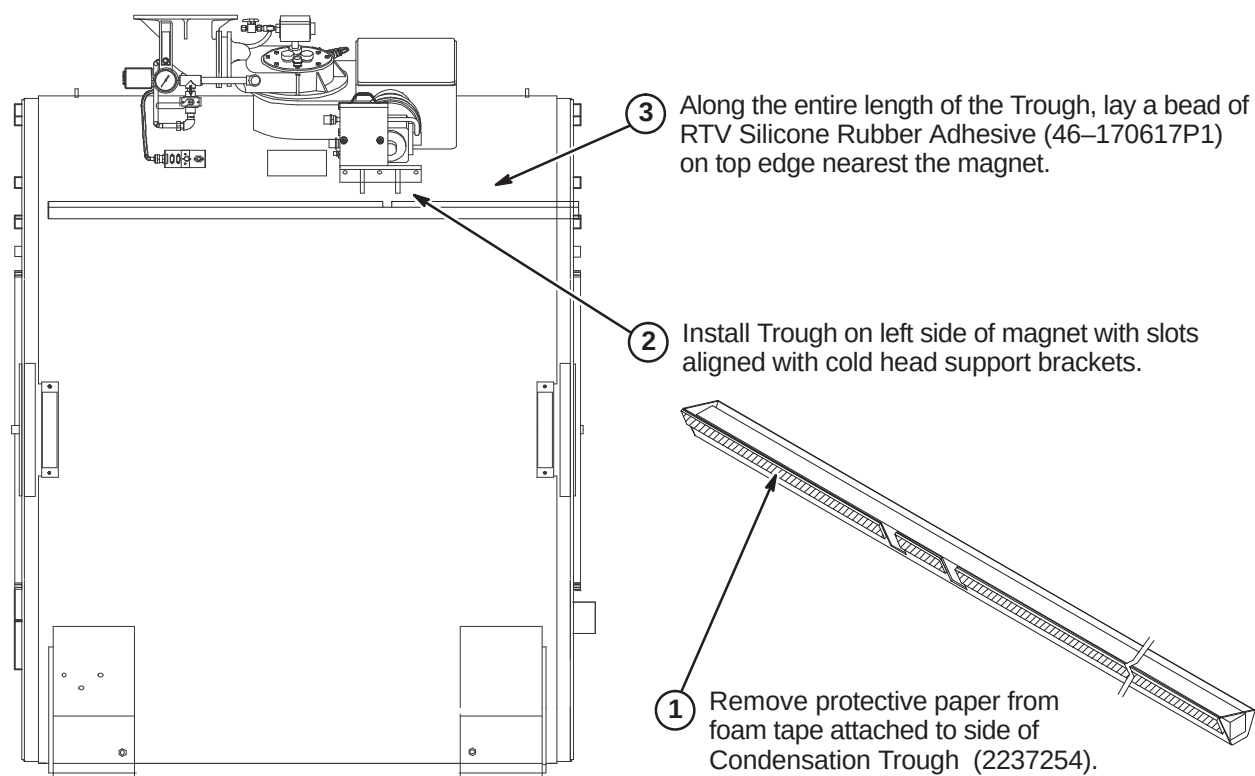
**Refer to MR Service Methods CD-ROM or Section 7 for instructions on :**

- REMOVING EXISTING BODY COIL
Note: Refer to Direction 5133315, *Signa EXCITE HD 1.5T Service Methods*, CD-ROM or the MR Service Engineering Web site for instructions to remove existing Body Coil using provided cradle and coil cart.
Path to instructions: Home>Replacements>Coil Replacements>Gradient Coil Replacement for Cx Magnet>Defective Gradient Removal Procedure
- INSTALLATION OF TwinSpeed END RINGS using Ring Alignment Tool, 2299496
Note: Refer to Section 7, Direction 2316316, *TwinSpeed Interface Ring Alignment*
- TRANSFERING TRM COIL FROM SKIDS INSTRUCTIONS
Note: Refer to Section 7, Direction 2326444, *TRM Coil Lifting Procedures*
- INSTALLATION OF NEW TRM BODY COIL
Note: Refer to Direction 5133315, *Signa EXCITE HD 1.5T Service Methods*, CD-ROM or the MR Service Engineering Web site for instructions to install the TRM Body Coil using provided cradle and coil cart.
Path to instructions: Home>Replacements>Coil Replacements>Twin Resonator Module (TRM) Replacement>Installing the TRM

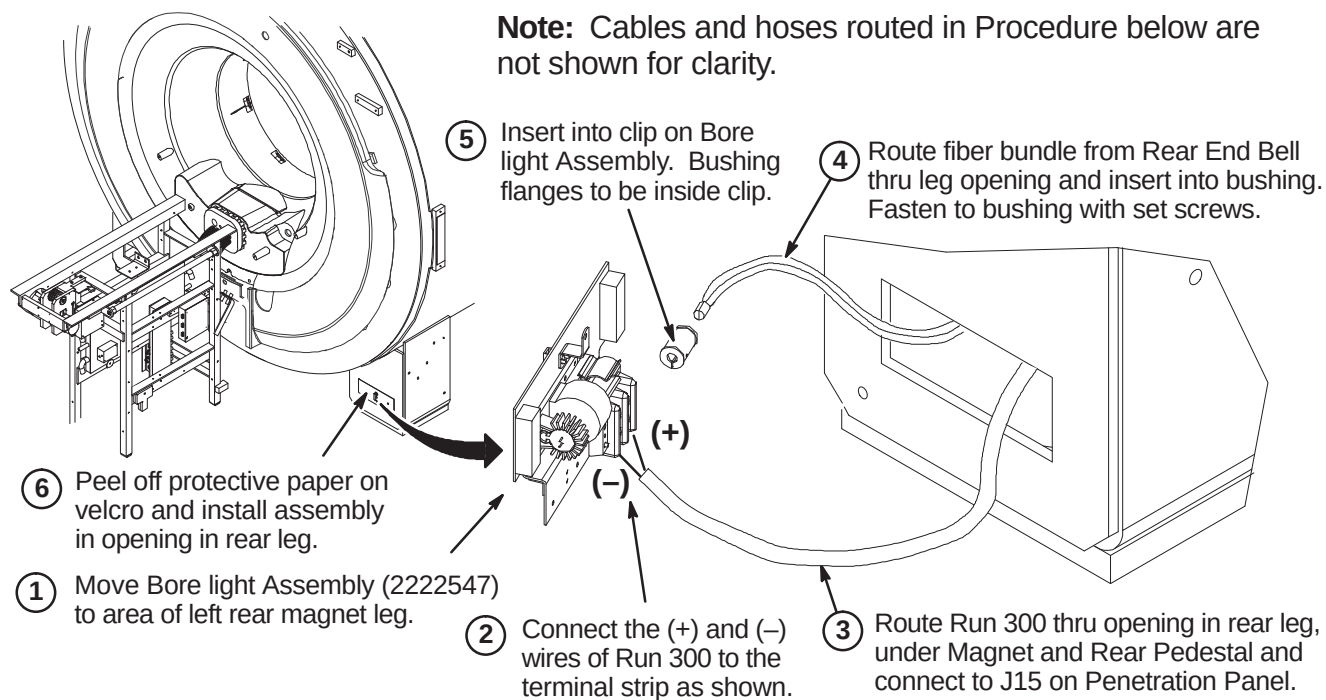
6-4 COOLING SYSTEM INSTALLATION

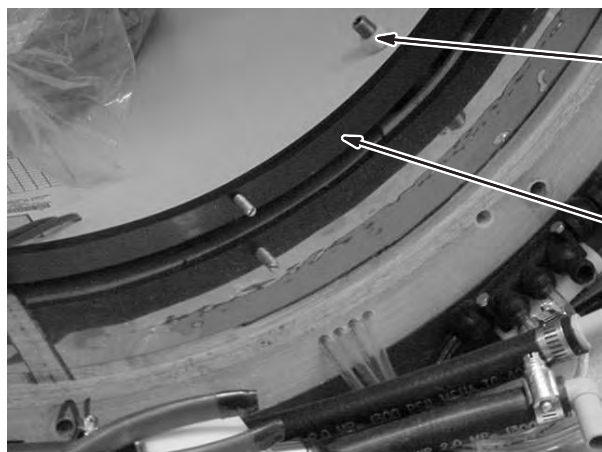
Make sure the Cooling Cabinet designated for this system has been installed and ready for start-up. Consult vendor supplied manual, that was delivered with Cooling Cabinet, for start-up and operating procedures. This will be required for the upcoming coolant leakage test.

6-5 INSTALL HORIZON TO TwinSpeed SPECIFIC MAGNET ENCLOSURE COMPONENTS**6-5-1 INSTALL LOWER CLAMPING BRACKETS****6-5-2 CONNECT RIGHT SIDE TOP FRAME AND WRAP COVER****6-5-3 CONNECT LEFT SIDE TOP FRAME AND WRAP COVER**

6-5-4 INSTALL RIGHT AND LEFT SIDE FRAME AND WRAP COVERS**6-5-5 INSTALL CONDENSATION TROUGH**

6-5-6 INSTALL BORE LAMP ASSEMBLY



6-6 INSTALL TwinSpeed MAGNET ENCLOSURE**6-6-1 INSTALL FRONT END BELL****6-6-1-1 Front End Bell Installation Preparation**

① Make sure RF Body Coil is centered in the gradient coil using the Feeler Gauge Tool provided with coil. Adjust as necessary.

② Install 6mm RF Tube-End Bell Gasket (2294358-2) onto studs of RF Coil. Make sure gasket is clean. DO NOT APPLY VACUUM GREASE.

Note: Nominal length between Front and Rear Alignment Ring surfaces is about 70-1/16 inch (1780mm).

If length is shorter, a thinner gasket may be required – 4mm gasket (2294358).

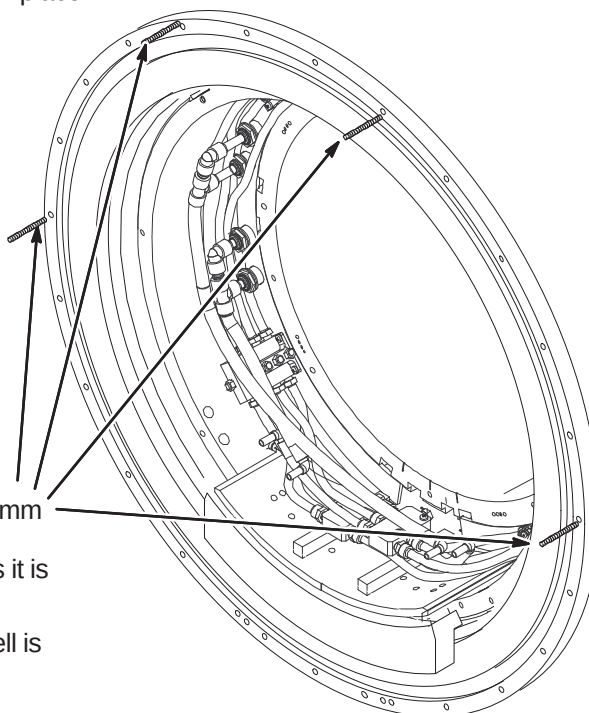
If length is longer, a thicker gasket may be required – 9.5mm gasket (2294358-3).



③ Apply Vacuum Grease (2297760) to the entire 3/4 inch diameter O-Ring (2293323) and install in the slot on the aluminum ring. The vacuum grease will hold it in place.

④ From the TRM Coil Insertion Tool Kit, get four M10 x 140mm studs and insert at four positions on outer bolt circle of aluminum ring. These will be used to support End Bell as it is being installed.

These must be returned to Insertion Tool Kit after End Bell is installed.

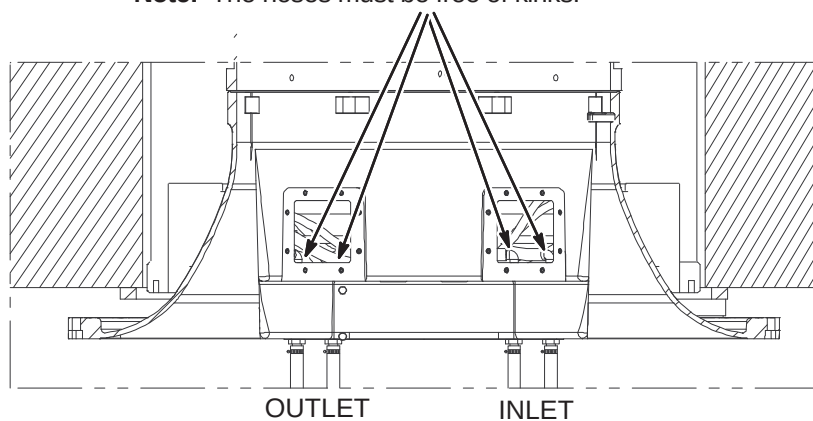


6-6-1-2 Position Front End Bell On Studs

- ① Hang the Front End Bell in place on studs and remove access covers for the coolant connections.

- ② Connect the four coolant hoses to the internal fittings on the End Bell using Safety Clip (2262660-9) on each fitting.

Note: The hoses must be free of kinks.



6-6-1-3 Attach Front End Bell

- ① Push the End Bell inward and align it to engage the studs on the RF Coil
- ② Install (24) M10 x 50mm Socket Head Cap Screws (2109867-27) around the perimeter of the Front End Bell. Remove the four studs after End Bell is secured and install three M10 x 50mm screws in their place.
- ③ Tighten to eliminate gap between End Bell and Magnet Ring. SEE **NOTE** BELOW.
- ④ Re-install the access panels over the water hose connections.

NOTE:

As Front End Bell is being tightened, make sure the RF Coil Gasket installed previously does not expand into the bore. This will make it difficult to install Front Cover Strap (2287881) in a later procedure.

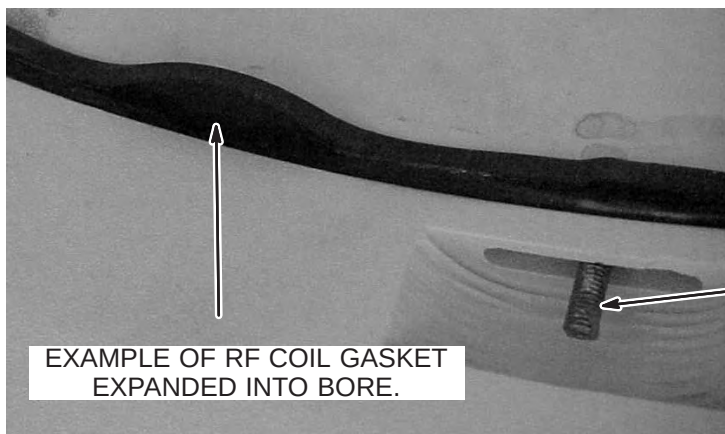
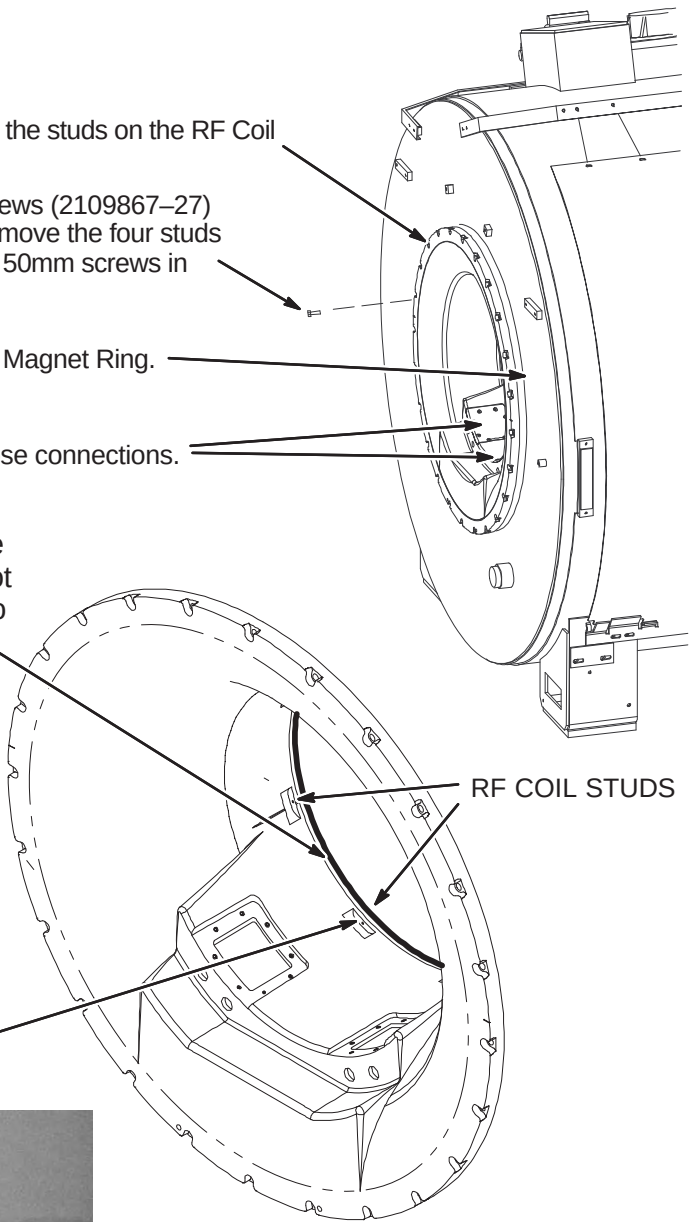
If RF Coil Gasket is too thin or too thick:

– Remove M10 x 50mm screws and slide End Bell away from RF coil so that RF Coil studs are disengaged. **END BELL DOES NOT HAVE TO BE COMPLETELY REMOVED.**

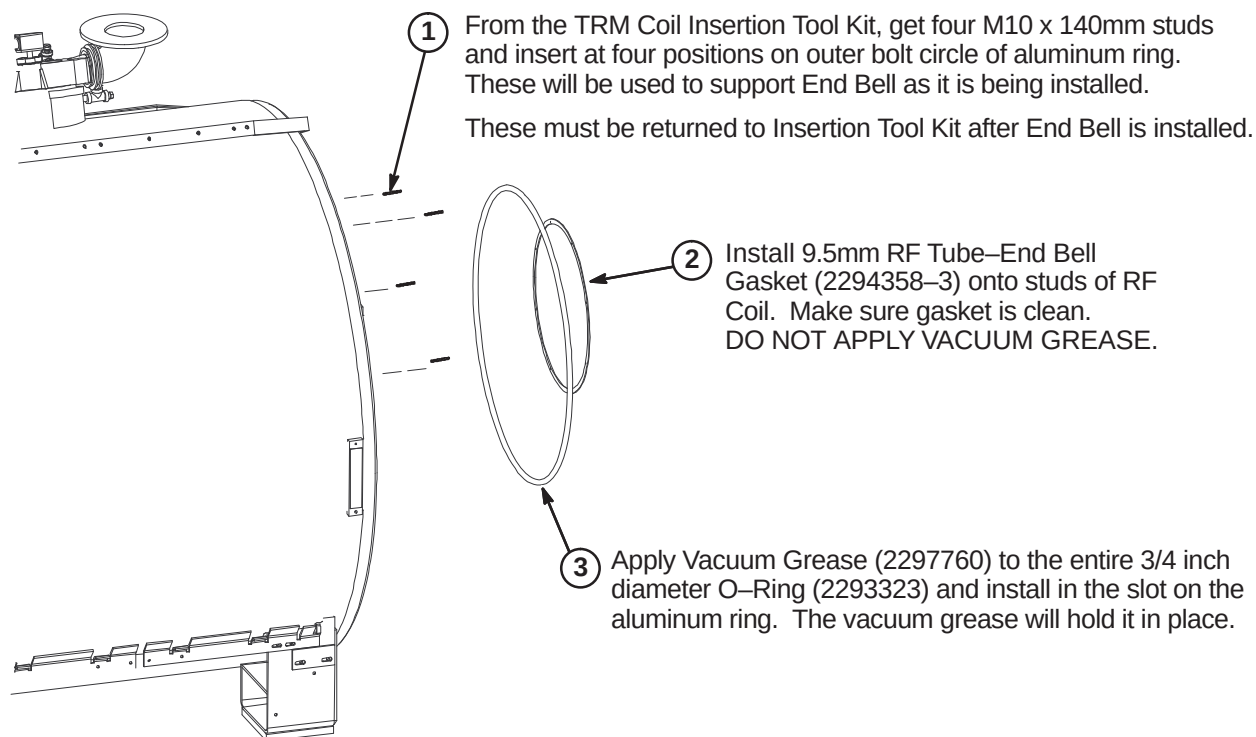
– If thinner gasket is required, install 4mm gasket (2294358).

– If thicker gasket is required, install 9.5mm gasket (2294358-3).

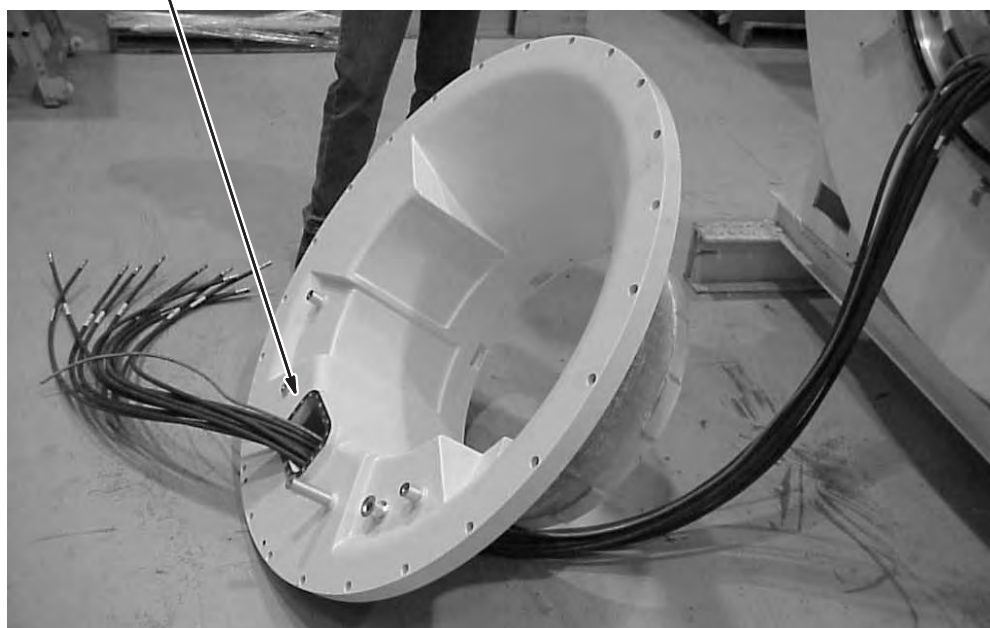
- ⑤ Fasten M6 Nylon nuts to each RF stud.

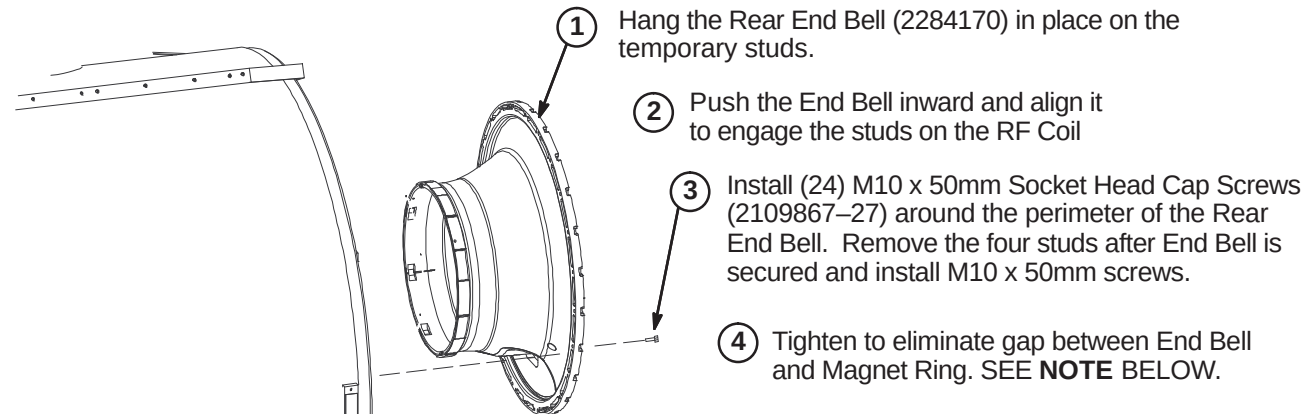


EXAMPLE OF RF COIL GASKET
EXPANDED INTO BORE.

6-6-2 INSTALL REAR END BELL**6-6-2-1 Install Rear End Bell Gaskets And Mounting Studs****6-6-2-2 Position Rear End Bells On Studs**

- 1 Place the Rear End Bell (2284170) at rear of magnet and route gradient cables thru opening.
- 2 Move End Bell into position, again hanging it on the four studs temporarily.



6-6-2-3 Rear End Bell Installation**NOTE:**

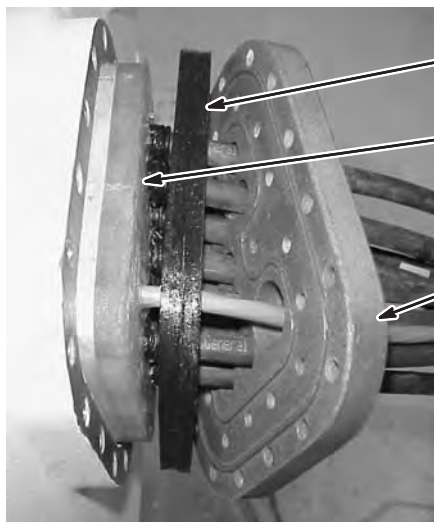
As Rear End Bell is being tightened, make sure the RF Coil Gasket installed previously does not expand into the bore. This will make it difficult to install Rear Cover Strap (2287881-2) in a later procedure.

If RF Coil Gasket is too thin or too thick:

– Remove M10 x 50mm screws and slide End Bell away from RF coil so that RF Coil studs are disengaged. **END BELL DOES NOT HAVE TO BE COMPLETELY REMOVED.**

– If thinner gasket is required, install 4mm gasket (2294358) or 6mm gasket (2294358-2).

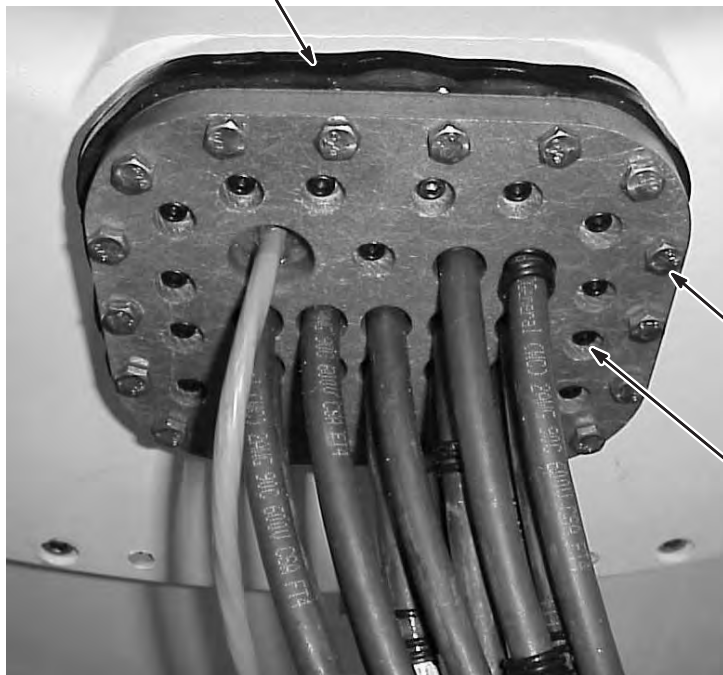
– If thicker gasket is required, add any combination of the remaining 4mm and 6mm gaskets with the existing 9.5mm until correct thickness is produced.

6-6-2-4 Gradient Block Gasket And Cover Installation

- ① Install the 12.7mm thick Gradient Block Gasket (2297540-3) over gradient cables.
- ② When in place, gasket should stick out just past end of Gradient Block. If adjustment is needed, add 4mm gasket (2297540) and/or 6mm gasket (2297540-2) or replace 12.7mm gasket with 4mm gasket (2297540) and/or 6mm gasket (2297540-2).
- ③ Thread the gradient cables thru the corresponding holes in the Gradient Block Cover (2285381).
- ④ Apply Vacuum Grease (2297760) to the sealing surfaces of the cover plate and gasket.

6-6-2-5 Tightening Of Gradient Block Gasket And Cover

① Determine total gasket thickness.



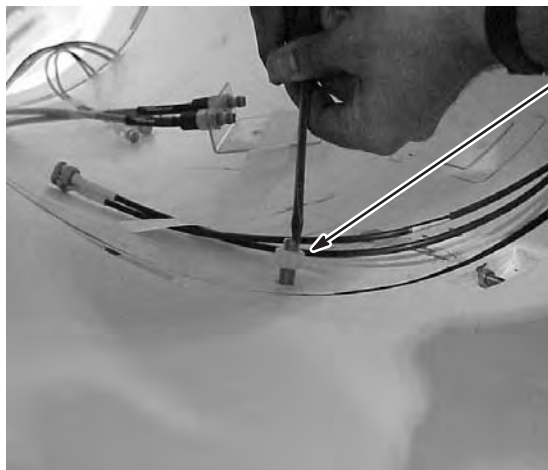
② Select the gasket thickness from the table below to determine which length of screw to use. Eighteen (18) screws of each length are supplied for this upgrade.

GASKET THICKNESS	SCREW LENGTH	PART NUMBER
4mm	25mm	2109873-27
6mm	25mm	2109873-27
10mm	30mm	2109873-28
12.7mm	35mm	2109873-29
16.7mm	35mm	2109873-29
18.7mm	40mm	2109873-30
22.7mm	50mm	2109873-31

③ After the correct length of screw is determined, insert the selected (18) M6 Pan Head Screws around the Block Cover and start them into the threads.

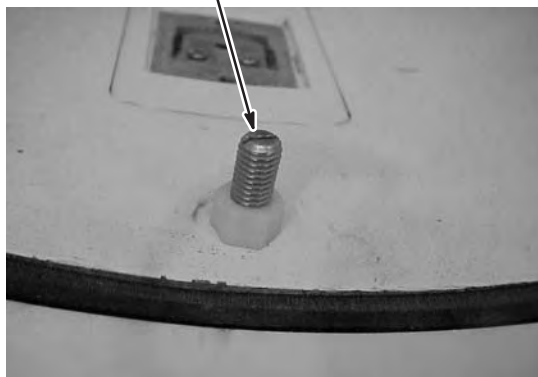
④ Insert (16) M4 x 25mm Pan Head Screws (2109873-18) in the inner holes of the Block Cover and start them into the threads.

⑤ Tighten all the inner and outer screws uniformly to compress the gasket and to form a leak-free vacuum seal.

6-6-2-6 Loosen RF Coil Alignment Set Screws

① Loosen all the set screws that were required for centering the RF Coil within the TRM Coil.
DO NOT REMOVE.

② Tighten the Nut to lock set screw in place.

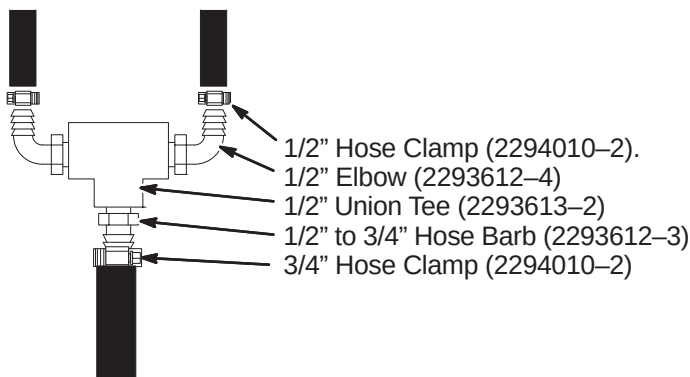


6-6-3 INSTALL GRADIENT COIL COOLING HOSES**6-6-3-1 Connect Coolant Hoses to Tees**

- ② Install the End Bell coolant lines to hose barbs on End Bell using (4) worm drive Hose Clamps (2294010):

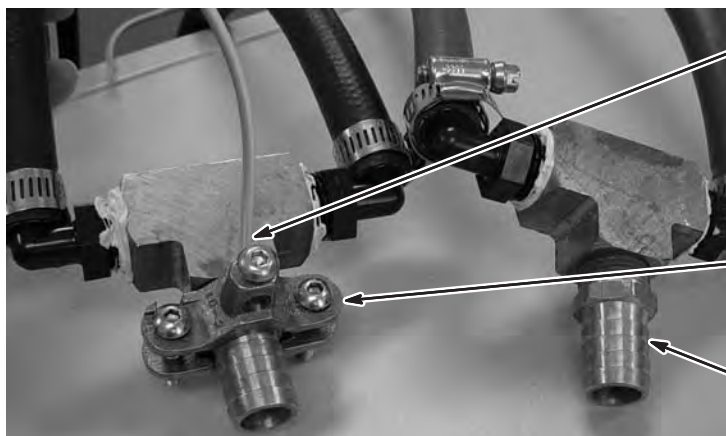
- Black hose (2293610-2) to be marked **OUTLET**
- Blue hose (2293610-3) to be marked **INLET**

- ③ Connect the 3/4" hose routed from the System Cooling Cabinet to the **INLET** and **OUTLET** hoses using:



- ① Route the **IN** and **OUT** Coolant hoses from the Cooling Cabinet underneath the magnet to the front of the enclosure.

Note: It is important that the **IN** and **OUT** hoses are connected to the respective **IN** and **OUT** ports on the Cooling Cabinet.

6-6-3-2 Install Ground Clamps on Tees

- ① Verify that one end of each Jumper Wire Asm (5126241) is stripped bare of insulation for 1/2 inch (12.7mm) at one end and insert stripped end into hole of clamp. Tighten the wire securing screw.
- ② Place Ground Clamp (5126242) over hex portion of 3/4 inch hose barb and tighten securely.
- ③ Repeat steps 1 and 2 for other coolant tee.

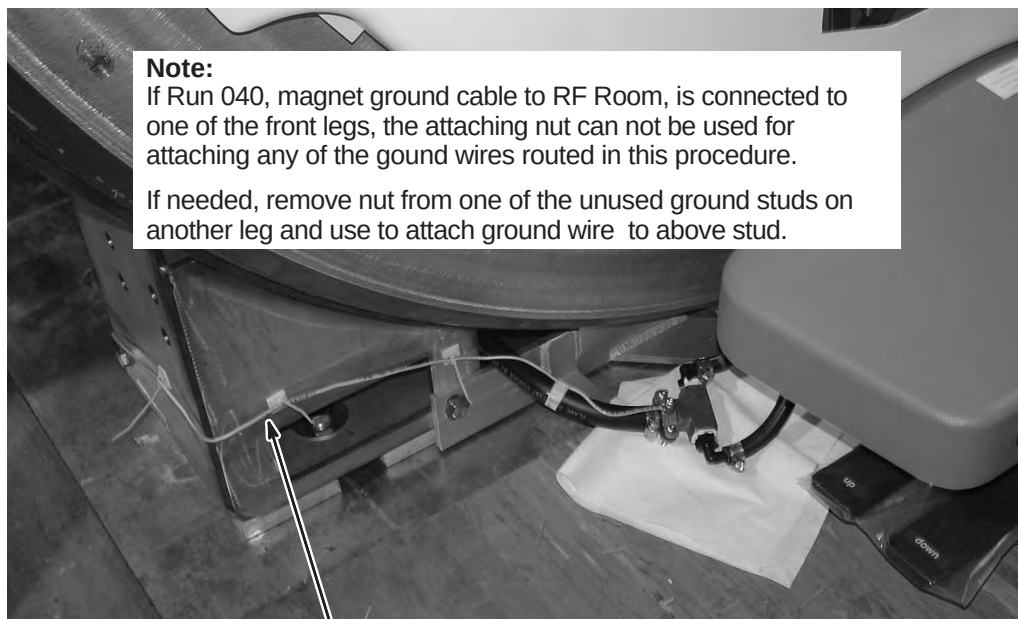
6-6-3-3 Route/Connect Ground Wires to Magnet**IMPORTANT !!!**

Make sure Jumper Wire Asm 5126241 is secure from motion along its entire length.

Note:

If Run 040, magnet ground cable to RF Room, is connected to one of the front legs, the attaching nut can not be used for attaching any of the ground wires routed in this procedure.

If needed, remove nut from one of the unused ground studs on another leg and use to attach ground wire to above stud.



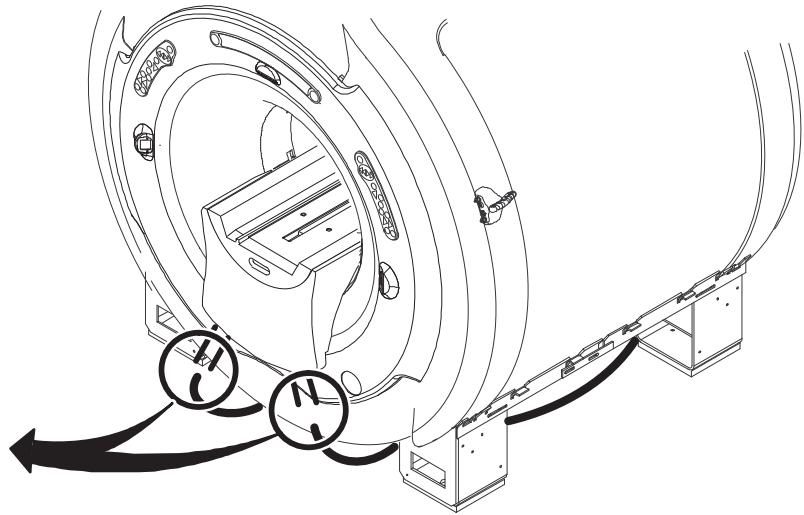
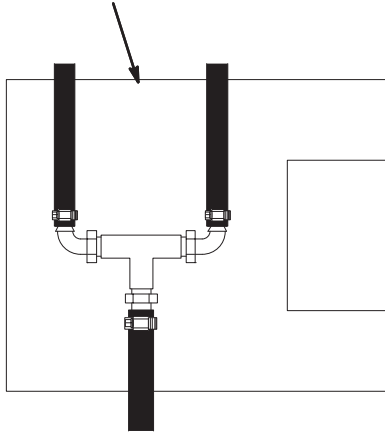
- ① Route one of the ground wires around the left leg to ground stud on inside of leg. Secure wire to leg with adhesive backed Tie Wrap Base (46-208747P1) and Tie Wraps (5112522). Securely fasten the ring terminal to ground stud.

- ② Route other ground wire to right leg ground stud. Secure and attach ground wire as shown.



6-6-3-4 Install Connector Wraps

- ① Wrap each Tee Connector with 177 x 200 Connector Wrap (2236130-2)

**6-6-3-5 Coolant Valve Installation**

Typical coolant Supply (**IN**) and Return (**OUT**) piping for outdoor cooling unit that is provided by customer in equipment room.

3/4" Tee Handle
(2293620)

3/4" Union Tee (2293613)
3/4" Hose Barb (2293612)

This end
towards
Cooling
Cabinet

INLINE DRAINAGE VALVES

This end
towards
TRM Coil

- ① The Inline Drainage Valves shown above are installed in coolant hoses near the Penetration Panel on the Equipment Room side. The valve end must be positioned towards the Cooling Cabinet and the Tee end towards the TRM coil.

- ② Connect **IN** and **OUT** coolant hoses to respective piping hose barbs, on outdoor or indoor unit.

6-6-4 COOLANT LEAK TEST

Consult applicable vendor supplied manual that was delivered with Cooling Cabinet for installation, start-up, and operating instructions and procedures. Manual can also be found on the MR Service Methods CD-ROM: Install > Coil Blower & Chiller Assemblies

Note: For Fixed sites only

- ① Verify Set Point is at 20 degrees C.
- ② At front of magnet, remove plexiglass inspection ports and check coolant line connections for leaks. Replace plexiglass when complete.
- ③ Check for leaks at shut off valves.

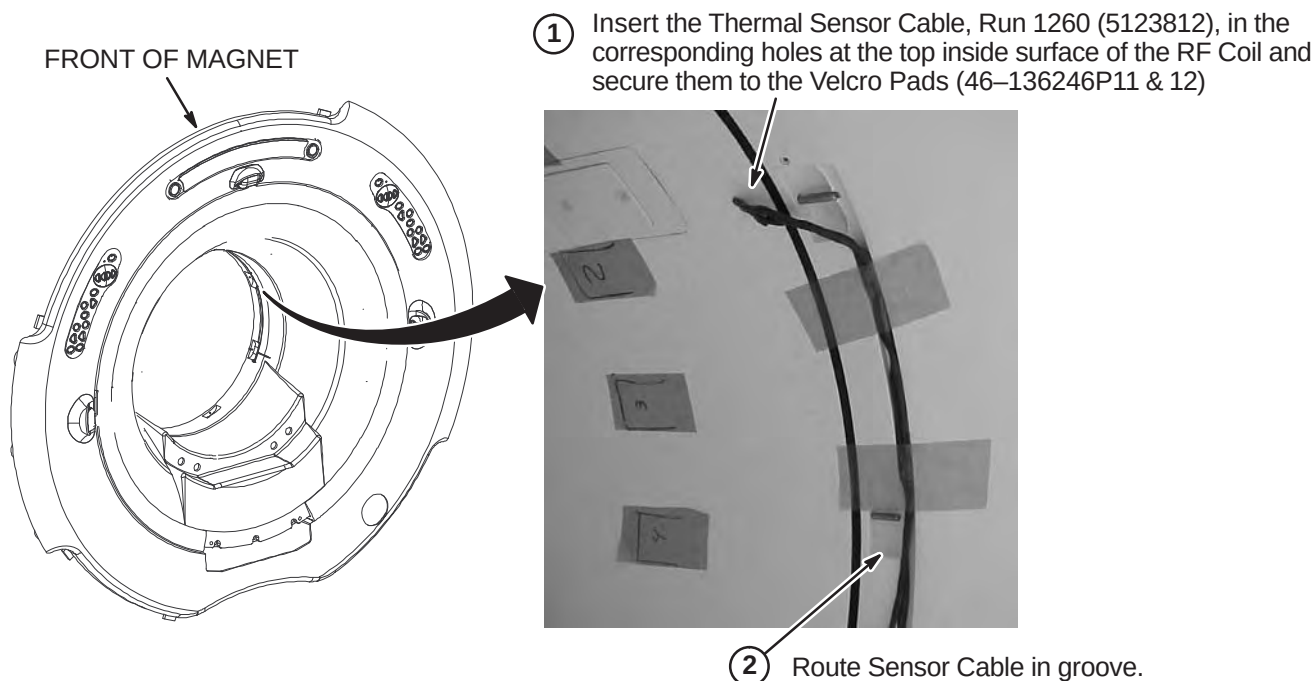
Note: If leak is found, determine source of leak and repair it. **DO NOT ADD MORE COOLANT UNTIL ALL LEAKS ARE FOUND AND REPAIRED.**

6-6-5 VACUUM LEAK TEST

Note: Vacuum Grease must be applied to all Centering Rings (2293608 & 2293608-2).

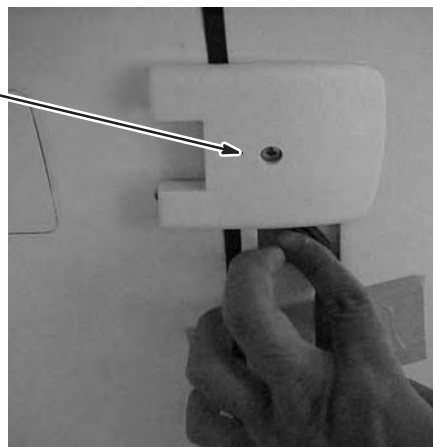
- ① Refer to EXCITE HD MR Service Methods CD-ROM: Twin Accessory Cabinet and complete TwinSpeed vacuum system Functional Checks (tafca3).
- ② To conduct vacuum leak test, refer to EXCITE HD MR Service Methods CD-ROM: Magnet Enclosures/ WideOpen Magnet Enclosures/Troubleshooting/Vacuum Enclosure Troubleshooting (me3tsa1).

6-6-6 INSTALL THERMAL SENSOR CABLE

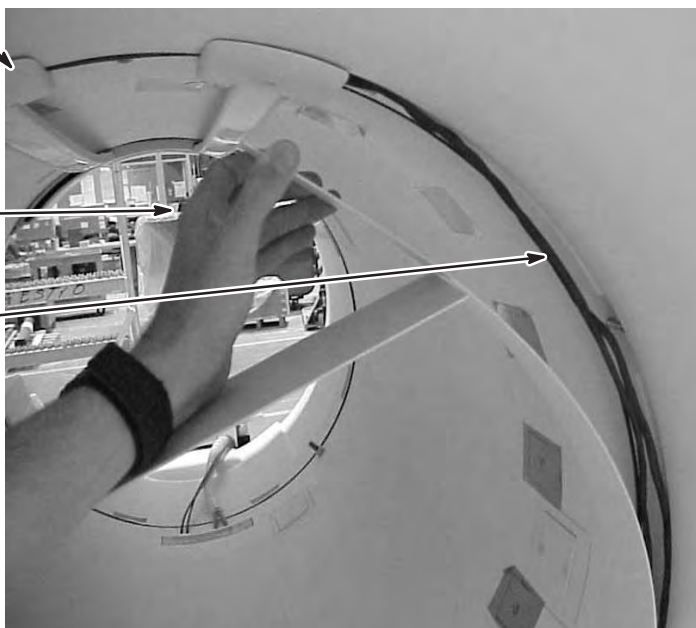


6-6-7 INSTALL FRONT END BELL COVER STRAP

- ① Retain the end of Sensor Cable by installing the Bore Light End Cap (2303834) using a M6 x 16mm Pan Head Screw (2109873-25).



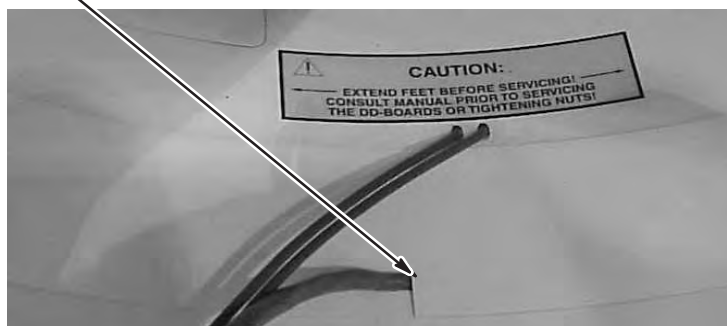
- ② Install the second Bore Light End Cap.



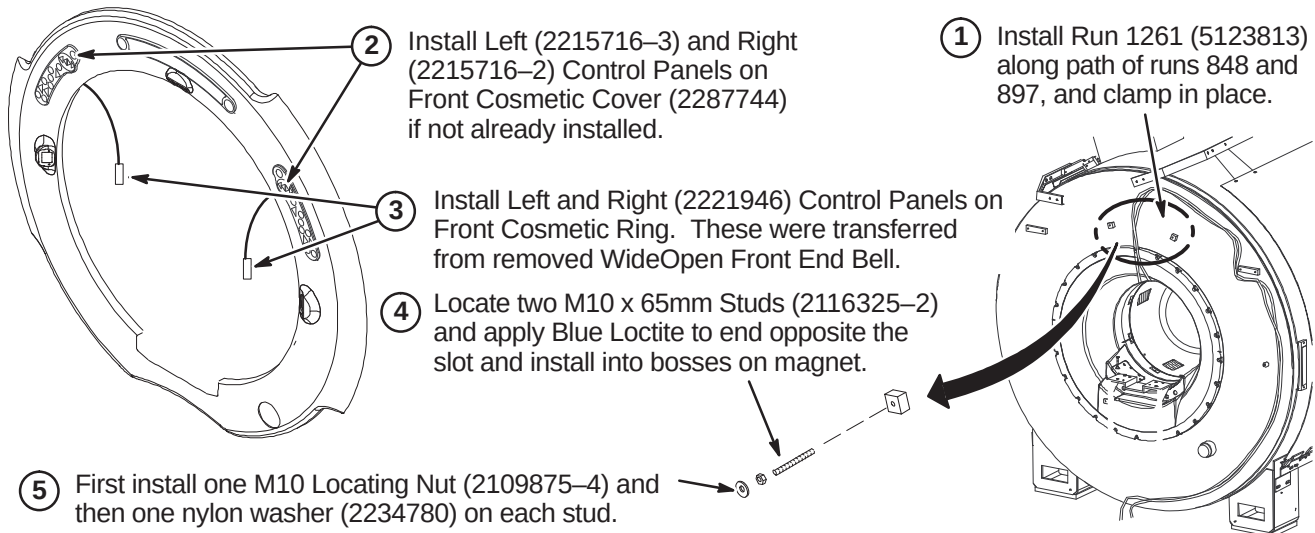
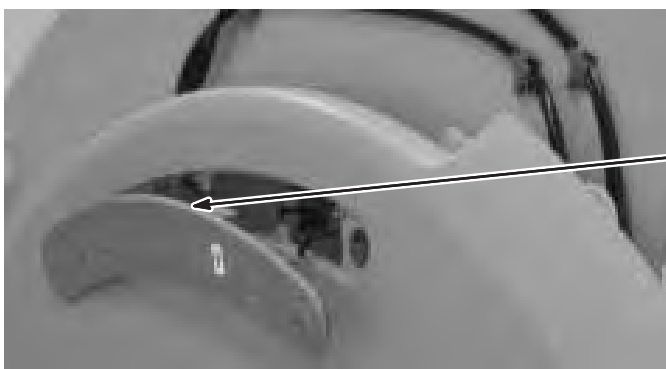
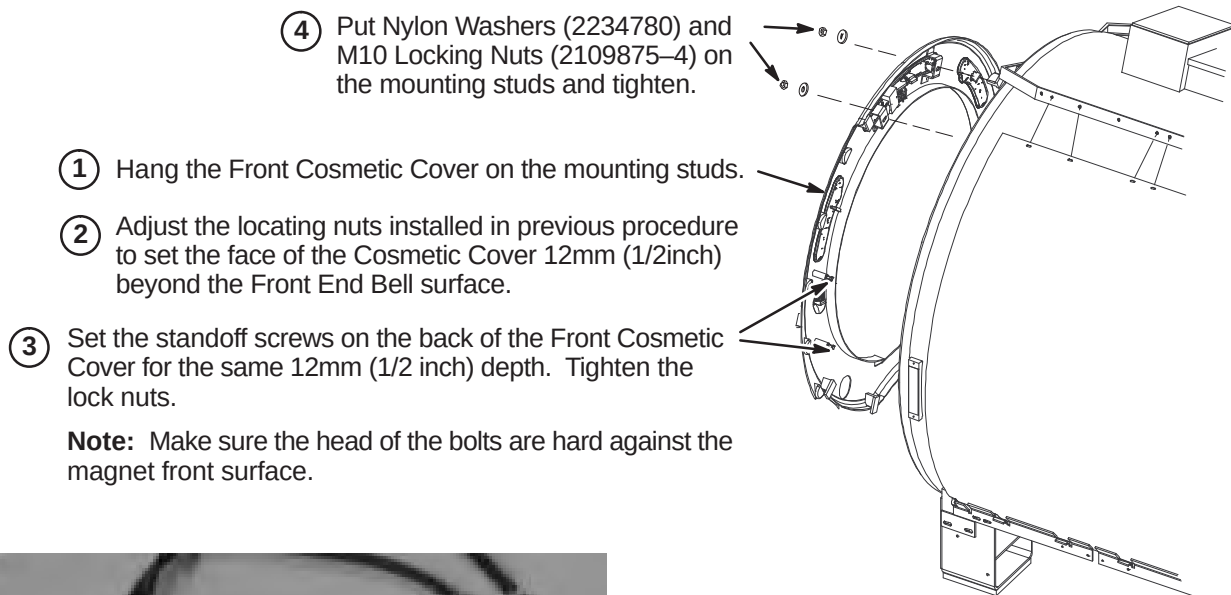
- ③ Install the Front Cover Strap (2287881) over the End Bell gasket capturing the Thermal Cable in the groove.

Note: Use tape to hold the cable in place temporarily.

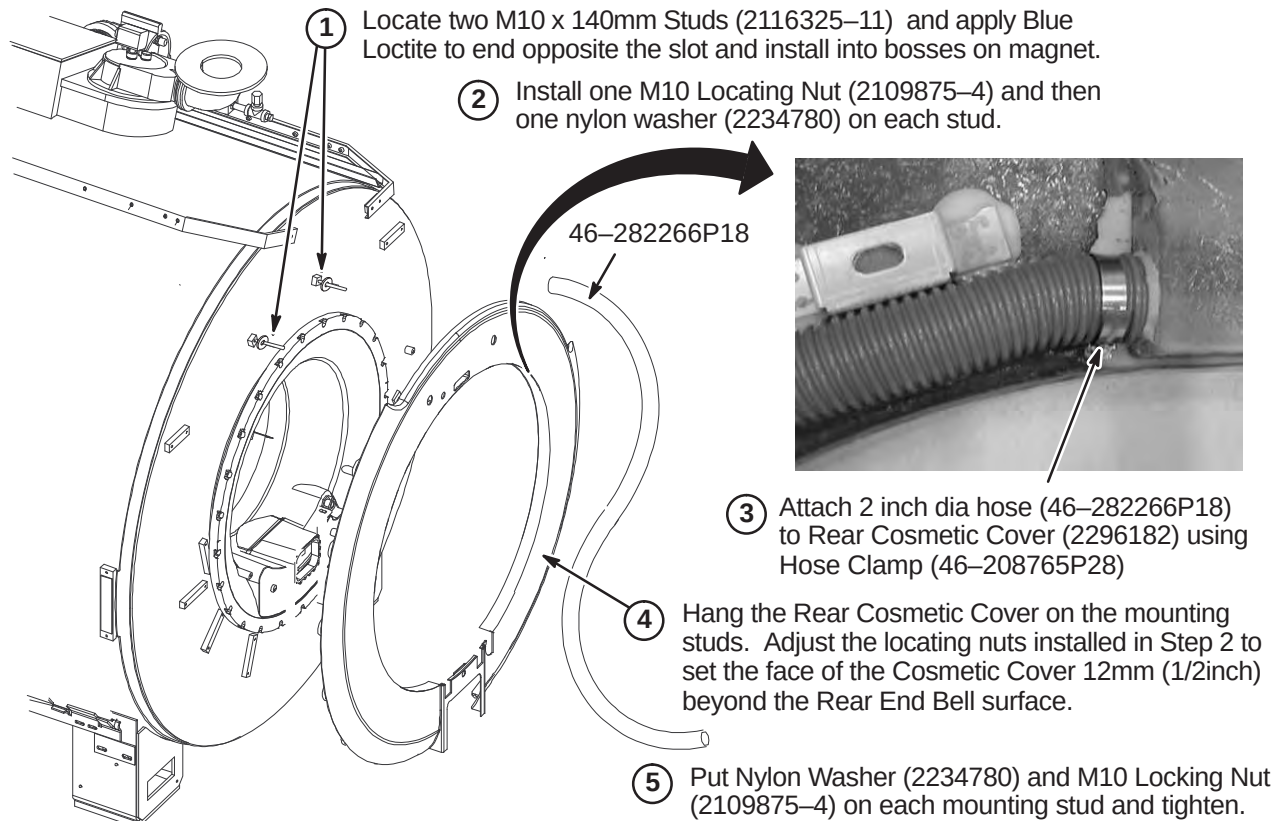
- ④ View at bottom of RF Coil showing Thermal Cable exiting from underneath the Cover Strap



- ⑤ Go to Section **6-6-18** to install Comort Module (2284175) and Bore Lights and Rear Cover Strap (2287881-2).
This must be completed before installing Bridge.

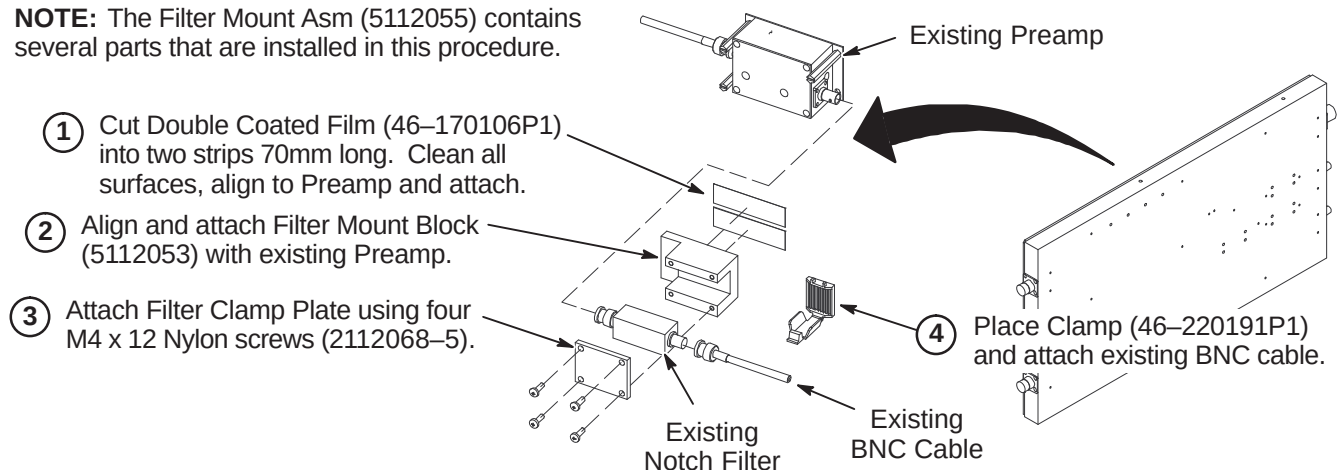
6-6-8 INSTALL FRONT COSMETIC COVER**6-6-8-1 Front Cosmetic Cover Installation Preparation****6-6-8-2 Front Cosmetic Cover Installation**

- ⑤ Carefully, using a small screwdriver, pry the Operator Display away from the front cover.
- ⑥ Connect new Run 1261 to J4 on back of Operator Display.
- ⑦ Re-install Operator Display.

6-6-9 INSTALL REAR COSMETIC COVER**6-6-10 INSTALL NEW COMPONENTS ON REAR PEDESTAL****6-6-10-1 Install Filter Mount Asm to Body Hybrid Splitter**

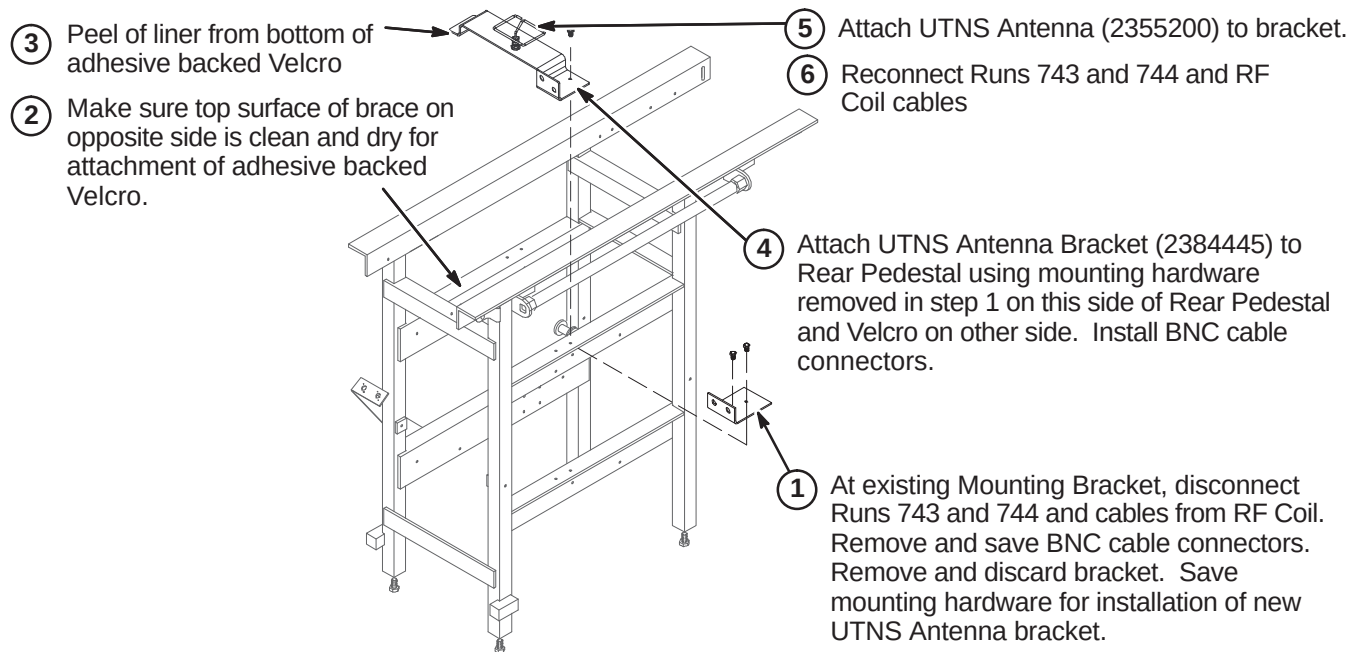
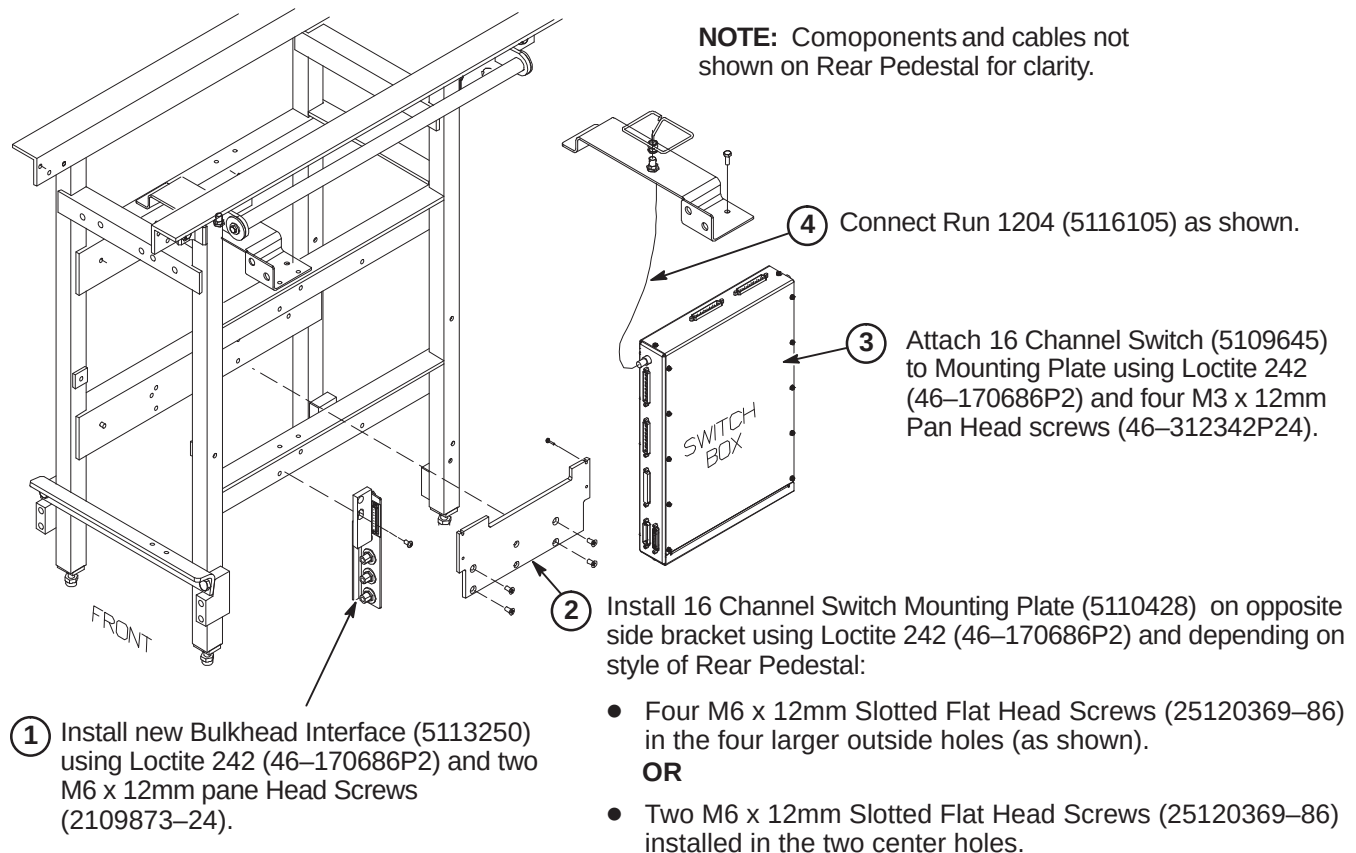
Depending on Rear Pedestal design, the Body Splitter may be in the horizontal or vertical position.

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



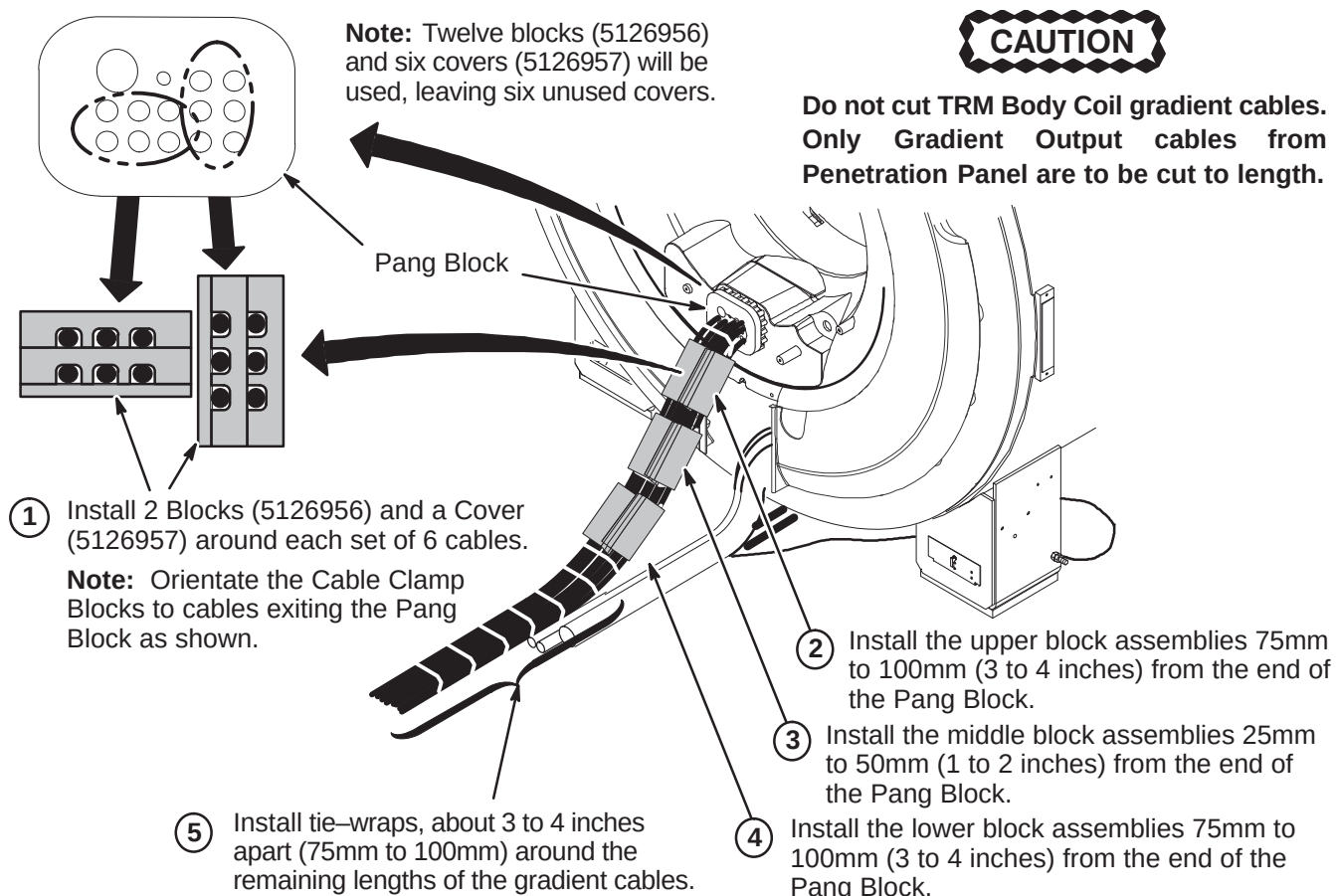
6-6-10-2 Install UTNS Antenna

Two mounting brackets are provided for this upgrade. Only one will be installed depending on magnet enclosure.

**6-6-10-3 Install 16 Channel Switch and Bulkhead Interface**

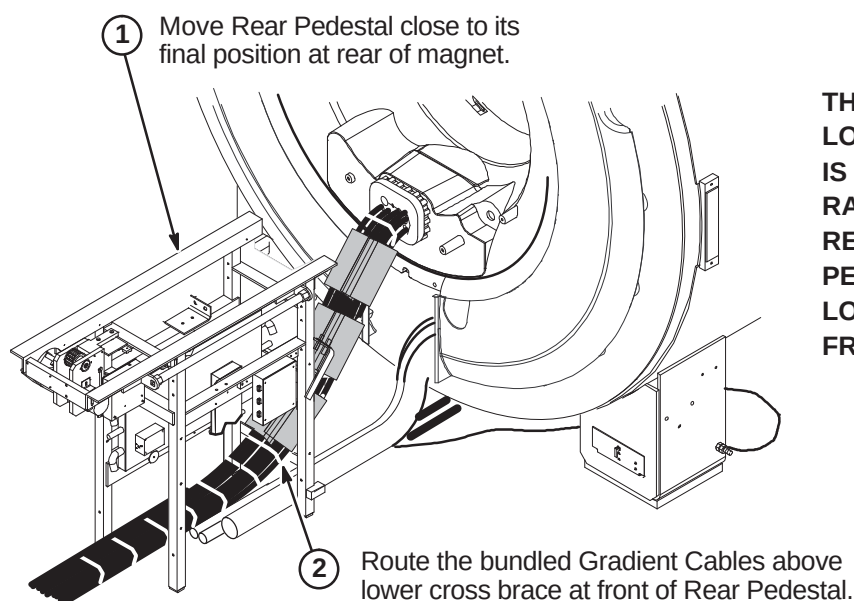
6-6-11 INSTALL GRADIENT CABLE BLOCKS

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.



6-6-11-1 Route Body Coil Gradient Cables Thru Rear Pedestal

Note: Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.

**WARNING!**

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

6-6-11-2 Connect ZM (Zoom) Cables to Penetration Panel

Nine cables have been delivered to site. Nine of the labeled pre-terminated leads will be connected to the Twin Gradient Filter on the Penetration Panel. The other three labeled pre-terminated ends will be connected to the TAC Cabinet. The other ends of all the cables will be cut to length, labeled, and terminated. The nine cables with supplied labels are as follows:

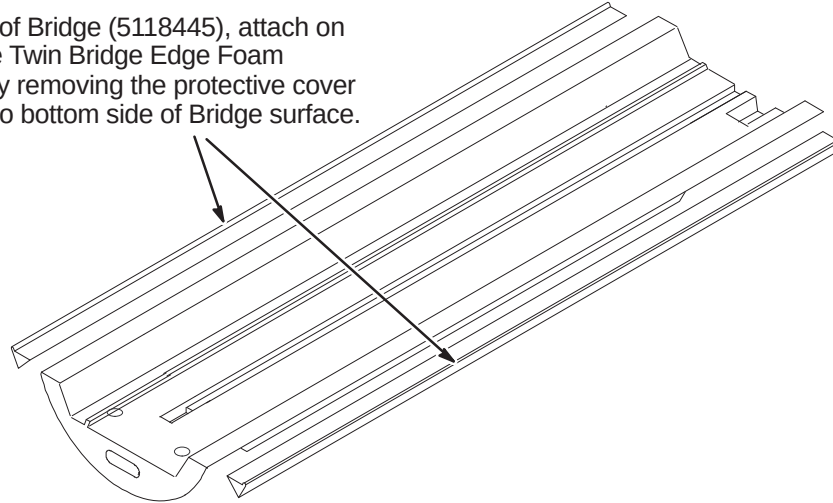
RUN	LABEL	SUPPLIED LABELS
976/979	PP1-A7-X	(1) RUN 976, (1) RUN 979, (2) X+, (1) X-, (1) GND, (1) XZM-, (1) XWB-
977/980	PP1-A7-Y	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y-, (1) GND, (1) YZM- (1) YWB-
978/981	PP1-A7-Z	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z-, (1) GND, (1) ZZM- (1) ZWB-
982	PP1-A7-Z _{ZM}	(1) RUN 982, (1) ZZM- (1) ZZM+
983	PP1-A7-Y _{ZM}	(1) RUN 983, (1) YZM- (1) YZM+
984	PP1-A7-X _{ZM}	(1) RUN 984, (1) XZM- (1) XZM+
985	PP1-A7-Z _{WB}	(1) RUN 985, (1) ZWB- (1) ZWB+
986	PP1-A7-Y _{WB}	(1) RUN 986, (1) YWB- (1) YWB+
987	PP1-A7-X _{WB}	(1) RUN 987, (1) XWB- (1) XWB+

TOOLS REQUIRED: 2134776 – Cable Crimper/Stripper Kit

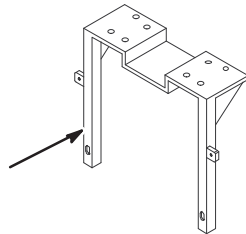
- ① The following instructions are for installing and connecting the Magnet Room gradient cable Runs 982, 983, 984, 985, 986, and 987.
- ② Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. **Do not cut Body Coil gradient cables.**
- ③ Locate furnished bag of parts that contain the terminals and labels required to be attached to the cables.

6-6-12 INSTALL FRONT SPLIT BRIDGE**6-6-12-1 Install Front Bridge Support onto Bridge**

- ① Starting at the front of Bridge (5118445), attach on each side one of the Twin Bridge Edge Foam (2293538) pieces by removing the protective cover and applying foam to bottom side of Bridge surface.



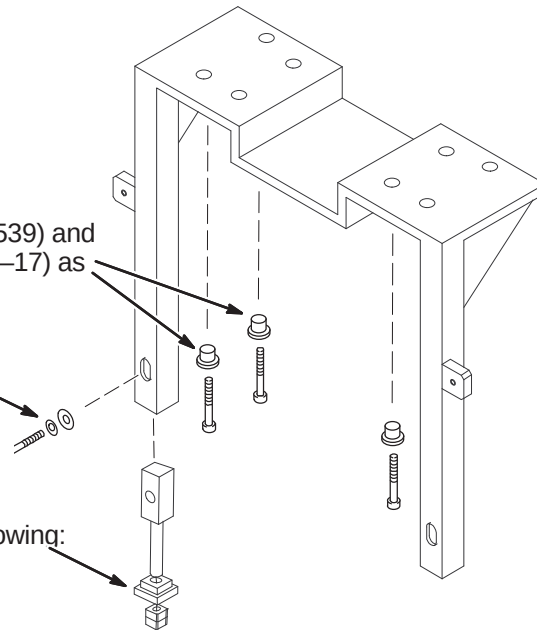
- ② Move Front Bridge Support Bracket (2293202) to front of Bridge.

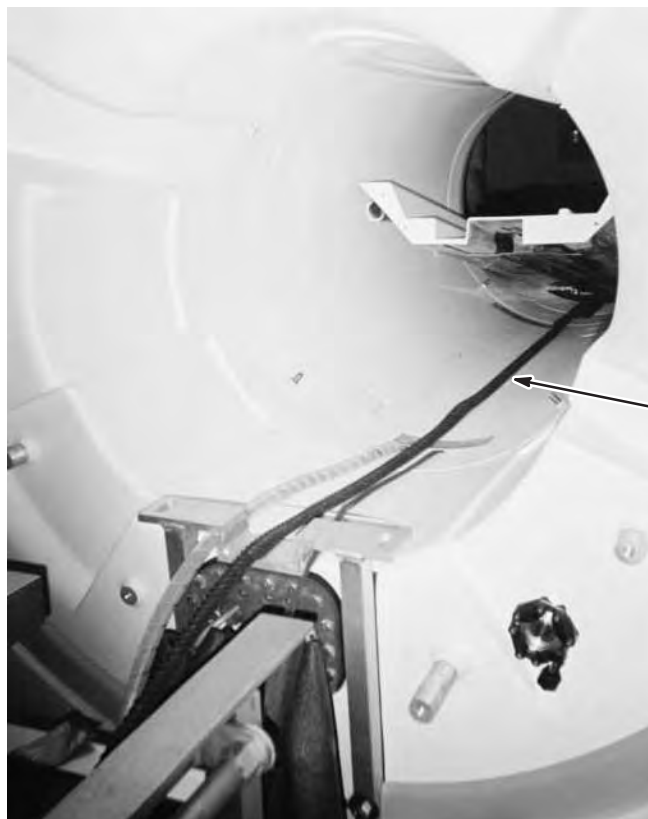


- ③ Install two Twin Bridge Support Bushings (2293539) and M6 x 35mm long Socket Head screws (2109867-17) as shown in opening at bottom of front of Bridge.

- ④ The Bridge Support will be attached to the front alignment ring on the magnet using on each leg: M10 x 100mm Hex Socket screw (2109866-17) Washer (2167321) Large Washer (2234780)

- ⑤ The Bridge Support will be leveled using the following:
Twin Support Lock Top (2293540)
Twin Bridge Support Slider (2293541)
(2) M10 Nylon Hex Nuts (2293956)

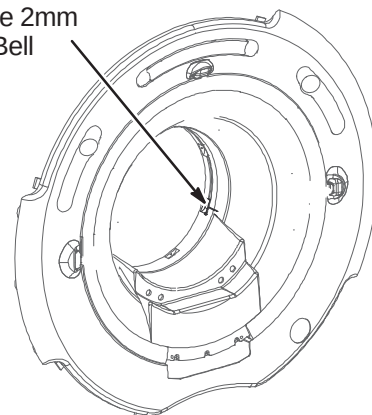


6-6-12-2 Installation of Bridge

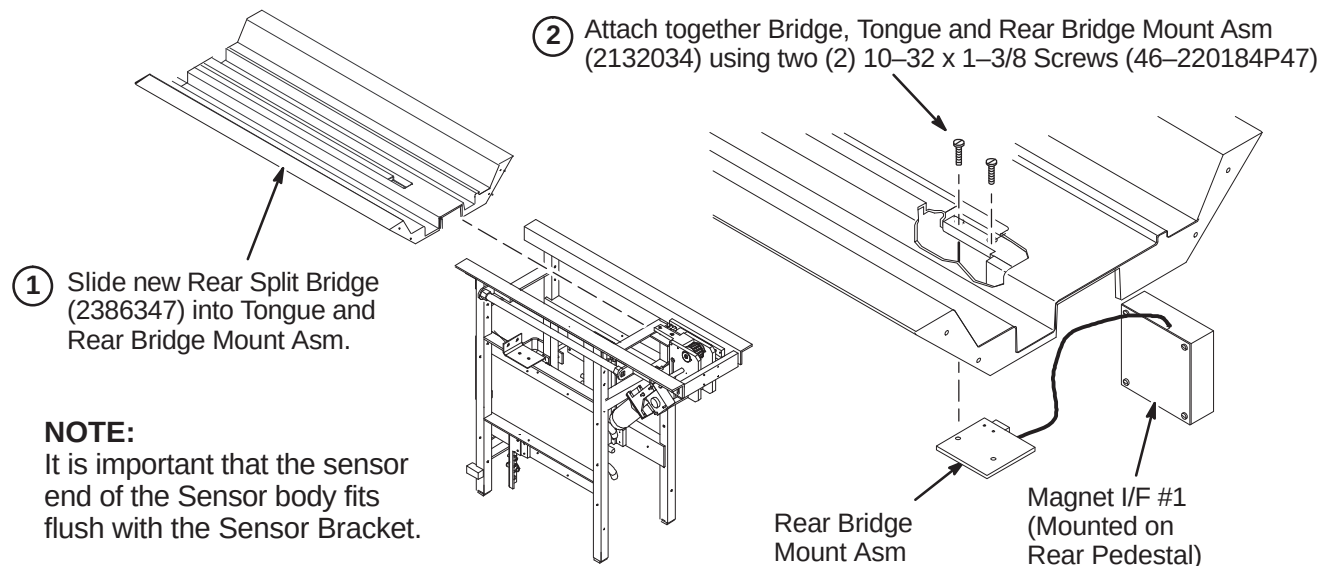
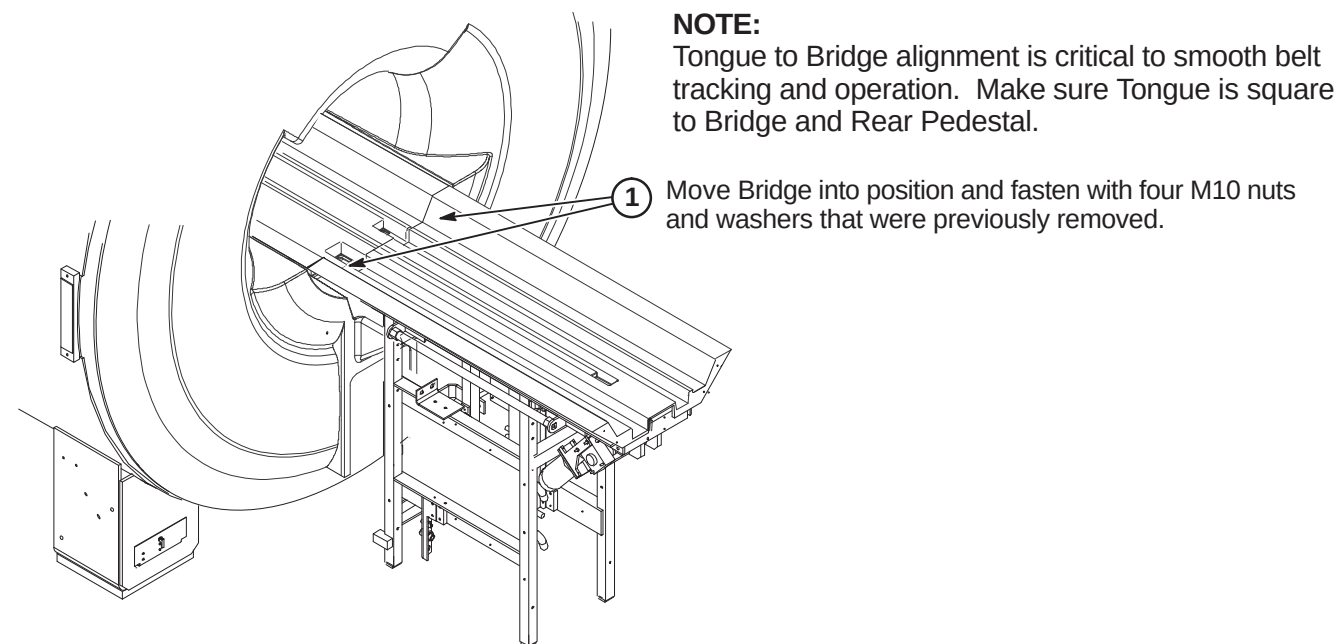
- ① Roll out belt prior to installation of Bridge.
- ② Guide the Bridge thru the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into Bridge.

6-6-12-3 Attaching Front of Bridge

- ① The final location of the top of front end of Bridge will be 2mm to 4mm above alignment marks on side of Front End Bell



- ② Align the Front Bridge Support screws (2109866-26) and washers (2234780 & 2167321) with the corresponding holes in the End Bell.
- ③ Snug the screws. Do not overtighten, since adjustments may be needed.

6-6-13 INSTALL REAR SPLIT BRIDGE**6-6-13-1 Install Rear Split Bridge on Rear Pedestal****6-6-13-2 Connect Rear Split Bridge to Front Split Bridge**

6-6-13-3 Level Bridge**NOTE:**

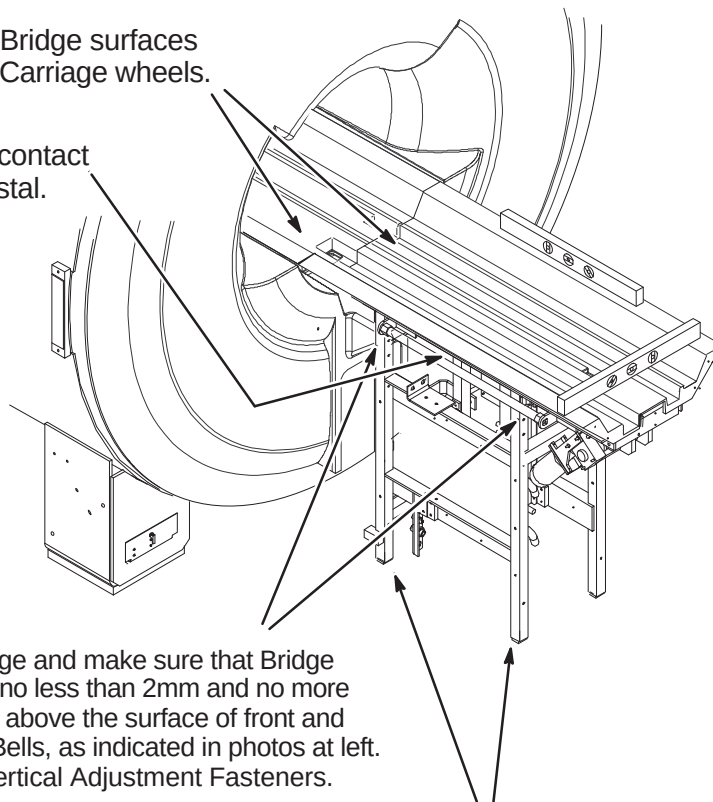
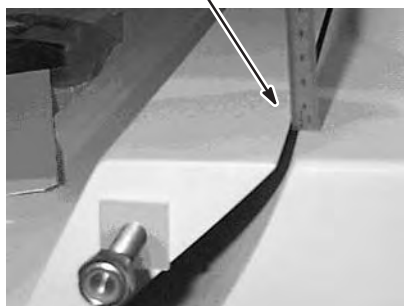
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:

Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

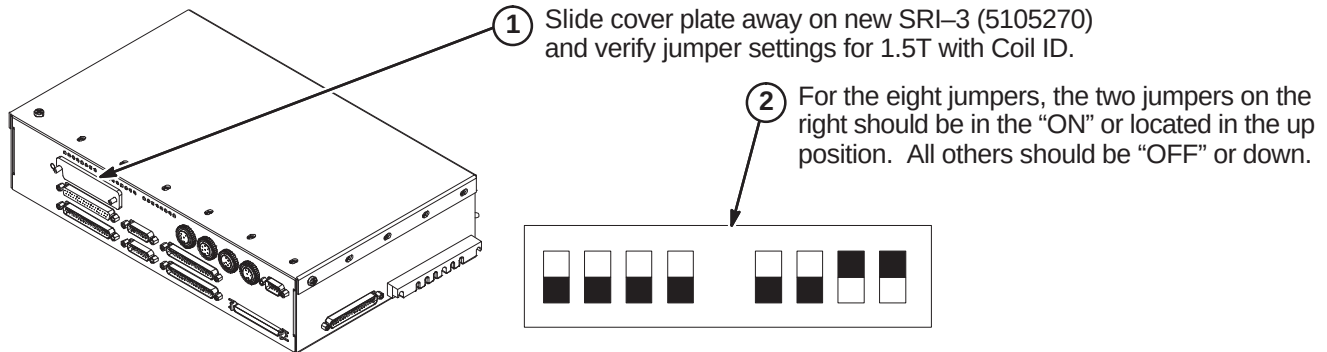
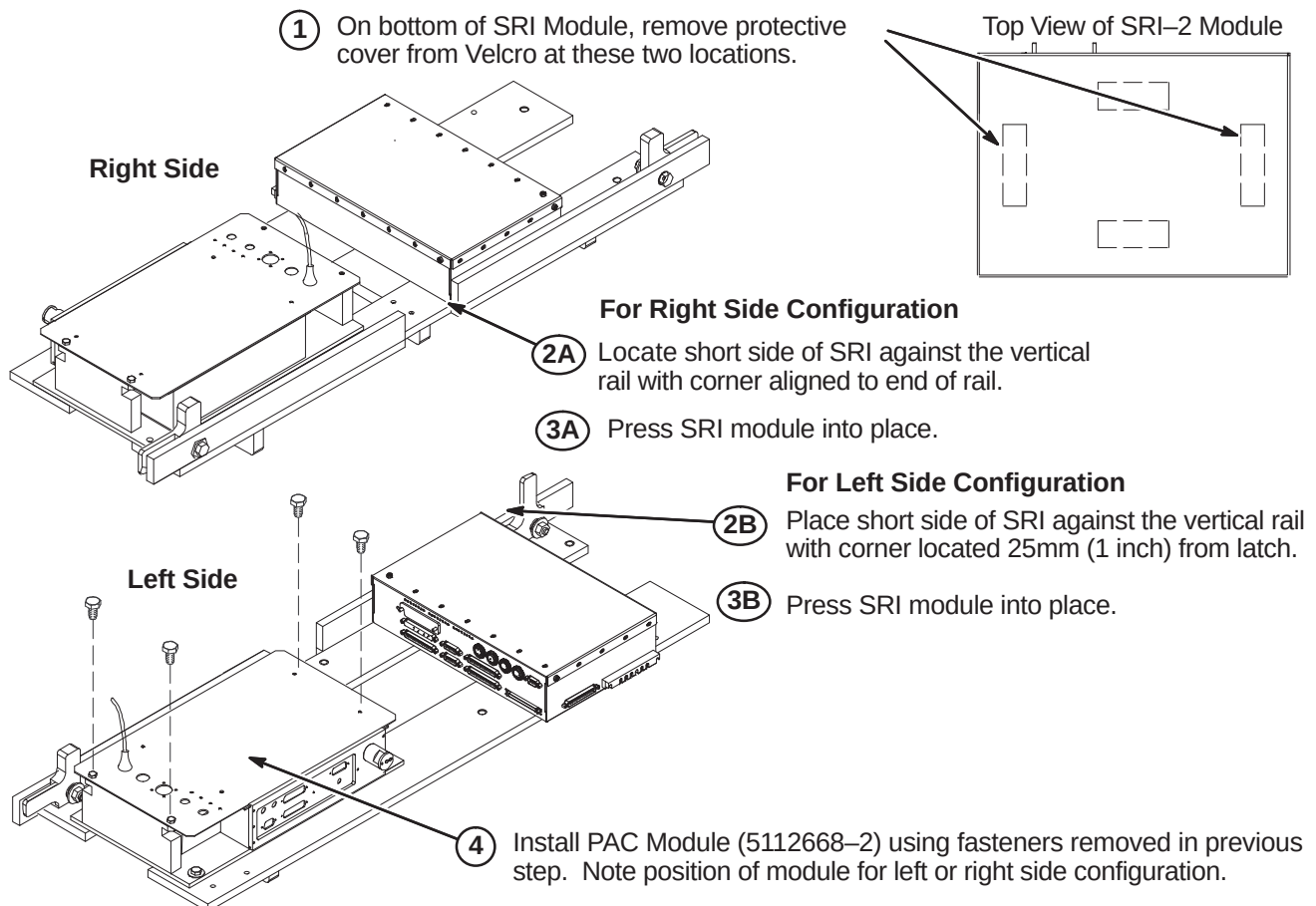


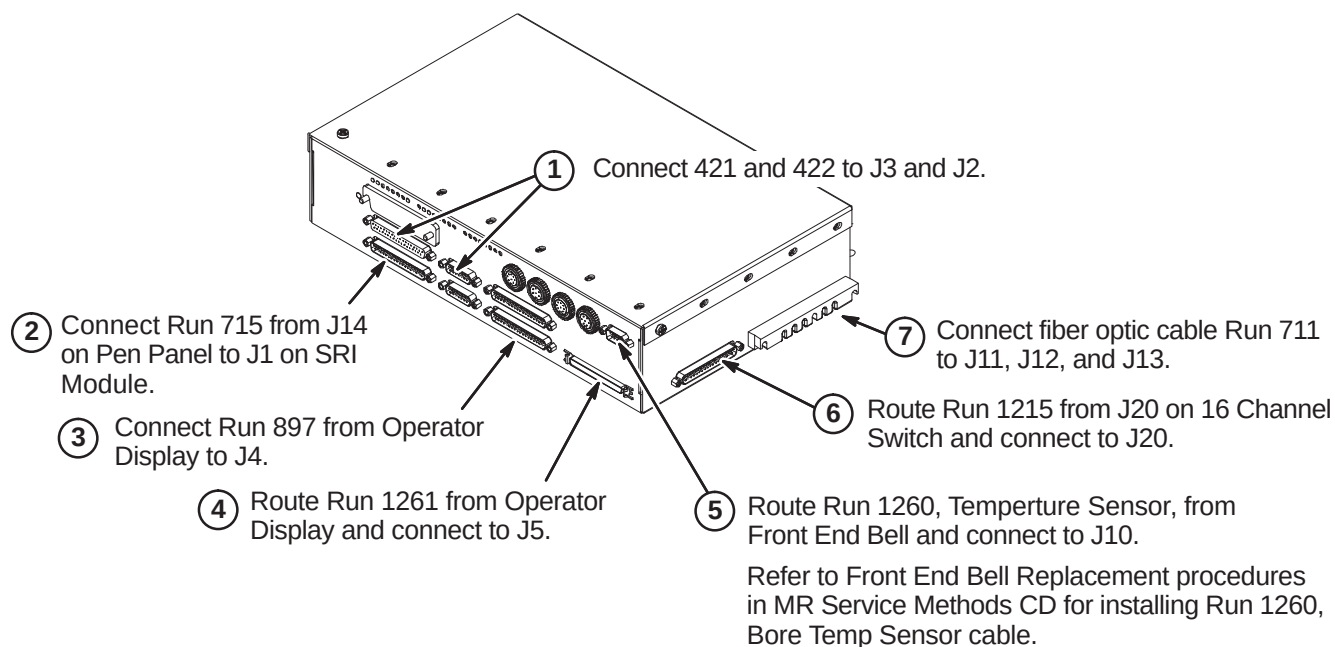
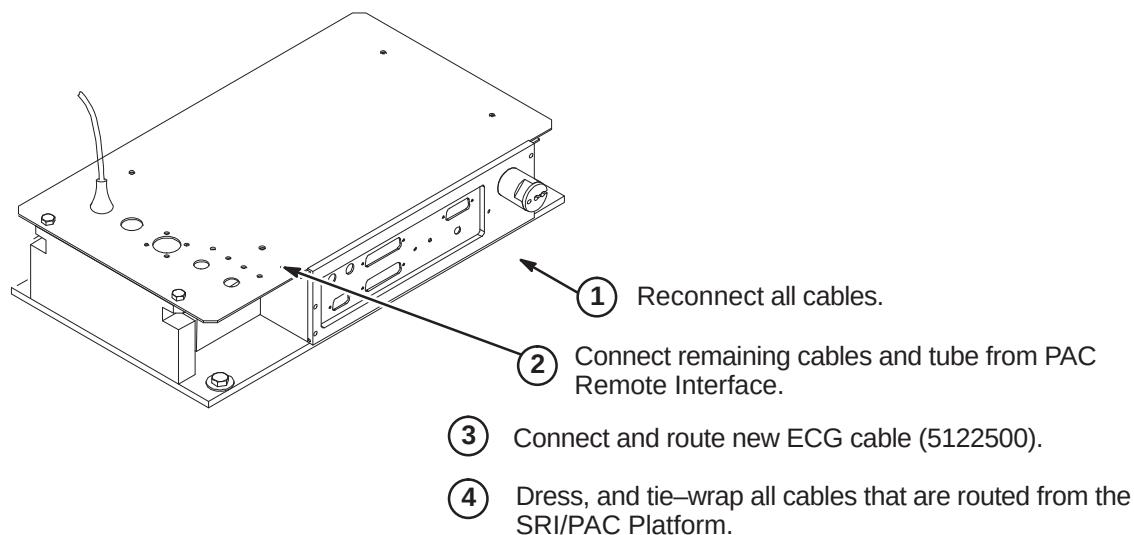
2mm to 4mm above End Bell surface.



- ① Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of front and rear End Bells, as indicated in photos at left. Tighten Vertical Adjustment Fasteners.

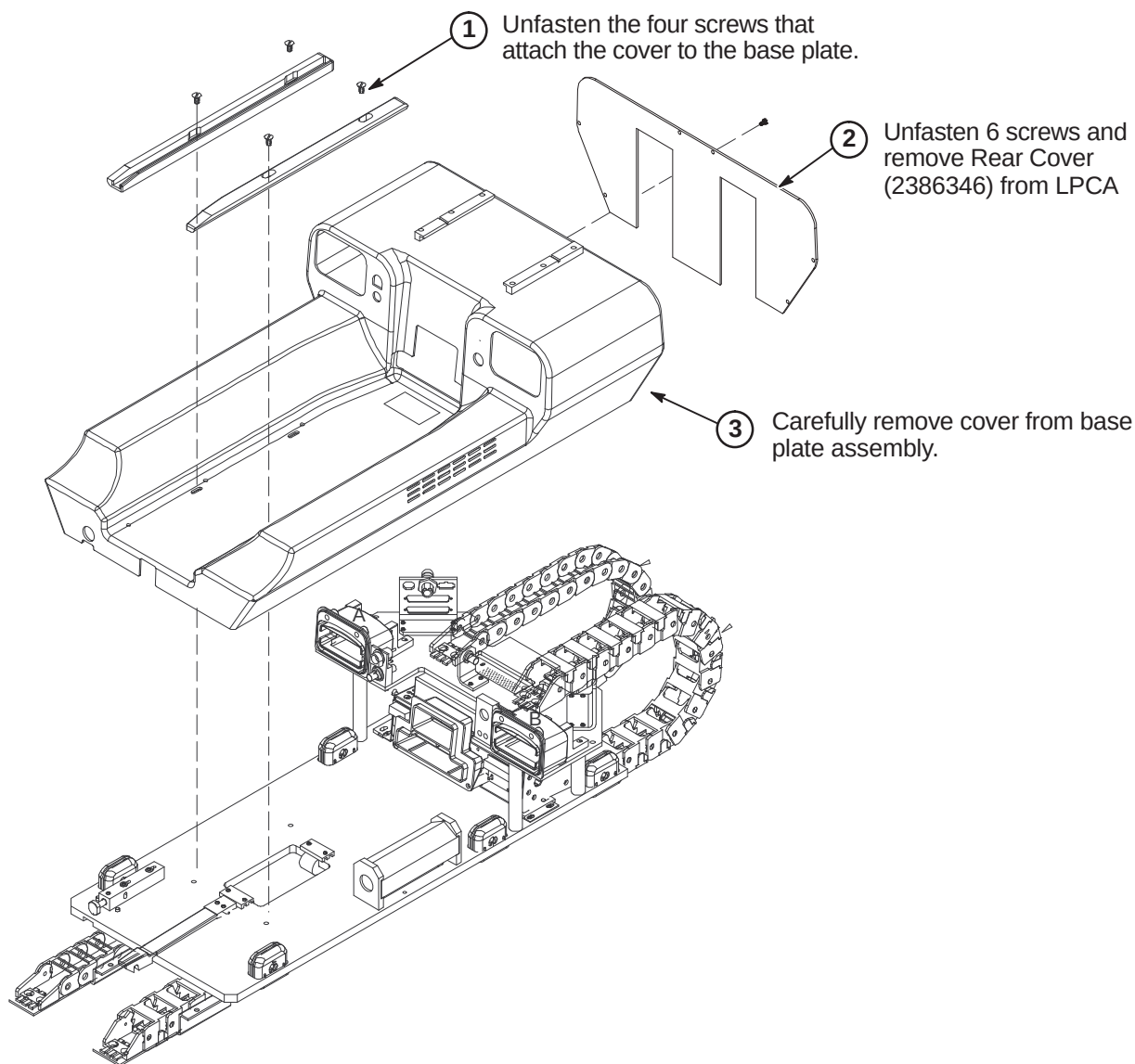
- ② Adjust leg screws on both sides as needed to level Rear Pedestal.

6-6-14 INSTALL PAC AND SRI MODULES**6-6-14-1 Verify Jumper Positions for SRI****6-6-14-2 Install New PAC and SRI On Right or Left Side of Enclosure**

6-6-14-3 Connect Cables to SRI-3 Module**6-6-14-4 Connect Cables to PAC Module**

6-6-15 INSTALL LPCA AND CABLE TRACKS

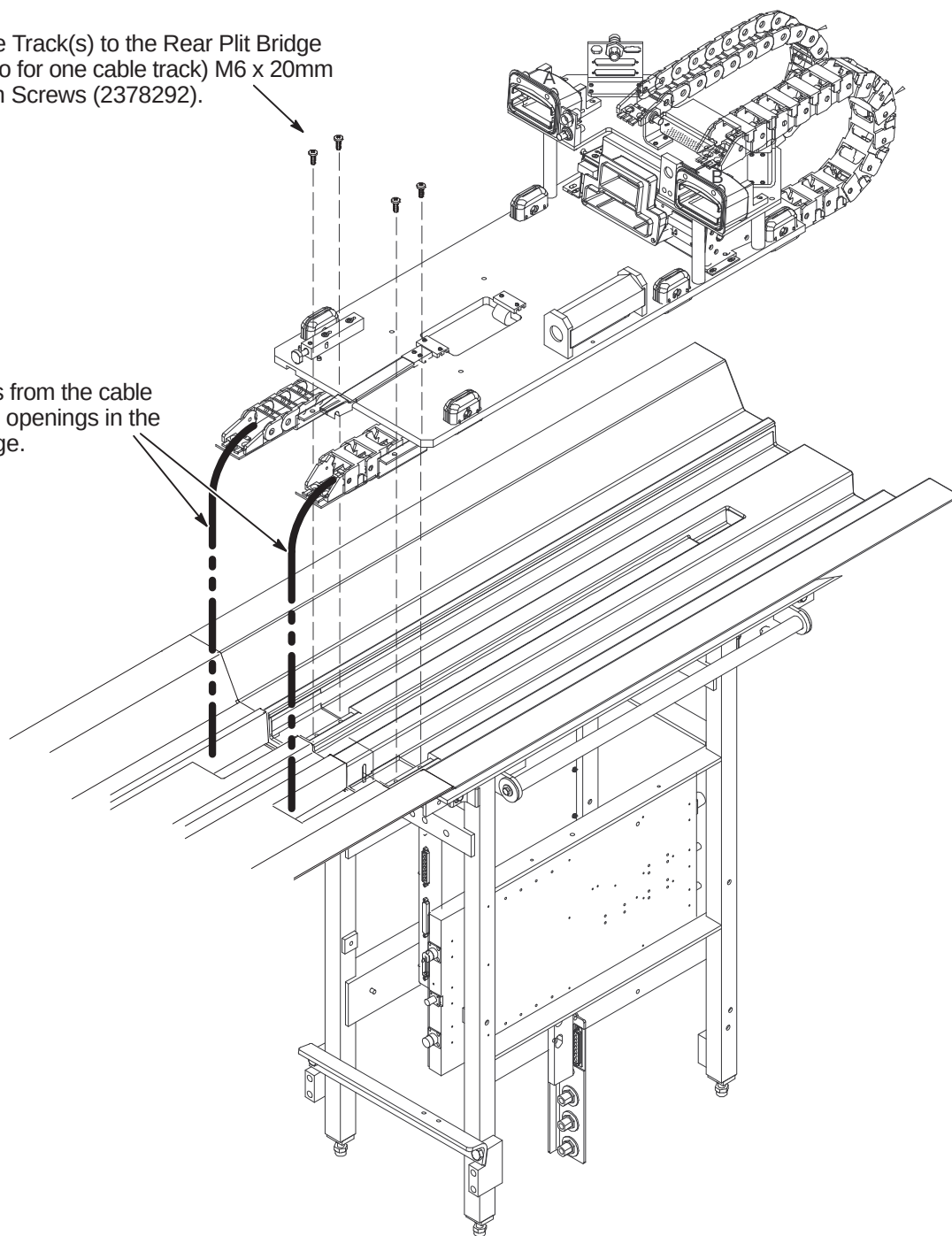
The WideOpen style enclosure will be upgraded to an LPCA with a dual cable track.

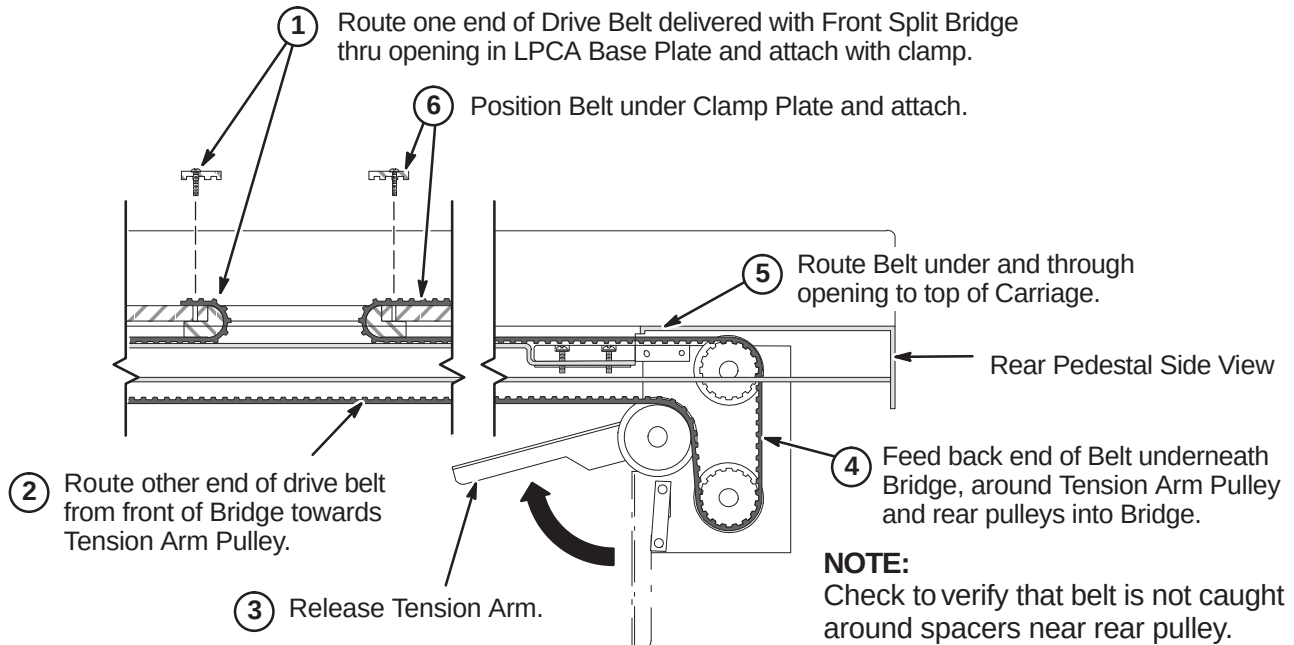
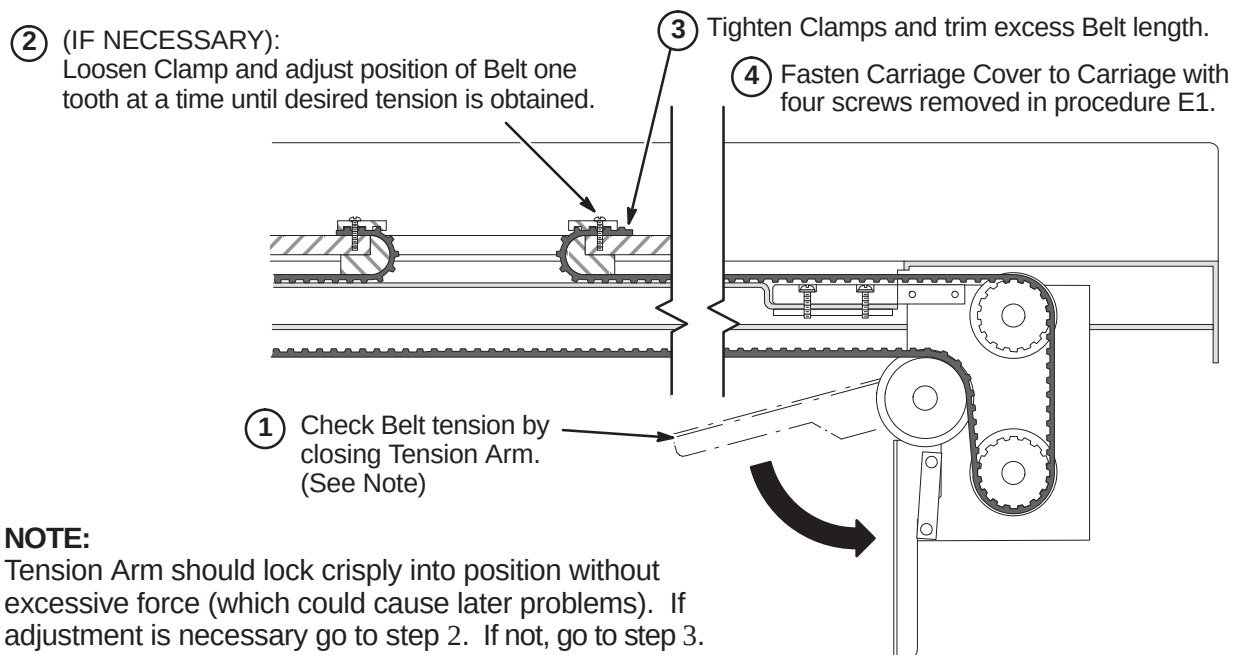
6-6-15-1 Remove LPCA Cover

6-6-15-2 Attach Cable Track(s) to Bridge

- ② Attach the Cable Track(s) to the Rear Plit Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

- ① Feed the cables from the cable track(s) thru the openings in the Front Split Bridge.



6-6-15-3 Route and Attach Drive Belt**6-6-15-4 Final Adjustment of Drive Belt**

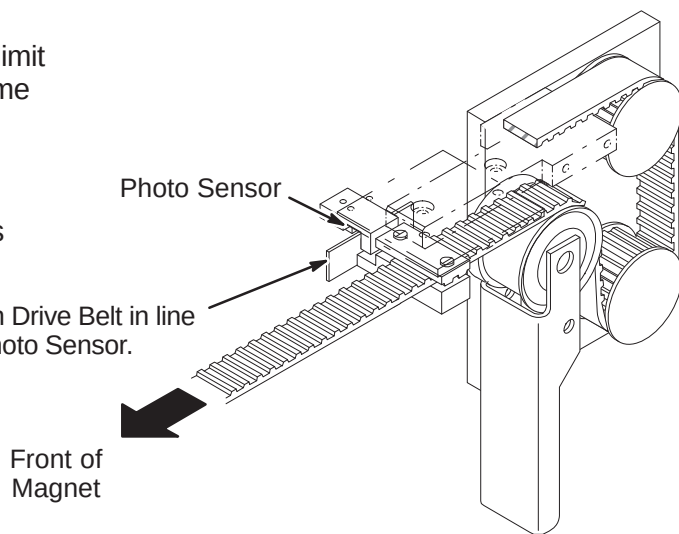
6-6-15-5 Installation of Sensor Flag**NOTE:**

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

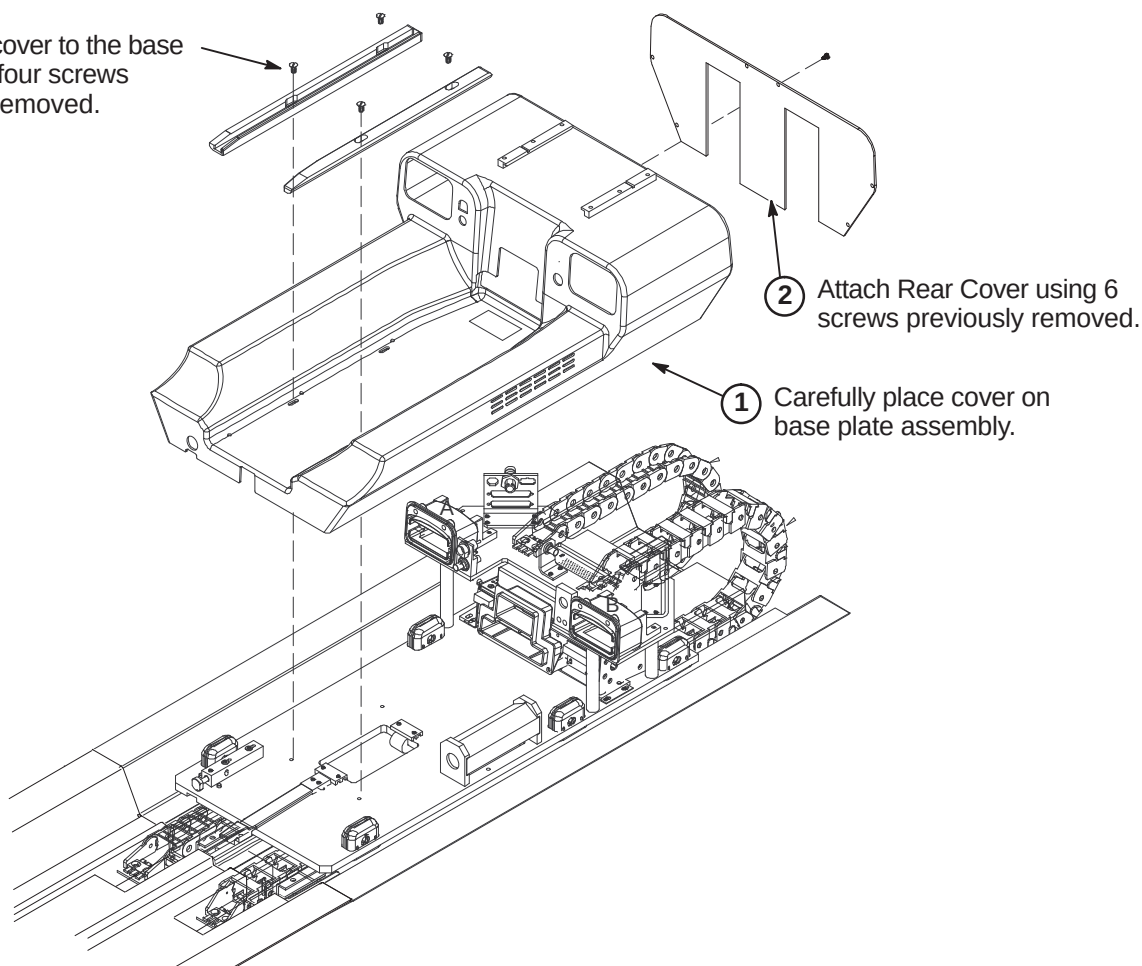
NOTE:

The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

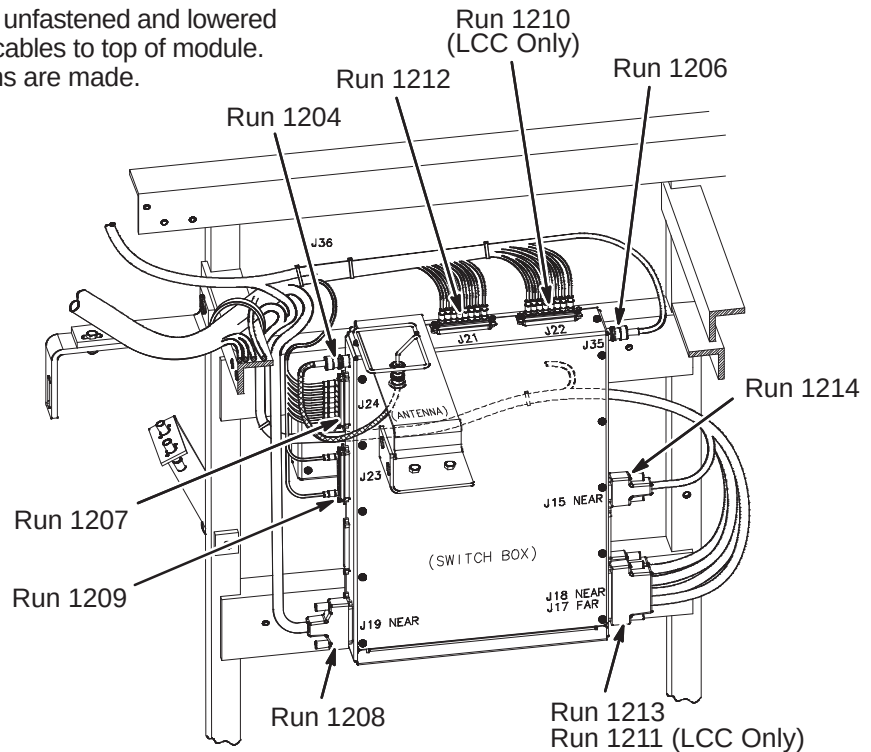
**6-6-15-6 Installation of LPCA Cover**

- ③ Attach the cover to the base plate using four screws previously removed.

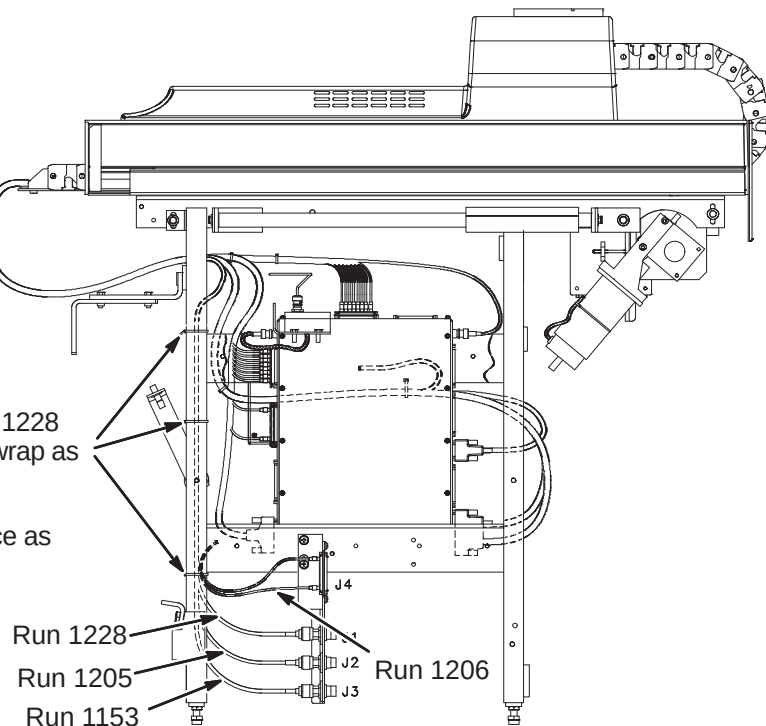


6-6-15-7 Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

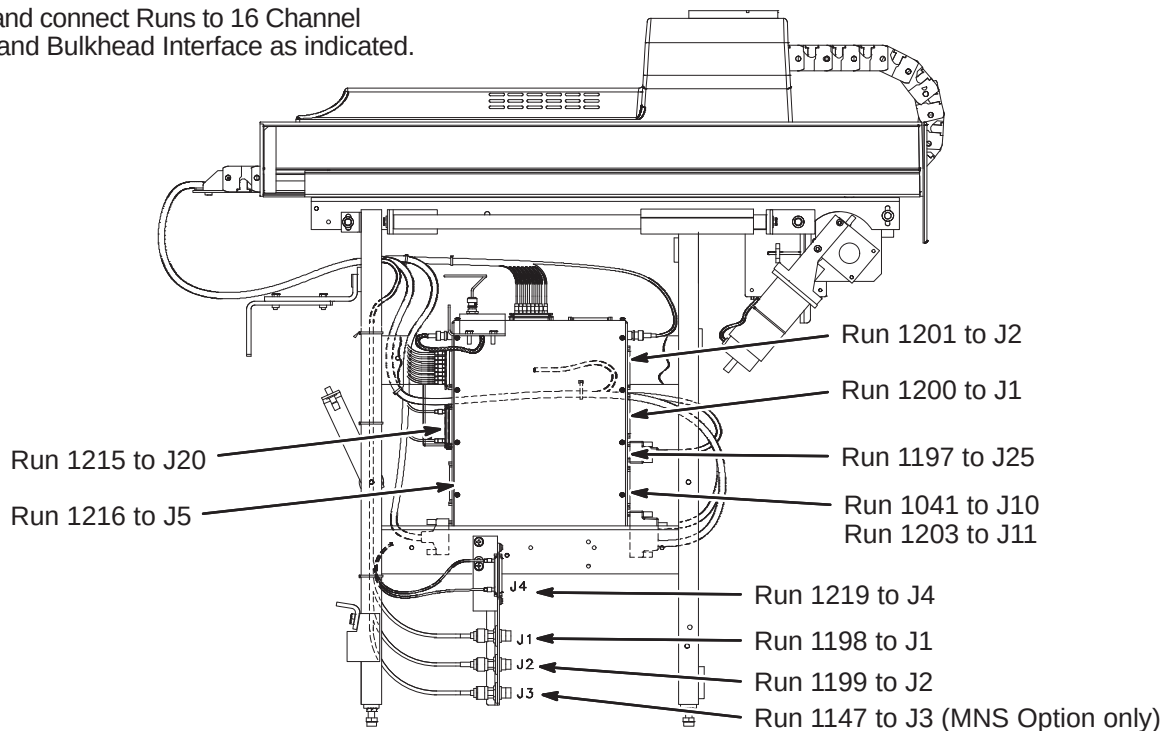
**6-6-15-8 Connect Cables from Cable Track to Bulkhead Interface**

- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.
- ② Connect Runs to Bulkhead Interface as indicated.

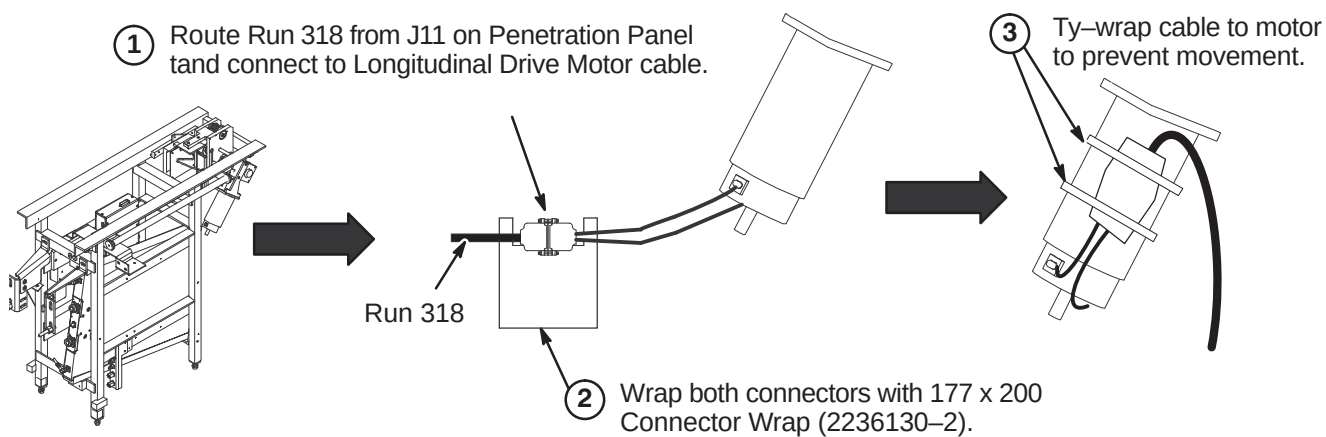


6-6-16 CONNECT REMAINING CABLES TO REAR PEDESTAL**6-6-16-1 Cable Connections to 16 Channel Switch and Bulkhead Interface**

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

**6-6-16-2 Route/Connect Run 318 to Longitudinal Drive Motor**

- ① Route Run 318 from J11 on Penetration Panel and connect to Longitudinal Drive Motor cable.

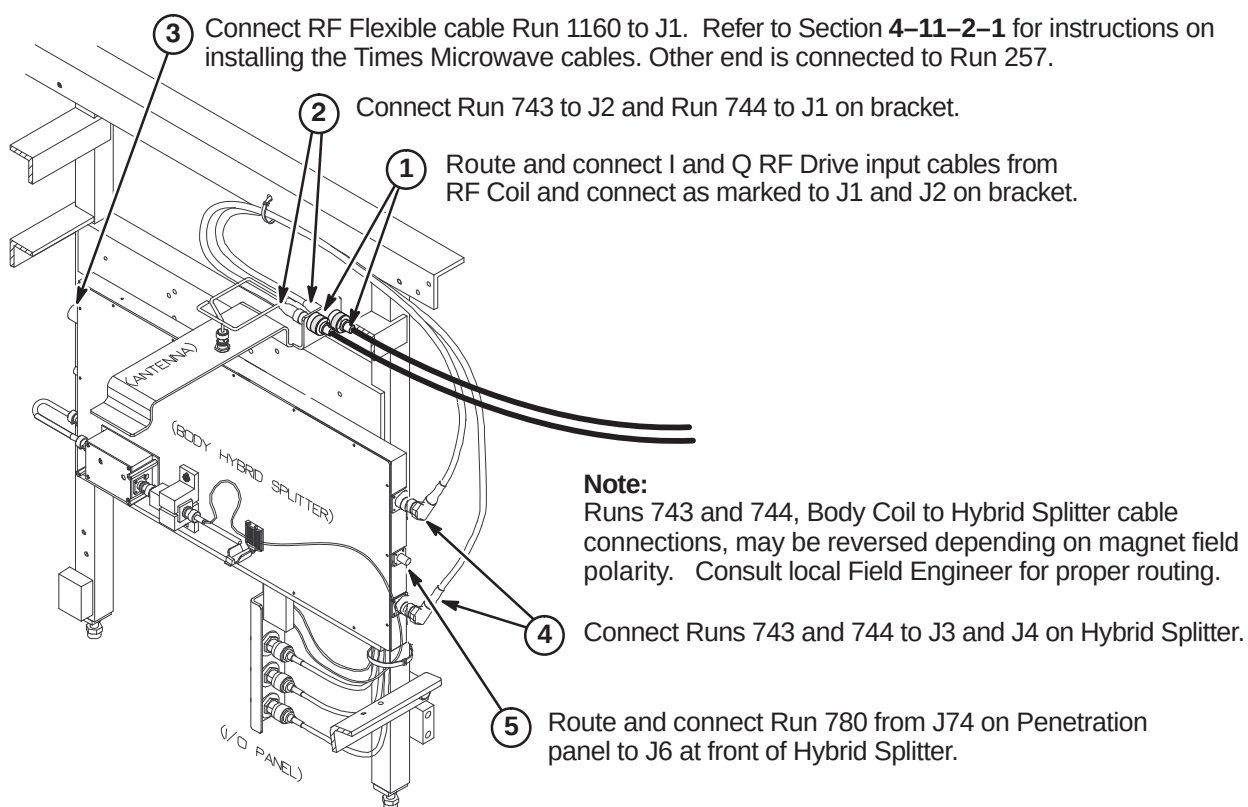


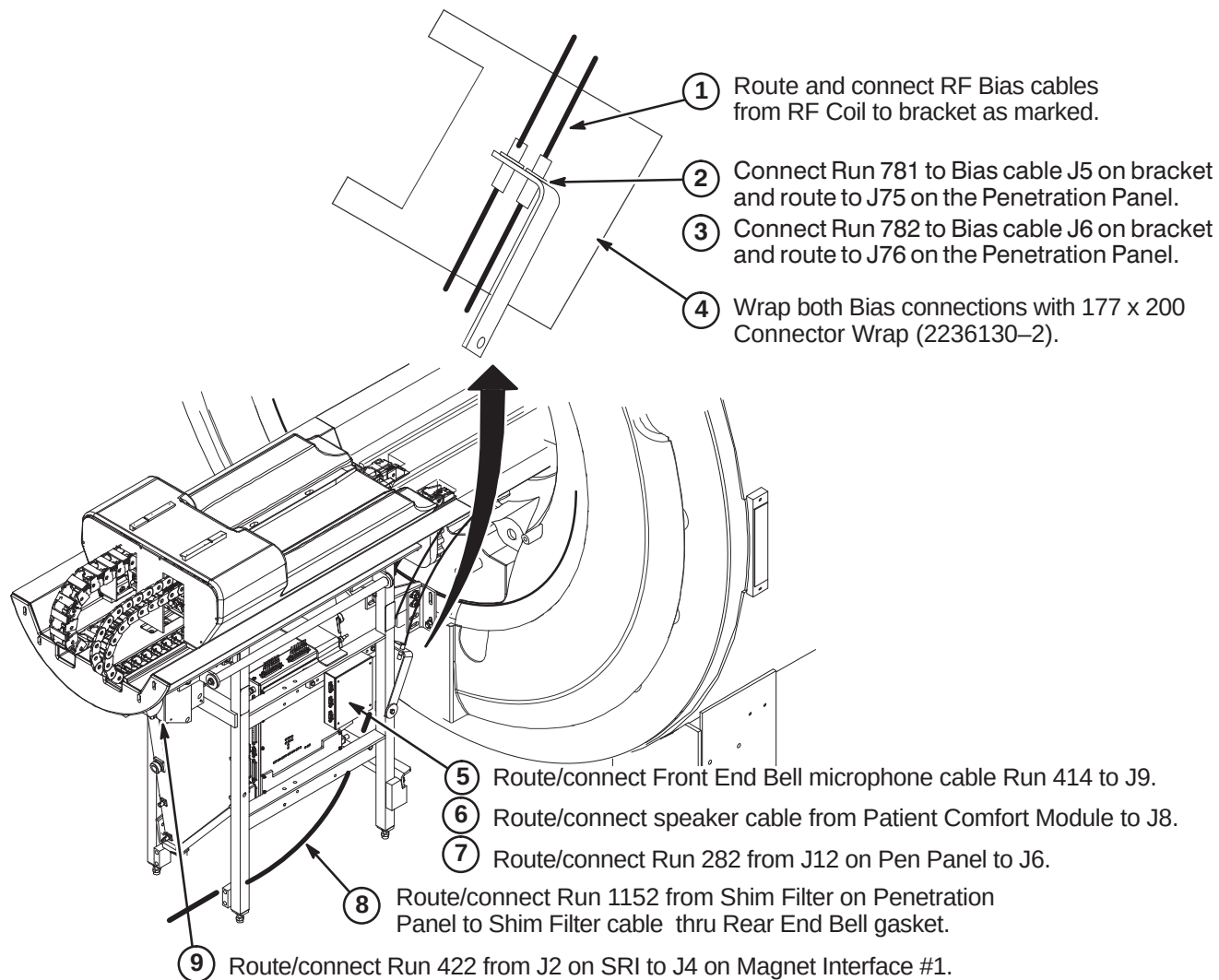
6-6-16-3 Connect Cables to Hybrid Splitter

CAUTION

The RF Drive Input cables utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cables to the RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



6-6-16-4 Route/Connect Runs 282, 414, 422, 781, 782, 1057, Multi-coil, & Front End Bell Speaker Cable

6-6-17 INSTALL VACUUM HOSE

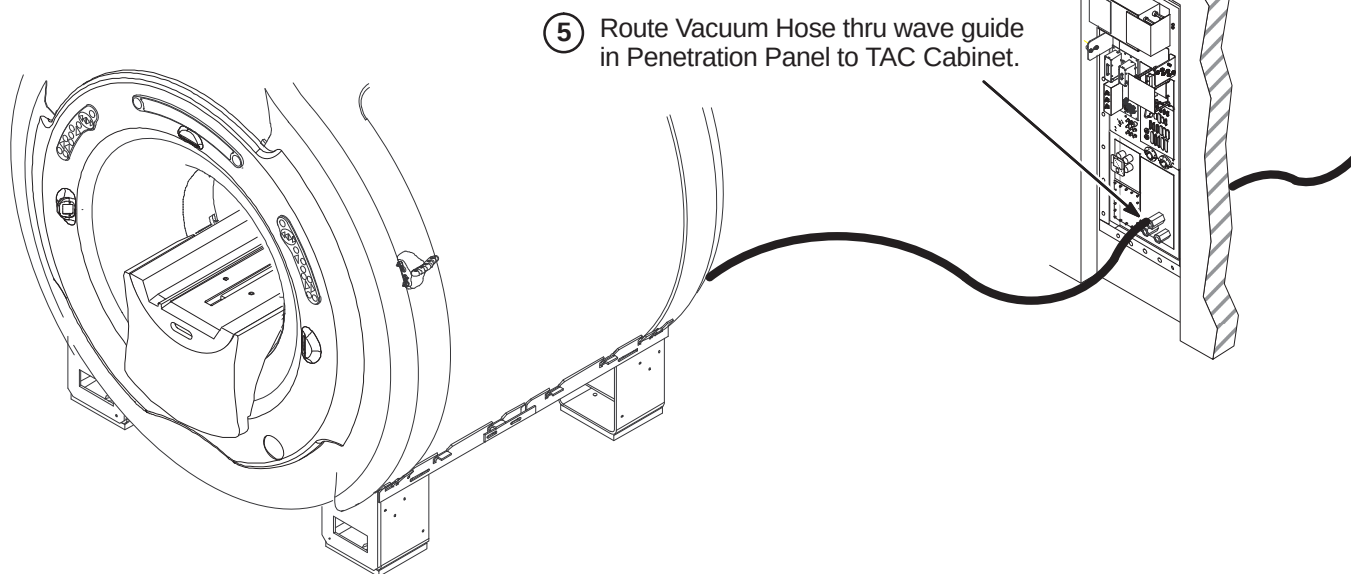
6-6-17-1 Route/Connect Vacuum Hose in Magnet Room

Note:

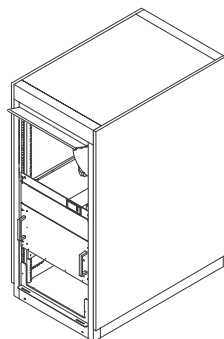
Vacuum Grease **must** be applied to all Centering Rings (2293608) and outside edge of vacuum connector flanges.



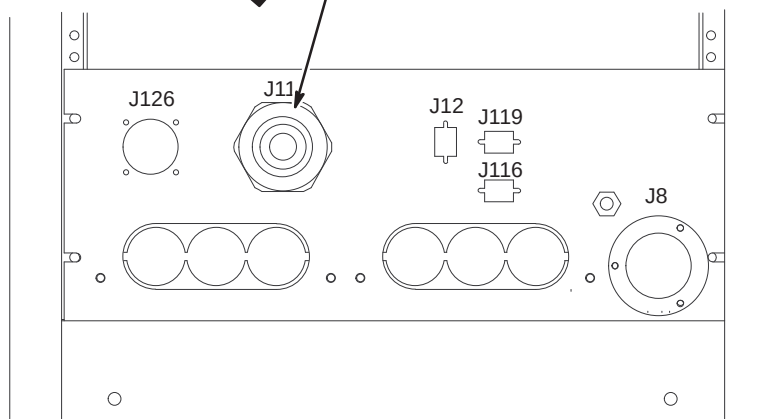
- ① Install Elbow (2293609) to End Bell using centering ring (2293608) and chain clamp (2293618-2). Tighten chain clamp to "finger tight".
- ④ Apply vacuum grease to outside edge of Hose Connector flanges and respective connector flanges on End Bell.
- ③ At end of vacuum hose insert Hose Connector (2293616), apply vacuum grease to outside edge of connector and fasten with two (2) Clamps (2294010-3).
- ② Route vacuum hose (2293607) underneath Rear Pedestal, under magnet end cover, and thru hole to Elbow on End Bell.

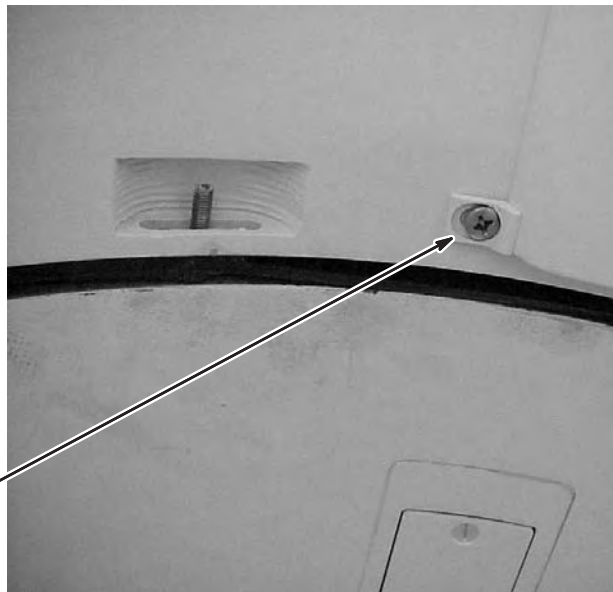
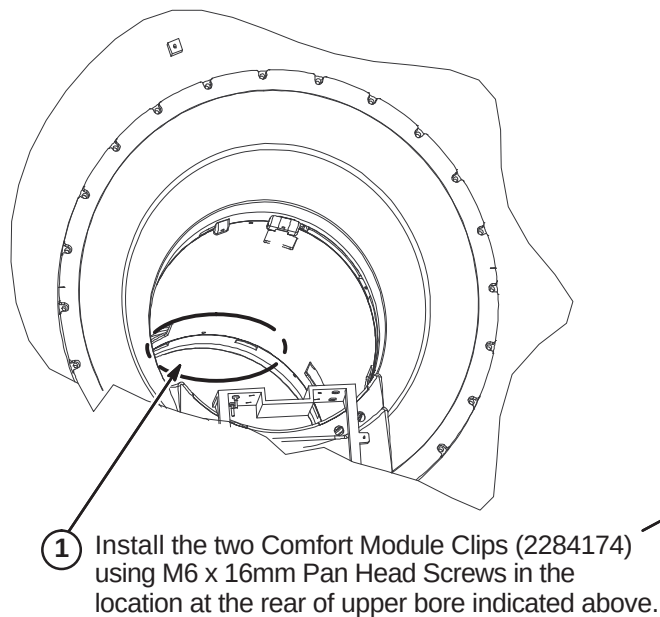
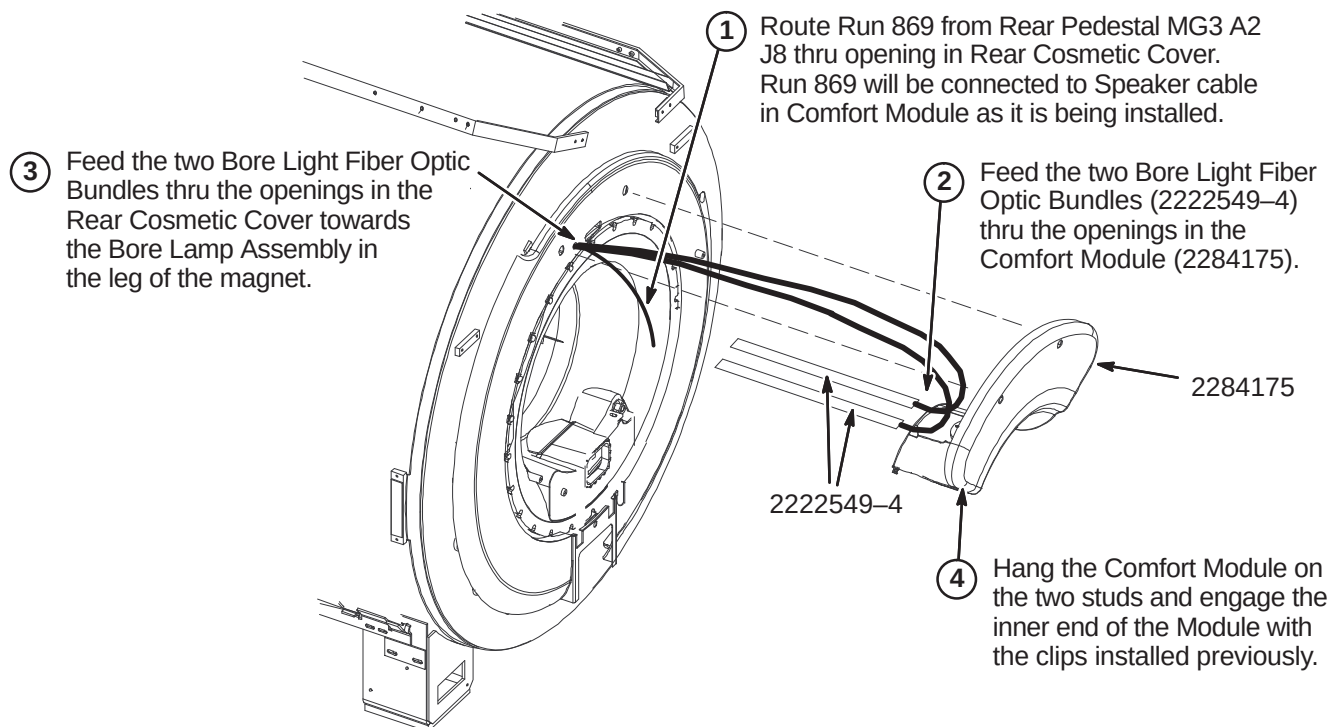


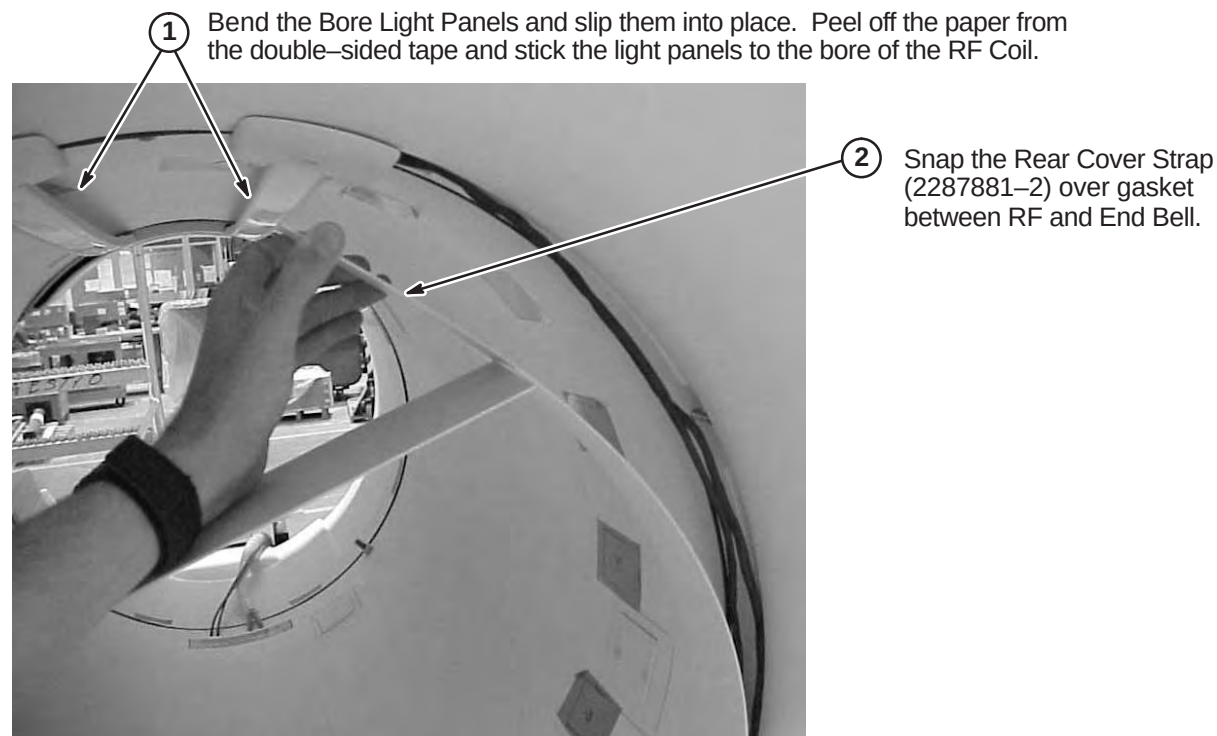
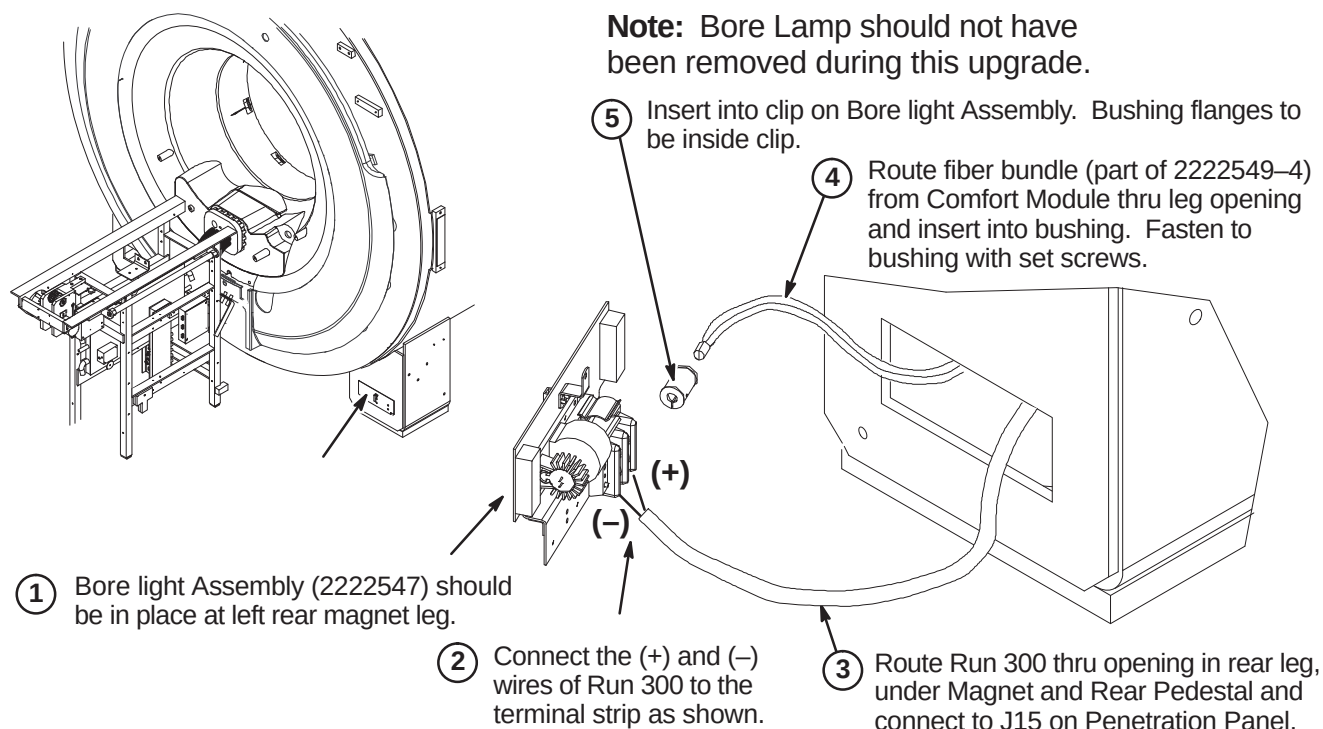
- ⑤ Route Vacuum Hose thru wave guide in Penetration Panel to TAC Cabinet.

6-6-17-2 Connect Vacuum Hose to TAC Cabinet

- ① Route vacuum tube to J11 at back of cabinet and cut to length.
- ② Install 1" Hose Connector (2293616) in the end of hose and attach with worm-drive clamp (2294010-3).
- ③ Apply vacuum grease to centering ring (2293608) and connect vacuum tube to J11 using supplied Chain Clamp (2293618-2).
- ④ Apply vacuum grease to outside edge of hose connector flanges and respective connector flanges on TAC Cabinet.

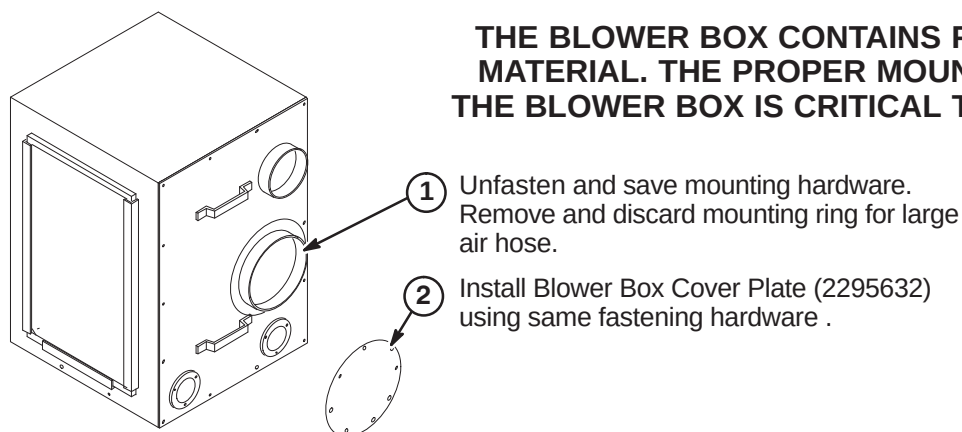
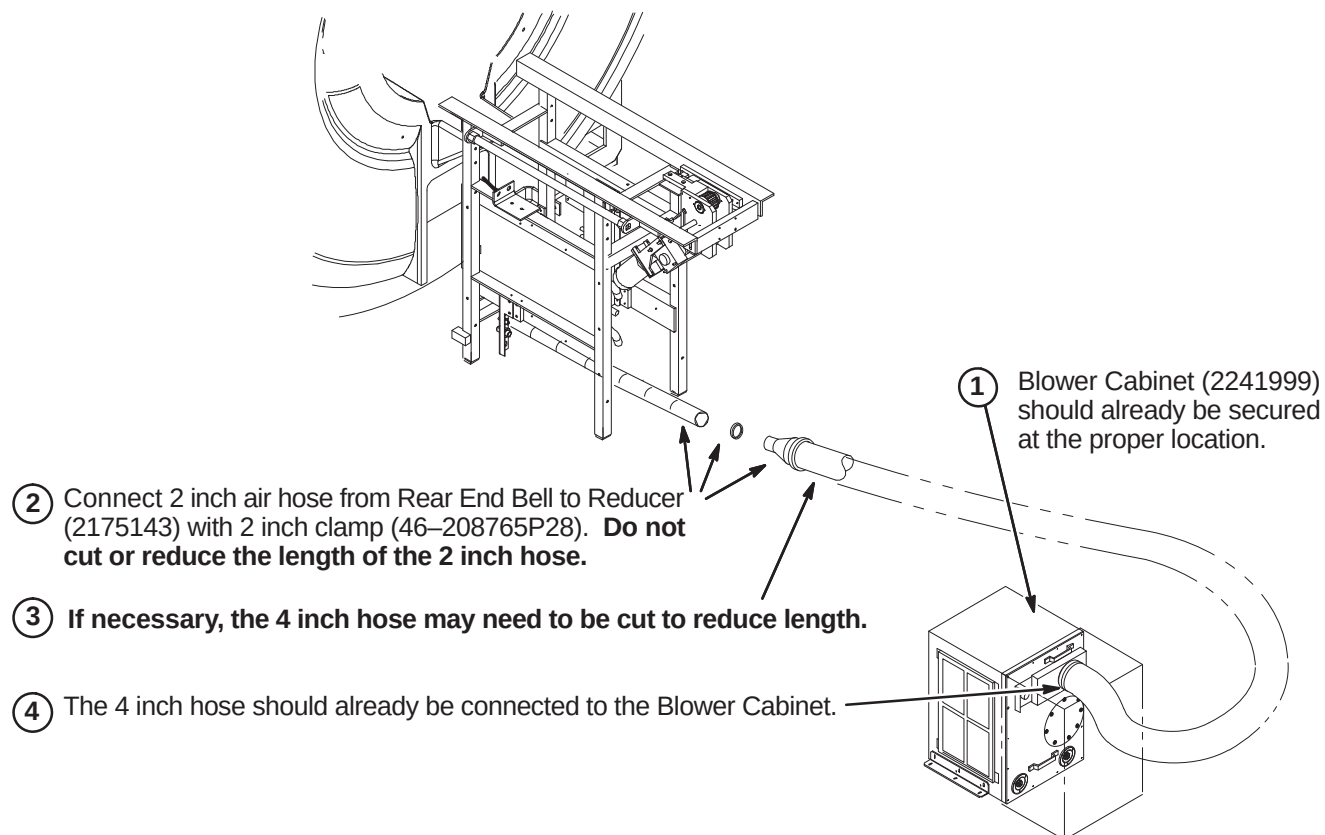


6-6-18 INSTALL COMFORT MODULE**6-6-18-1 Installation Of Comfort Module Clips****6-6-18-2 Installation Of Comfort Module**

6-6-18-3 Installation Of Bore Light Panels And Rear Cover Strap**6-6-18-4 Route/Connect Fiber Optic Bundle to Bore Lamp**

6-6-19 UPGRADE AIR HOSE SYSTEM**6-6-19-1 Remove Six Inch Hose Connector Ring From Blower Box****WARNING!**

THE BLOWER BOX CONTAINS FERROUS MATERIAL. THE PROPER MOUNTING OF THE BLOWER BOX IS CRITICAL TO SAFETY

**6-6-19-2 Connect Air Hose To Rear Of Magnet And Blower Box**

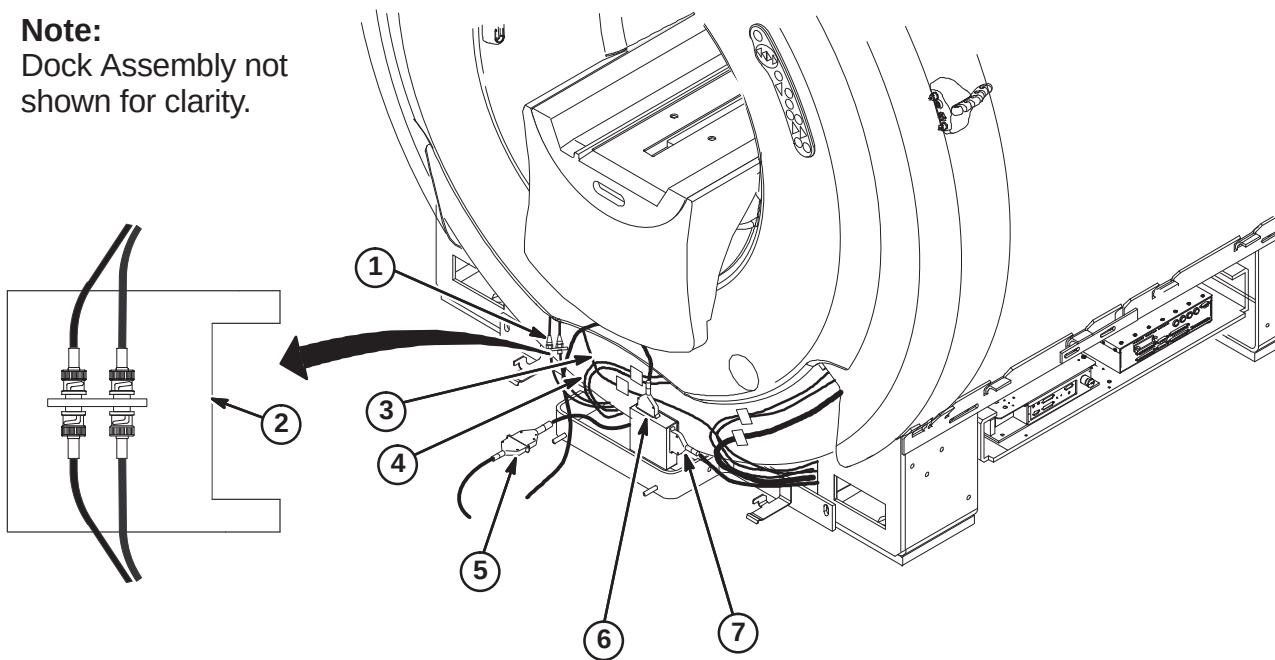
6-6-20 ROUTE AND CONNECT CABLES AT FRONT OF MAGNET ENCLOSURE

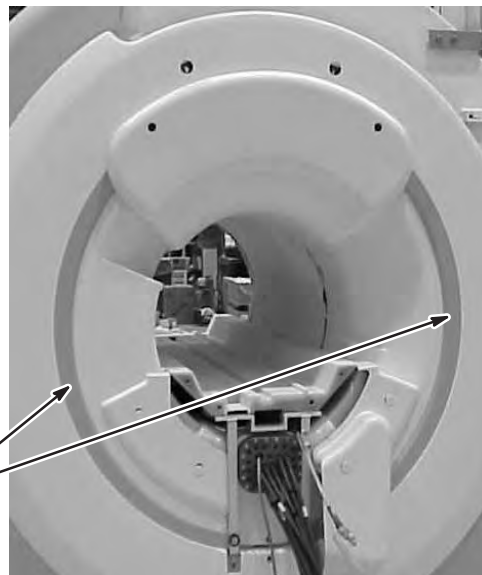
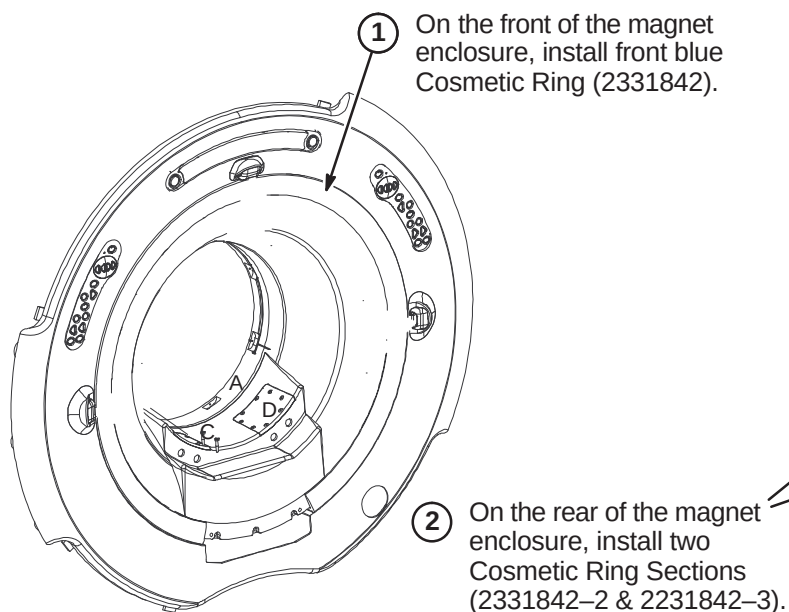
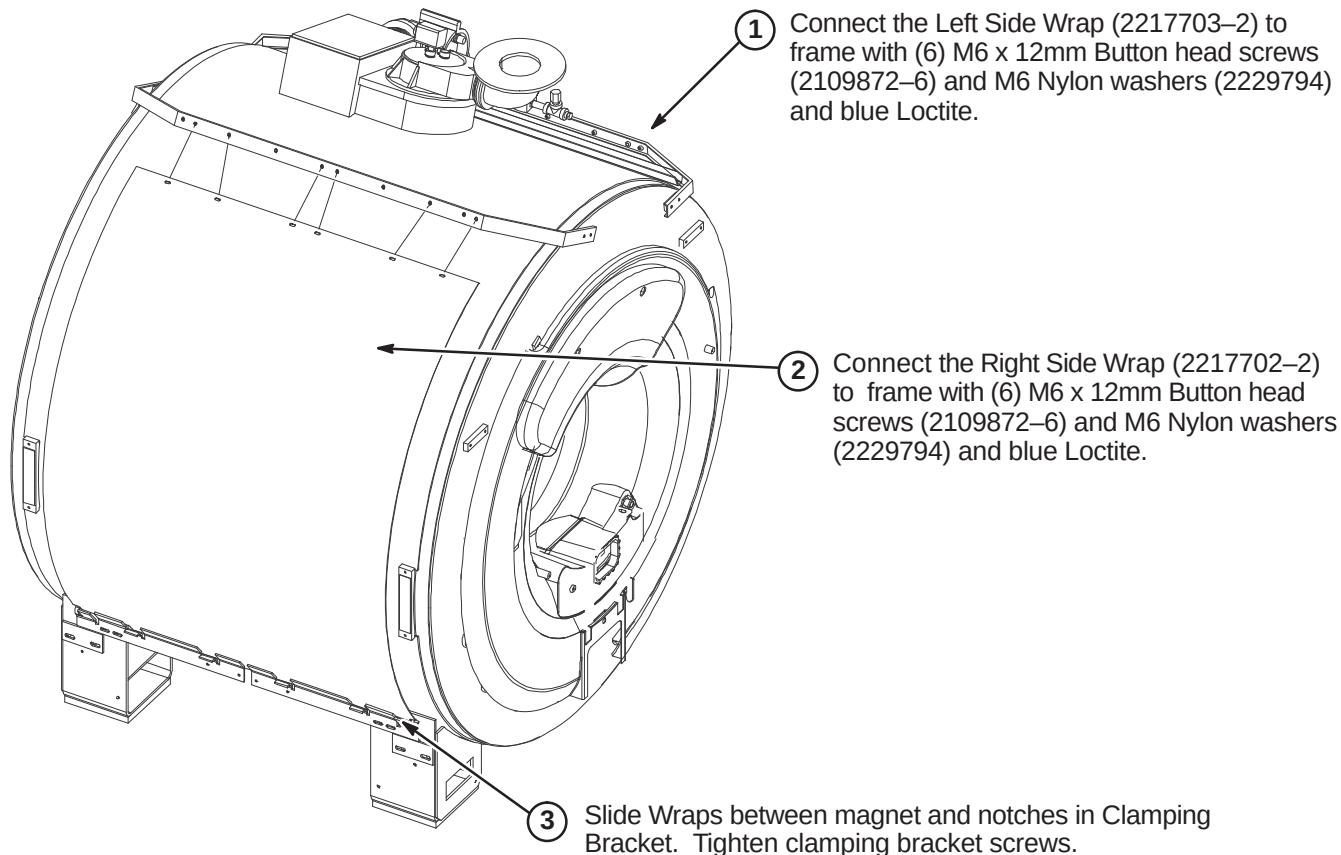
CAUTION

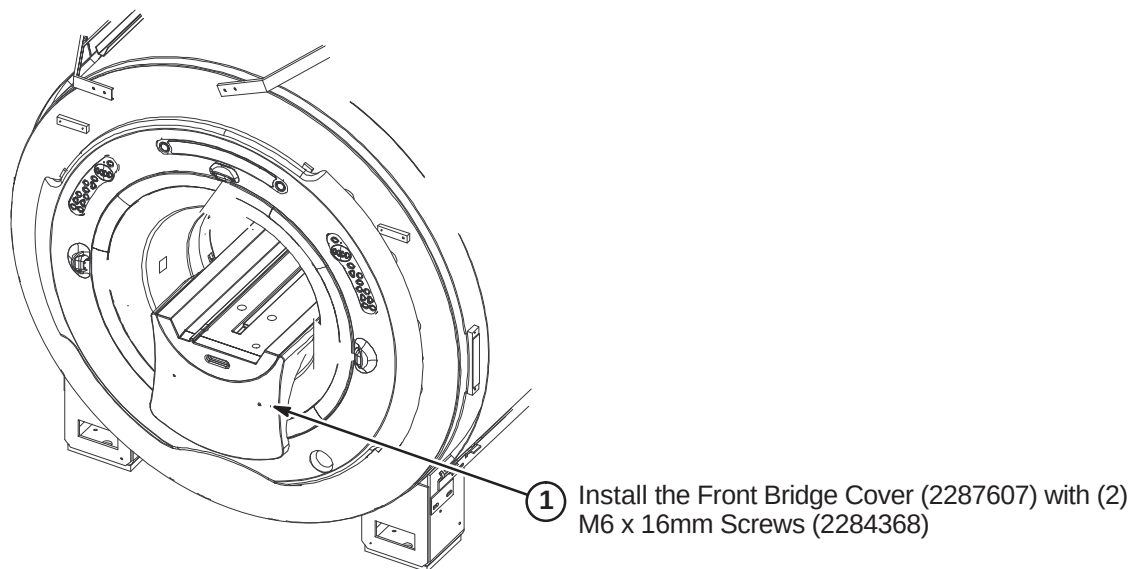
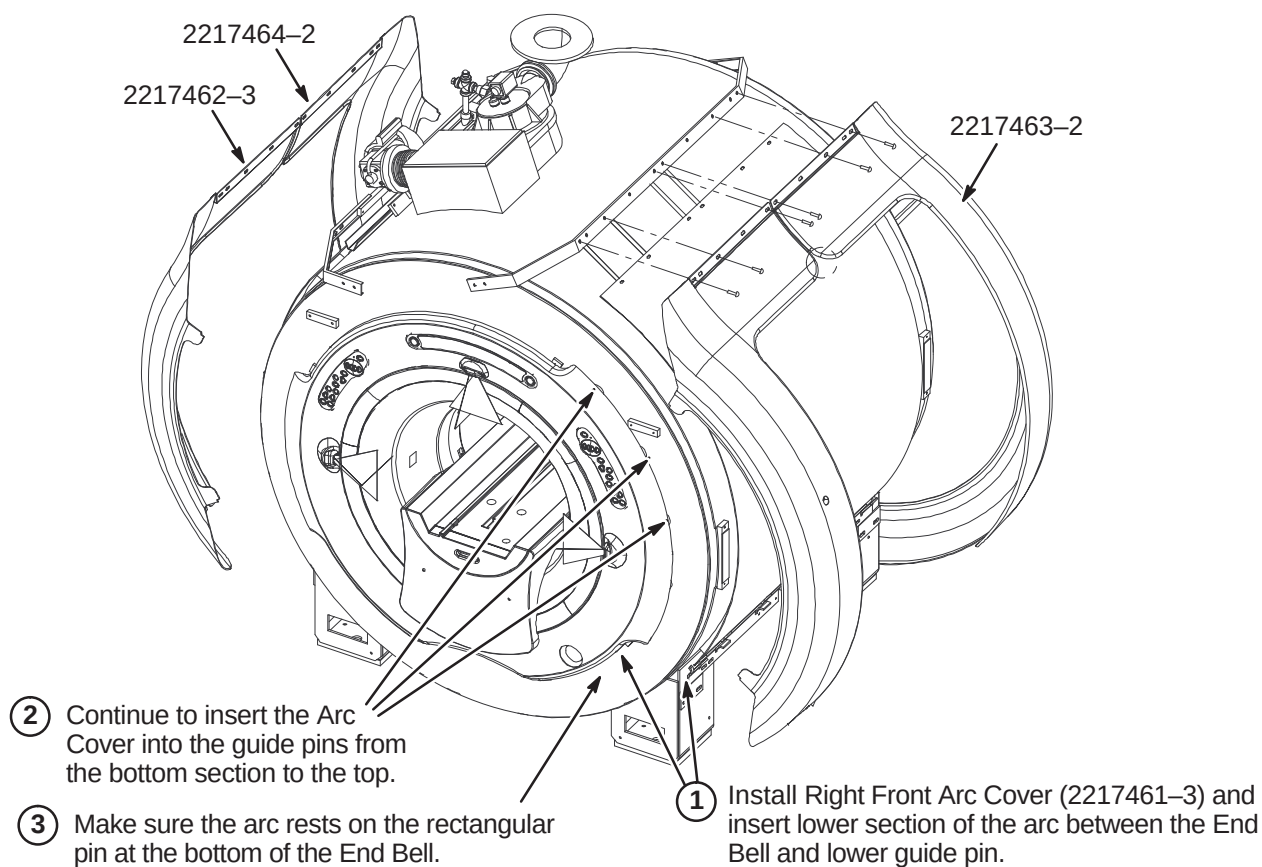
**No metal to metal contact is allowed for connectors to other areas.
i.e. connector touching the magnet.**

1. Route Runs 778 and 779 from J72 and J73 on Penetration Panel and connect to front RF Bias cables J3 & J4.
2. Wrap the RF Bias cable connections with one 177 x 200 Connector Wrap (2236130-2).
3. Route Run 1260, Bore Temperature Sensor, from Front End Bell and connect at J10 on SRI.
4. Route Front End Bell Microphone and Speaker cables underneath magnet to Rear Pedestal and connect to Patient Communication Box as marked.
5. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.
6. Connect Dock Cable to J2 on Dock Limit Box
7. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.

Note:
Dock Assembly not
shown for clarity.



6-6-21 INSTALL MAGNET ENCLOSURE COVERS**6-6-21-1 Install Front And Rear Blue Ring Sections****6-6-21-2 Install Right And Left Side Wraps**

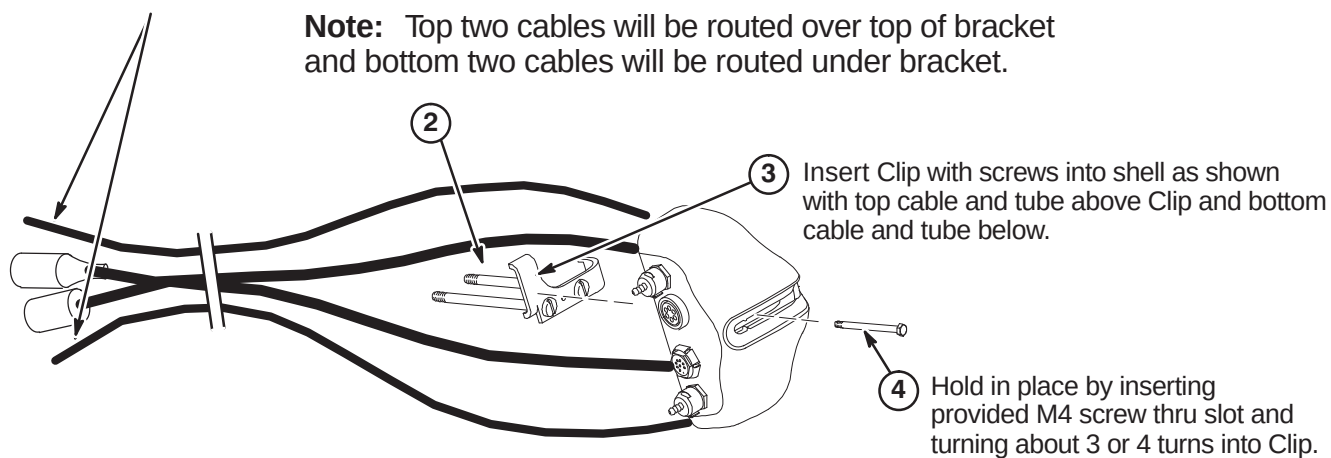
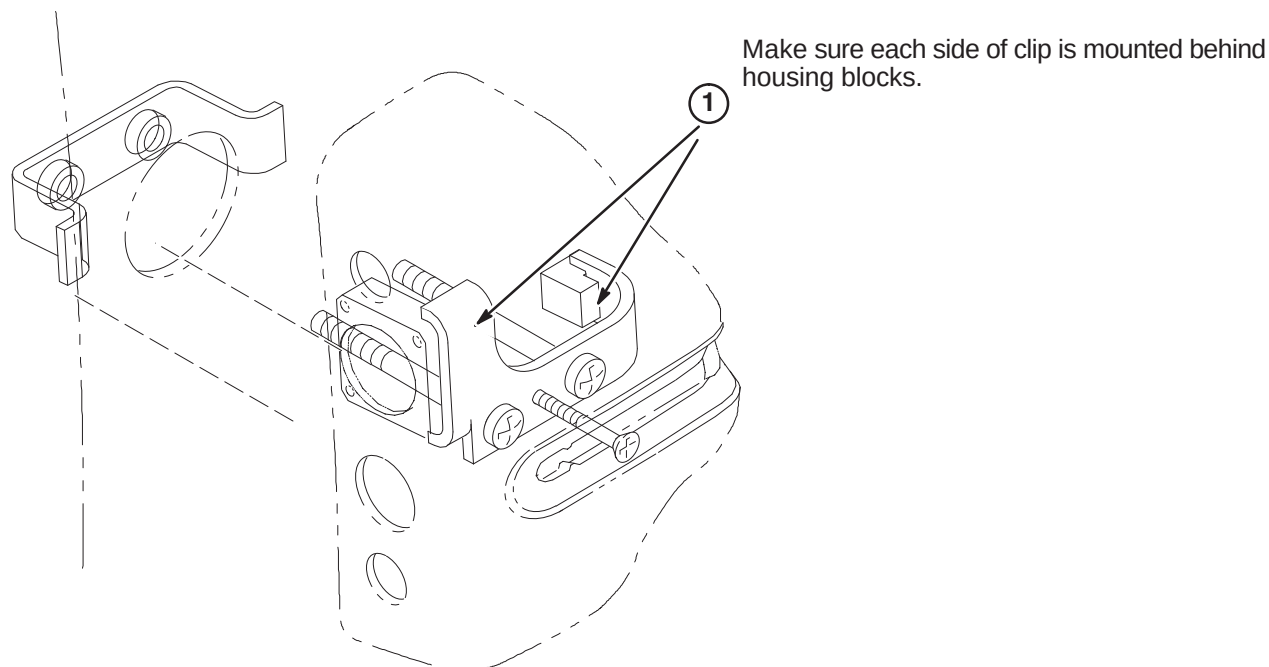
6-6-21-3 Installation Of Front Bridge Cover**6-6-21-4 Installation Of Arc Covers**

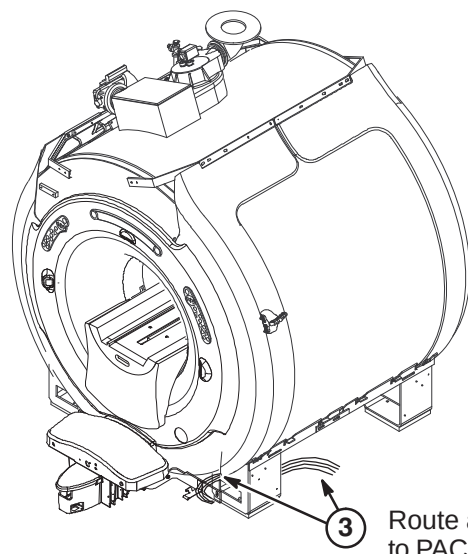
6-6-21-5 Prepare Cables And Tubes For Routing**Note:**

The Remote PAC Interface assembly must be on the same side as the PAC/SRI Platform.

- ① Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

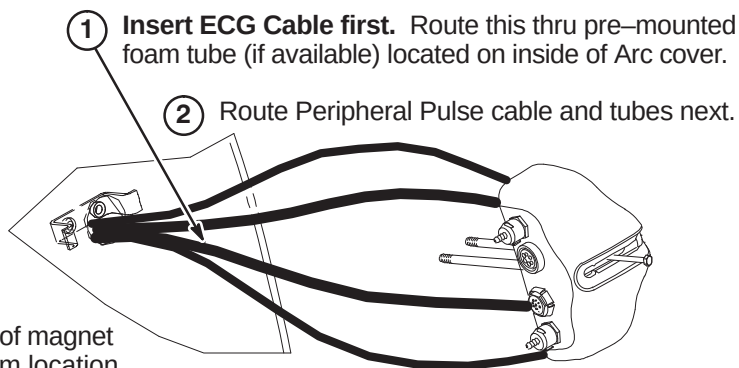
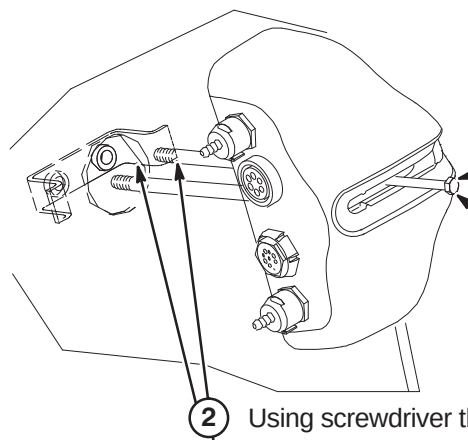
Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

**6-6-21-6 Position Remote PAC Interface Mounting Clip**

6-6-21-7 Route remote PAC Interface Cables And Tubes

Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

Note: Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.

**6-6-21-8 Installing Remote PAC Interface To ARC Cover**

Note:

Routed cables and tubes not shown for clarity.

① Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.

③ Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.

OR

Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in Section 6-6-21-9 into place.

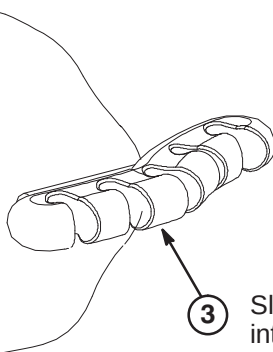
② Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place.

6-6-21-9 Install PAC Hanger Bar

① Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.

② Attach cables as indicated below.

Respiration
ECG
Peripheral Gating
Patient Call



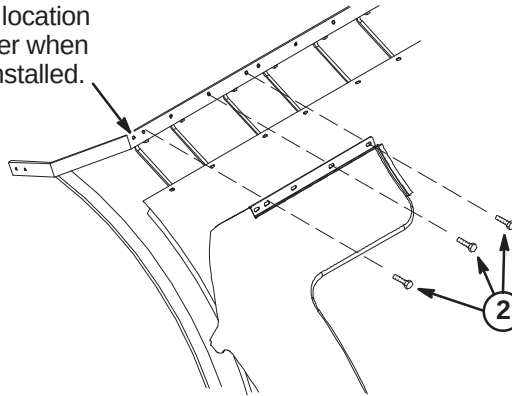
Left Side Label

Right Side Label

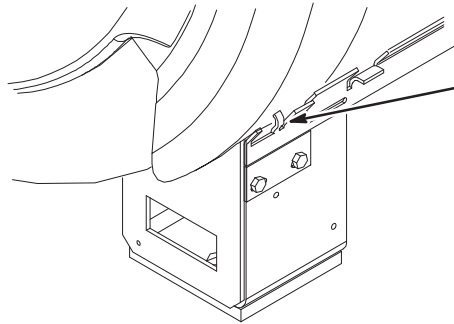
③ Slide PAC Hanger Bar (2221023) into position.

6-6-21-10 Fasten Top Of Arc Cover

- ① The screw for this location will be installed later when the Top Cover is installed.

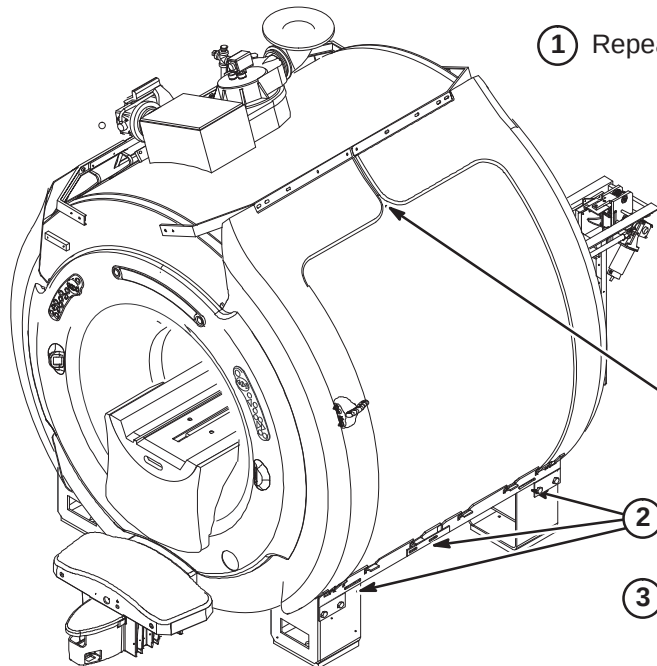


- ② Attach the Arc to the top frame using three (3) M10 x 25mm Hex Head screws (2109866-20) and Blue Loctite.

6-6-21-11 Adjust Bottom Of Arc Covers

- ① Adjust the bottom rails so that the finger fits into the slot in the bottom of the Arc Cover.

Note: The above process will help pull bottom of Rear End Bell flange closer to the magnet. Adjust Rear End Bell as required to keep distance between flange and magnet equal at top and bottom of End Bell.

6-6-21-12 Complete Arc Cover Installation

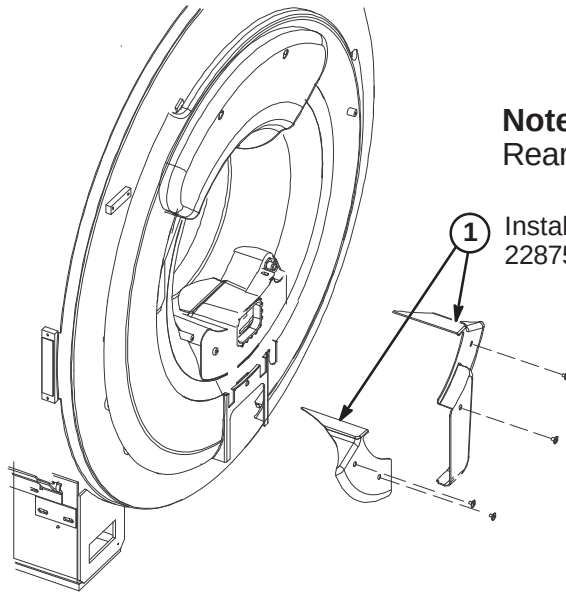
- ① Repeat process for the other three Arc Covers.

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). Exceptions can be made where esthetic visual enhancement of the component gap is required by industrial design.

Gap uniformity along any continuous seam shall be ± 0.0625 inch (1.59mm)

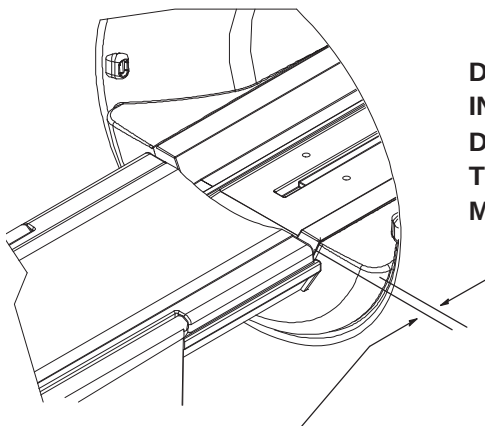
Note: This gap is not included in specification above.

- ② Tighten screws in bottom rails. First front rail, then middle connector, then rear of rail.
- ③ Visually inspect the gap between Arc Cover and End Bell, then the Arc Cover and side Wraps. Adjust as necessary.

6-6-21-13 Installation Of Rear Bridge Covers**Note:**

Rear Pedestal and Bridge are not shown for clarity.

- ① Install left and right Rear Bridge covers (2287511 & 2287512) using (4) Bridge Cover Screws (2284368).

6-6-22 DOCK ADJUSTMENT FOR PATIENT TRANSPORT GAP**Note:**

Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches $\pm$ 0.25 (19.0 $\pm$ 6.4mm). See Step 1 below.

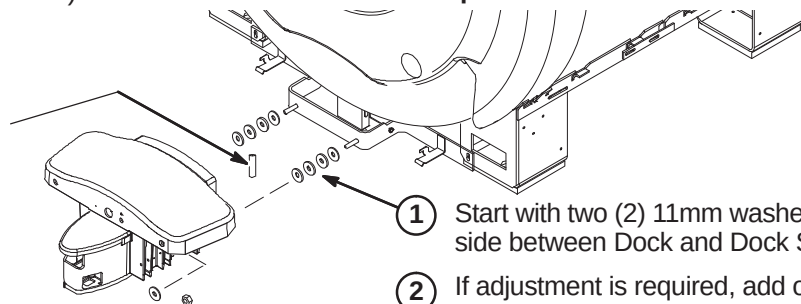
Note:

Floor stud installed by RF room vendor/mechanical contractor.

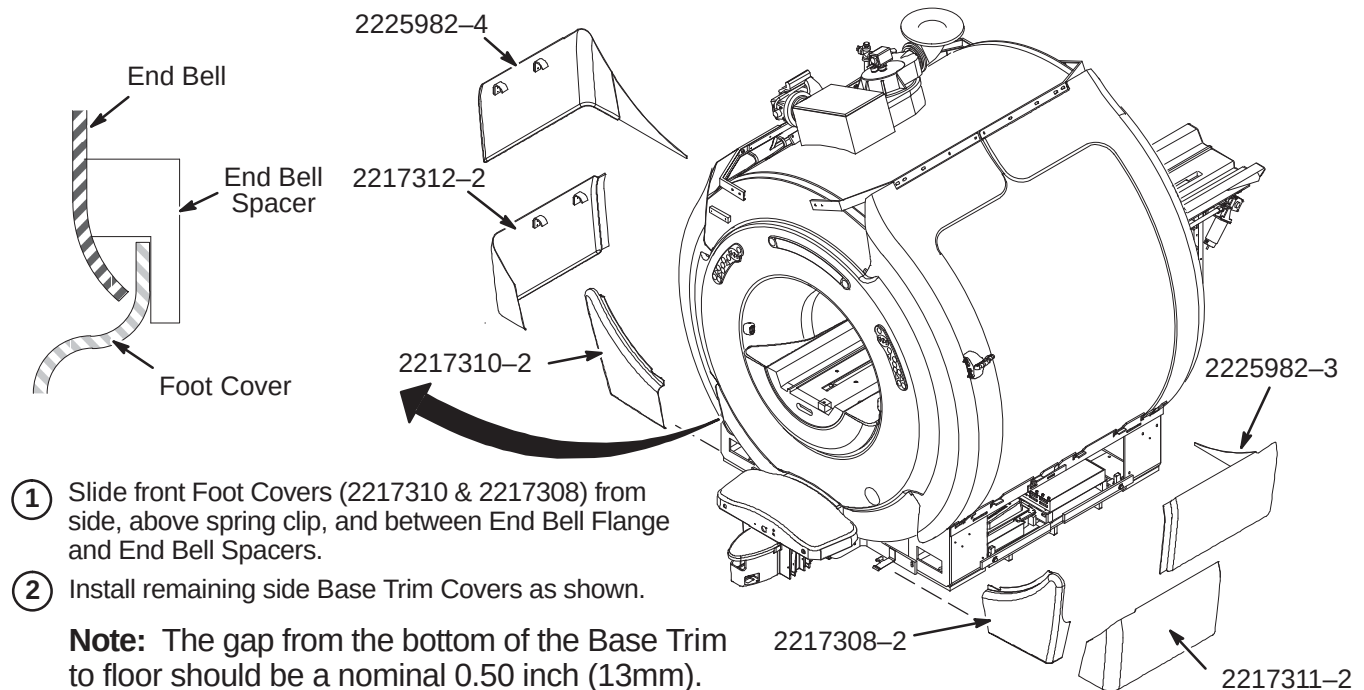
WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY.

DO NOT LIFT THE DOCK MORE THAN TWO INCHES (5 cm) OFF THE FLOOR. If the height (to clear the dock anchor) exceeds two inches (5 cm), the magnet must be ramped down prior to removing or installing the Dock as there is significant risk of the magnet attracting the Dock if it needs to be raised more than two inches (5 cm). For detailed Dock removal and installation procedures, refer to the *Signa EXCITE HD 1.5T Service Methods* CD-ROM, Direction 5133315: Replacements>Magnet Enclosure>LCC Dock Replacement.

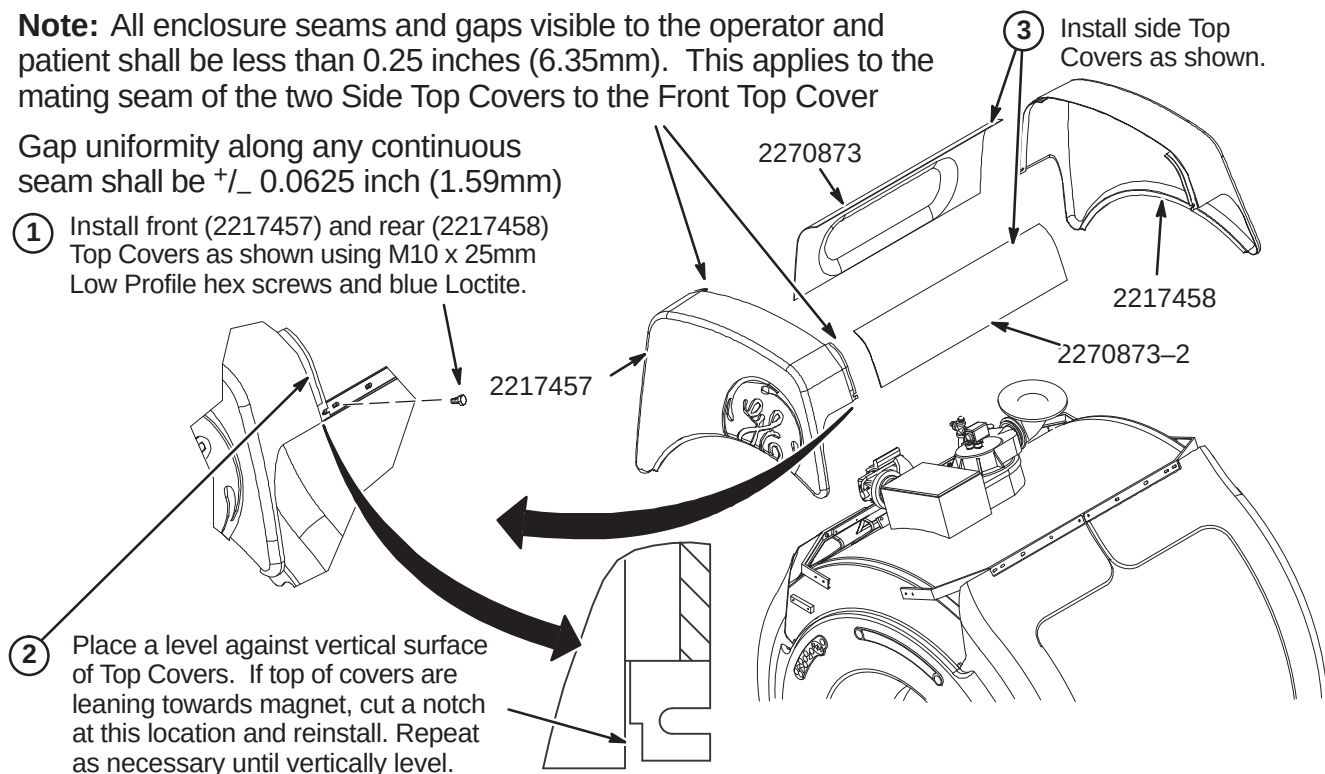


- ① Start with two (2) 11mm washers on each side between Dock and Dock Support.
- ② If adjustment is required, add or remove washers until proper gap is achieved.

6-6-23 FINAL ENCLOSURE COVER INSTALLATION**6-6-23-1 Install Enclosure Base Trim Covers****6-6-23-2 Install enclosure Top Covers**

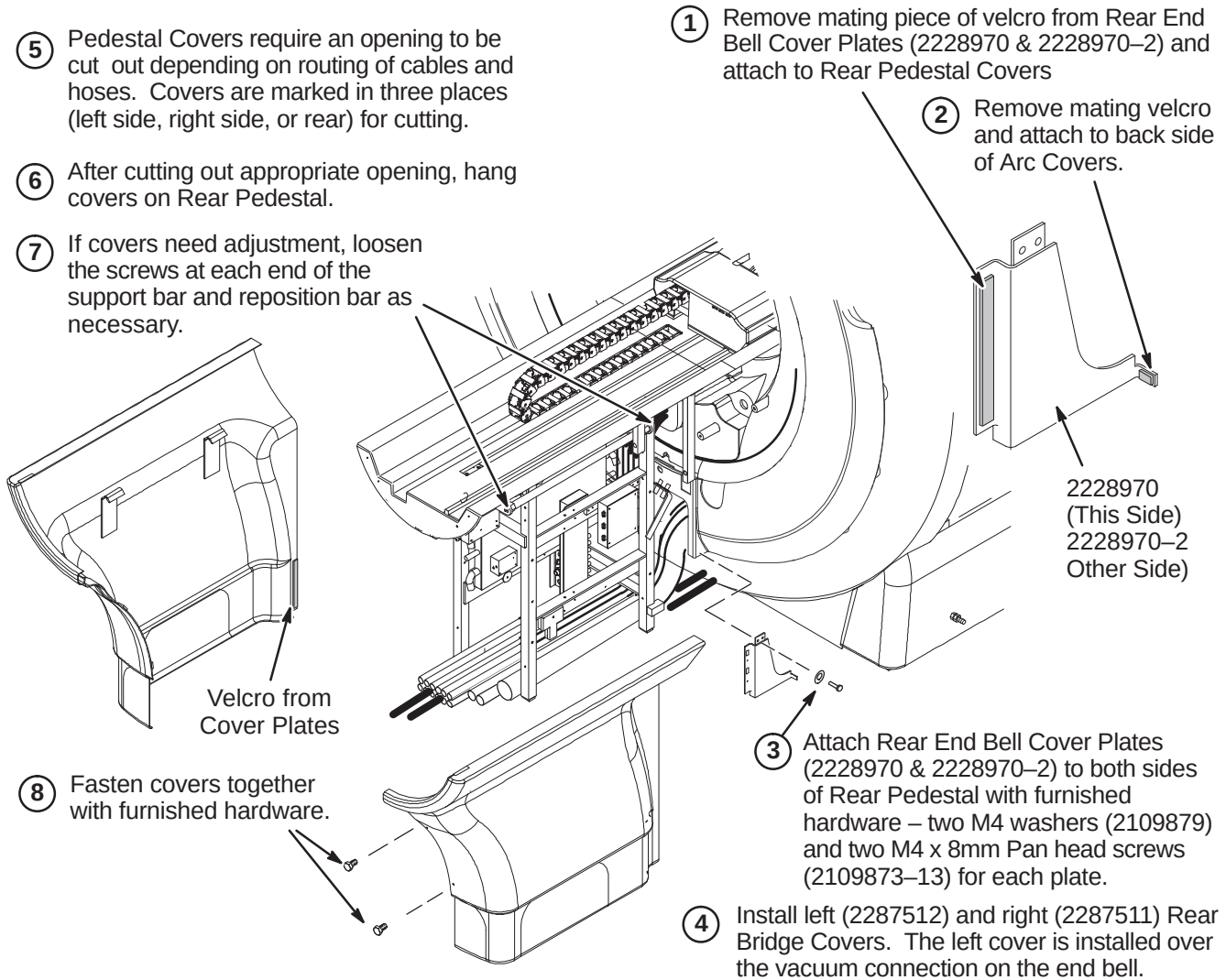
Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be ± 0.0625 inch (1.59mm)

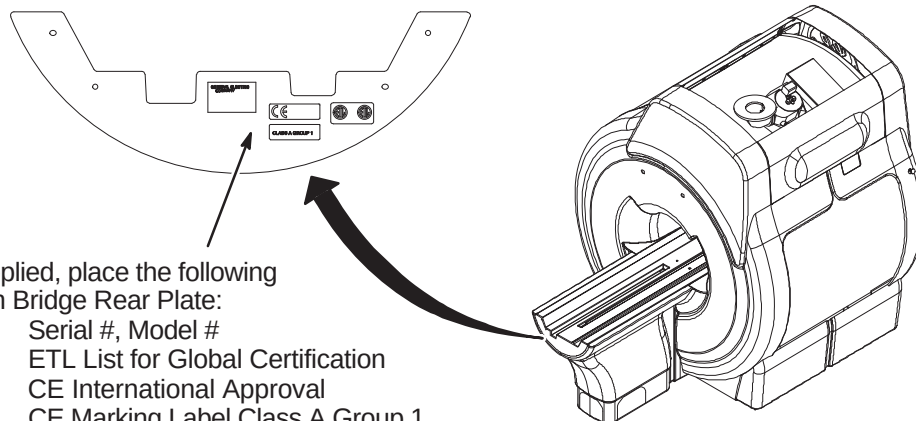


6-6-23-3 Install Rear Pedestal Covers

- ⑤ Pedestal Covers require an opening to be cut out depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting.
- ⑥ After cutting out appropriate opening, hang covers on Rear Pedestal.
- ⑦ If covers need adjustment, loosen the screws at each end of the support bar and reposition bar as necessary.
- ⑧ Fasten covers together with furnished hardware.
- ① Remove mating piece of velcro from Rear End Bell Cover Plates (2228970 & 2228970-2) and attach to Rear Pedestal Covers
- ② Remove mating velcro and attach to back side of Arc Covers.
- ③ Attach Rear End Bell Cover Plates (2228970 & 2228970-2) to both sides of Rear Pedestal with furnished hardware – two M4 washers (2109879) and two M4 x 8mm Pan head screws (2109873-13) for each plate.
- ④ Install left (2287512) and right (2287511) Rear Bridge Covers. The left cover is installed over the vacuum connection on the end bell.

**6-6-23-4 Installing rating Plates**

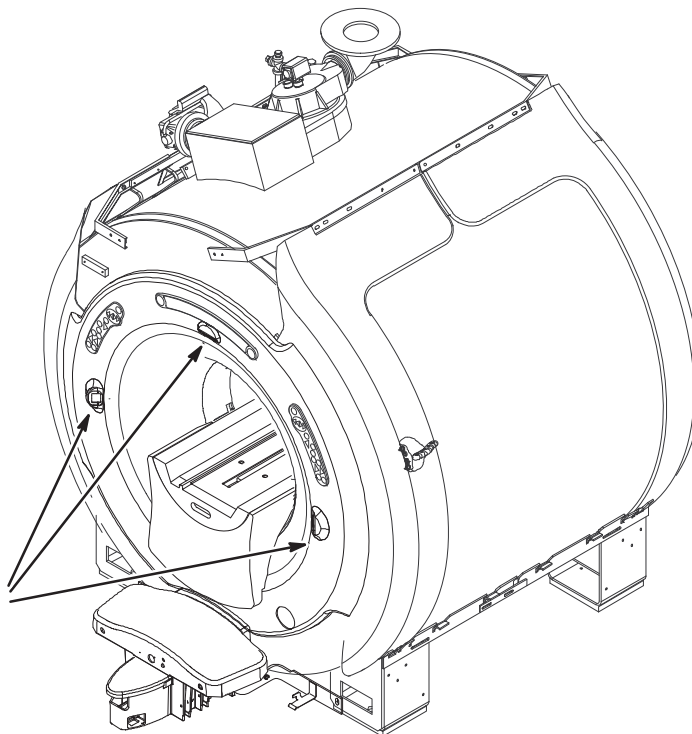
- ① If not already applied, place the following Rating Plates on Bridge Rear Plate:
 46-302200P7 Serial #, Model #
 2142857 ETL List for Global Certification
 2117390 CE International Approval
 2132620 CE Marking Label Class A Group 1



6-6-24 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.



- ① For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



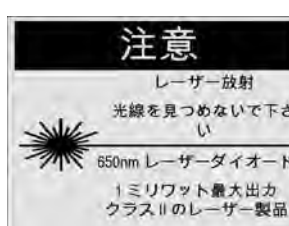
Spanish



Italian



Swedish



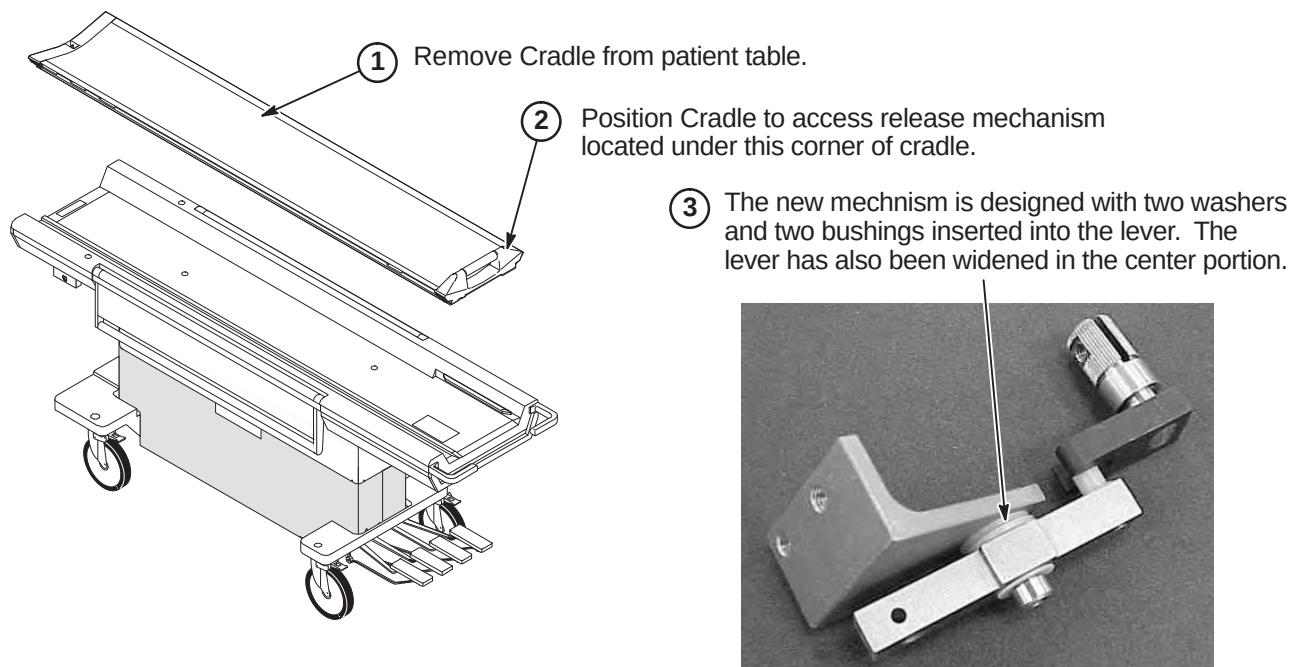
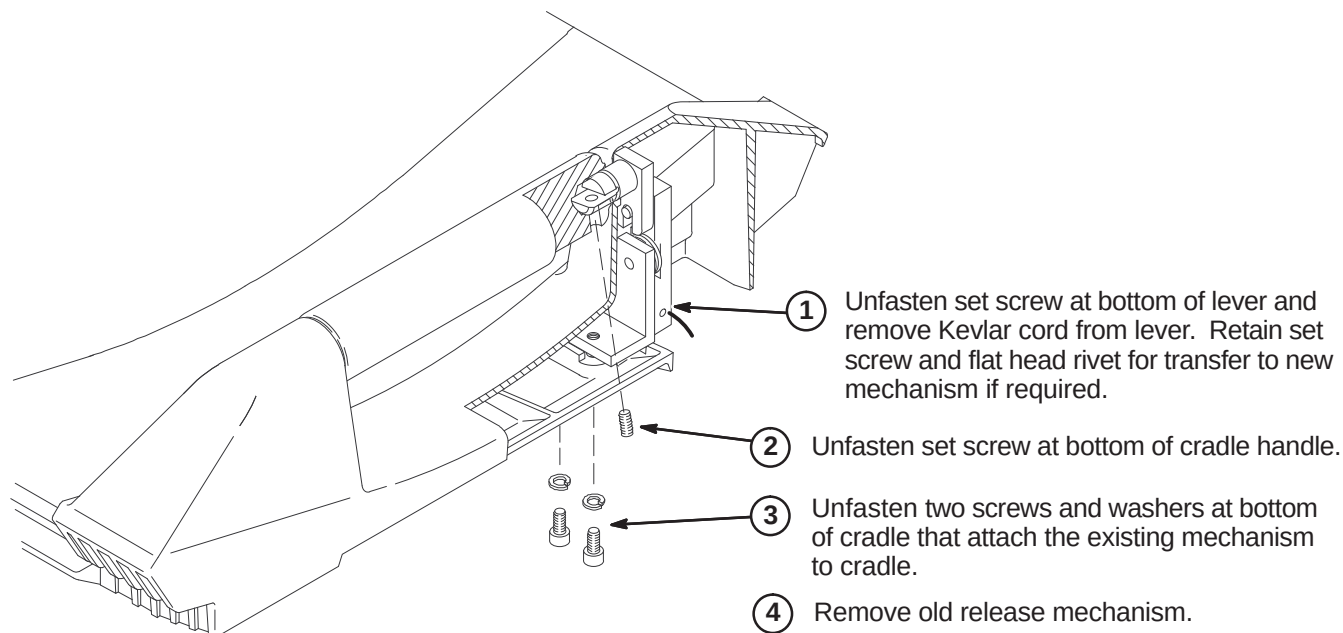
Japanese

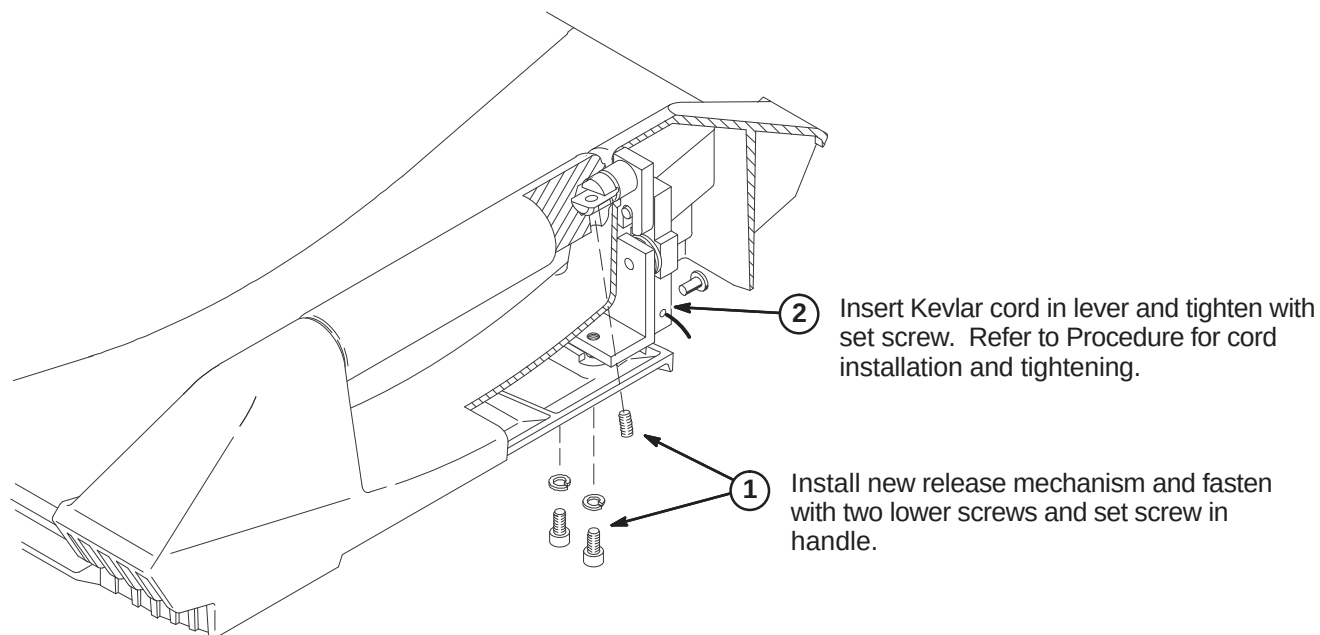
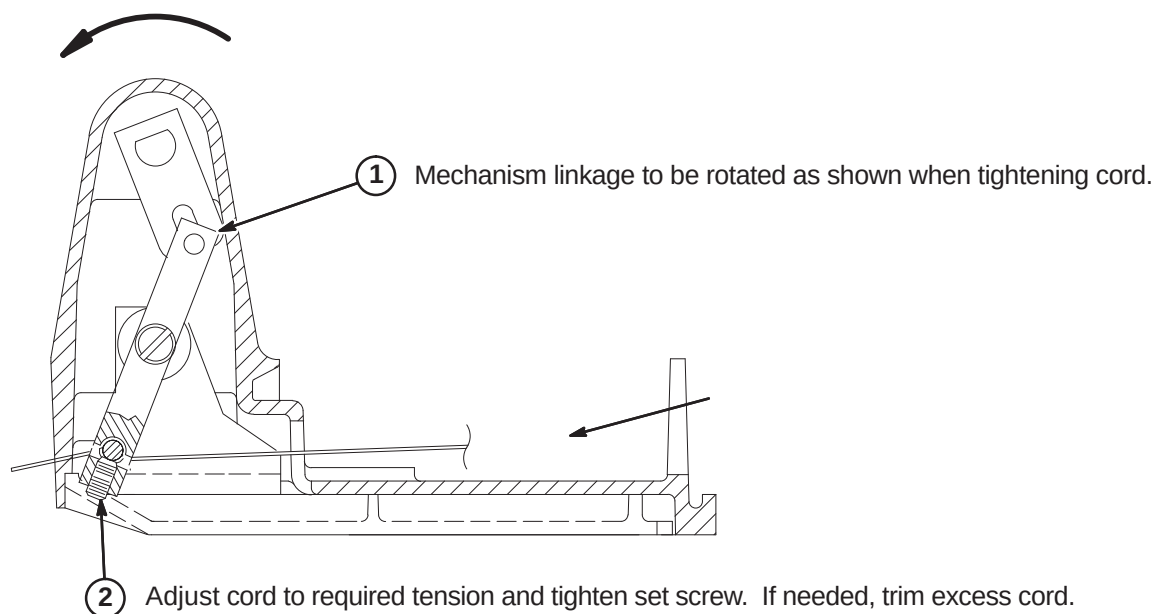


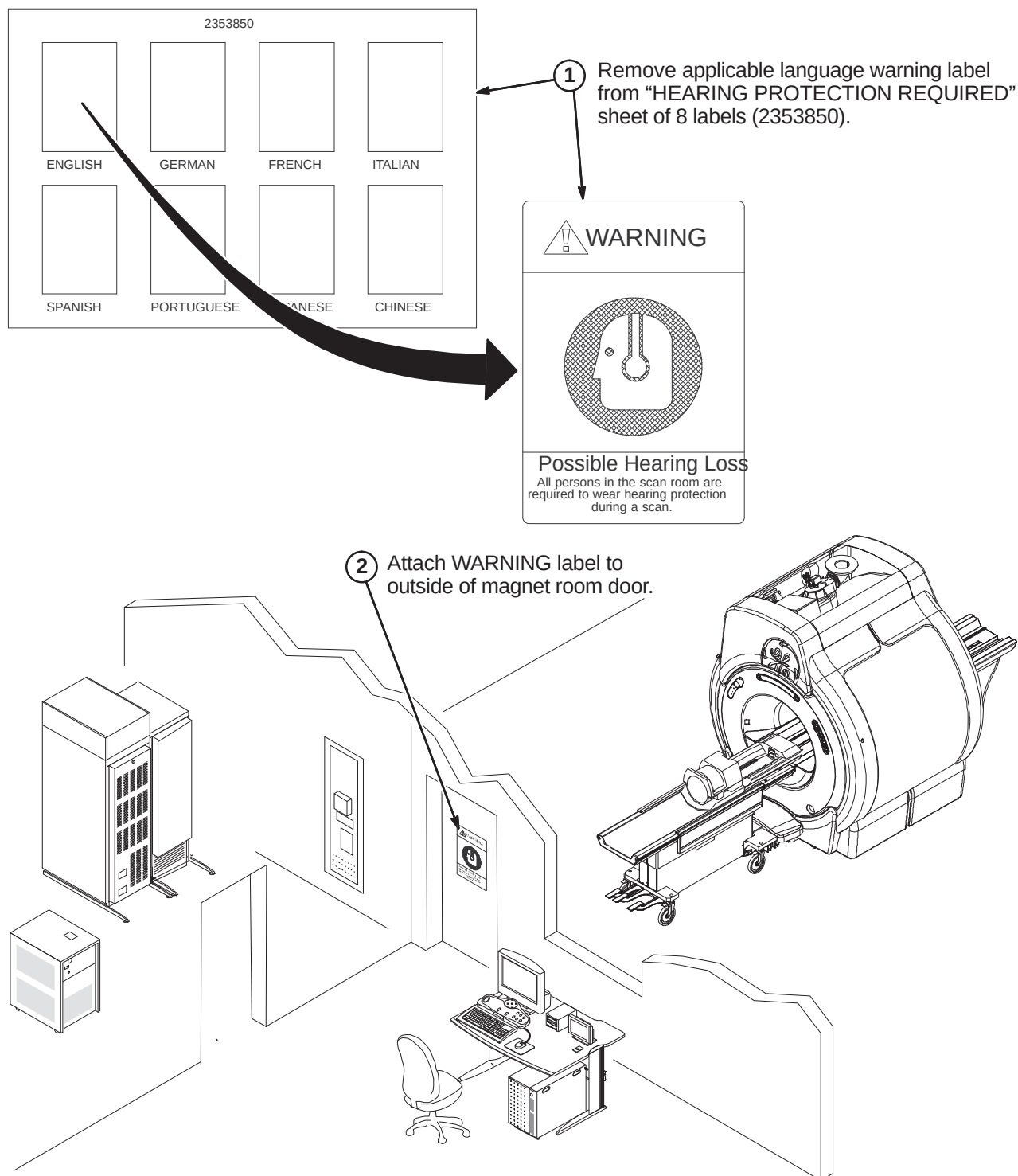
Chinese

6-6-25 PATIENT TABLE CRADLE RELEASE UPGRADE

This upgrade is performed to improve the release mechanism and improve image quality.

6-6-25-1 PREPARE CRADLE FOR UPGRADE**6-6-25-2 REMOVE EXISTING RELEASE MECHANISM**

6-6-25-3 INSTALL NEW RELEASE MECHANISM**6-6-25-4 ATTACH CORD**

6-6-26 INSTALL HEARING LOSS WARNING SIGN

SECTION 7 – BODY COIL

TABLE OF CONTENTS

<u>DESCRIPTION</u>	<u>DIRECTION NUMBER</u>
TRM LIFTING COIL PROCEDURE	2326444
TwinSpeed INTERFACE RING ALIGNMENT	2316316

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TRM Coil Rigging Using Forklift/Spreader Beam/Lifting Straps

Overview

The plan for TwinSpeed upgrades is to ship TRM Coils to the field on wooden shipping skids. This will make it necessary for the field team to work with riggers to remove the coil from the skid and place it on the aluminum service cart. This document provides information on the hardware used to perform the task and also shows the key elements required to perform the task safely.

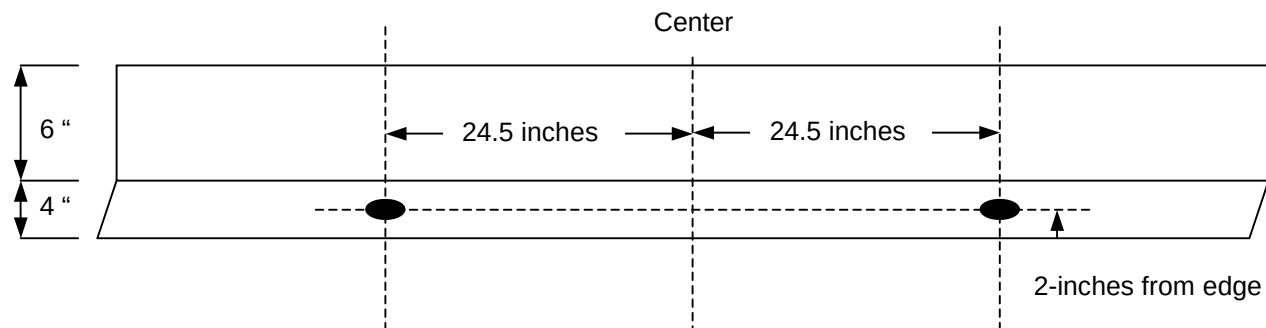
Hardware

Item	Qty	Description
1.	1	Modified - LCC magnet lifting beam – added two holes for attaching eye bolts (use as a spreader beam for vertical lift of the TRM Coil).
2.	1	Forklift (load capacity 4050 lbs with a center of gravity located 30-inches away from the truck with a maximum height of 180-inches).
3.	2	Nylon lift straps, pn 6EE1, twist tapered eye to eye, 12-feet long x 3-inches wide, load capacity – basket lift 6600 lbs, choke lift 2700 lbs.
4.	2	1-inch screw pin anchor shackle.
5.	2	3.5-inch eyebolts with flat washer, lock washer and nut.
6.	2	4-inch C-clamps.

Process

Note: Steps 1 to 4 refer to modifying one of the LCC magnet lifting beams to use as a spreader bar. This would not be required if rigger can provide a spreader bar that will support the 3000 lb. (1363 kg) TRM Coil. The position of the lifting straps on the coil is critical.

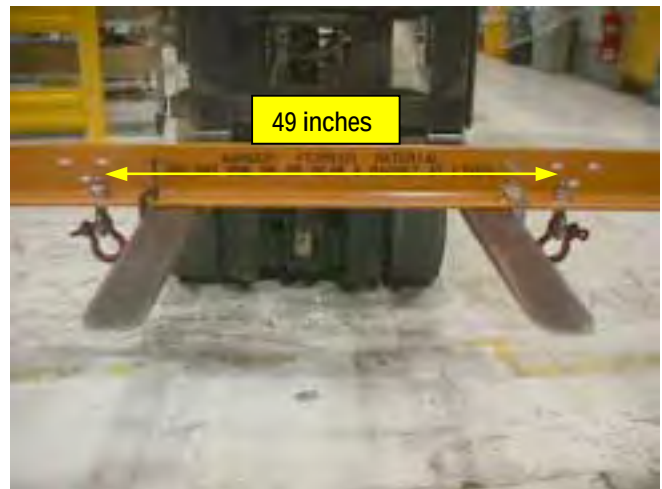
Step 1 – Modify one of the LCC magnet lifting beams. Add two-hole locations, 1-1/8 inch diameter on the 4-inch wide side of the angle beam as follows.



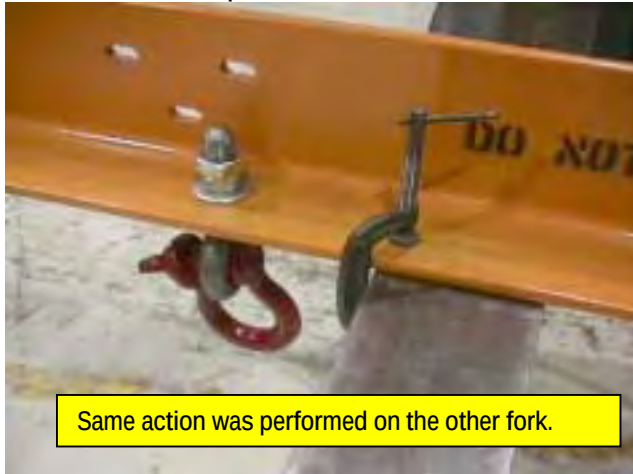
Step 2. Assemble eyebolt and shackle to lifting beam (49 inches on center).



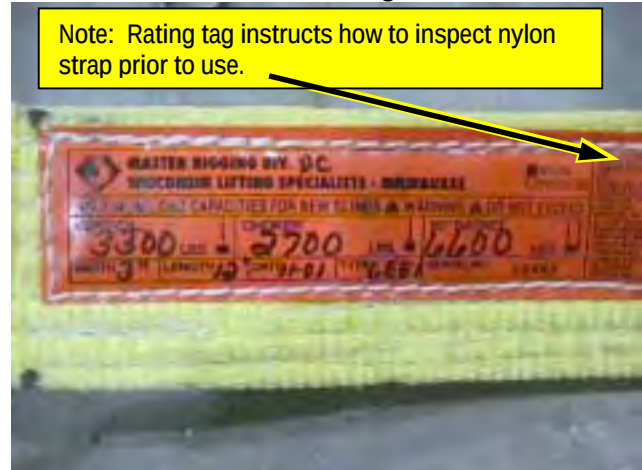
Step 3. Install beam on forks as shown below



Step 4. Secure beam to both forks with C-clamps.



Step 5. Inspect lifting strap for proper rating and wear / damage.



Step 6. Insert lifting straps through pockets on shipping skid.



Step 7. Attach lifting straps to shackles.



Step 8. Vertical lift of coil



A. The TRM coil should be lifted vertically without any sideways movement. Damage could occur to the TRM coil aluminum mylar shield, located under the cardboard wrap, where the straps are pressing against the coil.

B. Remove the wooden coil cradle.

Note: The distance from the top of the fork to the floor measured 98 inches. This height allowed for the removal of the skid and the installation of the aluminum cart with cradle.

Step 9. Move cart (with jacking assembly installed) under the coil.



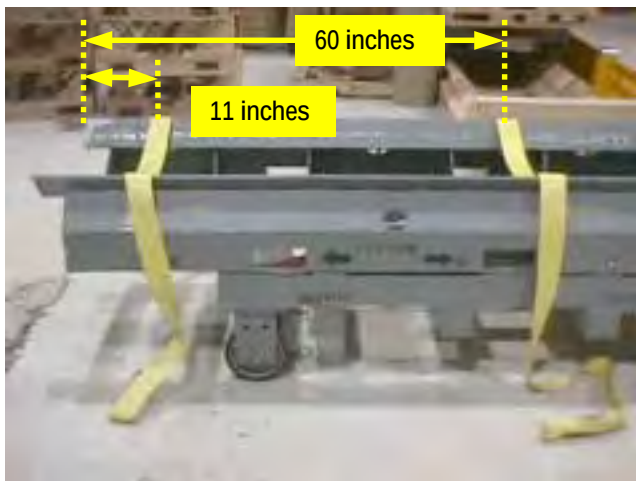
Step 10. Align the TRM coil bracket with end of the cradle, this allows enough room for the jack assembly at the other end of the cart.



Step 11. Disconnect straps from the shackles and transport to magnet room.



Note: At this point the cart would be moved to the magnet room where the Gradient Coil Insertion tool hardware is installed onto the coil and the magnet. The straps will be entrapped by the coil and cannot be removed until the coil has been lifted with the insertion tool.



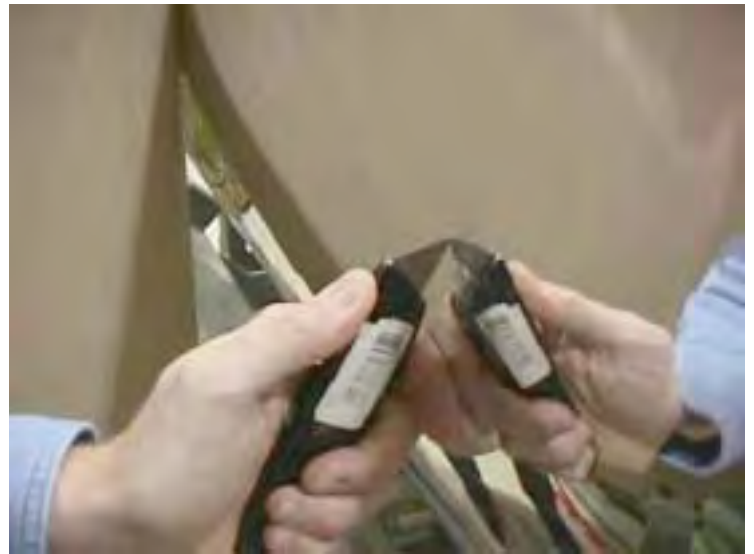
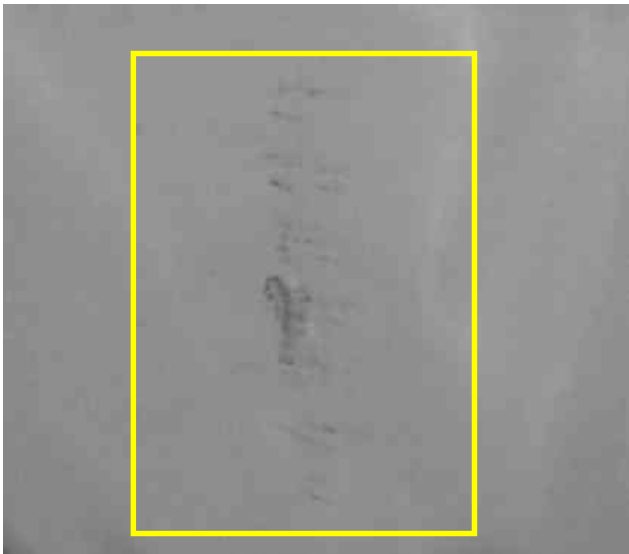
Note: When removing a coil from the magnet it will be necessary to pre-position the straps on the cart before extracting the coil from the magnet bore.

Inspection of Mylar Shield

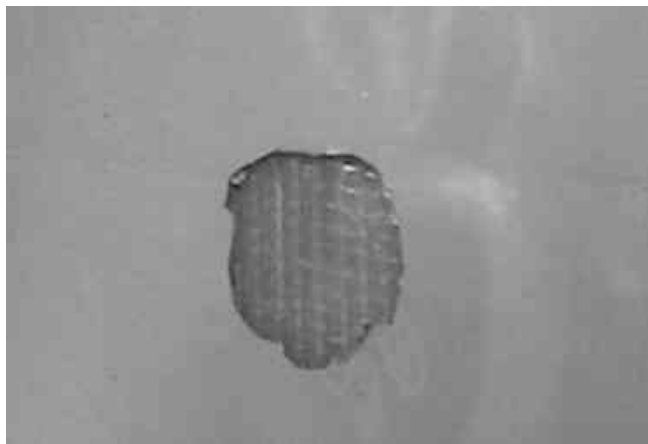
The following steps show a damaged section of shield and a repair process.

Step 1. As soon as the coil is suspended on the pipe and rollers, remove the protective cardboard wrap and inspect the aluminum Mylar shield for damage.

Step 2. Cut out damaged area of Mylar.
Note: Any cuts or punctures in the Mylar could become a spike noise source and must be repaired.



Step 3. Cut out the damaged area in a circle fashion not a square one that would result in sharp corners and cover the exposed area with Teflon tape.





GE Medical Systems

Technical Publications

Direction 2316316

Revision 2

TwinSpeed Interface Ring Alignment

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

Rev. 11/15/2000

Direction 2128126 - Language Policy For Service Documentation, Rev 0

W A R N I N G

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルは英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

<u>REV</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
0	Nov. 26, 2001	Initial version of TwinSpeed Alignment Procedure.
1	Feb 18, 2002	Added magnet flange check and procedures to address issues with magnet flange.

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
Title Page	0	Title Page	1	Title Page.....	2		
Damage in Trans.....	-	Damage in Trans....	-	Damage in Trans. ...	-		
Direction 2128126 ..0*		Direction 2128126 .0*		Direction 2128126..0*			
6 to 32	0	6 to 32	1	6 to 37	2		

* This revision/letter corresponds to the indicated document's revision control system.

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1- OVERVIEW

A new tool is required to align the TwinSpeed Vacuum Enclosure Interface Rings to each other for TwinSpeed upgrades (new systems are factory aligned). The vacuum end bells are mounted to the interface rings and then the RF coil is mounted to the end bells. The alignment of the interface rings is critical to the performance of the vacuum enclosure. The RF coil alignment to the gradient coils is critical to SNR performance of the system. Since the RF coil in a TwinSpeed system is mounted/supported by the enclosure end bells, the alignment of the RF coil is ultimately determined by the interface ring alignment.

2- REQUIRED TOOLS & EQUIPMENT

The tools listed below that are highlighted in gray are the parts required for the Magnet Interface Rings. The remaining parts are required for the Gradient Isolation Brackets. These parts are needed after the coil is installed in the magnet.

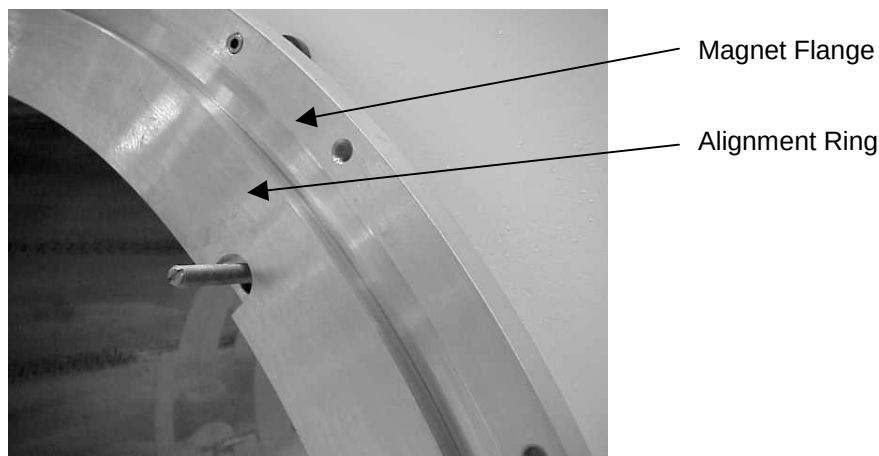
Description	Part Number	Qty.
Twin Centering Tool Kit (Order from Tool Depot) (Kit weighs approx. 78 Lbs./ 39 kg)	2299496	1
TRM LCC Gradient ISO Install Kit	2297692	1
SUPPORT MOUNTING, TRM	2272016	2
SPACER SHIM 1.5MM THK	2238908	2
SPACER SHIM 2.4MM THK	2238908-2	6
SPACER SHIM 16MM THK	2238908-3	2
RUBBER ISOLATION PAD	2241176	2
SCREW SET SCREW 10 MM 25 MM--Grub	2251295	2
SCREW SET SCREW 10 MM 30 MM--Grub	2251295-2	1
MTG PAD, 1/4" THICK NEOPRENE	2131953-5	4
SET SCREW- MTG.PAD	2293535	4
RADIAL SHIM SPACER 1.5MM THK	2293771	4
RADIAL SHIM SPACER 0.8MM THK	2293771-2	8
MAGNET RING	2306226	2
WEDGE RING	2306184	2
O-RING WEDGE RING	2310035	2
MAGNET RING GASKET	2246966	2
SCREW HEXAGON SOCKET 10 MM 25 MM	2109867-23	32
MAGNET RING BRACKET	2284172	1
SCREW PAN HEAD 6 MM 16 MM	2109873-25	2
CAP SCREW HEXAGON HEAD 10 MM X 1.5 X 35 MM	2109866-22	8
VACUUM GREASE MSDS# 8366442	2297760	1
SHOULDER WASHER NYLON	2308141-4	8
SCREW SET SCREW 10 MM 20 MM	2251294	4
RADIAL FOOT PADS	2310334	12

Field Supplied Tools

Disposable Gloves
Masking Tape
Tongue Depressor
Non Magnetic Tool Kit Metric/English
Allen Wrench 8 mm
Rubbing Alcohol
Rags

3- MAGNET FLANGE CHECK

There may be issues with the Magnet Flange that will prevent the O-rings and gaskets that are between Magnet Flange and Alignment Ring from maintaining proper vacuum. It is estimated that 10% of all LCC magnets may have issues that will need to be addressed. To check for Magnet Flange consistency, place the Alignment Ring against the Magnet Flange. See Illustration 3-1. Refer to Appendix C if gaps or protrusions are discovered.

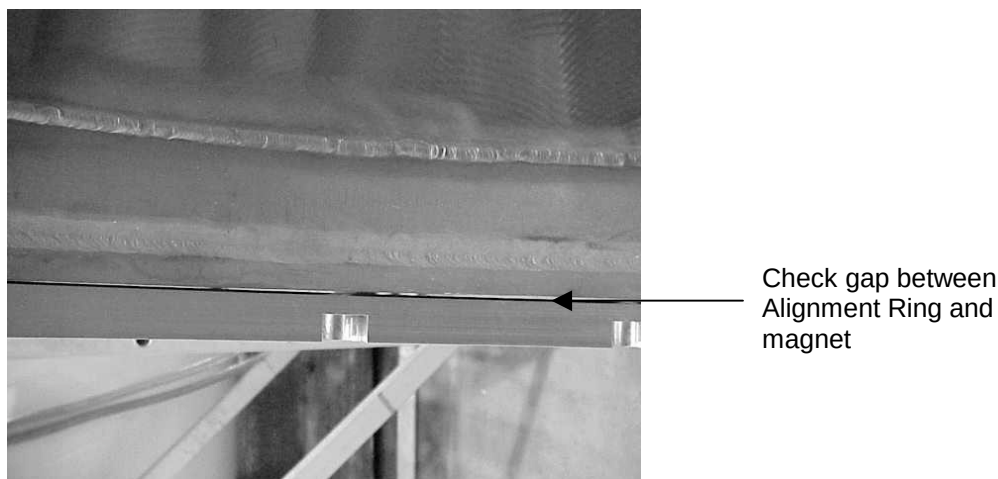


ALIGNMENT RING CHECK
ILLUSTRATION 3-1

Check for the following conditions:

- Protruding magnet weld
- Cavity in the flange
- Bolt pattern on the Magnet Flange does not match bolt pattern on the Alignment Ring
- Skewed Flange

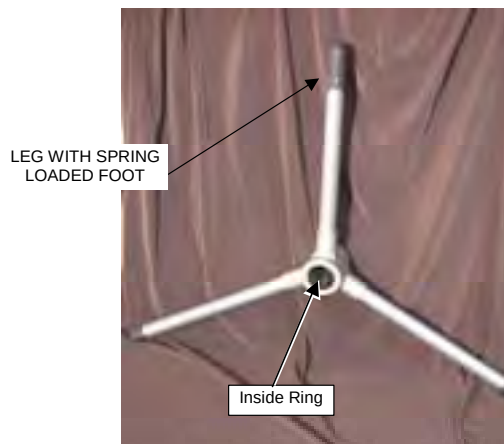
If any of the above conditions are noted, refer to Appendix C for the resolution needed.



GAP CHECK
ILLUSTRATION 3-2

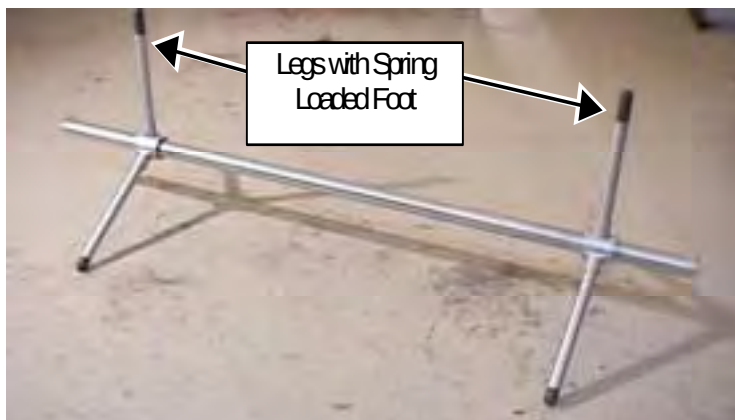
4- TOOL SETUP

1. Open the Twin Centering Tool case and assemble the parts for the two 3-legged spider assemblies as shown in Illustration 4-1. Note: each spider assembly gets one leg with Spring Loaded foot.



SPIDER ASSEMBLY
ILLUSTRATION 4-1

2. Wipe the center shaft and the inside ring of the Spider Assembly.
3. Assemble 3-piece center shaft and slide a spider assembly on at each end. The leg with the spring loaded foot should be pointed up.

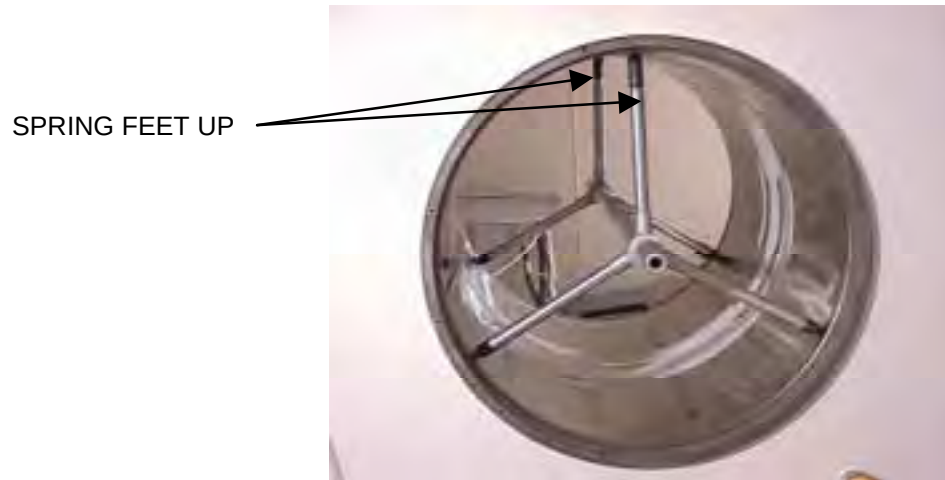


CENTER SHAFT WITH SPIDER ASSEMBLIES
ILLUSTRATION 4-2

4. Insert Center Shaft Assembly into magnet bore being careful not to drag across the passive shims. The legs with the spring loaded foot are pointed up. See Illustration 4-3.
5. After Center Shaft Assembly is in position, rotate the spring loaded foot of each Spider Assembly $\frac{1}{4}$ turn to activate the spring tension against the top of the magnet bore. *Hint:* Spider Assembly feet must maintain pressure against the magnet bore, outside of the passive shim area.

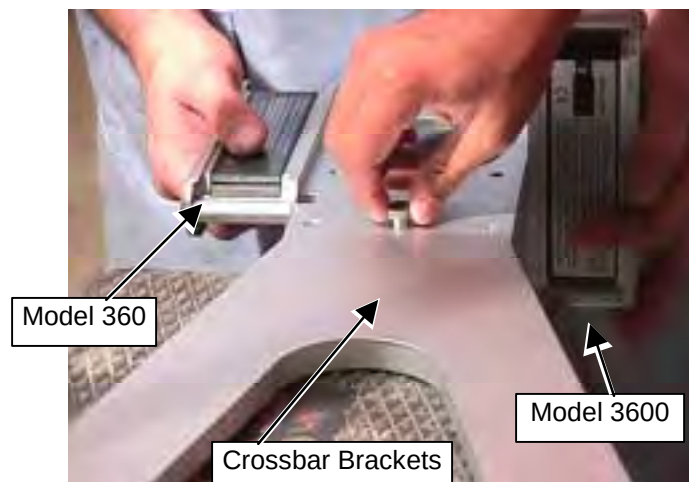
Note

It is important to place the spider assembly as close as possible to the passive shim location markers in the magnet bore and away from the weld seam.



CENTER SHAFT ASSEMBLY IN BORE
ILLUSTRATION 4-3

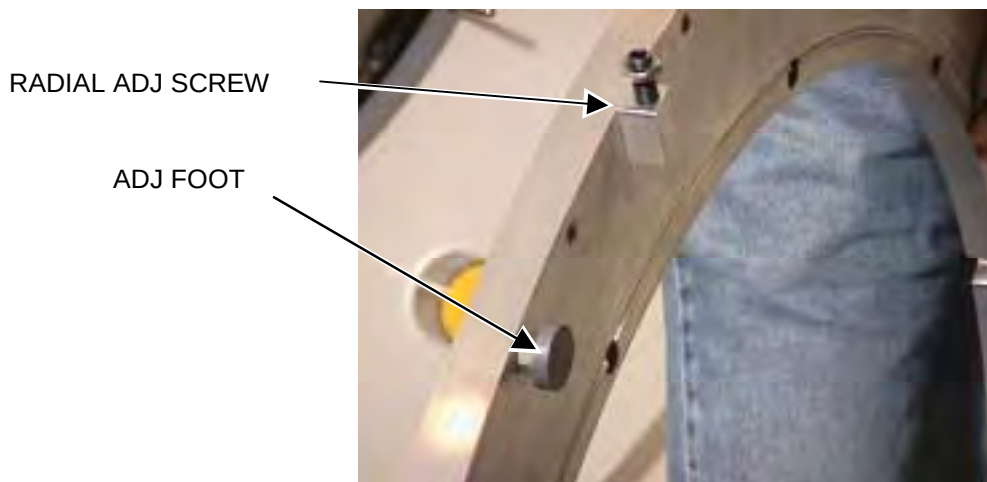
6. Connect the Digital Protractor Assembly to the crossbar brackets using the 6 allen-head capscrews and wrench provided in kit. Tighten snugly. See Illustration 4-4.



ASSEMBLE DIGITAL PROTRACTORS TO BRACKETS
ILLUSTRATION 4-4

7. Make sure the model 3600 protractor is mounted on the top and in line with the axial pipe and the model 360 protractor is mounted below and perpendicular to the axial pipe.
8. Wipe down both sides of the interface ring with alcohol.

9. Inspect magnet interface ring to ensure the adjustment feet and the radial screws are fully retracted. See Illustration 4-5. Note that in early production the foot screws are ferrous.



BACK OF INTERFACE RING – ADJUSTABLE SCREW/FOOT

ILLUSTRATION 4-5

10. Attach the Twin Centering Tool to the front of one of the enclosure interface rings using six 17 mm hex-head screws provided in kit (note: the front of the interface ring is opposite the side with the 4 round feet and the lower protractor readout faces out of the magnet, see Illustration 4-7). Tighten snugly. See Illustration 4-6.



TWIN CENTERING TOOL TO ENCLOSURE INTERFACE RING

ILLUSTRATION 4-6



READ DIRECTION OF LOWER PROTRACTOR

ILLUSTRATION 4-7

5- PREPARATION FOR INTERFACE RING

The Ring Alignment Tool is used to position the Vacuum Enclosure Interface Rings. There are some enclosure parts that must be installed before the interface rings.

1. The first enclosure part to be installed is the Wedge Ring that fits around the magnet's end ring. The Wedge Ring must sit flat against the magnet's end flange. It may be necessary to use non-ferrous tooling to remove any weld globs that inhibit the Wedge ring from sitting flat. See Illustration 5-1 showing removal of weld globs prior to installation of the Wedge Ring.

WARNING!

POSSIBLE PROPERTY DAMAGE AND IMAGE QUALITY ISSUES. THE WELD GLOBS ARE FERROUS MATERIALS AND MUST BE REMOVED FROM THE MAGNET ROOM TO PREVENT POTENTIAL WHITE PIXEL SOURCE.

WARNING!

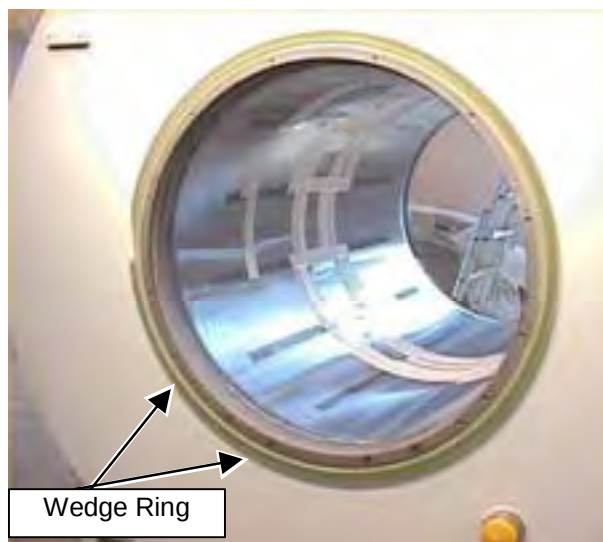
POSSIBLE PERSONAL INJURY. THE WELD GLOBS WILL COME LOOSE AND COULD INJURE EYES. WEAR PERSONAL PROTECTIVE EQUIPMENT SAFETY GLASSES WHILE REMOVING GLOBS.



PREPARATION FOR WEDGE RING
ILLUSTRATION 5-1

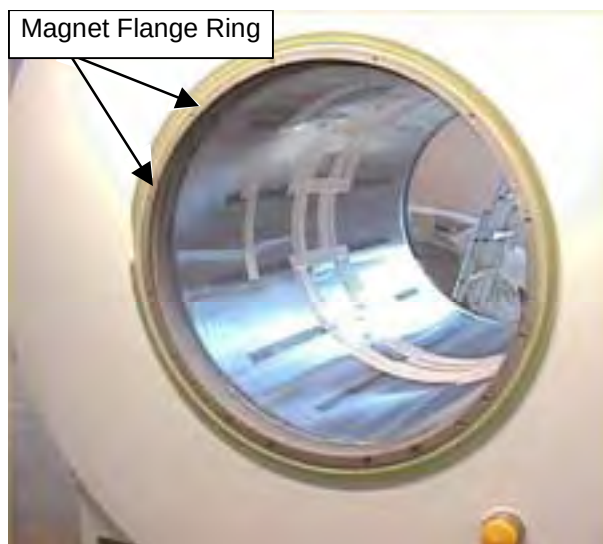
2. Wipe down the magnet end flange and magnet ring with alcohol before installing wedge ring.

3. Install Wedge Ring. See Illustration 5-2. There is a notch in the Wedge Ring. The position of the notch is not relevant for the front of the magnet. There is a weld in the rear of the magnet that will allow you to position the notch over the weld.



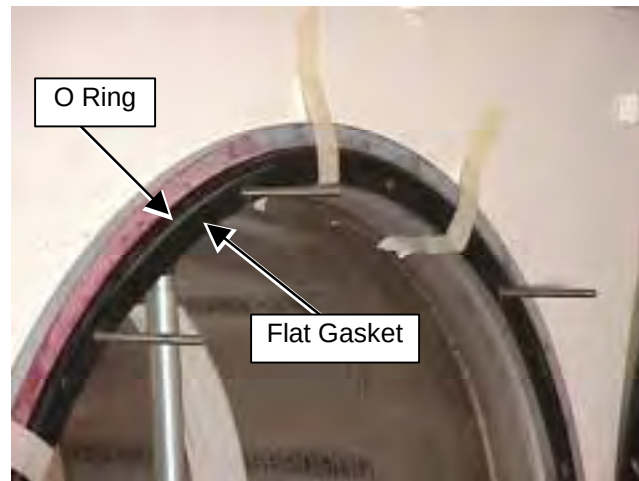
WEDGE RING INSTALLED
ILLUSTRATION 5-2

4. Using disposal gloves apply a thin film of vacuum grease (P/N 2297760) to the face of the magnet flange ring. See Illustration 5-3.



MAGNET FLANGE-RING LOCATION
ILLUSTRATION 5-3

5. Apply vacuum grease to the 6 threaded studs that came with the enclosure and install into threaded holes on magnet ring. See Illustration 5-4. Place 3 studs, as shown, around top half of magnet end ring and 3 around the bottom half.
6. Install flat gasket onto the studs installed in previous step. See Illustration 5-4.
7. Using disposal gloves apply vacuum grease (P/N 2297760) liberally to the round O-ring (5/8" thick), part number 2310035. Then install O-ring around the magnet end ring. See Illustration 5-4.



TAPE FLAT GASKET IN PLACE
ILLUSTRATION 5-4

8. Use masking tape or similar to hold the O-ring into place. See Illustration 5-4.
9. Use a tongue depressor to place vacuum grease on the bottom of the radial screws. See Illustration 5-5.



APPLYING GREASE TO RADIAL SCREWS
ILLUSTRATION 5-5

10. Place the nylon spacers on the end of the radial screw. See Illustration 5-6.



PLACING THE NYLON SPACER

ILLUSTRATION 5-6

11. Place the Interface Ring with the Twin Centering Tool over the studs on magnet end ring. See Illustration 5-7.



BOLT INTERFACE RING TO FRONT OF MAGNET

ILLUSTRATION 5-7

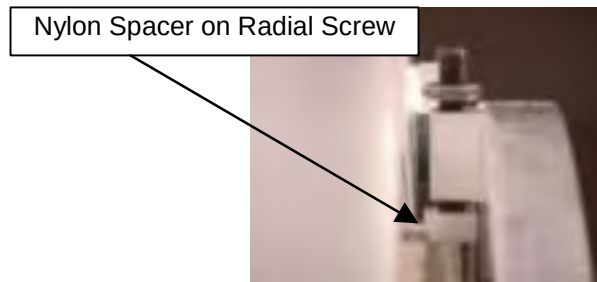
12. Remove the tape holding the flat gasket in place.

Note

In step 13, it may be necessary to use the nut with metal washer on the studs from the Gradient Insertion tool kit to start to tighten the interface ring to the magnet on both the front and rear rings of the magnet.

13. Put vacuum grease on threads before inserting the M10 x 25mm capscrews from the enclosure into the remaining holes around the interface ring. Just start the capscrew threads in the holes, do not tighten beyond that for now. After installing three screws, the threaded rods can be removed. They will be used when installing the rear Alignment Ring.

14. Inspect inside of magnet interface ring to ensure that the nylon spacers for radial screws are in place. See Illustration 5-8.



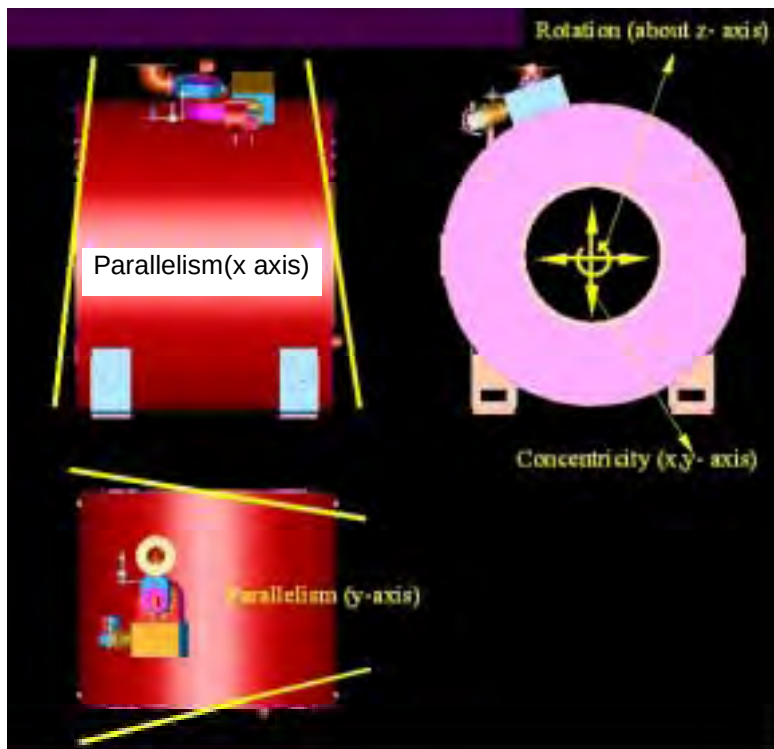
NYLON SPACER IN PLACE

ILLUSTRATION 5-8

15. Verify that the O-ring is in the proper place all around the wedge ring, otherwise damage to the O-ring could occur.

6- ALIGNMENT PROCEDURE

The alignment procedure requires patience and perseverance to be done correctly. The interface ring must be aligned in all physical planes to ensure proper alignment of the RF Coil to the TRM and vacuum system functionality. There are 4 alignment parameters. Concentricity alignment, rotational alignment, parallelism about x-axis and parallelism about y-axis, in the same order. See Illustration 6-1.



ALIGNMENT PARAMETERS
ILLUSTRATION 6-1

The alignments are completed for the front interface ring first and then repeated on the rear interface ring.

6-1 Rotational Alignment

The goal of the rotational alignment is to set the rotational position of the interface ring as close to zero degrees as possible (Note: The magnet should have been leveled at initial magnet installation). The Digital Protractor is used to set the reference.

1. Toggle ON/OFF switch on the Digital Protractor and check the reading. The arrow indicates which direction the interface ring must be moved. See Illustration 6-2.

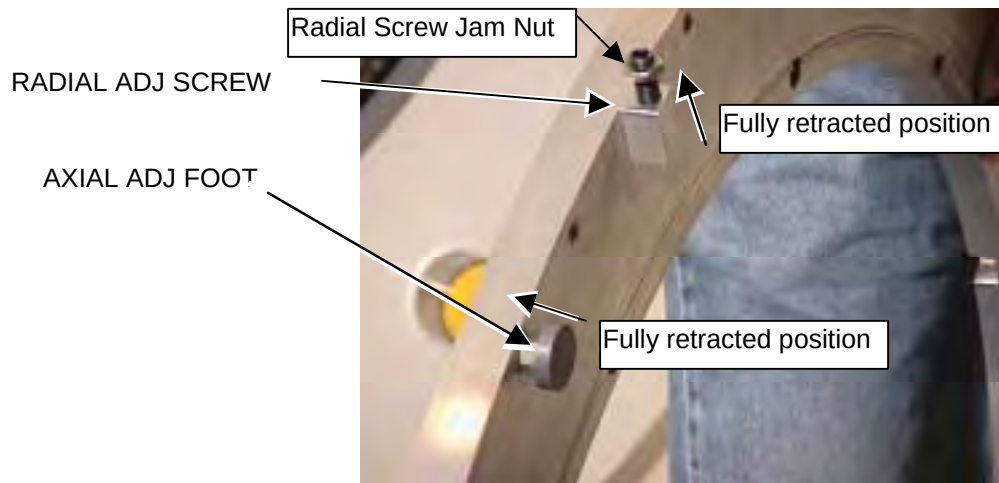


DIGITAL PROTRACTOR
ILLUSTRATION 6-2

2. Manually rotate the ring clockwise or counter clockwise to reach the goal of an alignment reading which is $\leq 0.5^\circ$.
3. Record final measurement.

6-2 X-Y Plane Concentric Alignment

The goal of the X-Y plane concentric alignment is to position the interface ring such that the center shaft can be inserted through the pivot ring of the Digital Protractor assembly. See Illustration 6-2 for pivot ring. **Do Not attempt to force the shaft into the pivot ring**; this could damage the pivot ring. The interface ring must be positioned using the radial screws on the interface ring and then the center shaft is inserted **by hand** through the pivot ring. See Illustration 6-3 of the backside of the interface ring.



BACK OF INTERFACE RING – ADJUSTABLE SCREW/FOOT
ILLUSTRATION 6-3

1. Position the pivot ring so it is perpendicular to the center shaft assembly, make sure the groove is even or aligned to the face of the plate (see Illustration 6-4). Slide the center shaft close to the pivot ring.

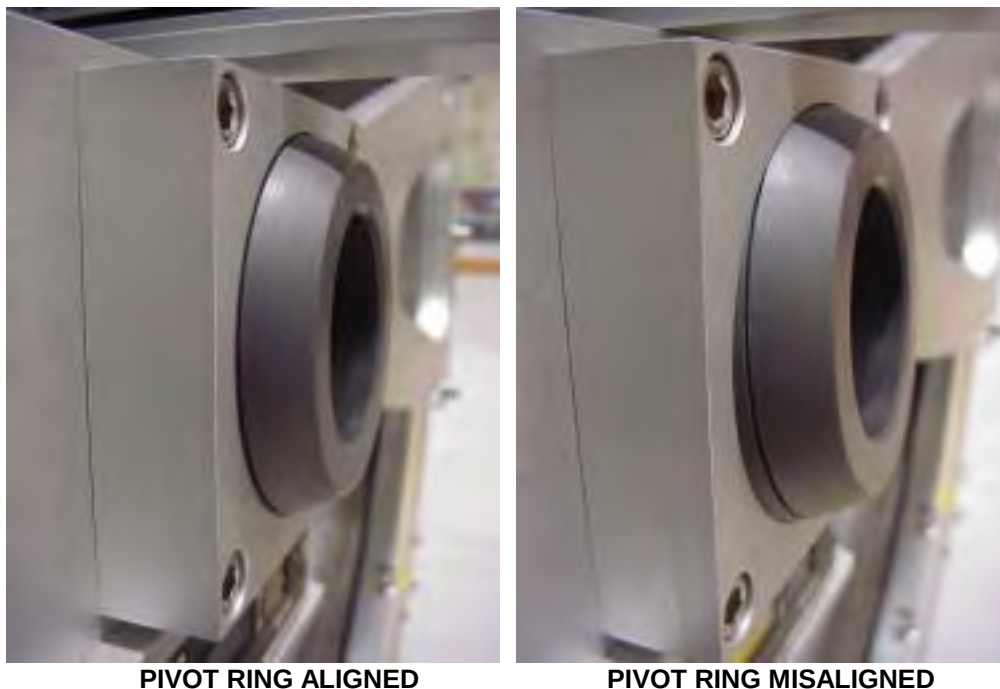
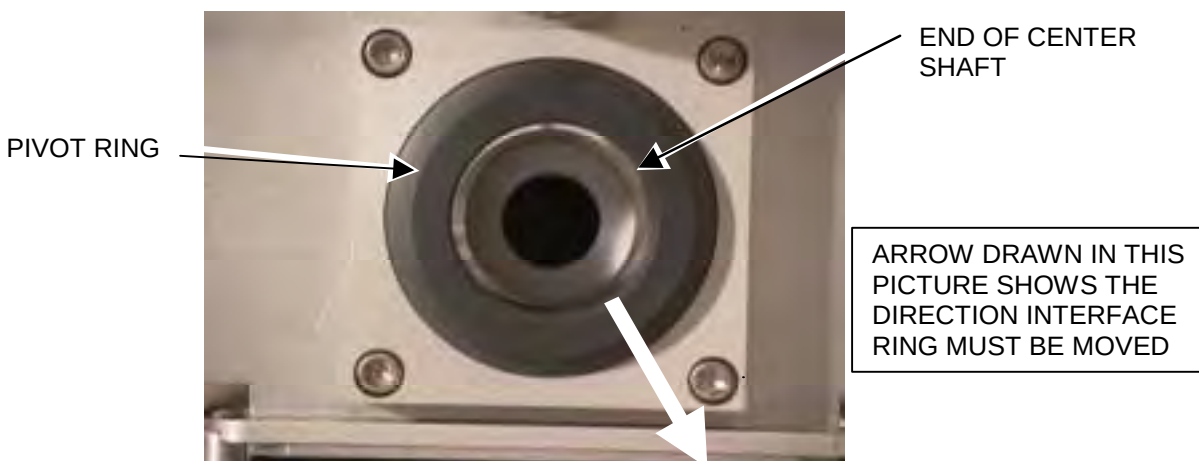


ILLUSTRATION 6-4

2. Look through the hole of the pivot ring to see alignment of center shaft. See Illustration 6-5. Determine direction the interface ring must move to align the center shaft to the pivot ring opening.



VIEW OF CENTER SHAFT THROUGH PIVOT RING

ILLUSTRATION 6-5

3. Use Allen wrench to turn the appropriate radial adjustment screws to move the interface ring into alignment with the center shaft.
4. Continue to make adjustments using the radial adjustment screws until the shaft can be pushed through the pivot ring using only the force of your hand. See Illustration 6-6.



CENTER SHAFT THROUGH PIVOT RING

ILLUSTRATION 6-6

Note

Due to the stack up of tolerances of the passive shim in the magnet bore; it may be necessary to shim the legs of the spider assembly. The 8mm shims work well for this. Then repeat steps 1 – 4 till center shaft moves through pivot ring.

5. Place all radial fingers snug to the wedge.

6. Evenly tighten 16 capscrews a $\frac{1}{2}$ turn using an alternating pattern maintaining a gap of 8mm between the wedge ring and the magnet interface ring. For example picture the magnet ring as the face of a clock, tighten the capscrew at ten o'clock position, followed by the capscrew at 4 o'clock, followed by the capscrew at 7 o'clock ...

Note

Use a 8mm allen wrench to check the gap between the wedge ring and the magnet interface ring.

6-3 X & Y-Axis Alignment

The goal of the Y-axis alignment is to adjust the axial position of the interface ring as close to zero degrees as possible using the Axial Digital Protractor.

Note

Start with the center shaft slid out of the Pivot Ring completely.

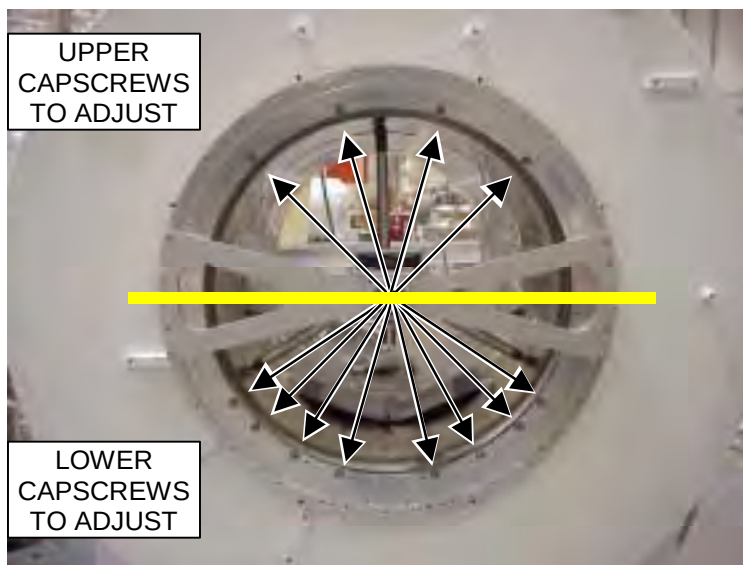
1. Toggle ON/OFF switch on the Axial Digital Protractor and check the reading. The arrow indicates which direction the interface ring must be moved. See Illustration 6-7.



AXIAL DIGITAL PROTRACTOR
ILLUSTRATION 6-7

6-3-1 Y-Axis Alignment

1. Using the directional arrow on the digital protractor, adjust the capscrews on the lower or upper half of the magnet interface ring to reach the goal alignment reading of $\leq 0.5^\circ$. See Illustration 6-8, the solid horizontal line divides the ring into the upper and lower halves.

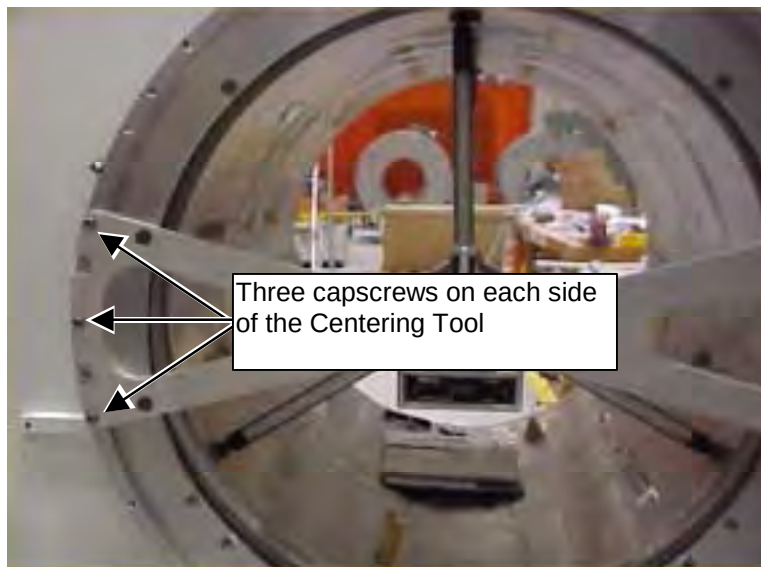


CAPSCREW ADJUSTMENT LOCATIONS
ILLUSTRATION 6-8

2. Record measurement.

6-3-2 X-Axis Alignment

1. Loosen the six (6) screws that hold the centering tool on the interface ring. This is done to remove any distortion caused by over tightening the centering tool. See Illustration 6-9.



SCREW LOCATION
ILLUSTRATION 6-9

2. Use the right angle T-square at the shaft and use the 0.25 mm gauge to make sure the gap does not exceed the gauge. To make the adjustment tighten the capscrews on the left or right side to adjust the gap which aligns the X-axis. See Illustration 6-10.



RIGHT ANGLE T-SQUARE MEASURING X-AXIS ALIGNMENT
ILLUSTRATION 6-10

3. After the axial alignment has met the alignment goal, extend all adjustment feet only finger tight to just make contact with magnet front face, and then lock with nut.
4. Tighten the radial screws evenly and then lock in place with the jam nut. See Illustration 5-9 for examples of radial screw locations.

6-4 Securing Adjustments

1. Check to ensure that the radial screws and axial feet are locked down.
2. Write down the final alignment numbers from the digital protractors.
3. Remove the 6 capscrews that hold the Centering Tool Assembly and install on the rear interface ring.

6-5 Rear Interface Ring Alignment

Repeat sections 4 and 5 for the Rear Interface Ring installation.

APPENDIX A – DIGITAL PROTRACTOR TESTING

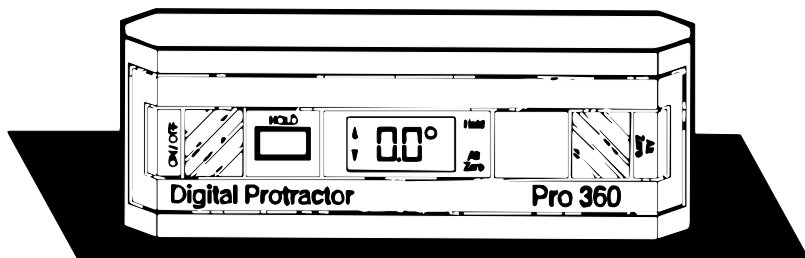
A-1 Recalibration

Like all measurement instruments the Pro 360/3600 must periodically be monitored for accuracy. The Pro 360/3600 should be tested daily with the simple procedure outlined below. If it is found to be out of calibration, it is easily recalibrated on site using Superset® - an eight-step procedure that realigns the unit through its entire 360° range and takes just a couple of minutes to perform. And Superset® doesn't require any special fixtures, tools, or expertise.

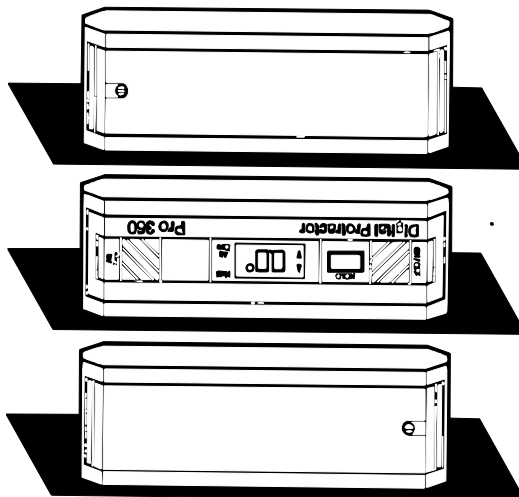
Accuracy Test

Perform this simple test each day before using the Pro 360/3600. Also perform it anytime the Pro 360/3600 has been dropped or is being used in an environment that varies more than 5° C (9° F) from the environment in which it was last calibrated. *If your Pro 360/3600 fails this test, you must recalibrate it using Superset®, before using the unit further to measure angles.*

1. Position the Pro 360/3600 with the display facing you on a clean, flat horizontal surface. It doesn't have to be exactly level. Wait 10 seconds so the unit is completely settled and note the angle on the display.



2. Rotate the unit end-for-end so the display is facing away from you. Be sure to set the Pro 360/3600 in exactly the same spot, and wait 10 seconds before reading the angle that's displayed.



3. Now roll the unit toward you so that the display is facing you, but the lettering on face of the unit is upside down. Wait 10 seconds, and note the angle on the display.
4. Finally, rotate the unit end-for-end so the display is facing away from you (the lettering should still be upside-down). Wait 10 seconds and note the angle on the display.

5. If any of the four measurements varies from any other by more than 0.1° , you must recalibrate (Superset[®]) your unit before using it further.

Superset[®]

Superset[®] recalibrates the Pro 360/3600 through its entire 360° range by electronically recording four horizontal and four vertical settings. It should be performed whenever the accuracy test shows a discrepancy of 0.1° or more.

How to Perform Superset[®]

1. Turn on the Pro 360/3600 and place it on a flat surface. You can use any horizontal surface within 10° of level and any vertical surface within 10° of plumb to perform Superset[®]. You must use the same surfaces throughout the entire process.

Note

Each time you reposition the Pro 360/3600 during Superset[®], wait a minimum of 10 seconds before pressing the **[HOLD]** button to advance to the next step.

2. Press and hold the **[HOLD]** and **[ALT ZERO]** buttons simultaneously. Keep them depressed for approximately three seconds.
3. Release the buttons when the symbol "SUP" appears. A "0" within flashing brackets will then appear. These brackets are composed of four horizontal and four vertical segments.



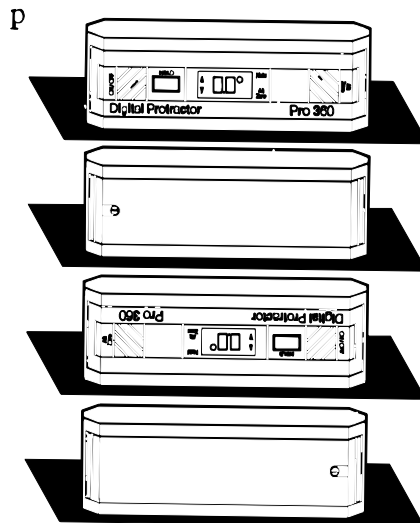
"0" within flashing brackets

Note

As you proceed through the next eight steps (4 – 12) of Superset[®], a new segment will hold steady after you complete each step.

Superset® - Horizontal Settings

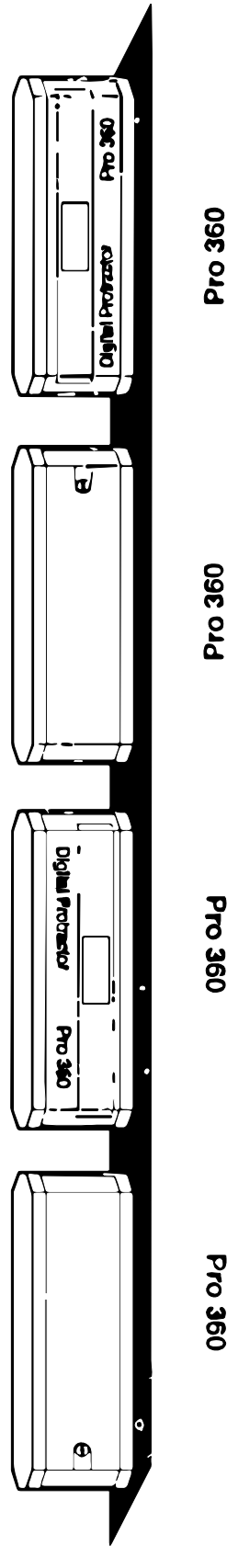
4. The unit faces you and the white lettering on face is right-side up. Align with an edge or line - wait 10 seconds Press [HOLD] button until 1 appears.
5. Rotate the unit so it faces away from you, the lettering should still be right-side up. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 2 appears.



6. Roll the unit so it faces you, the lettering should now be upside-down. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 3 appears.
7. Rotate the unit so it faces away from you, the lettering should still be upside-down. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 4 appears.

Superset® - Vertical Settings

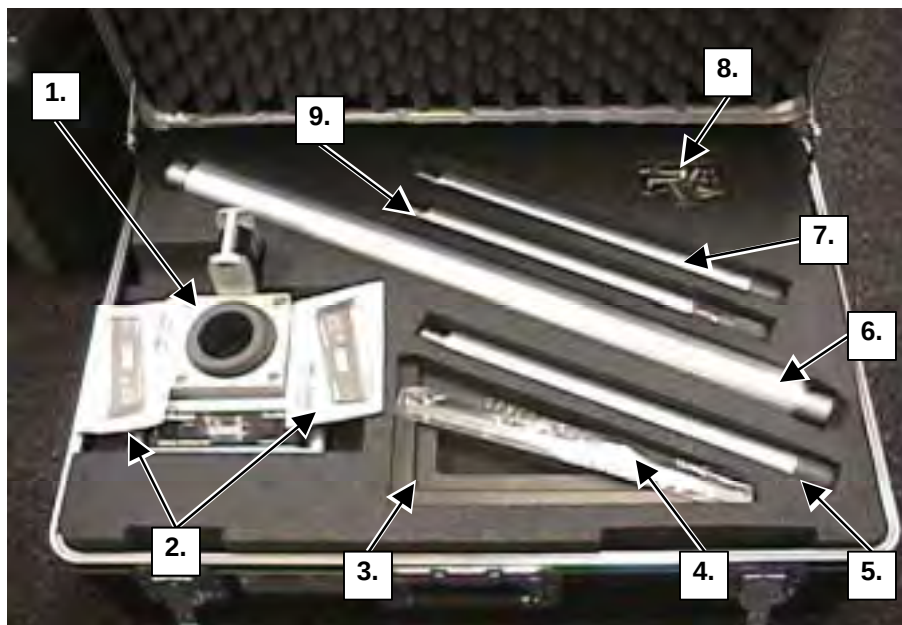
8. Place the unit against vertical surface so it faces you, the lettering on the face (Pro 360, etc.) should read from bottom-to-top. Align with an edge or line - wait 10 seconds. Press [HOLD] button until 5 appears.



9. Roll the unit so it faces away from you, the lettering should still read from bottom-to-top. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 6 appears.
 10. Rotate the unit end-for-end so it faces you, the lettering should now read top-to-bottom. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 7 appears.
 11. Roll the unit so it faces away from you, the lettering should still read top-to-bottom. Align with same edge or line - wait 10 seconds. Press [HOLD] button. An 8 will very briefly appear, followed immediately by regular angle measuring.
 12. Your Pro 360/3600 has been Superset[®] back to manufacturer's spec's.
- Canceling Superset[®]
- You may cancel Superset[®] at any time during the process by turning the unit off.

APPENDIX B – TWIN CENTERING TOOL KIT

The Twin Centering Tool Kit is packed in the case as follows.



TOP LEVEL OF KIT
ILLUSTRATION B-1

Item No.	Description
1.	Pivot Ring and Digital Protractors
2.	Instruction Manuals for Digital Protractors
3.	Right Angle
4.	.25 mm Feeler Gauge
5.	Leg for Spider Assembly
6.	Center Shaft
7.	Leg for Spider Assembly
8.	Part Box containing: 6 Hex head screws, 17 MM 1 Allen wrench 6 Bolts 6 Washers
9.	Leg for Spider Assembly with Spring loaded Foot



SECOND LEVEL OF KIT
ILLUSTRATION B-2



THIRD LEVEL OF KIT
ILLUSTRATION B-2

Item No.	Description
1.	Cross Bar Assembly
2.	Cross Bar Assembly



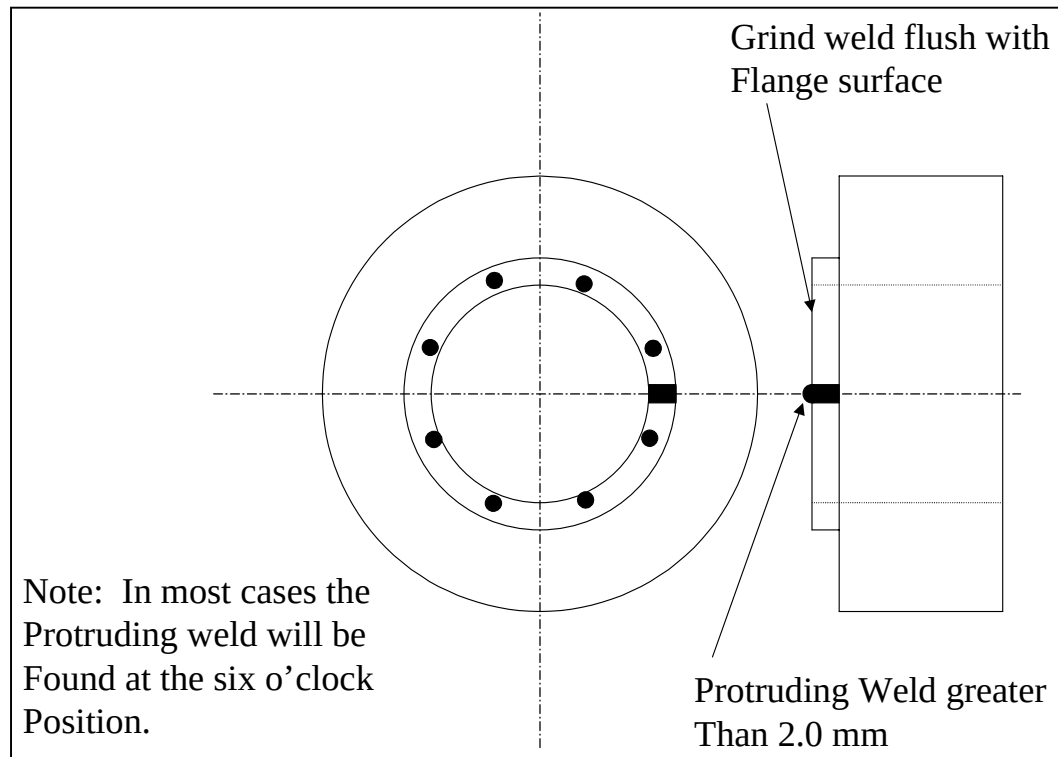
BOTTOM LEVEL OF KIT
ILLUSTRATION B-4

Item No.	Description
1.	Legs for Spider Assembly
2.	Leg for Spider Assembly with Spring loaded Foot
3.	Center Shaft Ends
4.	Spider Assemblies

APPENDIX C – MAGNET FLANGE ISSUES

If any of the following conditions are observed additional actions must be performed to allow proper fit between the Alignment Ring and the magnet flange. Worst case is a protruding magnet weld. The magnet must be ramped down for this situation.

C-1 Protruding Magnet Weld



If the Protruding weld is greater than 2.0mm, you must ramp the magnet down and grind off the weld.

IMPORTANT

Inform the customer that the Upgrade will take longer than expected.

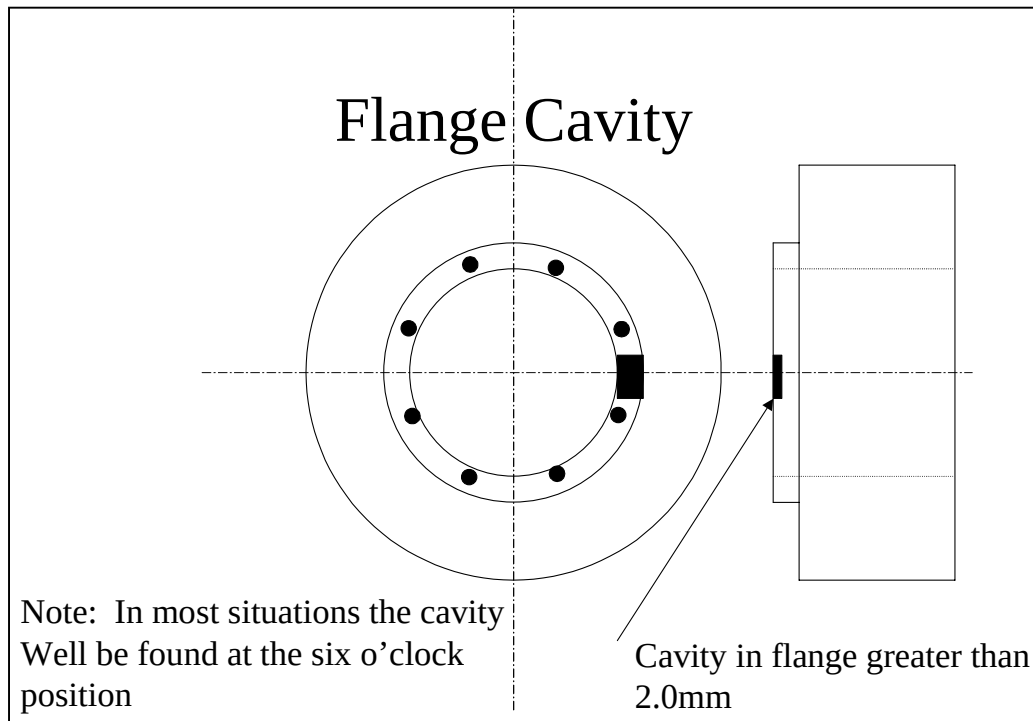
Tools needed:

- Ramp supply, Shim supply and associated equipment needed to ramp magnet.
- Die Grinder/Electric Drill
- Abrasive media for above selected tool
- Shop Vacuum or similar
- Drop Cloth or similar
- Tape
- Rags
- Alcohol

Procedure:

1. Ramp magnet down using standard procedure.
2. Remove the gradient coil from the magnet.
3. Use Tape and Drop Cloth, cover the area around the protruding weld of the magnet.
4. Prior to grinding turn the Shop Vac on to capture the debris from the grinding action.
5. Grind off the weld so the magnet flange surface is flush within .5mm.
6. Remove tape and drop cloth, then clean the magnet with rags and alcohol.
7. Install the TRM gradient coil.
8. Ramp the magnet.
9. Complete the Upgrade.

C-2 Cavity in Magnet Flange



If any Cavity on the Magnet Flange is greater than 2mm, it must be filled in with RTV.

Tools Needed:

- RTV
- Level or Straight Edge

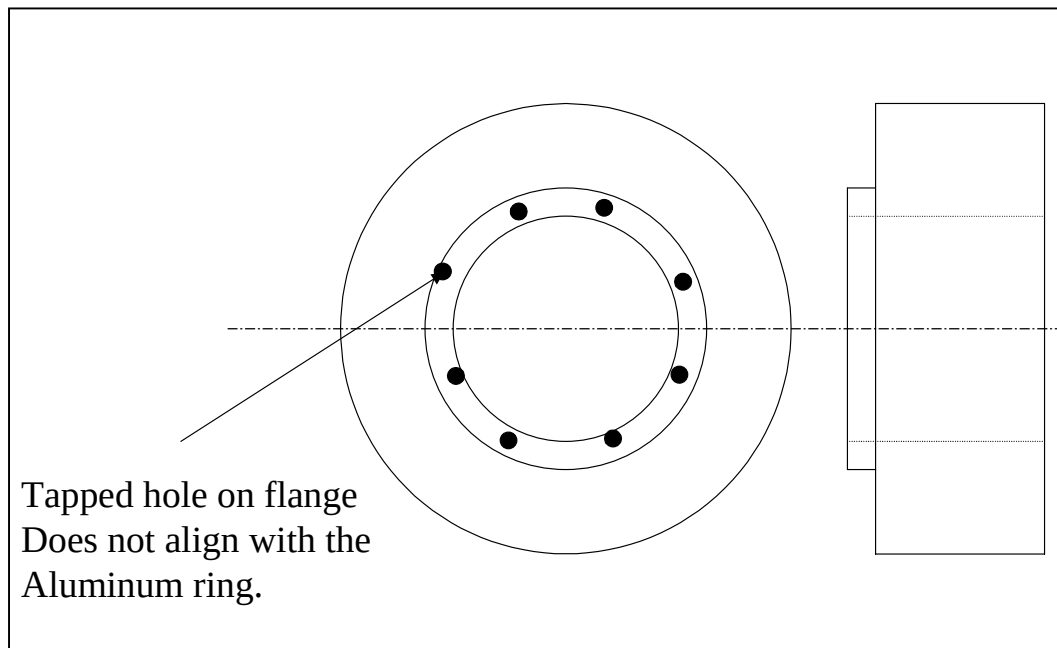
Procedure:

1. Using a level determine the amount of RTV needed to fill in the cavity.
2. Fill in the cavity with RTV using a straight edge.

IMPORTANT

The RTV must be supported to prevent it from being pulled into the vacuum space.

C-3 Miss-aligned Hole in Magnet Flange



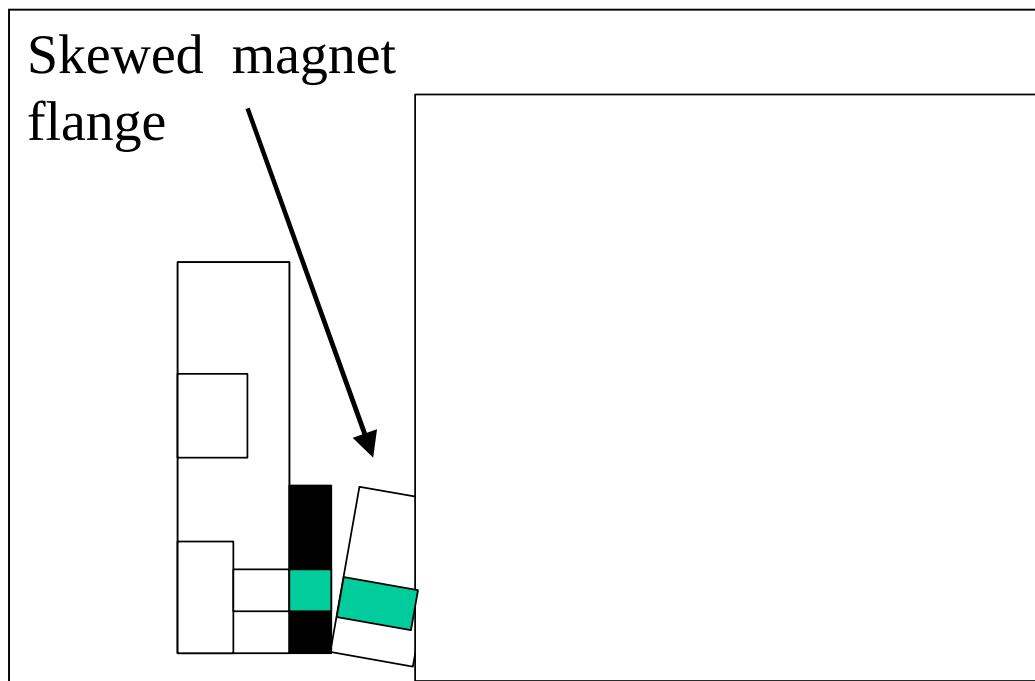
Tools needed:

- RTV

Procedure

1. Remove the plastic bushing in the Aluminum Alignment Ring. Try to install the bolt with the bushing removed. If the bolt can be threaded into the magnet, use Electrical tape to isolate the bolt from the Alignment Ring.
2. One bolt can be left out per flange, provided that the bolt does not support the TRM coil. Downside to this method will be a vacuum leak. The open bolthole must be filled with RTV to avoid a vacuum leak.
3. The Aluminum Alignment Ring may need to be drilled out. Determine new diameter of hole and take alignment ring to a machine shop to be drilled out. Use a non-conductive washer to isolate the Alignment Ring from the bolt.

C4 Skewed Magnet Flange



Tools Needed:

- RTV
- Level or Straight Edge
- Square

Procedure:

1. Using a straight edge or square, determine amount and location of RTV needed on the magnet to produce a flat surface.
2. Apply RTV to the magnet flange to create a parallel surface for mounting of the Alignment Ring.

SECTION 8 – OPERATOR WORKSPACE

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8-1 INTRODUCTION

The equipment de-installed for the EXCITE HD upgrade depends on the existing site configuration. The general guideline for removal of equipment is:

- De-install indicated modules on Operator Workspace table and Computer cabinet.
- Determine if DAT and/or MOD drives are to be transferred to new GOC Operator Workspace. Refer to Section 8-3.

Refer to applicable de-installation instructions in next section for removal of obsolete equipment.

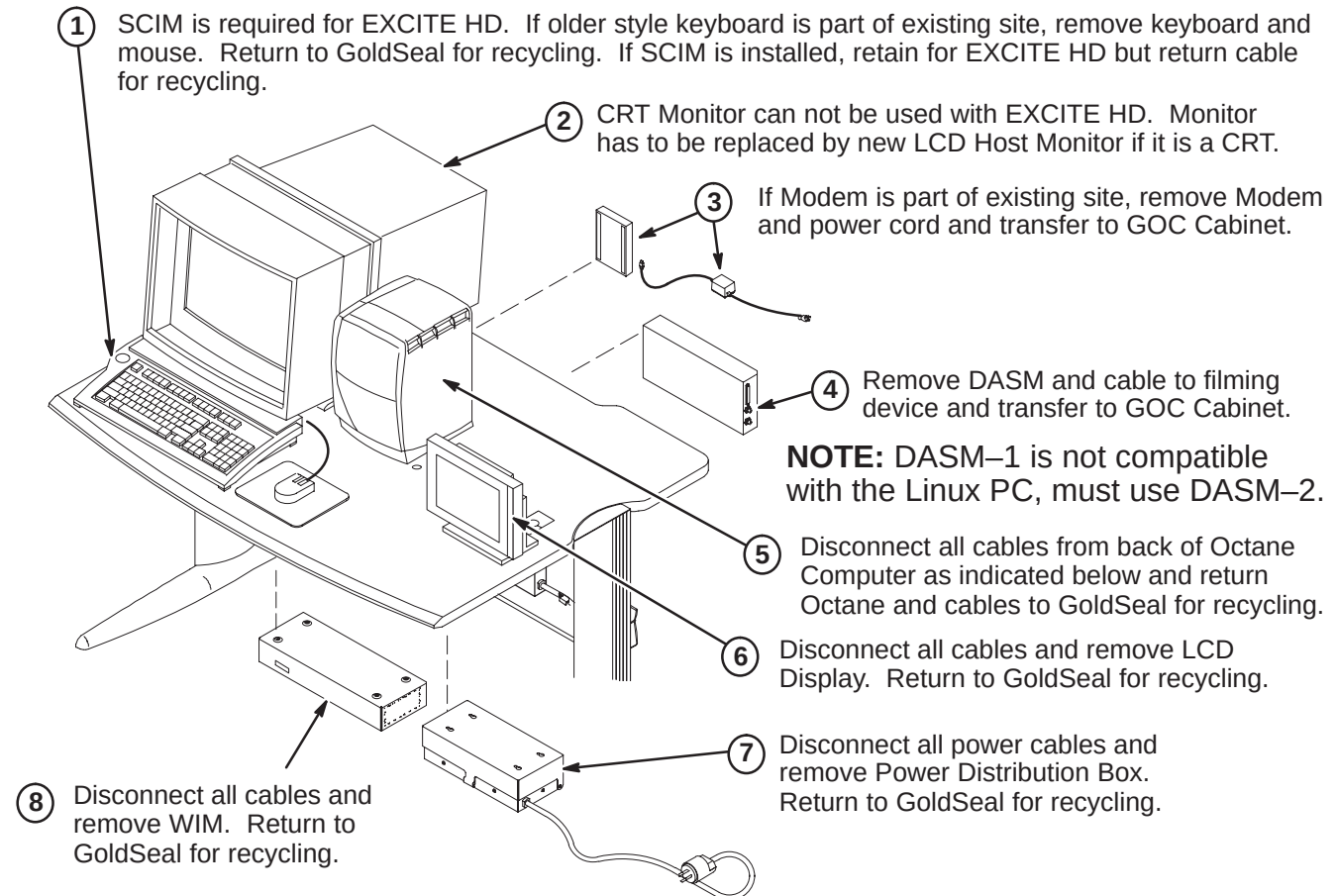


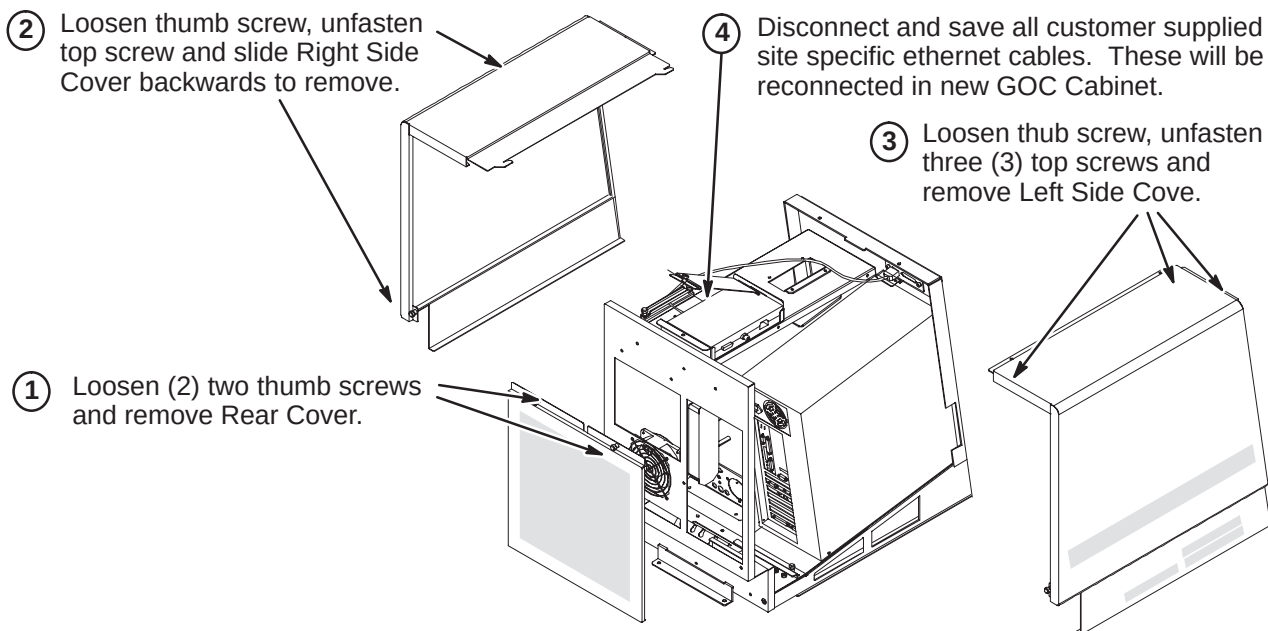
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

8-2 DE-INSTALLATION OF OPERATOR WORKSPACE

Confirm that all cables are disconnected from OperatorWorkspace and Computer. Perform the following procedure for return of the OW modules and Computer. Table will remain as part of upgrade.

8-2-1 DE-INSTALL OCTANE OW TABLE MODULES



8-2-2 DE-INSTALL OCTANE DUPLEX COMPUTER CABINET

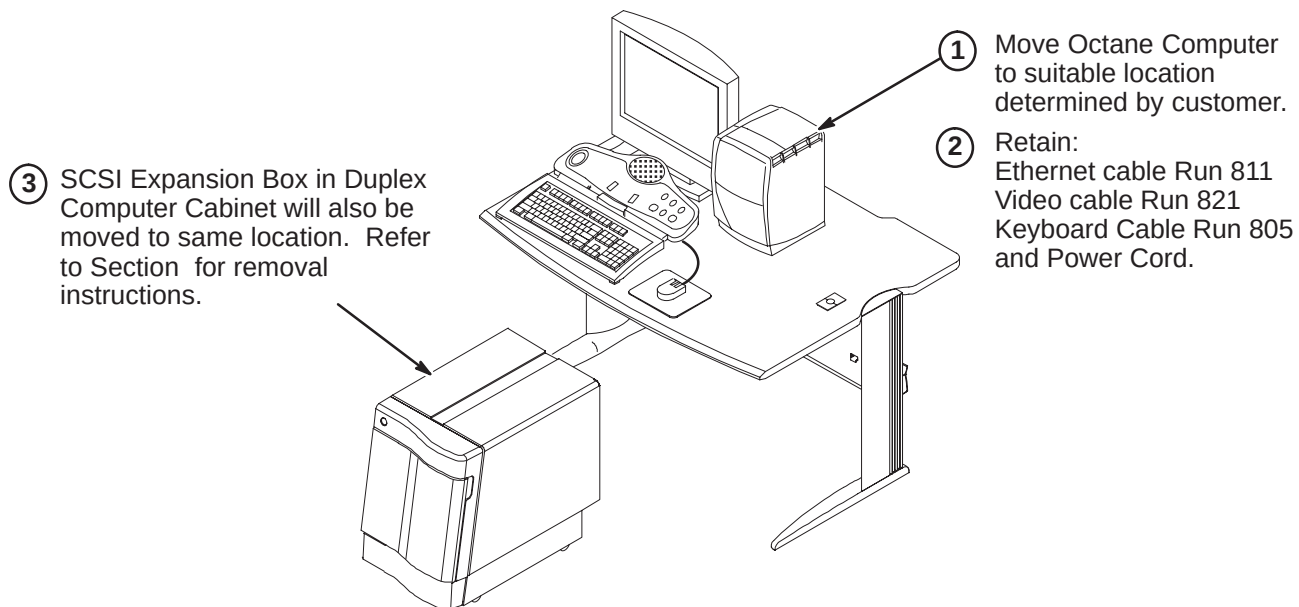
- ⑤ Refer to Section **8-3-3** for instructions on transferring Legacy Archive modules to EXCITE HD.
- ⑥ The remaining Duplex cables will not be used on the new GOC Cabinet and must be returned for recycling to Gold Seal.
- ⑦ The Duplex must be returned for recycling to Gold Seal. Duplex can be inserted onto skid that was used to ship the GOC by removing the white foam on the bottom that supported the GOC.

8-3 LEGACY ARCHIVE MEDIA SUPPORT

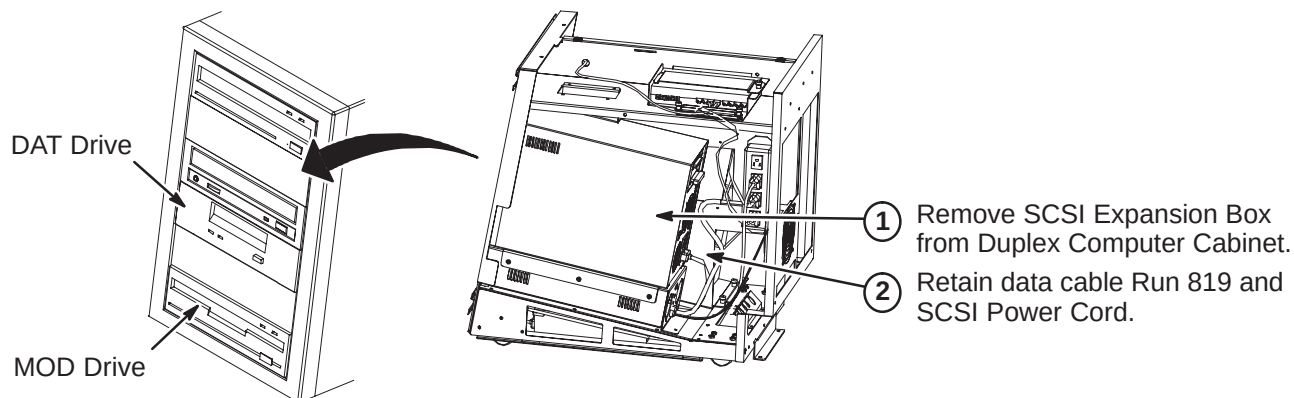
EXCITE HD does NOT support Legacy Archive Media, such as DAT and Pioneer MOD Archive Devices. Complete the applicable procedure for transfer of Legacy Archive Media for use with EXCITE HD.

LX sites with Octane computers will retain their Octane computer and legacy Media Devices. This upgrade would require a Monitor (M1000NT) and Keyboard (M1000MN).

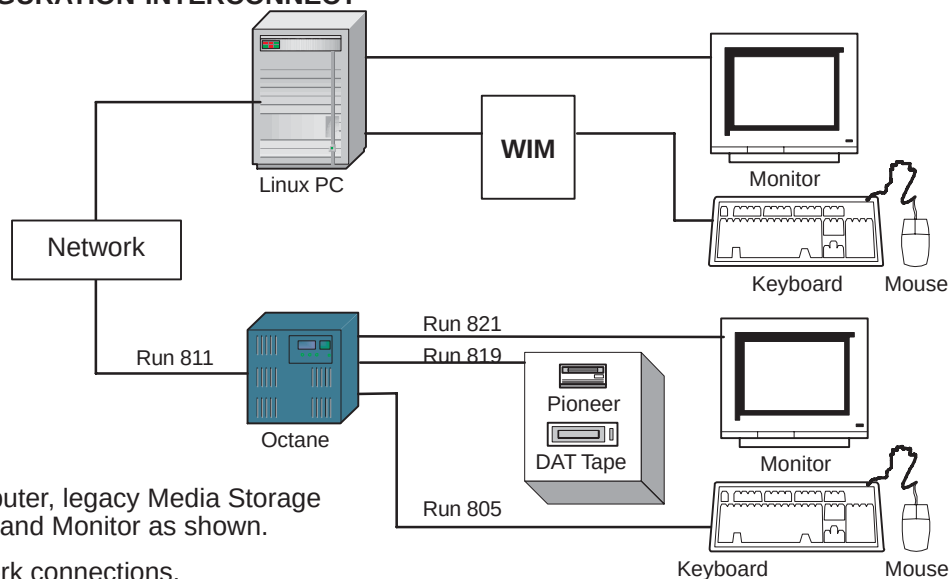
8-3-1 DE-INSTALL OCTANE COMPUTER



8-3-2 DE-INSTALL SCSI EXPANSION BOX AND CABLES



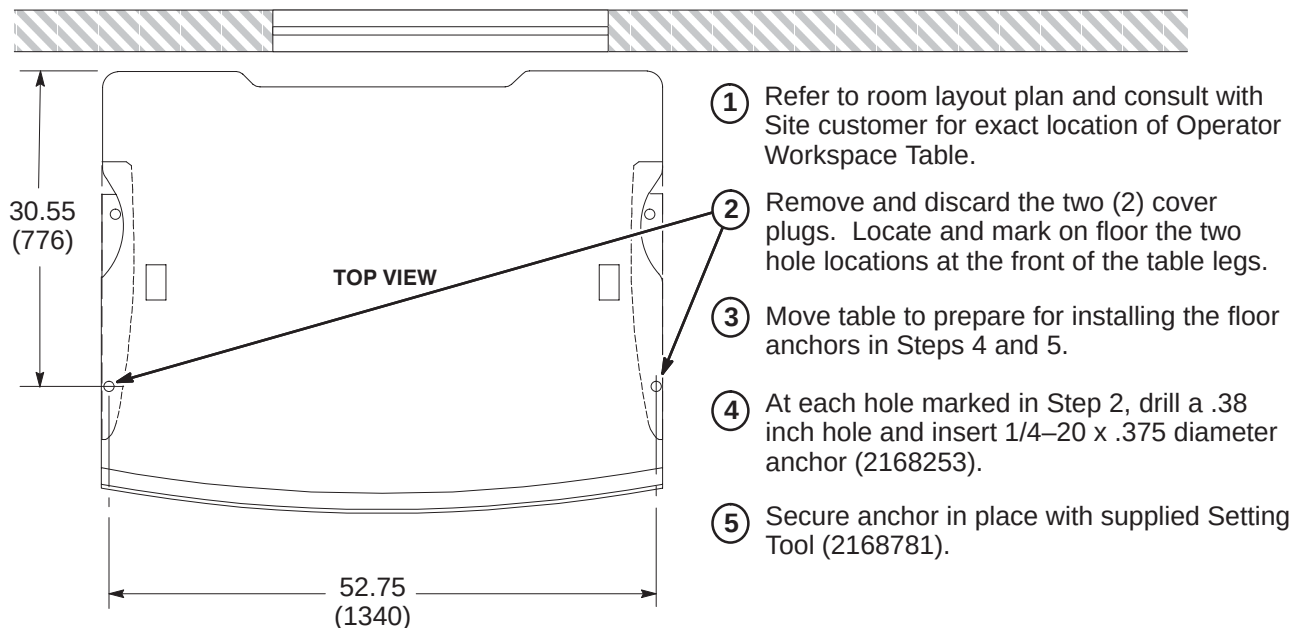
8-3-3 HARDWARE CONFIGURATION INTERCONNECT



- ① Connect the Octane Computer, legacy Media Storage Device, Keyboard, Mouse and Monitor as shown.
- ② Connect power and Network connections.
- ③ After Octane computer is powered up and switched ON:
 1. Login as 'root'
 2. `cd /export/home`
 3. `touch .SHOW_MODE`
 4. Logout
 5. Login as 'signa'
- ④ The 'Octane' system will now boot up in **SHOW_MODE** which will enable to further setup the DICOM Network to the different hosts to transfer images.

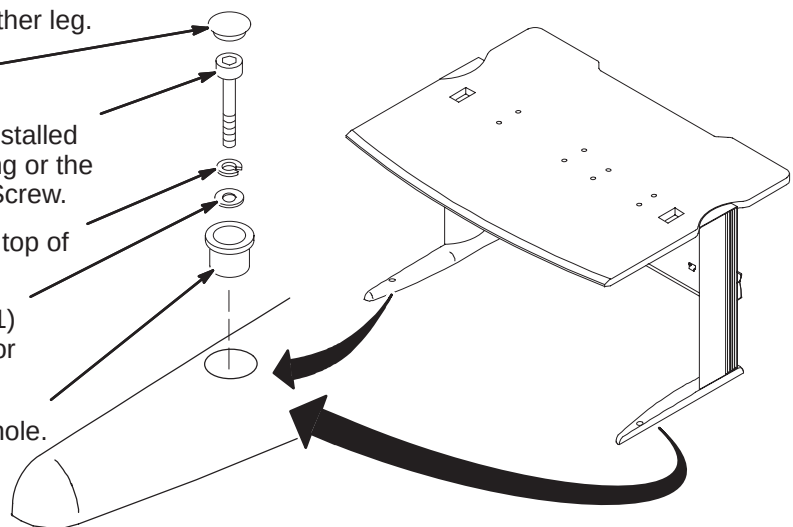
8-4 OPERATOR WORKSPACE UPGRADE FOR EXCITE HD**8-4-1 OPERATOR WORKSPACE TABLE INSTALLATION (If Needed)**

A pre-LX site will not have an installed Operator Workspace Table. Refer to room layout plan and consult with Site customer for exact location of new Operator Workspace Table. If the site is an LX system, the table will be retained for this upgrade.

8-4-1-1 Operator Workspace Table Anchor Installation**8-4-1-2 Securing Operator Workspace Table to Floor**

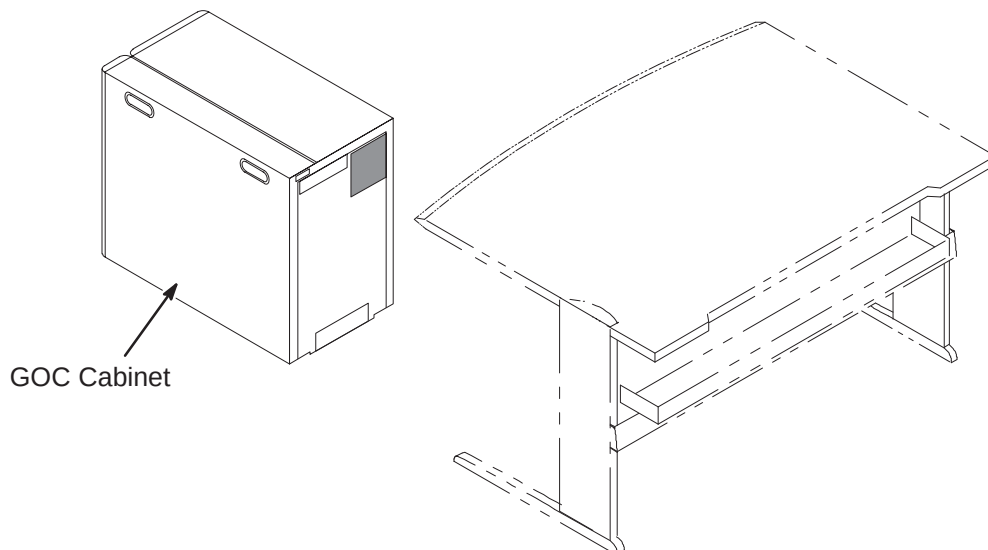
Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

- ⑥ Repeat anchoring process below for other leg.
- ⑤ Install black nylon Plug (2168254).
- ④ Depending upon depth of previously installed anchor, install either the 1-3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
- ③ Insert one Lock Washer (2168252) on top of Flat Washer(s).
- ② Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
- ① Insert Bushing (2151454) in table leg hole.

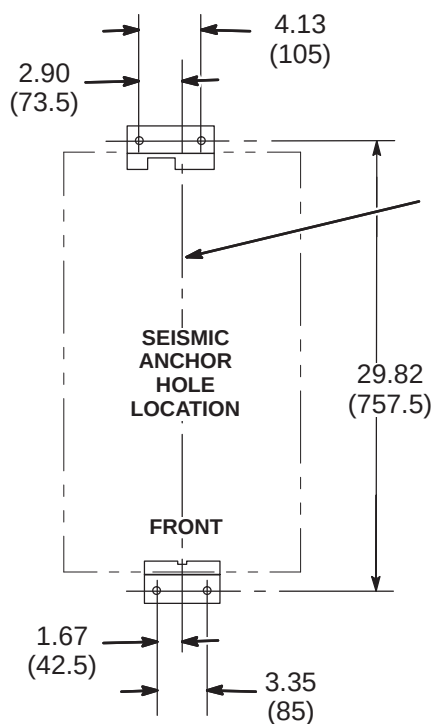


8-4-2 POSITION NEW GOC COMPUTER CABINET

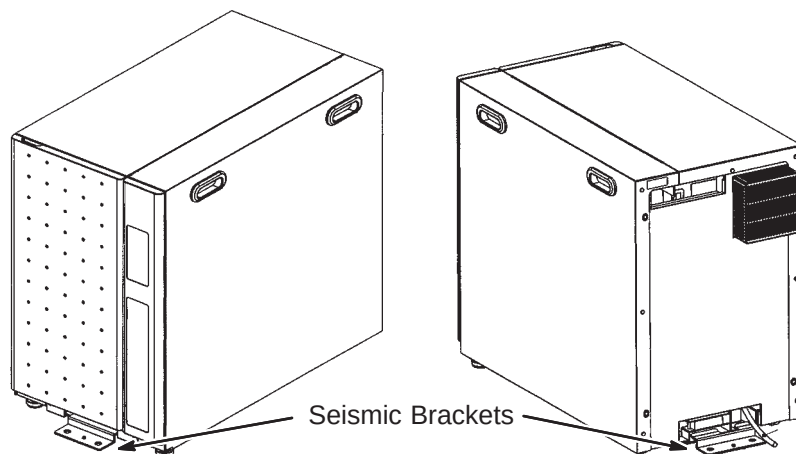
Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.

**8-4-3 SEISMIC ANCHORING (IF REQUIRED)**

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets.

**NOTE:**

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.

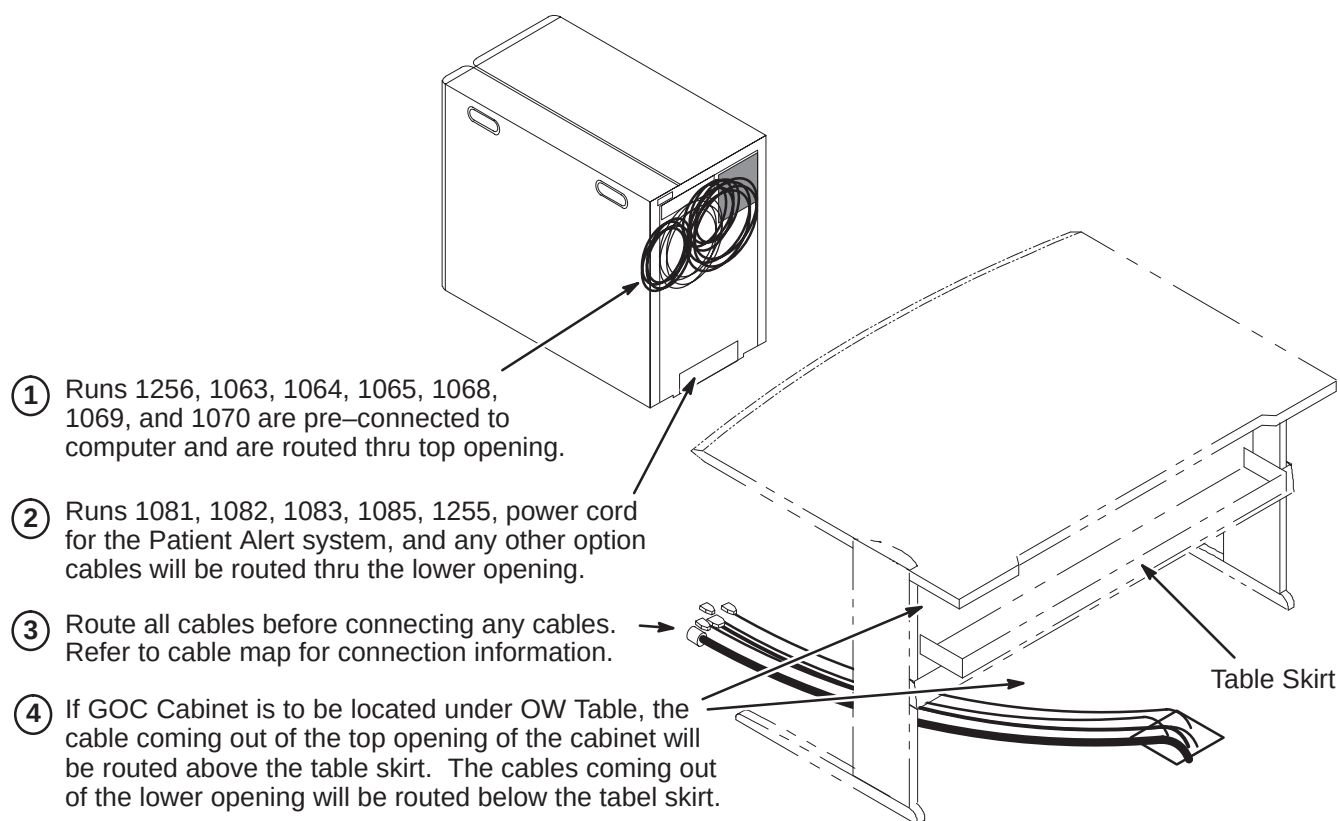


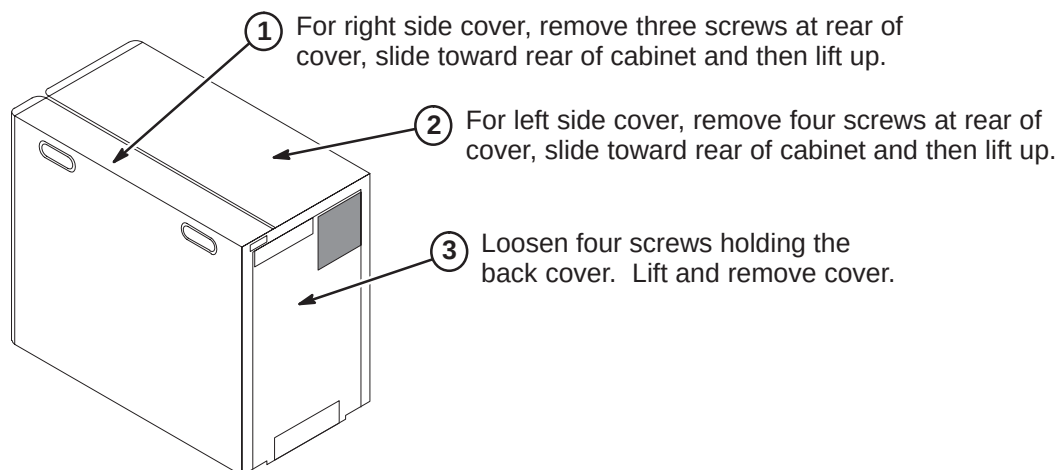
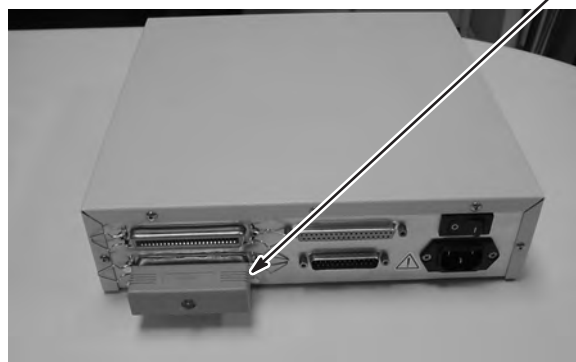
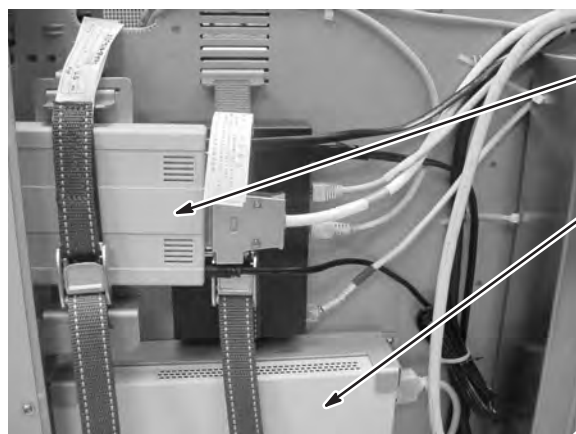
8-4-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

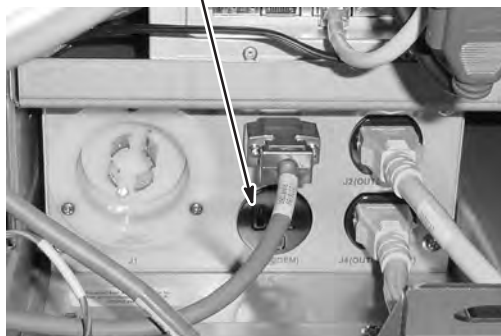
- Three power cords
- Two video output cables
- Keyboard output cable
- SCSI output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.



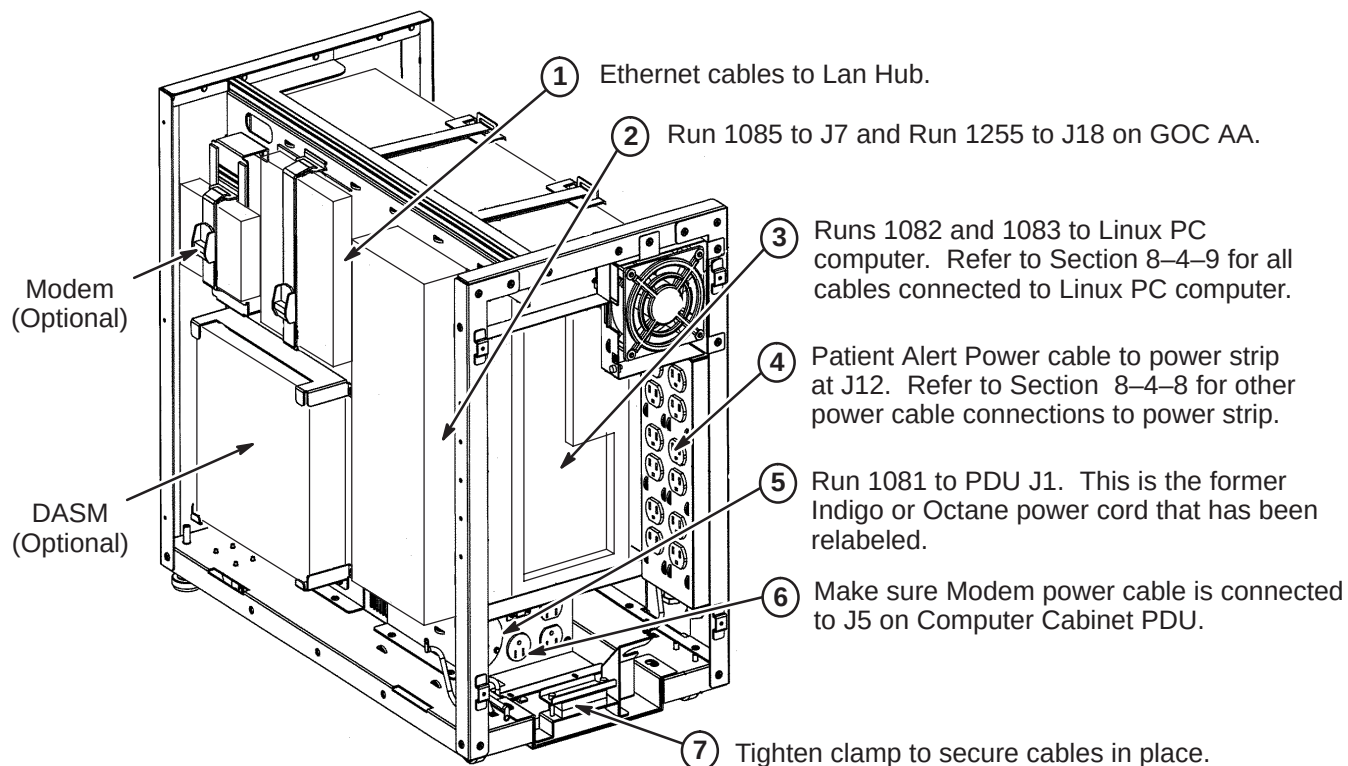
8-4-5 GOC CABINET COVER REMOVAL**8-4-6 INSTALL MODEM AND DASM IN GOC CABINET (If Applicable)**

- ⑤ Verify that the **Active SCSI Terminator** located on the bottom of the DASM is in the **OFF** position.
- ⑥ Modem power cord must be connected to J5 on GOC PDU Module

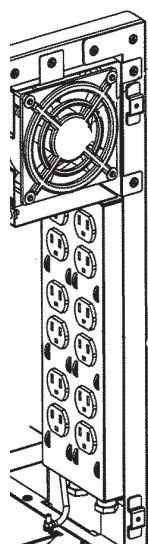


8-4-7 ROUTE CABLES FOR LOWER OPENING

Refer to cable map for information on cables to be connected thru lower opening.

**8-4-8 POWER STRIP CONNECTIONS**

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



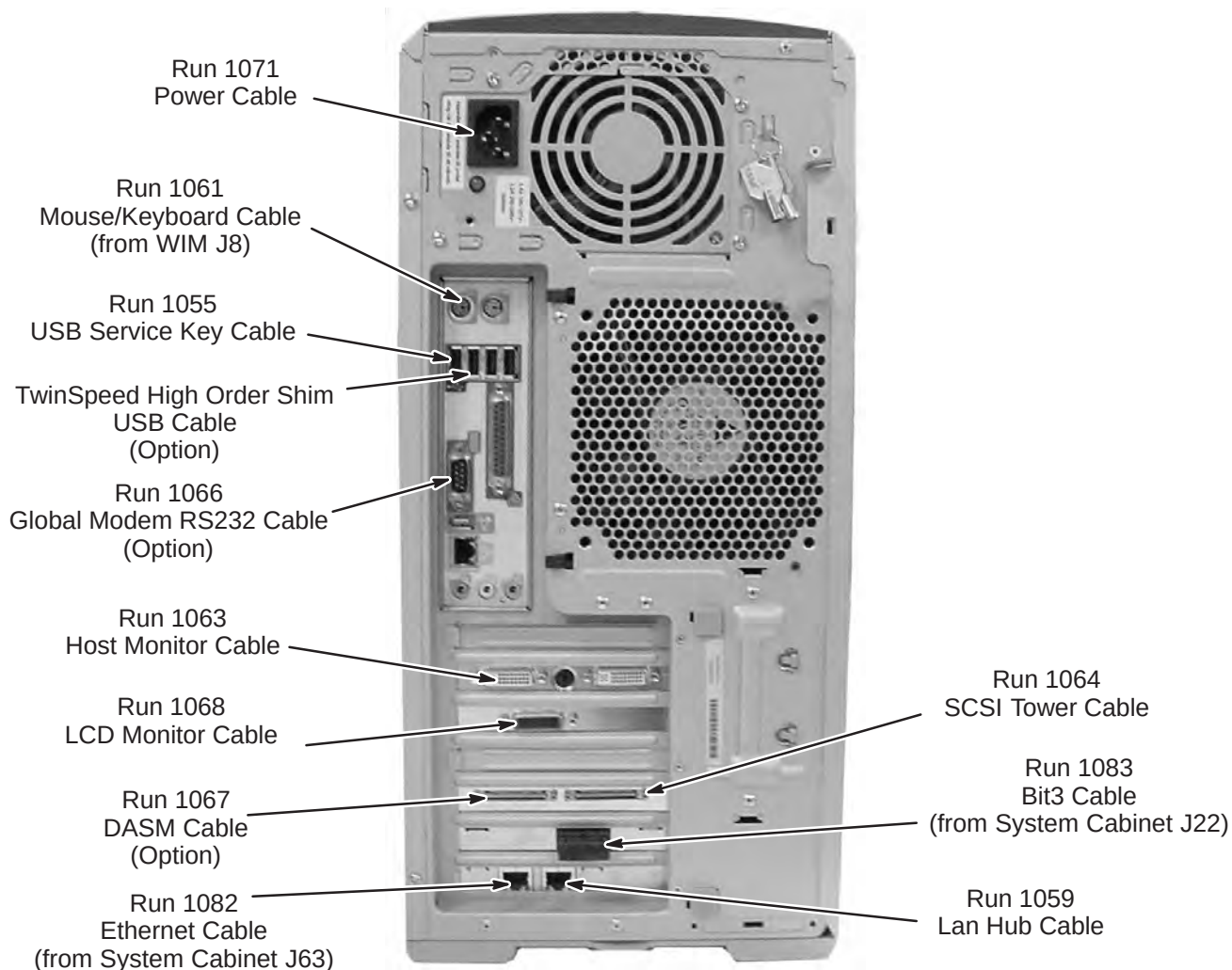
J1		J7
J2		J8
J3		J9
J4		J10
J5		J11
J6		J12

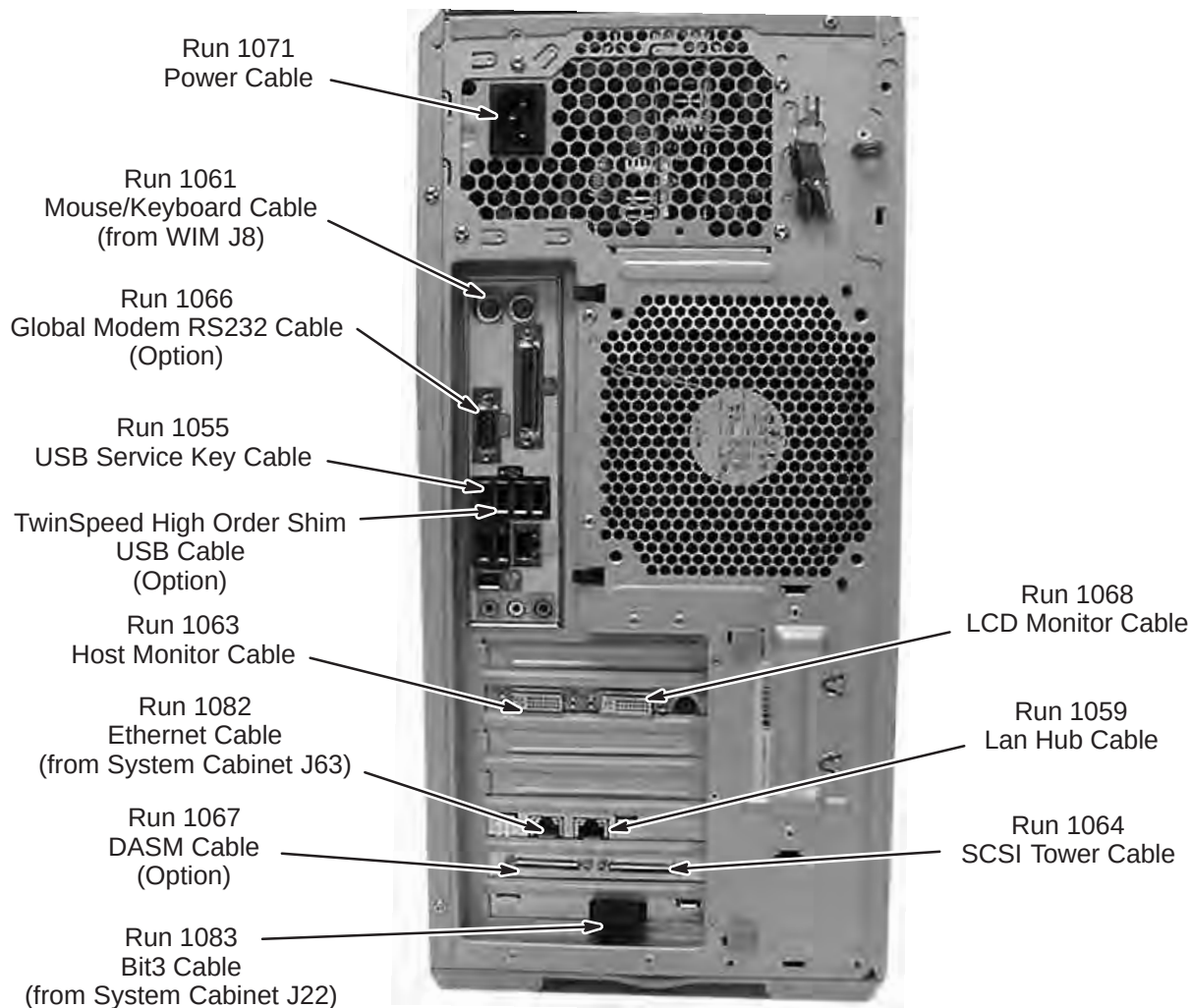
POSITION	NAME	MAX AMP	
		CIRCUIT 1	CIRCUIT 2
J1	Host Comp	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	LAN Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

8-4-9 LINUX PC CABLE CONNECTIONS

The connection points of cables routed to the HP 8000 and the HP 8200 PC's are described in the following two sections. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site. Refer to the applicable PC for your site.

8-4-9-1 HP 8000 Connection Points



8-4-9-2 HP 8200 Connection Points

8-4-10 INSTALL SCIM/KEYBOARD

The Scan-Control Intercom Module (SCIM) is the current keyboard on all Signa LX and OpenSpeed systems. Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

1. Remove the SCIM, Data Cable, and Keyboard From the box.



SCIM Module

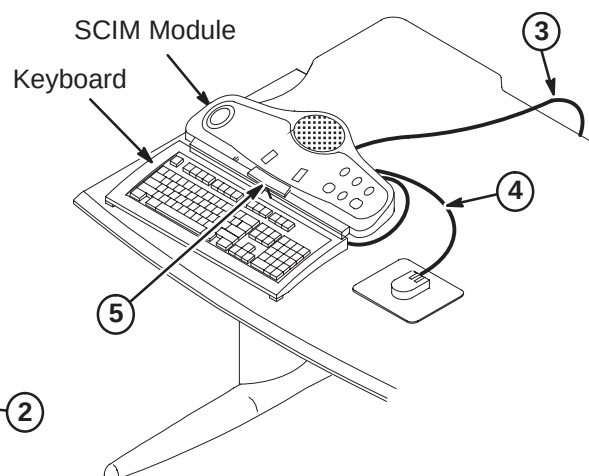
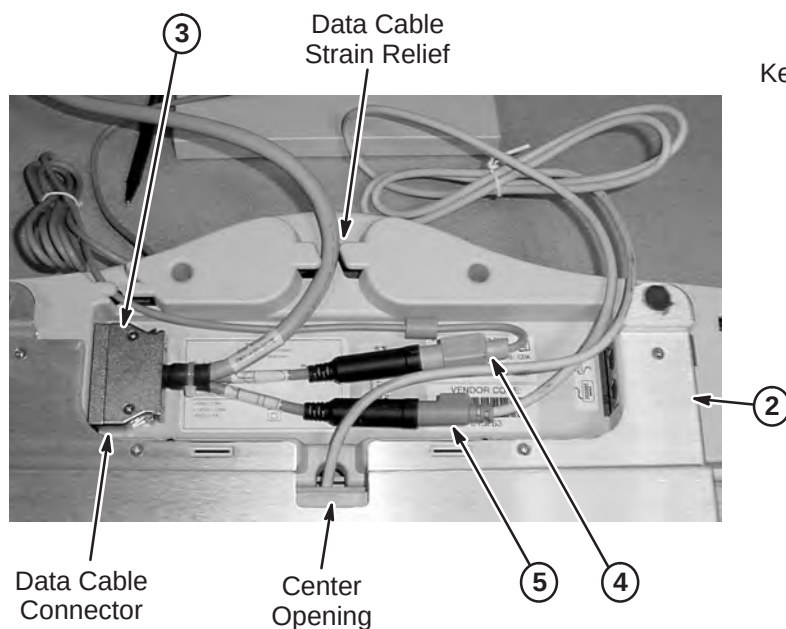


Data Cable



Keyboard

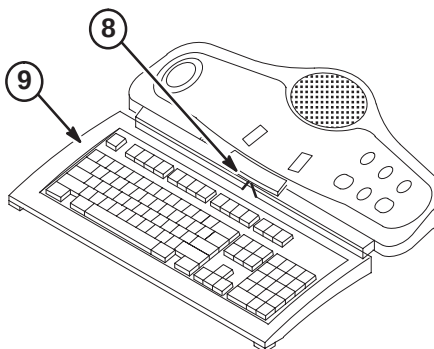
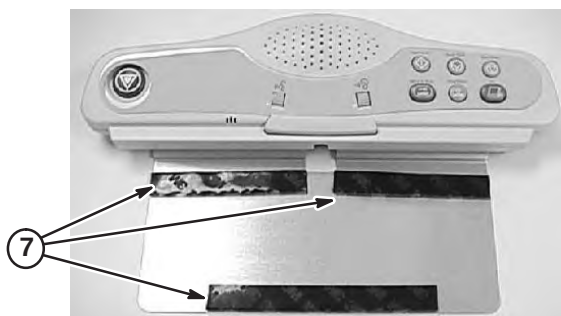
2. Turn the SCIM unit up side down exposing the cable connections.
3. Route data cable Run 1262, preconnected to the GOC Cabinet and is located on back side of cabinet, from Computer Cabinet to bottom side of SCIM Module. Route thru data cable strain relief and connect as marked. (Run 805, supplied with SCIM Kit (2307175) is not used for EXCITE)
4. Attach Run 1262 cable marked with mouse icon to the Mouse. Route the cable thru the strain relief notches at the rear of the SCIM.
5. Guide the keyboard cable thru the center opening and attach to the jack on Run 1262 with the keyboard icon. (Store excess cable in the base of the SCM by tucking it in the depressed area).



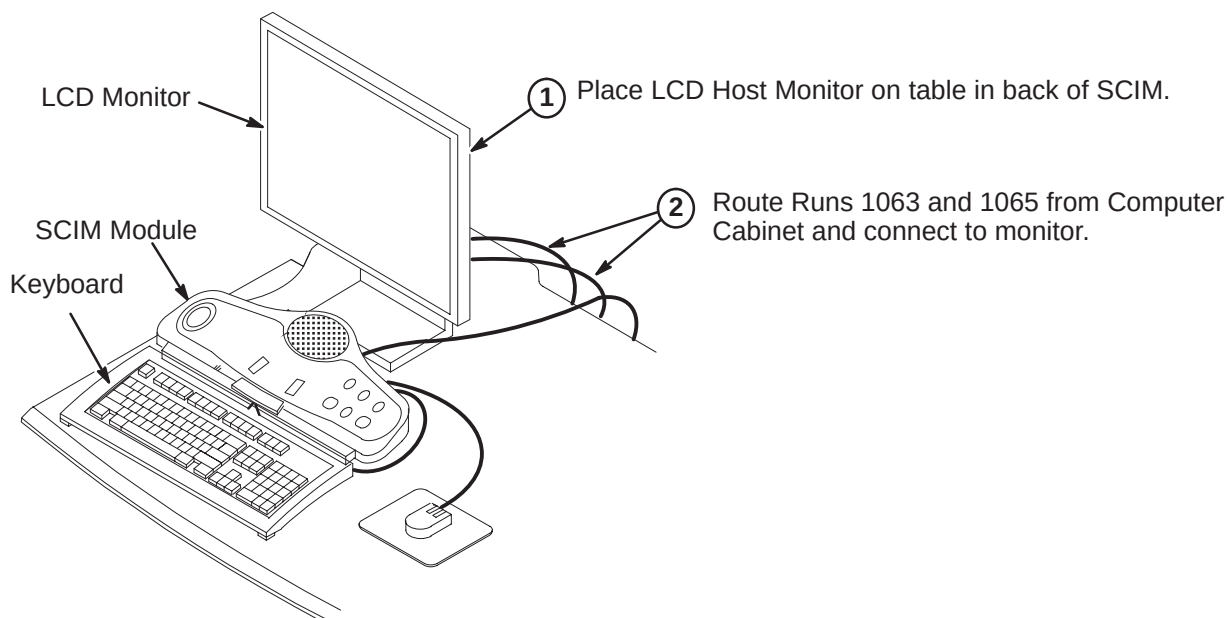
6. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
7. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).

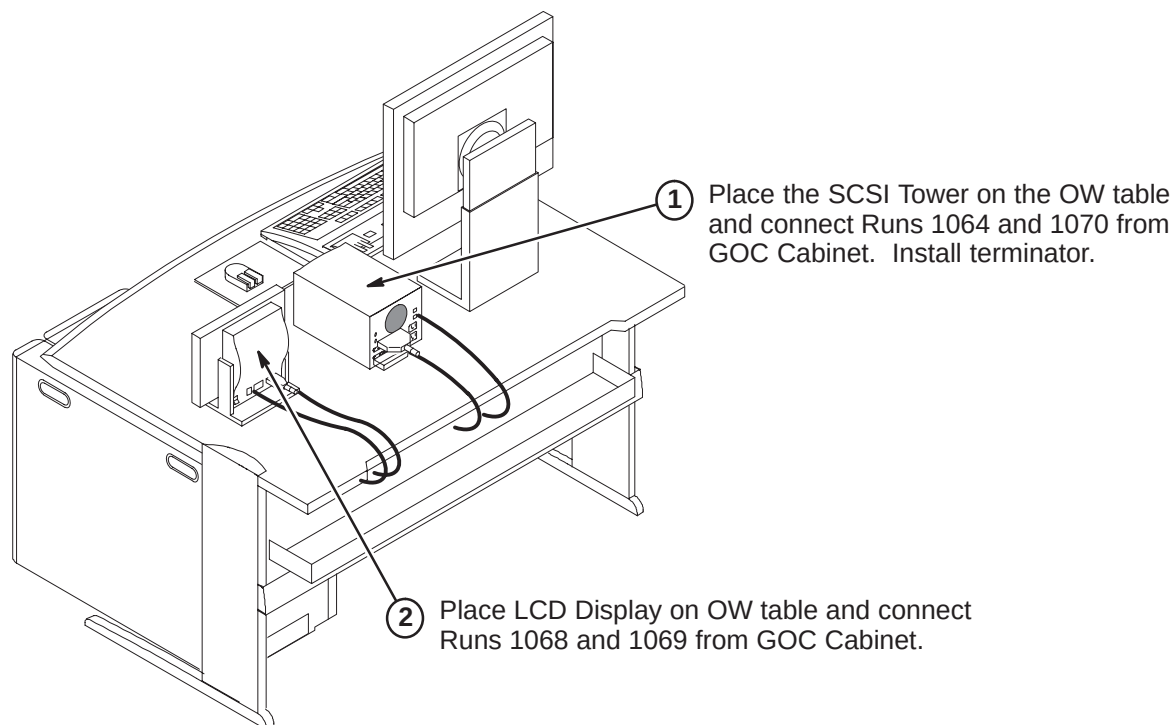
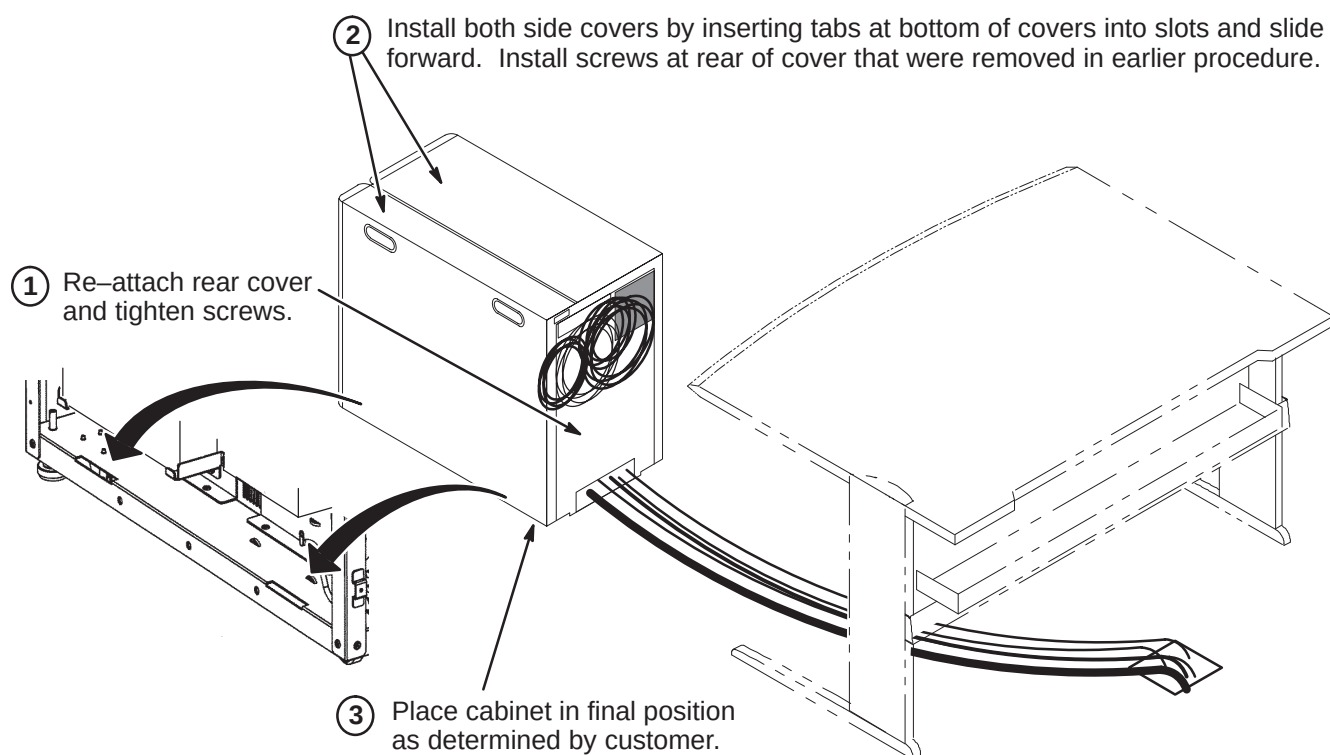
8-4-10 INSTALL SCIM/KEYBOARD (Continued)

8. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
9. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

**Note**

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

8-4-11 INSTALL MONITOR

8-4-12 INSTALL SCSI TOWER AND LCD DISPLAY**8-4-13 FINAL POSITIONING OF GOC CABINET**

8-5 RETURN OF REPLACED PARTS

Refer to Section 1-7 for instructions on returning discarded parts.

1. Remove the white Styrofoam from the bottom of the GOC skid. The Duplex will now fit onto the bottom of the GOC skid. Place the removed Duplex onto the GOC Skid. Properly secure the top cardboard box onto the skid using straps. Return to GoldSeal.
2. Return the Octane system to GoldSeal.
3. Return all cables that have been replaced by this upgrade to GoldSeal.

8-6 POWER ON INFORMATION**IMPORTANT**

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the WIM and Modem power.

SECTION 9 – HFD/PDU CABINET

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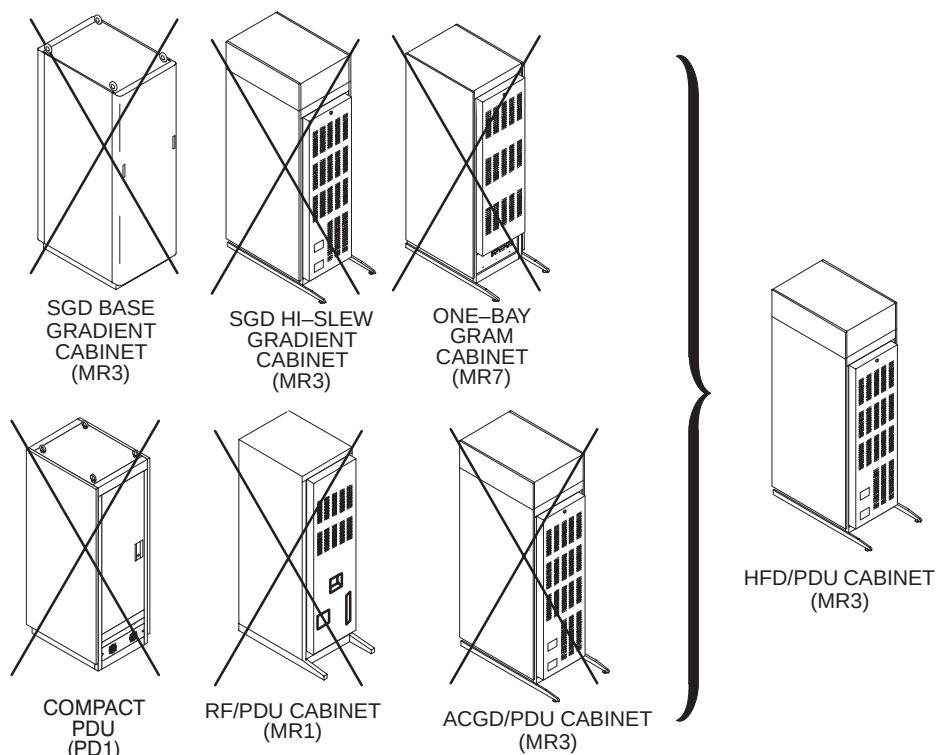
<u>SECTION</u>	<u>PAGE</u>
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9-2 POSITION HFD/PDU CABINET	9-3
9-3 INSTALL FAN ASSEMBLY AND ATTACH CABINET IDENTIFICATION MAGNET	9-4
9-4 INPUT VOLTAGE SELECTION	9-5
9-5 CIRCUIT BREAKER DIP SWITCH SETTINGS	9-5
9-6 CABLE CONNECTIONS TO HFD/PDU CABINET	9-6

9-1 DE-INSTALL CABINET



LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

The HFD/PDU Cabinet will replace one or more of the cabinets shown below.



All de-installed cabinets are to be returned for recycling. Refer to applicable cable removal map and applicable column in cable removal Table 4-1 for disconnection and removal of designated cables for system being de-installed.

9-2 POSITION HFD/PDU CABINET

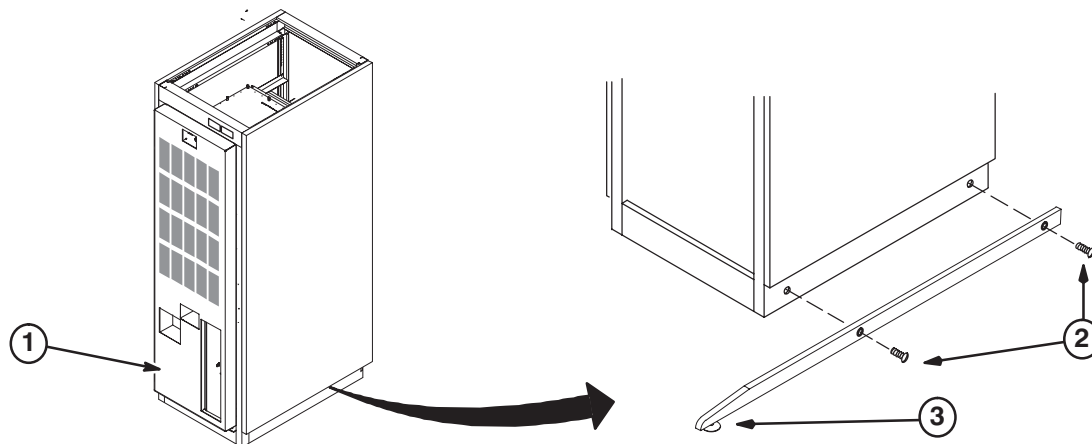
Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

WARNING!

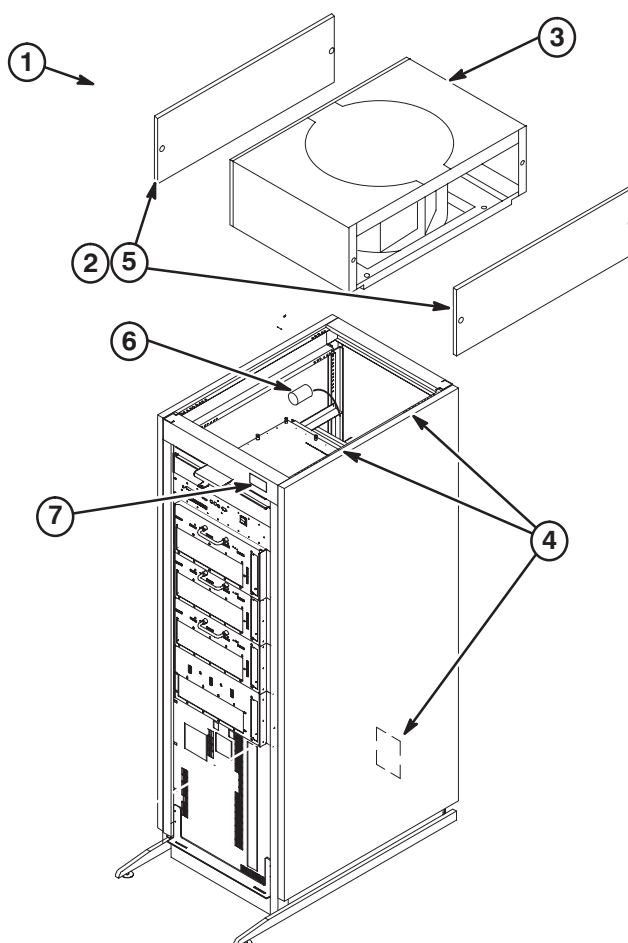
DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

1. Move cabinet to final position as indicated by site layout drawings.
2. Attach the anti-tip bars to cabinet. (Anti-tip bars are shipped separately)
3. Adjust leveling pads to level cabinet.



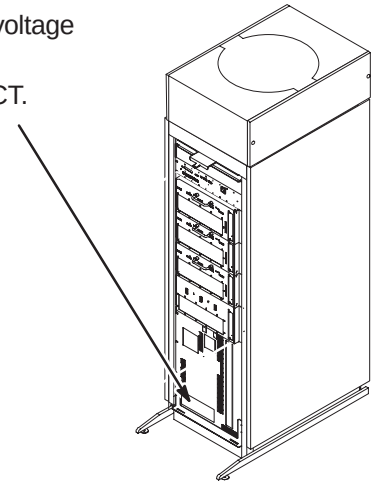
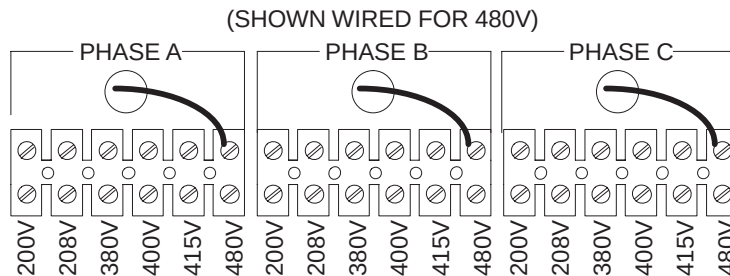
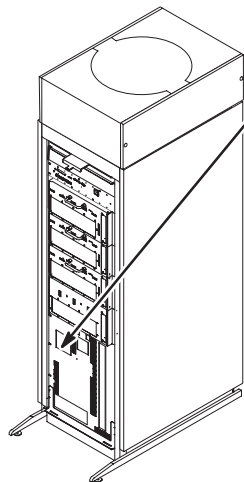
9-3 INSTALL FAN ASSEMBLY AND ATTACH CABINET IDENTIFICATION LABEL

1. Remove Fan Assembly from shipping container.
2. Remove two side trim panels (two screws from each).
3. Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
4. Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet. The fasteners are provided in a plastic bag located in PDU module. Rear PDU covers will need to be removed for access to plastic bags with fasteners.
5. Install two side trim panels with previously removed screws.
6. Connect cable from SGA Power Supply to the fan connector.
7. Make sure the "GRADIENT/PDU" cabinet magnet, part of the Cabinet ID Label Sheet (5126718), is attached to the front of cabinet as shown.



9-4 INPUT VOLTAGE SELECTION

- ① Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
- ② Remove the plate from the front of the PDU marked INPUT VOLTAGE SELECT.
- ③ For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.

**9-5 CIRCUIT BREAKER DIP SWITCH SETTINGS**

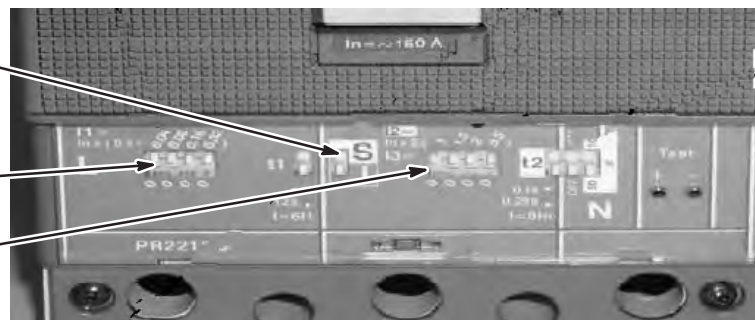
- ① Remove the plate from the front of the PDU marked CURRENT ADJUST.
- ② Table below shows the position of the DIP switches for each input voltage selection. Set DIP switches according to input voltage. The white portion indicates the switch.

INPUT VOLTAGE	200	208	380	400	415	480
"L" Switch Setting						
"I" Switch Setting						

- ③ The "S/I" switch must be in the "I", down position

DIP SWITCH
"L" SETTING

DIP SWITCH
"I" SETTING

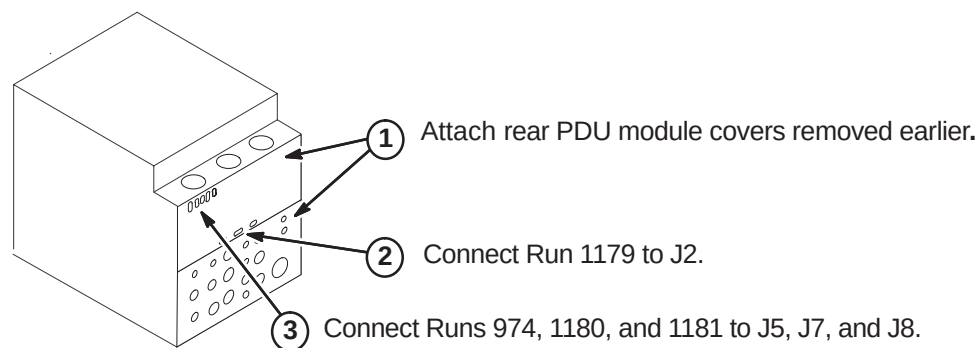


9-6 CABLE CONNECTIONS TO HFD/PDU CABINET

Refer to the following for additional information on connecting cables to cabinet:

- Section 4-6 Cable “added” map for information on connecting new and retained cables to cabinet.
- Section 4-8 Power Cable Installation in Equipment Room
- Section 4-9-1 . . Gradient Cable Installation in Equipment Room
- Section 4-10-1 . . Fiber Optic Installation in Equipment Room

Connect remaining cables as indicated below.



SECTION 10 – RFS CABINET

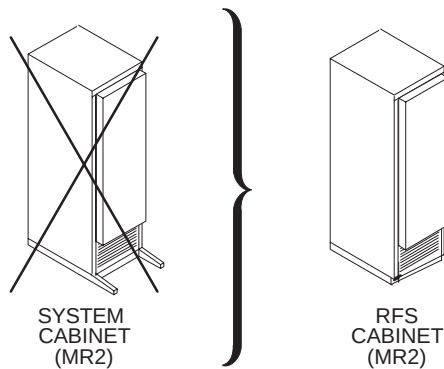
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10-2 POSITION RFS CABINET	10-2
10-3 CABLE CONNECTIONS TO RFS CABINET	10-3

10-1 DE-INSTALL CABINET

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

The RFS Cabinet will replace the System cabinet shown below.



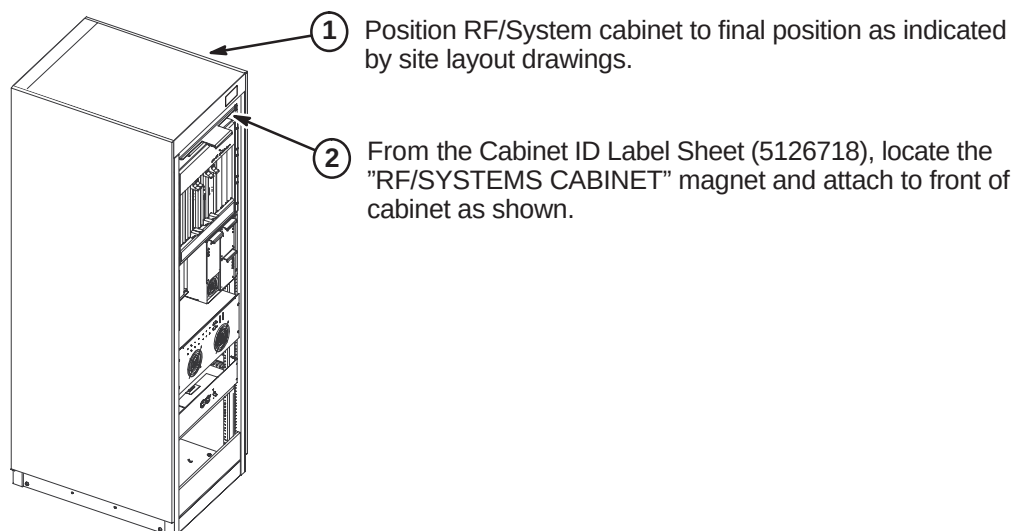
The de-installed System cabinet is to be returned for recycling. Refer to cable removal map in Section 4 for disconnection and removal of designated cables.

10-2 POSITION RFS CABINET

Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.



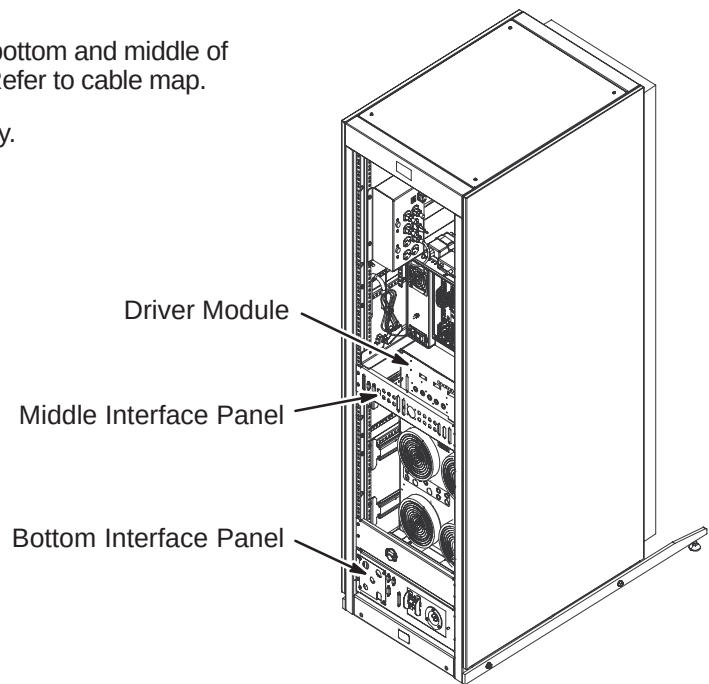
10-3 CABLE CONNECTIONS TO RFS CABINET

Refer to the following for additional information on connecting cables to cabinet:

- Section 4-6 . . . Cable “added” map for information on connecting new and retained cables to cabinet.
- Section 4-8-2 . Power Cable Installation in Equipment Room
- Section 4-10-1 Fiber Optic Installation in Equipment Room
- Section 4-11 . . . Times Microwave RF Cable Installation in Equipment Room

Connect remaining cables as indicated below.

- ① Connect routed cables to Interface Panels at bottom and middle of cabinet and to the Driver Module as marked. Refer to cable map.
- ② Dress and secure cables. Adjust as necessary.



SECTION 11 – TAC CABINET

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11-1 POSITION TAC CABINET	11-2
11-2 CABLE CONNECTIONS TO TAC CABINET	11-2



LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

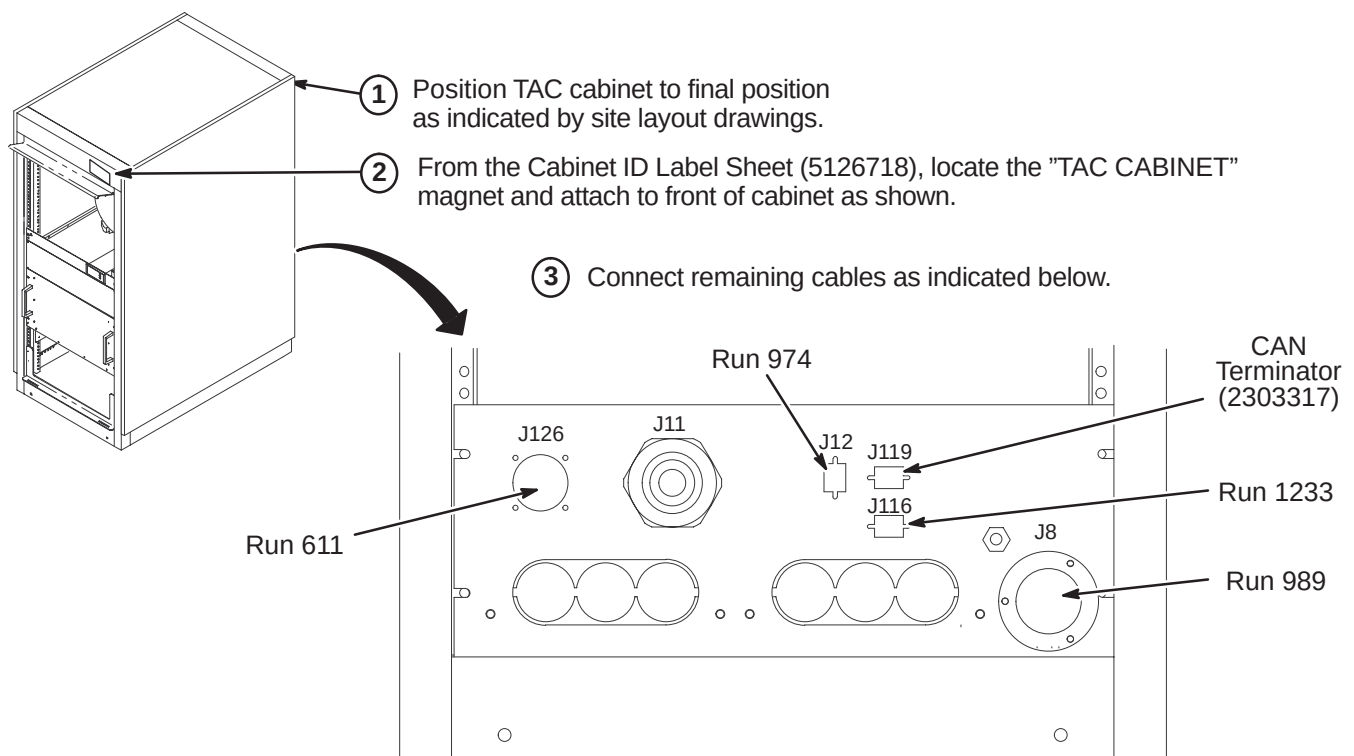
11-1 POSITION TAC CABINET

Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for:

- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

11-2 CABLE CONNECTIONS TO TAC CABINET



Refer to the following for additional information on connecting cables to cabinet:

- Section 4-6 Cable "added" map for information on connecting new and retained cables to cabinet.
- Section 4-9-1 . Gradient Cable Installation in Equipment Room
- Section 6-6-17 Vacuum Hose Installation

SECTION 12 – COMPLETION

12-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Have the on-site GE Healthcare Field Engineer, Service Engineer, or Project Manager of Installation (PMI) to confirm that the blower box has been anchored properly and installed outside the minimum service area.
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 12-2, 12-3, and 12-4.

12-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 12-3. Recycling Operation will route them to appropriate location.

12-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

12-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 11.x has been made before returning system to the users. Turn site over to applications who will instruct users.

12-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.

SECTION 13 – APPENDIX

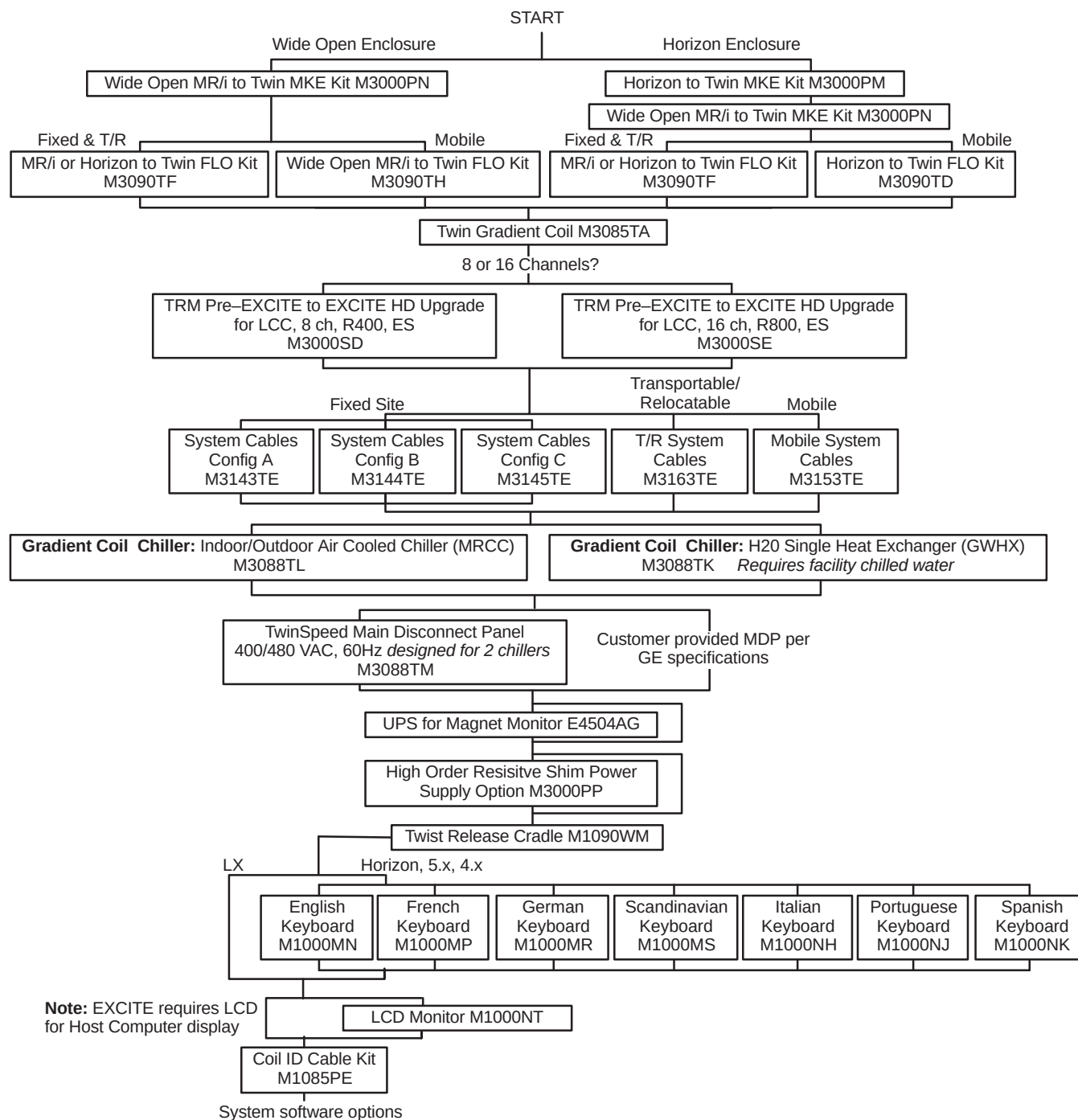
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13-3 FIBER OPTIC TERMINATION INSTRUCTIONS	13-16
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13-1 SYSTEM CONFIGURATION

Signa LX with 1.5T LCC Magnet upgrade to Signa EXCITE HD TwinSpeed catalogs are listing in Illustration 10-1. Illustrations 10-2 and 10-3 shows the equipment/cabinets modified, replaced, and added by the listed catalogs.

13-1-1 BASIC SYSTEM



Signa LX WITH LCC MAGNET TO SIGNA EXCITE HD TWINSPED UPGRADE CATALOGS

ILLUSTRATION 10-1

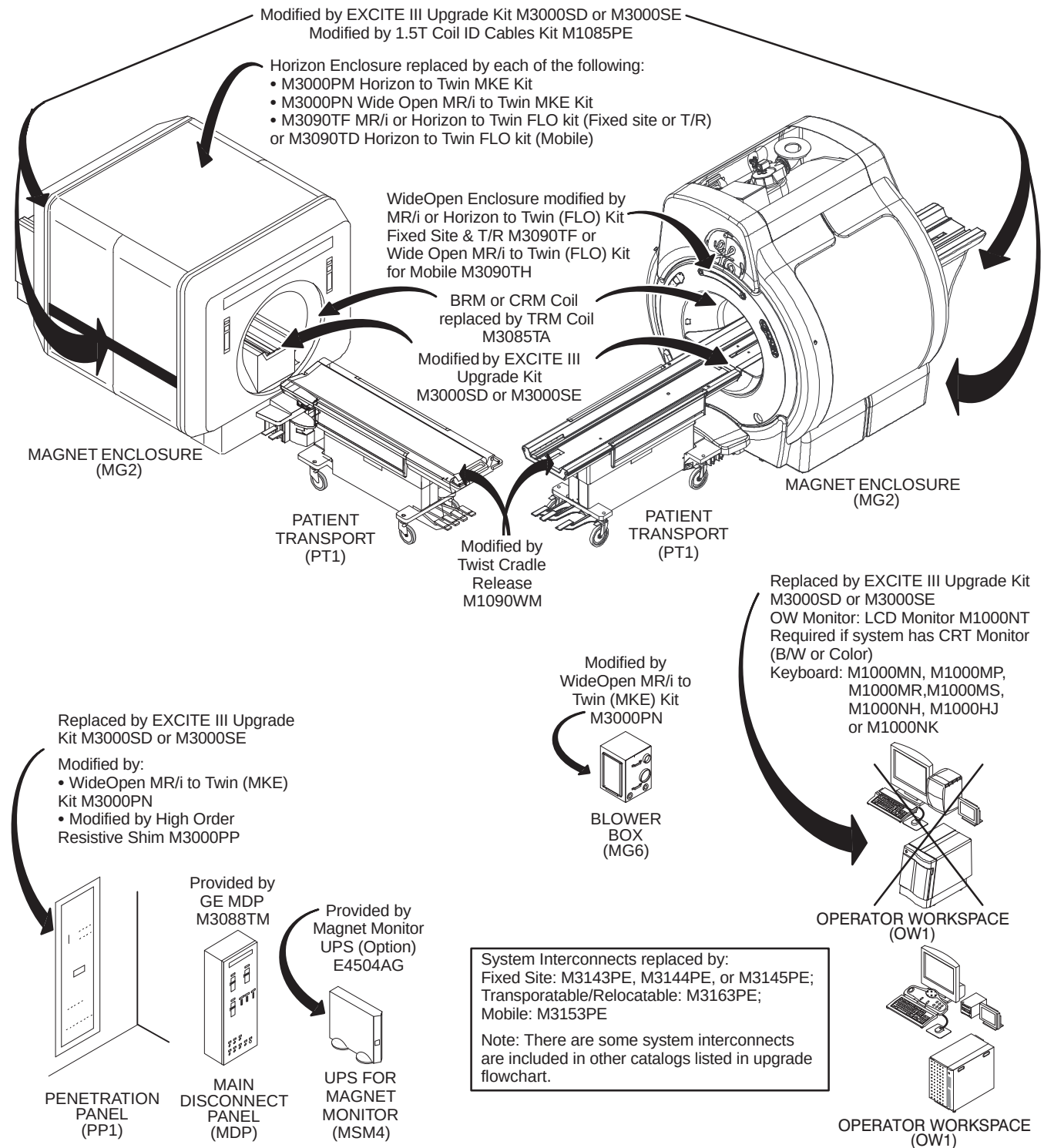
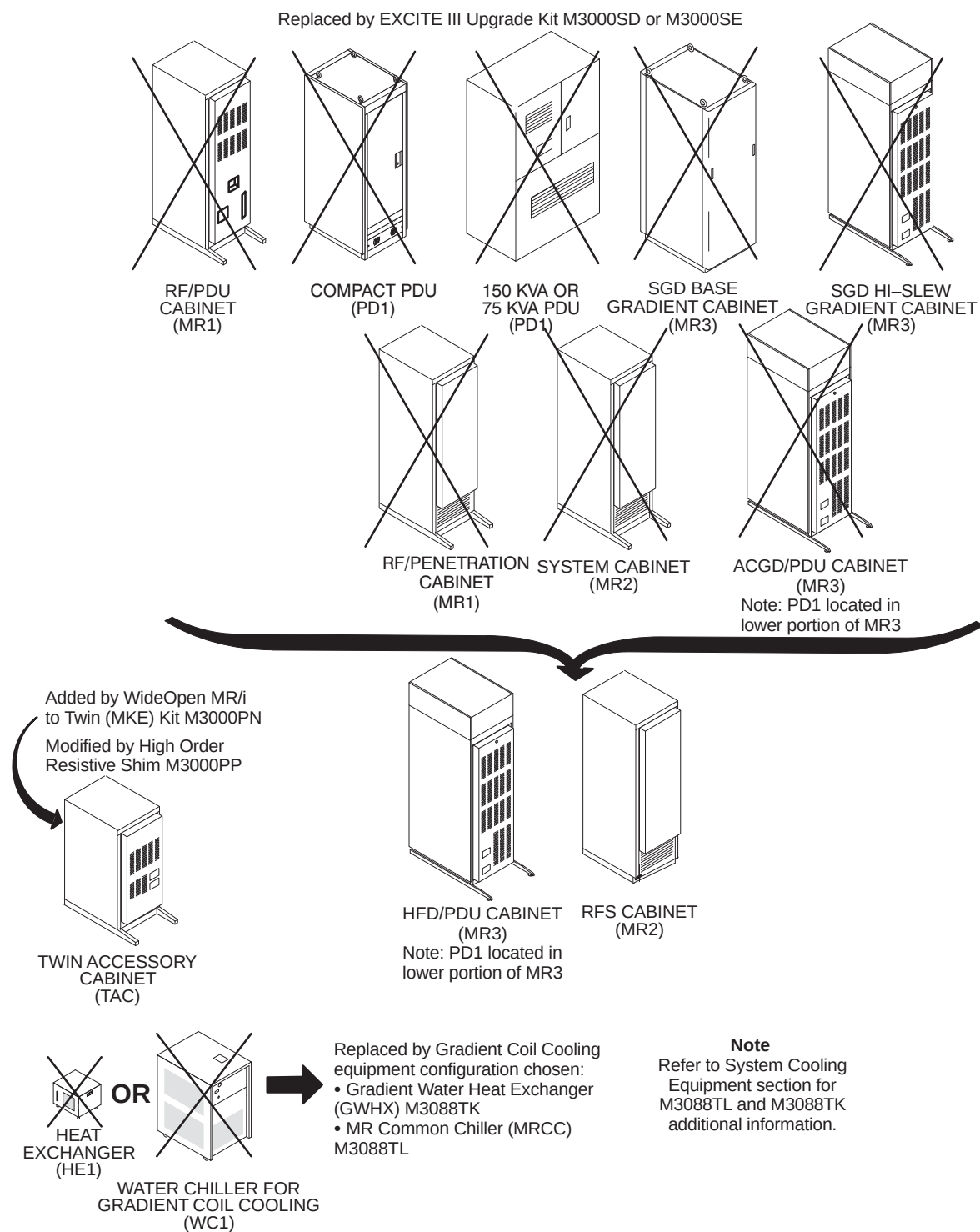
10-1-1 BASIC SYSTEM (Continued)
SIGNA MR/i & SIGNA LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 10-2

10-1-1 BASIC SYSTEM (Continued)



SIGNAL MR/i & SIGNAL LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 10-3

13-1-2 SYSTEM COOLING

The Signal EXCITE HD 1.5T TwinSpeed system requires water cooling for the Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. The following system cooling equipment is available with the MR system:

Note

See Illustration 1-1 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.

For water cooling specifications and requirements for Site Provided Water Cooling, refer to **Direction 5133302, Signal EXCITE HD 1.5T TwinSpeed Pre-Installation** manual.

- **MR Common Chiller (MRCC)**
The MRCC is a single-loop 10KW 50/60 Hz water chiller that can be used to cool either Shield/Cryo Cooler Compressor or the TwinSpeed Gradient Coil. The MRCC can be installed either indoors or outdoors. There is a Remote Panel to be installed in the MR system Equipment Room that can stop or restart the chiller and can display supply water temperature, chiller setpoints and alarms. The MRCC power is supplied by the MR system Main Disconnect Panel (MDP). This chiller will be serviced by a third party vendor.
- **Gradient Water Heat Exchanger (GWHX)**
The GWHX is a 10KW 50/60 Hz water-to-water heat exchanger that is used for cooling the TwinSpeed Gradient Coil. The GWHX is for indoor use only. The GWHX requires facility provided chilled water. The GWHX power is supplied by the MR system Power Distribution Unit (PDU). This heat exchanger will be serviced by GE Service. When the GWHX is used then the customer must supply water cooling for the Shield/Cryo Cooler Compressor.

Note

Gradient Coil water cooling must be supplied by cooling equipment (MRCC or GWHX) supplied with the MR system to prevent contamination/damage to the coil and for proper image quality.

There are several possible scenarios which a customer may utilize to meet the system water cooling requirements for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil.

- Two MRCC installed outdoors where one unit provides Shield/Cryo Cooler Compressor cooling and the other provides Gradient Coil cooling. The 2 MRCC Remote Control Panels to be installed in the MR system Equipment Room.
- Two MRCC installed indoors where one unit provides Shield/Cryo Cooler Compressor cooling and the other provides Gradient Coil cooling. The 2 MRCC Remote Control Panels to be installed in the MR system Equipment Room. Customer supplied air cooling is required for the MRCC units.
- One GWHX installed indoors provides water cooling to the Gradient Coil (GWHX requires facility water cooling) and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet.
- One MRCC installed outdoors provides water cooling to the Gradient Coil and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. The MRCC Remote Control Panel to be installed in the MR system Equipment Room.
- One MRCC installed indoors provides water cooling to the Gradient Coil and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. The MRCC Remote Control Panel to be installed in the MR system Equipment Room. Customer supplied air cooling is required for the MRCC unit.

13-1-2 SYSTEM COOLING (Continued)

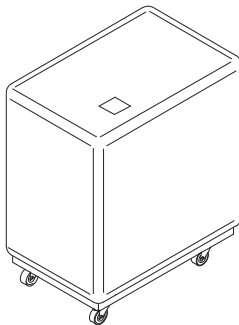
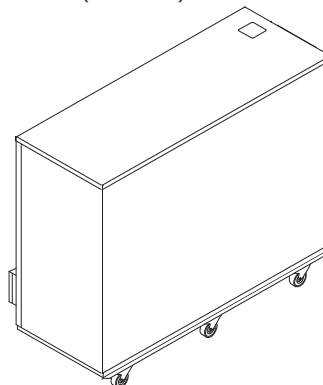
TABLE 18-7

TWIN SYSTEM COOLING CABINET (SEE TABLE NOTE NO TAG)

CATALOG	DESCRIPTION
M3088TK	Gradient Water Heat Exchanger (GWHX) See Note 1
M3088TL	MR Common Chiller (MRCC) See Note 2
Note 1 When GWHX is selected, customer supplied water cooling is required for the LCC Magnet Shield/Cryo Cooler Compressor. Refer to Direction 5133302, Signa EXCITE HD 1.5T TwinSpeed Pre-Installation for water cooling specifications. 2 Two M3088TL installed outdoors is standard cooling equipment configuration of Transportable/Relocatable Mobile systems.	

Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.

GWHX
(M3088TK)MRCC
(M3088TL)REMOTE CONTROL
PANEL
(RCP)

SYSTEM COOLING EQUIPMENT SELECTIONS

ILLUSTRATION 10-4

13-2 CABLE IDENTIFICATION

Cables delivered to site for installation have specifications dependent on its usage. The following tables contain information on cable runs for this system as it was originally installed.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
"FROM" & "TO"	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continuously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material may be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material's ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable's use in the MR system	

13-2-1 PDU POWER AND GROUND CABLES

The cables listed in Table 18-8 are connected at one end to the PDU module in the HFD/PDU Cabinet.

TABLE 18-8
PDU POWER AND GROUND CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW-1	System Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW-1	Magnet Ground Cable
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW-1	Blower Cabinet Separate ground
968	MR1 A12 J1	MR3 PD1	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	RFS Cabinet Power
970	WC1	MR3 PD1	82 ft Power (3-#10)	STO	600	208	90	FT-4	HFD Water Chiller Power
988	PP1 A7 GND	TAC GND	60 ft. Ground (#8)	SO	600	0	80	VW-1	TAC to PP1 ground (3)
989	MR3 PD1	TAC J8	25 ft. Power (3-#10)	SO	600	120	90	FT-4	TAC Power
1081	OW1 A19 J1	MR3 PD1	65.5 ft Power (4-#10)	SO	600	120	75	FT-4	Operator Workspace Power
1182	PP1 PHPS PWR	MR3 PD1	61 ft. Power (3-#10)	AWM	600	120	90	FT-4	PHPS Module Power
1234	MR2 RF AMP	MR3 PD1	35.5 ft Power (4-#8)	SO	600	208	90	FT-4	RF Amplifier Power
1254	MR3 PD1	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW-1	Magnet Ground Cable
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5-#8) With 30A Plug and Receptacle						Cable for PDU to GE Magnet Power Supply or SC Shim Supply Power

13-2-2 MAGNET ROOM CABLES NOT CUT TO LENGTH

The cables listed in Table 18-9 are routed and connected in the Magnet Room and are not cut to length.

TABLE 18-9
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Table Elevation motor power
414	MG3 A35	MG3 A2 J9	30 ft. 9-pin Sub-D	CI3	300	<30	80	FT-4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Long Drive Encoder/Limit Sw
458	PP1 J17	OM3	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional Remote O2 Sensor
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	PAC Power
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT-4	Patient Cooling Blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub-D/rt. angl	CI3	300	<30	80	FT-4	Operator Display to PAC II
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Operator Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Operator Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW-1	Splitter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub-D/26 pin	CI3	300	<30	80	FT-4	Operator Display to SRI

13-2-2 MAGNET ROOM CABLES NOT CUT TO LENGTH (Continued)

TABLE 18-9 (CONT'D)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1152	PP1 A5 J1	MG2 A12 A3 P1	29.5 ft. 37 pin CPC to 16 pin	PLTC	300		105	FT-4	Shim Filter Cable
1197	PP1 RFTXSW J150	MG3 16CHSW J25	49 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	
1200	PP1 J200	MG3 16CHSW J1	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1201	PP1 J201	MG3 16CHSW J2	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1202	PP1 J202	MG3 A17 J3	2-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1203	PP1 MC SW FLTR J203	MG3 16CHSW J11	49 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	
1215	MG3 16CHSW J20	MG2 SRI J20	37 pin Sub-D	CI3	300	<30	80	FT-4	
1216	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
1217	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	SRI Power
1250	MG3 A3 J1	CONNECT TO RUN 257							
1260	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9-Sub D	CI3	300	<30	105	FT-4	Bore Temperature Sensor
1261	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80-#28)	CI2	300	<30	80	VW-1	SRI Splitter to Oper Display

13-2-3 MAGNET MONITORING CABLES

The Magnet Monitoring cables listed in Table 18-10 are delivered with the Magnet.

TABLE 18-10
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8-#22) & Data Cable (4-#22)	CI3	300	<30	80	FT-4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT-4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8-#24)	CI3	300	<30	80	FT-4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8-#22)	CM	300	<30	80	FT-4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6-#24)	CM	300	<30	80	FT-4	Compressor to Compressor I/F
939	MDP UPS	MSM4	6 ft. Power (5-#15)	SO	600	120	85	VW-1	MDP to UPS
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor to Modem
1077	MSM1 J5	OW1 A2 A4 PRT7	80 ft. Cat 5 Ethernet	CI3	300	<30	80	FT-4	Monitor to OW Ethernet
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

13-2-4 GE MAGNET CABLES

The Magnet cables listed in Table 18-11 arrive on site with the magnet and are installed at the time of Magnet delivery.

TABLE 18-11
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12-#22)	CI3	150	60 VDC	80	FT-4	Switch Heaters
606	MS4	MS1 A3 A1	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Emergency Run Down Unit
609	MS7 OR MS8	PDU	Power Cable (5-#16)	SO	300	208Y	105	FT-4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Cold Head Control Cable

13-2-5 FIBER OPTIC CABLES

The Fiber Optic cables listed in Table 18-12 are routed and connected in the Magnet Room and the Equipment Room.

TABLE 18-12
FIBER OPTIC CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
711/ 712	PP1-J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC-II
1045	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1083	MR2 A11 J22	OW1 A15 XMT1	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1179	MR2 A11 J17	MR3 A11 J2	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic cables

13-2-6 EQUIPMENT ROOM AND MAGNET ROOM CABLES AND HOSES CUT TO LENGTH

The Gradient cables and the RF Head and Body cables listed in Table 10-16 will be routed, cut, and terminated at designated cable ends according to instructions found in the CABLE INSTALLATION section of this manual.

TABLE 10-16
EQUIPMENT ROOM AND MAGNET ROOM CABLES & HOSES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
235	CONNECT TO RUN 1259	PP1 J44	Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	1.5T Body Coil Run created from kit
257	PP1 J44	CONNECT TO RUN 1250	Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	1.5T Body Coil Transmit
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
976	MR3 CABI/F J3	TAC $\pm$ X	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Power
977	MR3 CABI/F J4	TAC $\pm$ Y	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Power
978	MR3 CABI/F J5	TAC $\pm$ Z	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Power
979	TAC X+,WB-,ZM-	PP1X+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Filter
980	TAC Y+,WB-,ZM-	PP1Y+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Filter
981	TAC Z+,WB-,ZM-	PP1Z+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Filter
982	PP1 ZZM+,ZM-	MG2 A12 A1 7/8	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Zzm Gradient Output
983	PP1 YZM+,ZM-	MG2 A12 A1 9/10	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Yzm Gradient Output
984	PP1 XZM+,ZM-	MG2 A12 A1 11/12	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Xzm Gradient Output
985	PP1 ZWB+,WB-	MG2 A12 A1 1/2	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin ZWB Gradient Output
986	PP1 YWB+,WB-	MG2 A12 A1 3/4	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin YWB Gradient Output
987	PP1 XWB+,WB-	MG2 A12 A1 5/6	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin XWB Gradient Output
1198	PP1 RFTXSW J151	MG3 PEDI/F J1	Times Microwave LMR-600FR	CATV R	4000 VDC		85	FT-4	RF Head Transmit
1199	PP1 RX TX SW J153	MG3 PEDI/F J2	Times Microwave LMR-600FR	CATV R	4000 VDC		85	FT-4	RF Head Transmit
1238	PP1 J4	CONNECT TO RUN 1249	Times Microwave LMR-600FR	CATV R	6000 VDC		85	FT-4	RF Head Transmit

13-2-7 EQUIPMENT AND OPERATOR ROOM CABLES

The cables listed in Table 10-17 are routed and connected between Penetration Pnael, cabinets in equipment room and Operator Workspace.

TABLE 10-17
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional O2 Monitor cable
611	TAC J22	PP1 SHIM							Resistive Shim P.S. Option
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	RF Door switch
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	PDU Status (ACGD)
974	MR3 PD1 J5	TAC J12	49 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Twin GIP/GP
1056	OW1 A17 PRT6	OW1-A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Laptop Access cable
1059	OW1 A17 PRT2	OW1-A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Lan Hub cable
1060	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	GOCAA Power cord
1061	OW1 A15 KYBD/MS	OW1 A21 J8	2.9 ft. 25 pin Sub-D 2 PS/2s	CI3	300	<30	90	CL-2	Keyboard/Mouse cable
1063	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW-1	Host LCD Display Video cable
1064	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW-1	SCSI Tower Data cable
1065	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host LCD Display Power cable
1066	OW1 A15	OW1 A13	3.9 ft. 25/9 pin Sub-D	CM	300	<30	80	FT-4	Modem I/O cable
1067	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW-1	DASM I/O cable
1068	OW1 A15 Video2	OW1 A3 LCD	9.9 ft. Data (15-#28)	CI3	300	<30	80	VW-1	LCD Display Video cable
1069	OW1 A11 J8	OW1 A3 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	LCD Display Power cable
1070	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	SCSI Tower Power cable
1071	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host Computer Power cable
1074	OW1 A19 J2	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1075	OW1 A19 J4	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1082	OW1 A15 ETH SLOT 2	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT-4	EXCITE Ethernet C'cable
1085	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Intercom
1180	MR3 J7	MR2 A11 J56	26 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1181	MR3 J8	PP1 PHPS J154	65 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1184	MR2 A11 J125	PP1 PHPS J32	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	
1185	MR2 A11 J200	PP1 J200	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1186	MR2 A11 J201	PP1 J201	8-Coax 8W8 Combo D	CATV E	1500	<30	85	FT-4	
1187	MR2 A11 J202	PP1 J202	2-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	

13-2-7 EQUIPMENT AND OPERATOR ROOM CABLES (Continued)TABLE 10-17 (CONT'D)
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1188	MR2 DRV J3	PP1 A1 J77	65 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	16 Channel SW Cntrl Power
1189	MR2 DRV J5	PP1 A17 J78	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1190	MR2 DRV J10	PP1 J203	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1191	MR2 DRV J21	PP1 A11 J92	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1192	MR2 DRV J22	PP1 A11 J93	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1193	MR2 DRV J23	PP1 A11 J94	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1194	MR2 DRV J24	PP1 A11 J95	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1195	MR2 DRV J25	PP1 A11 J96	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1196	PP1 A17 J1	PP1 PHPS J35	2 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	Repeater Board Power
1233	TAC J116 (J5)	PP1 PHPS J155	30 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1249	MR8 J3	CONNECT TO RUN 1238	5 ft. Times Microwave LMR	CATV R	6000 VDC		85	FT-4	RF Extension cable
1255	OW1 A21 J18	MR2 A11 J26	60 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	E-stop
1256	OW1 A20, A7, A8	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub-D	CI3	300	<30	80	VW-1	SCIM, Mouse, Keyboard cable
1259	MR2 RF AMP J4	CONNECT TO RUN 235							

13-2-8 CABLE SPARES (FRU'S)TABLE 10-18
EQUIPMENT & OPERATOR ROOM RUNS

CABLE PART NUMBER	CABLE RUN
46-244710G30	Run 1227 Run 1236
46-244712G30	Run 1188
46-244713G30	Run 1181
46-244714G30	Run 1180 Run 1181 Run 1233
46-258877G30	Run 1189 Run 1190

13-2-9 EXCITE 4KW AND 8KW MNS CABLES (Optional)

TABLE 10-19
EXCITE 4KW AND 8KW MNS RUNS (optional)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1129	CPC J8	MR3 PD1	35 ft. Power (3-#10)	STO	600	208	90	FT-4	4KW CPC Amp Power cable
1130	CPC J3	MR2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	4KW MNS cable
1136	CPC J9	MR2 A11 J4	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	4KW MNS cable
1137	RUN 1158	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	4KW MNS cable
1138	MR6 J12	MR3 PD1	65 ft. Power (5-#8)	SO	600	208Y	75	FT-4	BB RF Amp Power Cable
1139	AMT J3	Mr2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1140	MR1 J123	MR6 J1	30 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	8KW MNS cable
1141	MR1 J102	AMT J5	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1142	MR1 J103	AMT J6	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1143	MR1 J104	AMT J7	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1144	MR1 J105	AMT J8	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1145	MR2 A11 J4	AMT J11	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1146	RUN 1159	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	8KW MNS cable
1147	MG3 A17 J3	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	4KW & 8KW MNS cable
1158	RUN 1137	MR1 CPC J2	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT-4	4KW MNS Extension cable
1248	RUN 1146	MR9 J4	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT-4	8KW MNS Extension cable

13-3 FIBER OPTIC TERMINATION INSTRUCTIONS

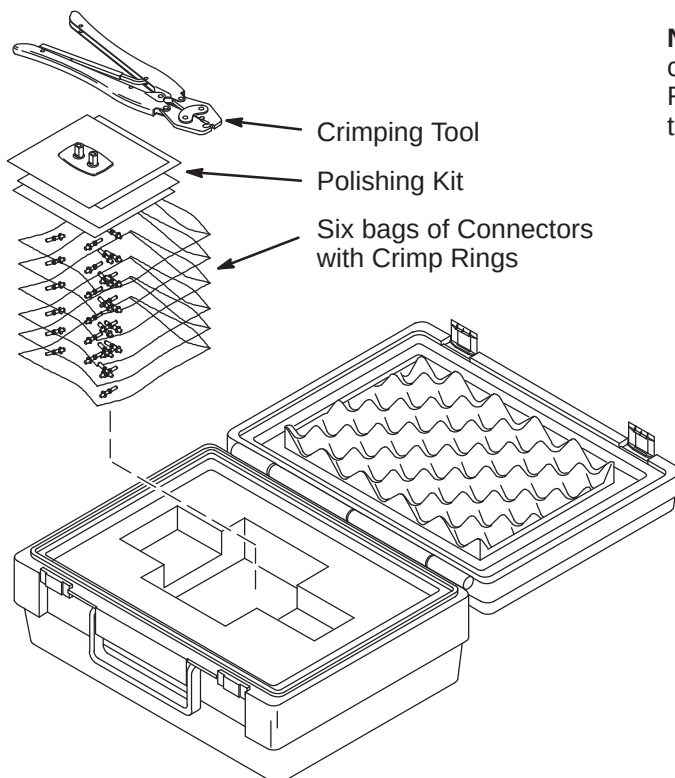
WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

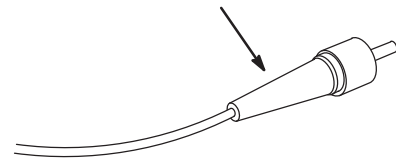
13-3-1 FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.

1. The Fiber Optic Repair Kit, 46-301450G1 shown below, contains all the necessary materials to replace a damaged connector.
2. The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.
3. Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.



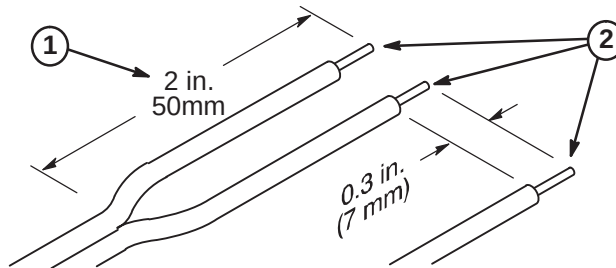
Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



13-3-2 STRIPPING FIBER OPTIC OUTER JACKET

The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

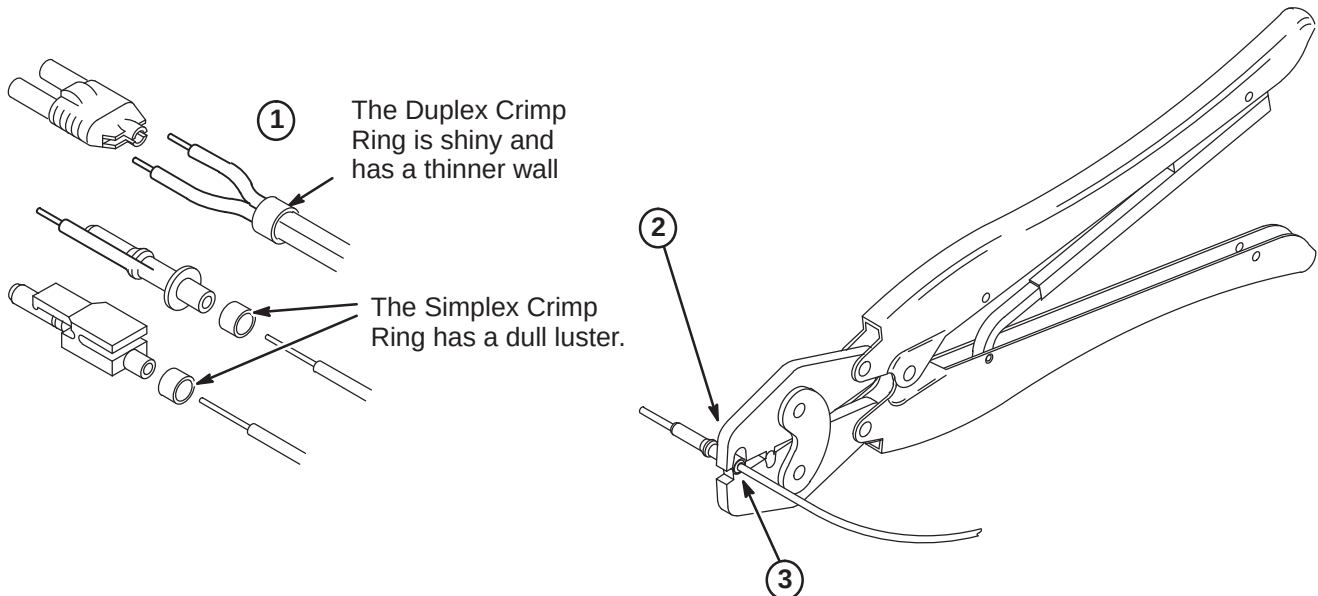
1. For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



13-3-3 CRIMPING CONNECTOR TO FIBER OPTIC CABLE

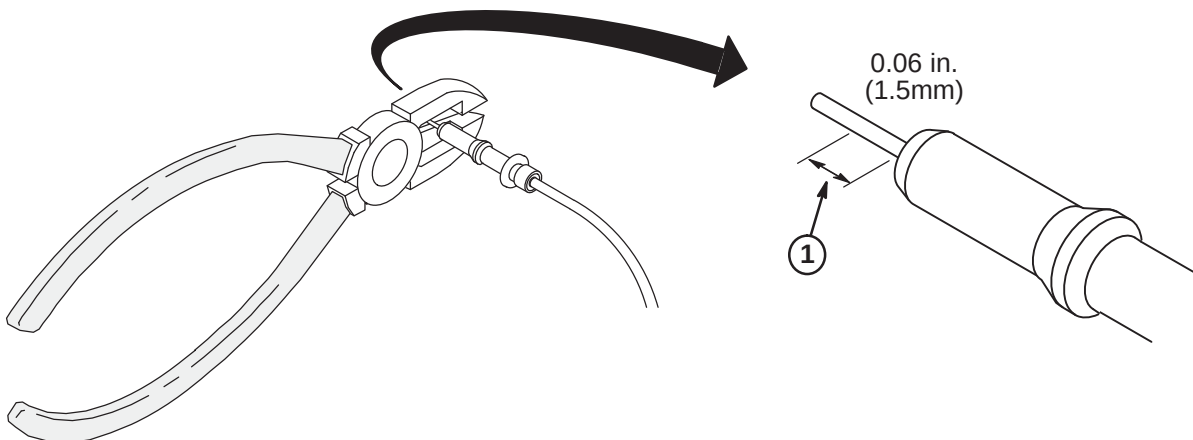
The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

1. Place the Crimp Ring and Connector over the end of the cable.
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.
3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.



13-3-4 TRIMMING EXCESS FIBER FROM CONNECTOR

1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 inch (1-1/2mm).

**13-3-5 POLISHING FIBER OPTIC END**

The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

